

THE SHARP QUANTITATIVE FABER–KRAHN INEQUALITY: TWO PROOFS

ABSTRACT. For every open set $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$ the sharp quantitative Faber–Krahn inequality $\mathcal{A}(\Omega)^2 \leq C_{\text{FK}}(n) \delta_{\text{FK}}(\Omega)$ holds, with $C_{\text{FK}}(n)$ depending only on the dimension. We give two independent proofs of this inequality, written with constants tracked through every step. The first proof follows the Brasco–De Philippis–Velichkov selection-principle route, including the graph-entry / Schauder bootstrap, the BDV nearly-spherical closure, bounded reduction, and the Kohler–Jobin transfer. The second proof is structurally different: an exact level-set deficit identity expresses the Saint–Venant deficit as an integral over the volume-radius foliation of two pointwise non-negative defects; a finite-perimeter metric first-variation in the quotient space $\mathcal{X}$ of sets modulo translation, an oriented L^1 -radial trace lemma at Fusco–Julin oscillation centres, a good/bad level decomposition, and a boundary-layer transfer convert this integral identity into the same global sharp Saint–Venant inequality and, after Kohler–Jobin, into the same final Faber–Krahn estimate with identical dimensional constant. The two proofs are presented in parallel so that their common closing step and the precise place where they diverge are explicit.

1. INTRODUCTION

1.1. The sharp quantitative Faber–Krahn inequality. Let $n \geq 2$ and let $\Omega \subset \mathbb{R}^n$ be an open set with $0 < |\Omega| < \infty$. Write $\lambda_1(\Omega)$ for the first Dirichlet eigenvalue of $-\Delta$ on Ω and B^* for the open ball centred at the origin with $|B^*| = |\Omega|$. The Faber–Krahn inequality [23, 41, 42] asserts

$$(1) \quad \lambda_1(\Omega) \geq \lambda_1(B^*),$$

with equality if and only if Ω is a ball (up to a Lebesgue negligible set). The scale-invariant deficit

$$(2) \quad \delta_{\text{FK}}(\Omega) := \left(\frac{|\Omega|}{\omega_n} \right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq 0$$

vanishes only at balls; the Fraenkel asymmetry

$$(3) \quad \mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^n} \frac{|\Omega \triangle (B^* + x_0)|}{|B^*|} \in [0, 2)$$

measures the distance from Ω to a ball in L^1 . Both proofs in the present paper close the following sharp quantitative statement, in the canonical normalisation fixed in §2.

Theorem (Sharp quantitative Faber–Krahn). *There is a constant $C_{\text{FK}}(n) > 0$, depending only on the ambient dimension $n \geq 2$, such that for every open*

$\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,

$$(4) \quad \mathcal{A}(\Omega)^2 \leq C_{\text{FK}}(n) \delta_{\text{FK}}(\Omega).$$

The exponent 2 on $\mathcal{A}(\Omega)$ is sharp, in the sense that no power greater than 2 on the left-hand side can hold with a constant depending only on n .

The theorem is the main result of Brasco–De Philippis–Velichkov [15] (BDV) and was reproved by Allen–Kriventsov–Neumayer [2] via the Alt–Caffarelli–Friedman monotonicity formula. Our contribution is to give *two independent proofs*, each with full quantitative tracking of constants, written so that they can be read in parallel and compared step-by-step. We refer to these as *Plan 1* and *Plan 2*; both proofs end at the identical theorem statement (4), with the same explicit dimensional constant

$$(5) \quad c_{\text{FK}}(n) = 1/C_{\text{FK}}(n) = \min \left\{ \frac{4L_n c_{\text{SV}}^{\text{glob}}(n)}{(n+2)T_n}, \frac{L_n(2^{2/(n+2)} - 1)}{4} \right\},$$

where $L_n = \lambda_1(B_1)$ and $T_n = T(B_1) = \omega_n/(n(n+2))$ are dimensional (cf. Theorems 1.3 and 1.4 below).

The two proofs differ *only in how the bounded sharp Saint–Venant inequality is obtained*: Plan 1 follows the selection-principle route of [15], supplemented with detailed graph-entry, Schauder interpolation, and nearly-spherical closure; Plan 2 follows the level-set / metric-quotient route inspired by Nicola–Tilli [49, 34] and uses Fusco–Julin oscillation [29] together with a finite-perimeter first-variation identity in a metric quotient space, an oriented radial-trace estimate, and a boundary-layer transfer. From the global sharp Saint–Venant inequality onwards, the two proofs share the same Kohler–Jobin transfer [40, 13] and bounded reduction (BDV Lemma 5.3). The aim is to make both routes explicit, with constants tracked, so that the reader sees clearly what each route requires and what each route delivers.

1.2. History I: Rayleigh’s conjecture, Faber, Krahn. The inequality (1) was conjectured by Lord Rayleigh in his *Theory of Sound* (1894/96) on physical grounds: among homogeneous membranes of given area and uniform tension, the disc produces the lowest fundamental tone. In two dimensions Rayleigh’s conjecture was proved by Faber [23] in 1923 by Steiner-symmetrisation of the Rayleigh quotient: rearranging the trial function $u \in H_0^1(\Omega)$ as the radially decreasing equimeasurable u^* on B^* decreases $\int u^2$ at fixed level distribution and decreases $\int |\nabla u|^2$ by the Pólya–Szegő principle [50], so the Rayleigh quotient does not increase. In 1925 Krahn [41] extended the proof to $n = 2$ (in a slightly different form) and in 1926 [42] to all dimensions $n \geq 3$, again by symmetrisation. The argument is the foundational application of decreasing rearrangement to spectral geometry; modern textbook treatments are given by Henrot [36], Bucur–Buttazzo [18], Bandle [9], Kawohl [37] and Kesavan [38].

The rigidity statement — equality in (1) forces Ω to be a ball — is also classical. In the modern setting of sets of finite perimeter (introduced by De Giorgi [22]), rigidity in the isoperimetric inequality is the key input ([46, Chapter 19]); see also Brothers–Ziemer [17] for the equality case in the Pólya–Szegő rearrangement principle.

1.3. History II: the Saint–Venant analogue. A parallel functional inequality of the same vintage governs the torsional rigidity of an elastic bar. For an open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$, the (variational) torsion function $u_\Omega \in H_0^1(\Omega)$ is the unique minimiser of

$$v \longmapsto \frac{1}{2} \int_{\Omega} |\nabla v|^2 dx - \int_{\Omega} v dx,$$

equivalently $-\Delta u_\Omega = 1$ in Ω with $u_\Omega = 0$ on $\partial\Omega$ in the trace sense, and $u_\Omega \geq 0$ a.e. The torsional rigidity is

$$(6) \quad T(\Omega) := \int_{\Omega} u_\Omega dx = -2 \int_{\Omega} E_\Omega dx,$$

where, following the BDV normalisation of the notation register, the Saint–Venant energy is

$$(7) \quad E(\Omega) := -T(\Omega)/2 = \min \left\{ \frac{1}{2} \int |\nabla v|^2 - \int v : v \in H_0^1(\Omega) \right\} \leq 0.$$

Pólya and Szegő [50] established the Saint–Venant analogue of Faber–Krahn:

$$(8) \quad T(\Omega) \leq T(B^*), \quad E(\Omega) \geq E(B^*),$$

with equality only if Ω is a ball (up to translations and negligible sets). Again the proof is by decreasing rearrangement; we adopt the BDV-normalised deficit

$$(9) \quad \delta_T(\Omega) := E(\Omega) - E(B^*) \geq 0,$$

which vanishes only at balls. Modern accounts of the Saint–Venant inequality and its rearrangement proof are in [37, 9, 38, 36, 50]. In the BDV literature [15, §2], [13, §2], [14, §2], the quantitative form of (8) is the natural Saint–Venant companion of Faber–Krahn, and is the form actually proved by both of our routes before transferring to the eigenvalue side.

1.4. The quantitative stability problem and the sharp exponent.

The qualitative rigidity statement — equality in (1) or (8) forces Ω to be a ball — raises the quantitative question: for Ω *close* to a ball, how does $\delta_{\text{FK}}(\Omega)$ (or $\delta_T(\Omega)$) compare to $\mathcal{A}(\Omega)$?

A volume-preserving ellipsoidal one-parameter family Ω_ε of perturbations of B^* satisfies $\mathcal{A}(\Omega_\varepsilon) \sim \varepsilon$ while $\lambda_1(\Omega_\varepsilon) - \lambda_1(B^*) \sim \varepsilon^2$ and $E(\Omega_\varepsilon) - E(B^*) \sim \varepsilon^2$ as $\varepsilon \rightarrow 0$; see [14, Introduction] and [15, §2]. Hence the best possible quantitative form of Faber–Krahn / Saint–Venant is

$$(10) \quad \mathcal{A}(\Omega)^2 \leq C_{\text{FK}}(n) \delta_{\text{FK}}(\Omega), \quad \mathcal{A}(\Omega)^2 \leq C_{\text{SV}}(n) \delta_T(\Omega);$$

the exponent 2 is sharp. The first quantitative stability results in the eigenvalue case (1) are due to Hansen–Nadirashvili [35], Melas [47], and Bhat-tacharya [12], all with non-sharp exponents. Fusco, Maggi, and Pratelli [32], building on the sharp quantitative isoperimetric inequality of [31], obtained

$$(11) \quad \mathcal{A}(\Omega)^4 \leq C(n) \delta_{\text{FK}}(\Omega) \quad \text{and} \quad \mathcal{A}(\Omega)^4 \leq C(n) \delta_T(\Omega);$$

see [32, Theorems 1.1, 1.4]. The sharp form (10) is the content of [15] for the ball case in any dimension $n \geq 2$; see also the survey of Brasco–De Philippis [14] and the alternative proof of Allen–Kriventsov–Neumayer [2]. The

sharp stability of the second eigenvalue under the natural class constraint is due to Brasco–Pratelli [16].

The Kohler–Jobin inequality [40, 13] bridges the two sides of the problem: any sharp δ_T -stability translates into a sharp δ_{FK} -stability with the same exponent on $\mathcal{A}$. Both of our proofs first close the sharp Saint–Venant inequality (10) and then transfer through Kohler–Jobin. We make this bridge explicit in Sections 1.7 and 1.11 below.

1.5. The selection-principle / BDV route. The BDV selection principle [15] was inspired by the prototype of Cicalese–Leonardi [21] for the isoperimetric inequality; related second-variation selection mechanisms appear in Acerbi–Fusco–Morini [1]. The structure of the argument in [15] is: among open sets $U \subset B_R$ with $|U| = \omega_n$ and prescribed lower bound on $\mathcal{A}$, minimise a penalised functional

$$J_{\Omega_0}(U) := E(U) + f_{\hat{\eta}}(|U|) + \sqrt{\varepsilon^2 + \tau^2(\alpha_{\text{BDV}}(U) - \varepsilon)^2}$$

in which α_{BDV} is the BDV smooth asymmetry of [15, Def. 4.1] and $f_{\hat{\eta}}$ is a volume penalty. The selected minimiser U_* inherits a regularity package ([15, Lem. 4.6, 4.9, 4.12]: density estimates, Lipschitz non-degeneracy, smooth Bernoulli coefficient) and via the BDV flatness theorem ([15, Thm 4.18], in the spirit of Alt–Caffarelli [3]) enters the $C^{1,\gamma}$ *graph regime*: ∂U_* is the graph of a function g over a sphere. A Kinderlehrer–Nirenberg hodograph step [39], followed by a Lieberman oblique Schauder bootstrap [43, 44, 33, 11], brings g into C^{2,γ_0} with controlled norm. The BDV nearly-spherical theorem ([15, Thm 3.3], the modern descendant of the convex-stability estimate of Fuglede [27]) then closes the chain:

$$E(U_*) - E(B_1) \geq \frac{1}{32n^2} \|g\|_{H^{1/2}(\partial B_1)}^2$$

on nearly-spherical sets. After unwinding the asymmetry chain ([15, Lem. 4.2]) and combining with the far-regime $\mathcal{A}^4$ -estimate ([15, Lem. 5.1, 5.2], itself based on the FMP quantitative isoperimetry of [31]), one obtains a *bounded sharp Saint–Venant* statement

$$(12) \quad \delta_T(\Omega) \geq c_{\text{SV}}(n, R) \mathcal{A}(\Omega)^2, \quad \Omega \subset B_R, \quad |\Omega| = \omega_n.$$

The dependence on the confining radius R is then removed by the constructive BDV truncation lemma [15, Lem. 5.3], applied to a truncate $\tilde{\Omega} \subset B_{R_*(n)}$ whose deficit and asymmetry are controlled by those of Ω . The output is a global sharp Saint–Venant inequality $\delta_T(\Omega) \geq c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2$, which feeds into Kohler–Jobin.

This is the load-bearing Plan 1 route. The argument is conditional only in one respect: the BDV nearly-spherical theorem [15, Thm 3.3] is invoked as a black-box quantitative identity on the $H^{1/2}$ -norm of g , and the regularity bootstrap that prepares U_* for that theorem requires the full BDV regularity package. None of these inputs is conditional in the sense of being unproved in the literature; they are simply non-trivial.

Remark 1.1 (Bernoulli spectral alternative). A conceptually different conditional approach to the bounded sharp Saint–Venant inequality (12) would invoke a spectral / Bernoulli linearisation of the BDV selected minimiser around the ball, using the eigenvalues of the Bernoulli operator on ∂B_1 . This

route is sketched in some of the local source notes (cf. `Plan 1/route-A-summary.md`) but is *not* the main Plan 1 proof of this manuscript: the load-bearing route is the BDV selection principle / nearly-spherical / FMP far-regime chain above. The Bernoulli alternative is mentioned only as a remark and its conditional status (it requires a spectral gap statement we do not prove here) is made explicit.

1.6. Quantitative isoperimetry background. The Plan 1 far-regime estimate [15, Lem. 5.1, 5.2] and several intermediate steps in Plan 2 rely on the quantitative isoperimetric toolbox developed in the last twenty years. The cornerstone is the sharp quantitative isoperimetric inequality of Fusco–Maggi–Pratelli [31]: there is $C(n) > 0$ such that for every set $E \subset \mathbb{R}^n$ of finite perimeter with $|E| = |B^*|$,

$$(13) \quad \mathcal{A}(E)^2 \leq C(n) \frac{P(E) - P(B^*)}{P(B^*)},$$

where P is the De Giorgi perimeter. The exponent 2 is sharp. An alternative mass-transport proof of (13) is due to Figalli–Maggi–Pratelli [26]; a Sobolev-type strengthening is [30]; an analogous sharp Sobolev / Wulff stability appears in [20, 48]; see the surveys of Fusco [28] and Maggi [45] for context.

The strong form of Fusco–Julin [29] adds an oriented oscillation index: for every $E \subset \mathbb{R}^n$ of finite perimeter with $|E| = |B^*|$,

$$(14) \quad \mathcal{A}(E)^2 + \beta(E)^2 \leq C(n) \frac{P(E) - P(B^*)}{P(B^*)}, \quad \beta(E) := \inf_{z \in \mathbb{R}^n} \left(\int_{\partial^* E} \left| \nu_E(x) - \frac{x-z}{|x-z|} \right|^2 d\mathcal{H}^{n-1} \right)^{1/2}.$$

Both the asymmetry and the normal-oscillation defect are controlled by the isoperimetric defect, with a sharp choice of centre. This is the load-bearing isoperimetric input for Plan 2: it is (14), not (13), that supplies the levelwise oscillation centres along the foliation $\rho \mapsto E_\rho = \{u_\Omega > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$.

Two technical inputs that come with the quantitative isoperimetric literature will be used throughout. First, the Figalli–Maggi Sobolev–Sard theorem [25, App. A, Thm A.1]: for the torsion function $u_\Omega \in W_{\text{loc}}^{2,p}$ with $-\Delta u = 1$, $\mathcal{H}^{n-1}(\partial^* E_t \cap \{|\nabla u| = 0\}) = 0$ for $\mathcal{L}^1$ -a.e. level t ; equivalently, the normal velocity $V_\rho(x) = -t'(\rho)/|\nabla u(x)|$ is well-defined $\mathcal{H}^{n-1}$ -a.e. on $\partial^* E_\rho$ for $\mathcal{L}^1$ -a.e. ρ . Second, the finite-perimeter / BV toolbox of Maggi’s book [46] (coarea, divergence theorem, slicing, structure of $\partial^* E$, rigidity in the isoperimetric inequality) and Ambrosio–Fusco–Pallara [4] (Reshetnyak lower semicontinuity, BV chain rule, Vol’pert slicing). The measurable selection statement we use for the levelwise oscillation centre on good levels is Federer [24, §2.13].

1.7. The Kohler–Jobin bridge. The Faber–Krahn and Saint–Venant inequalities are not independent. The Kohler–Jobin inequality [40] states that

$$(15) \quad T(\Omega) \lambda_1(\Omega)^{(n+2)/2} \geq T(B^*) \lambda_1(B^*)^{(n+2)/2}$$

for every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$, with equality if and only if Ω is a ball. The original proof [40] is by a rearrangement of the eigenfunction along the level sets of the torsion function. The modern statement used

here is Brasco's [13, Thm 3.3], which packages the rearrangement as a single comparison of the functionals $T \cdot \lambda_1^{(n+2)/2}$ across all open sets of finite volume. Inverting the dependence at fixed volume, one obtains the pointwise transfer

$$(16) \quad \lambda_1(\Omega) \geq \lambda_1(B^*) \left(\frac{T(B^*)}{T(\Omega)} \right)^{2/(n+2)},$$

and after the elementary inequality $(1-s)^{-p} - 1 \geq ps$ for $s \in [0, 1]$, $p \geq 0$, the Saint-Venant deficit $\delta_T(\Omega)$ becomes a multiple of $\delta_{\text{FK}}(\Omega)$ on the small-deficit regime, with explicit dimensional multiplier $2L_n/((n+2)T_n)$ (cf. [13, §3]). On the large-deficit regime $T(\Omega) \leq T(B^*)/2$, a direct evaluation of (16) provides a one-sided bound of size $L_n(2^{2/(n+2)} - 1)/4$ on $\delta_{\text{FK}}(\Omega)/\mathcal{A}(\Omega)^2$. Taking the minimum of the two multipliers yields the explicit constant (5) for both proofs.

The Kohler-Jobin transfer is therefore the common closing step in the manuscript: *any* sharp δ_T -stability gives a sharp δ_{FK} -stability, and the constants on the two sides are linked by an explicit dimensional formula. Both Plan 1 and Plan 2 produce the same input $\delta_T(\Omega) \geq c_{\text{SV}}^{\text{glob}}(n)\mathcal{A}(\Omega)^2$ to feed into (15), and so the final constants $c_{\text{FK}}(n)$ on both sides of (5) agree verbatim.

1.8. Nicola-Tilli / rearrangement / level-set viewpoint. A conceptually different mechanism underlies Plan 2. The starting point is the observation, made precise by Nicola-Tilli [49] in the context of the short-time Fourier transform Faber-Krahn inequality and developed quantitatively in the stability companion of Gómez-Guerra-Ramos-Tilli [34], that a Faber-Krahn-type deficit admits an *exact integral identity* over the level sets of the relevant function. In our setting, the level sets of the torsion function u_Ω are parametrised by volume, $E_\rho := \{u_\Omega > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$, $\rho \in [0, 1]$, and the Saint-Venant deficit decomposes as

$$(17) \quad \delta_T(\Omega) \simeq \int_{[\rho_*, 1]} (D_H(t(\rho)) + D_I(t(\rho))) w(\rho) d\rho,$$

where $D_H(t) = \alpha(t)\beta(t) - P(E_\rho)^2 \geq 0$ is a Cauchy-Schwarz defect along the level set, $D_I(t) = P(E_\rho)^2 - c_n m(t)^{2-2/n} \geq 0$ is the squared-perimeter isoperimetric defect, $\alpha(t) = \int_{\partial^* E_t} |\nabla u|^{-1} d\mathcal{H}^{n-1}$, $\beta(t) = \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1}$, and $w(\rho)$ is an explicit non-negative weight. The identity is proved in §4.1.11 below; it builds on the Pólya-Szegő / Talenti rearrangement identities [50, 51, 37, 9, 10, 17] and uses the BV chain rule of Bénéilan-Brezis-Crandall [10] in the form needed by the level-set foliation.

The identity (17) is fully integral: it expresses the global deficit as an integral over the foliation of two pointwise non-negative quantities, D_H controlling the oscillation of $|\nabla u|$ on the level set, D_I controlling the isoperimetric oscillation of the level set itself. This is the opposite philosophy from Plan 1: instead of selecting a single near-extremal set and analysing it as a graph over a sphere, one analyses *every* level ρ of the foliation, classifies levels as good or bad according to their isoperimetric defect, and transfers the integrated defect bound to the metric quotient $\mathcal{X} = \mathfrak{M}/\mathbb{R}^n$ in which sets are identified with their translates.

1.9. What Plan 1 proves and why it is included. Plan 1 follows the BDV route [15] step-by-step, with the level of detail demanded by §2 of MANUSCRIPT_ASSEMBLY_AGENTS.md. The skeleton is

- selection principle (§3.1, Agent 5)
- graph entry $C^{1,\gamma}$ (§3.2, Agent 6)
- Schauder/interpolation C^{2,γ_0} (§3.2, Agent 6)
- BDV nearly-spherical closure (§3.3, Agent 7)
- far-regime $\mathcal{A}^4$ glue (§3.3, Agent 7)
- bounded sharp Saint–Venant (§3.3, Agent 7)
- bounded reduction (§3.4, Agent 8)
- Kohler–Jobin transfer (§3.4, Agent 8)
- sharp quantitative Faber–Krahn (Theorem 1.3).

The bounded sharp Saint–Venant estimate

$$\delta_T(\Omega) \geq c_{\text{SV}}(n, R) \mathcal{A}(\Omega)^2, \quad \Omega \subset B_R, |\Omega| = \omega_n$$

is the substance of the BDV argument; bounded reduction is the constructive truncation of [15, Lem. 5.3]; Kohler–Jobin is [13, Thm 3.3].

The reasons for including Plan 1 in this manuscript are threefold. First, it is the canonical proof: any honest discussion of the sharp quantitative Faber–Krahn inequality must reproduce the BDV chain, because the structure of the rest of the literature ([14, 2, 16, 28, 45]) is organised around it. Second, the constant $c_{\text{SV}}(n, R)$ produced by Plan 1 is fully explicit at every step (selection thresholds, regularity radii, nearly-spherical constant $c_*(n) = 1/(32n^2 C_3(n))$, glue constants), and the chain shows exactly how dimensional and confinement dependences arise. Third, both proofs converge at the same global Saint–Venant statement and the same Kohler–Jobin transfer; presenting Plan 1 in full makes the common closing step explicit and makes the comparison with Plan 2 honest.

1.10. What Plan 2 proves and why it is structurally different. Plan 2 produces *the same* global sharp Saint–Venant inequality $\delta_T(\Omega) \geq c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2$ along a structurally different route. The mechanism is a five-step chain in the metric quotient $\mathcal{X}$:

- Level-set deficit identity (§4.1, Agent 9)
- Metric framework / first variation (§4.2, Agent 10)
- Fusco–Julin centres and oriented radial trace (§4.3, Agent 11)
- Integrated action / weighted metric trace (§4.4, Agent 12)
- Boundary-layer transfer + global assembly (§4.5, Agent 13).

Each ingredient is necessary, and none is conditional. We summarise the role of each.

(1) Exact level-set deficit identity. Theorem 4.13 (Agent 9, §4.1.11) records the identity (17) as a finite-perimeter equality, on the reduced boundary $\partial^* E_\rho$, with $\alpha(t)$, $\beta(t)$, $D_H(t)$, $D_I(t)$, and $w(\rho)$ defined precisely in the notation register. The proof uses the BV coarea theorem ([46, Thm 13.1], [4,

Thm 3.40]), a weak divergence-flux identity for the gradient of the torsion function truncated by a smooth cut-off of the radial vector field $x/|x|$, and the BV chain rule for monotone absolutely continuous inverses ([10, Thm 0.4]; see also [4, Thm 3.99]). The identity isolates two non-negative defects at every level: D_H (a Cauchy–Schwarz defect, controlling oscillation of $|\nabla u|$ on $\partial^* E_t$) and D_I (a squared-perimeter isoperimetric defect, controlling oscillation of the level set itself).

(2) Metric framework and first variation. §4.2.10 (Agent 10) embeds the family $F_\rho = \rho^{-1}E_\rho$ in the quotient metric space $(\mathcal{X}, d_{\mathcal{X}}) = (\{\mathbf{1}_A\}/\mathbb{R}^n, d_{L^1}/\sim)$ of indicator functions modulo translation, and proves absolute continuity of $\rho \mapsto [F_\rho]$ with metric derivative in the Ambrosio–Gigli–Savaré sense [6, Def. 1.1.1, Thm 1.1.2]. The smooth first variation is computed under a smooth flow (with normal velocity $V_\rho = -t'(\rho)/|\nabla u|$ — well-defined $\mathcal{H}^{n-1}$ -a.e. on $\partial^* E_\rho$ for a.e. ρ by [25, Thm A.1]), then passed to the finite-perimeter setting through Reshetnyak lower semicontinuity [4, Thm 2.38]. The output is an upper bound on the metric derivative $|\dot{F}_\rho|_{\mathcal{X}}$ by an explicit Lebesgue norm of V_ρ on $\partial^* E_\rho$.

(3) Fusco–Julin centres and oriented radial trace. §4.3.4 (Agent 11) chooses, on every *good* isoperimetric-defect level (i.e., where $D_I(t(\rho))$ is below a threshold), a Borel measurable oscillation centre $z_\rho \in \mathbb{R}^n$ extracted from the Fusco–Julin inequality (14) [29], using the Federer measurable selection principle [24, §2.13]. The asymmetry and normal-oscillation defects of E_ρ at centre z_ρ are then both controlled by $D_I(t(\rho))^{1/2}$ and $D_I(t(\rho))$ respectively. The load-bearing lemma is the *oriented L^1 -radial trace estimate*

$$(18) \quad T_\rho = \int_{\partial^* E_\rho} \left| \frac{|x-z_\rho|}{\rho} - 1 \right| d\mathcal{H}^{n-1} \leq C \left(|E_\rho \triangle B_\rho(z_\rho)| + \int_{\partial^* E_\rho} \left| \nu_\rho - \frac{x-z_\rho}{|x-z_\rho|} \right|^2 d\mathcal{H}^{n-1} \right).$$

The proof of (18) is a finite-perimeter divergence identity applied to the vector field $X(x) = \left| \frac{|x-z_\rho|}{\rho} - 1 \right| \frac{x-z_\rho}{|x-z_\rho|}$, truncated near the centre to handle the L^1_{loc} singularity at z_ρ (Maggi [46, Thm 16.3]). The Fusco–Julin inequality controls both terms on the right of (18) by powers of D_I , and the level-set deficit identity converts the integrated trace into a metric kinetic estimate.

(4) Integrated action and weighted metric trace. §4.4.7 (Agent 12) decomposes the foliation into good and bad isoperimetric-defect levels, controls the bad set by Markov’s inequality applied to D_I , and reassembles

$$\int_{[\rho_*, 1]} (d_{\mathcal{X}}([F_\rho], [B_1])^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) \mu(d\rho) \leq C(n, R, \rho_*) \delta_T(\Omega),$$

where $\mu(d\rho) = -t'(\rho) d\rho$. The abstract weighted metric trace lemma (an arc-length argument in $(\mathcal{X}, d_{\mathcal{X}})$ with respect to the measure μ) then produces a radius $\hat{\rho} \in [\rho_*, 1]$ such that $d_{\mathcal{X}}([F_{\hat{\rho}}], [B_1])^2 \leq C_* \delta_T(\Omega)$ and $\hat{\rho}$ is close to the heuristic value $\rho_\delta = 1 - \kappa \sqrt{\delta_T(\Omega)}$. The Talenti bad-set Lebesgue estimate ([51], used in combination with the volume-radius parametrisation, [38, Thm 1.3.2]) is what converts weighted kinetic energy on the bad- $-t'$ set into Lebesgue kinetic energy on $[\rho_*, 1]$ without losing the δ_T rate.

(5) Boundary-layer transfer. §4.5.6 (Agent 13) returns from $F_{\hat{\rho}}$ to Ω . The removed boundary layer $\Omega \setminus E_{\hat{\rho}}$ has measure $O(\sqrt{\delta_T(\Omega)})$, and the transfer

of asymmetry from $E_{\widehat{\rho}}$ to Ω squares this $\sqrt{\delta_T}$ without producing a rate loss. The result is the bounded sharp Saint-Venant inequality (12) in the Plan 2 form. Bounded reduction ([15, Lem. 5.3], imported through Plan 1's **BoundedReduction**) removes the R -dependence, and Kohler-Jobin [13] closes the chain.

Remark 1.2 (Auxiliary centroid / space-time route). A separate centroid bound and a related space-time Π -identity (§4.6, Agent 14) compute, for the bulk centroid $\bar{z}_\rho$, the signed radial moment $\Pi(E_\rho, \bar{z}_\rho)$ and a Fubini-density identity relating $\Pi(E_\rho, \bar{z}_\rho)$ across ρ . These quantities are auxiliary diagnostics: the repaired Plan 2 chain above uses only the *integrated* forms of D_H and D_I through the identity (17) and the oriented radial trace of (18), not pointwise $\Pi(E_\rho, \bar{z}_\rho)$ -control. The Π -route is presented in the manuscript for the record and would become load-bearing only if a separate integrated Π -control estimate were proved (which we do not). Up to that hypothetical upgrade, the centroid/ Π -route is not used in any of the load-bearing steps of Plan 2.

The reasons for including Plan 2 are: (i) it is structurally different from Plan 1, in that it does not select a single near-extremal set and does not use a graph parametrisation; (ii) it exposes the deficit as an integral over the level-set foliation, so that the role of isoperimetric stability (14) is direct rather than indirect; (iii) it uses only finite-perimeter regularity throughout, without a smoothness or graph assumption on $\partial\Omega$; (iv) the oriented L^1 -radial trace lemma (18) is, to our knowledge, the first place in the Faber-Krahn / Saint-Venant quantitative literature where Fusco-Julian oscillation is used in this form to transfer levelwise control to the metric quotient $\mathcal{X}$.

1.11. Precise statement of the two final theorems. Both proofs in this manuscript close the same theorem, in the same normalisation, with the same explicit dimensional constant (5). We state both forms here for the record; Plan 1 produces the proof in §3 and Plan 2 in §4.

Theorem 1.3 (Sharp Faber-Krahn via Kohler-Jobin; Plan 1, upper form). *There is a dimensional constant $C_{\text{FK}}(n) > 0$ such that, for every open set $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,*

$$(19) \quad \mathcal{A}(\Omega)^2 \leq C_{\text{FK}}(n) \delta_{\text{FK}}(\Omega),$$

where

$$C_{\text{FK}}(n) := \frac{1}{c_{\text{FK}}(n)} = \max \left\{ \frac{(n+2)T_n}{4L_n c_{\text{SV}}^{\text{glob}}(n)}, \frac{4}{L_n (2^{2/(n+2)} - 1)} \right\},$$

with $L_n = \lambda_1(B_1)$, $T_n = T(B_1) = \omega_n/(n(n+2))$, and $c_{\text{SV}}^{\text{glob}}(n)$ the global sharp Saint-Venant constant defined in (185) of §3. The deficit $\delta_{\text{FK}}(\Omega)$ is as in (2) and $\mathcal{A}(\Omega)$ is the Fraenkel asymmetry (3). Equality cases: $\delta_{\text{FK}}(\Omega) = 0$ iff Ω is a ball (up to translation and a Lebesgue negligible set); in that case $\mathcal{A}(\Omega) = 0$.

Theorem 1.4 (Sharp quantitative Faber-Krahn; Plan 2 final form). *Let $n \geq 2$. There exists a constant $c_{\text{FK}}(n) > 0$, depending only on n , given*

by the explicit formula (5) above, such that for every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,

$$(20) \quad \delta_{\text{FK}}(\Omega) = \left(\frac{|\Omega|}{\omega_n} \right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2.$$

The exponent 2 is sharp.

The two statements are identical up to inversion of the constant ($C_{\text{FK}}(n) = 1/c_{\text{FK}}(n)$). The proofs end at the same place because they share the bounded reduction [15, Lem. 5.3] and the Kohler–Jobin transfer [13, Thm 3.3]. The constants are identical because both routes feed the same global Saint–Venant constant $c_{\text{SV}}^{\text{glob}}(n)$ into the Kohler–Jobin transfer, and the dimensional factors L_n, T_n, ω_n are universal.

1.12. Related work and contributions. The sharp inequality (4) was first proved by Brasco–De Philippis–Velichkov [15] in 2015. A companion proof, structurally distinct in the use of the Alt–Caffarelli–Friedman monotonicity formula, was obtained by Allen–Kriventsov–Neumayer [2]. The sharp stability of the second Dirichlet eigenvalue is the result of Brasco–Pratelli [16]. Pre-BDV non-sharp results include Hansen–Nadirashvili [35], Melas [47], Bhattacharya [12], and the $\mathcal{A}^4$ -bound of [32]. A modern survey of the quantitative Faber–Krahn / Saint–Venant literature is in Brasco–De Philippis [14]; for the broader quantitative isoperimetric viewpoint see Fusco [28] and Maggi [45].

On the isoperimetric side, the sharp quantitative inequality [31] admits the mass-transport proof of [26], the BV-Sobolev strengthening of [30], the sharp Sobolev companion of [20], and the Wulff strengthening of [48]. The strong form of Fusco–Julin [29] is what makes Plan 2 work: the oscillation index controls a quadratic centre-dependent average of $|\nu - e_r|^2$ with the same exponent as the asymmetry. The selection-principle mechanism that underlies Plan 1 originates in Cicalese–Leonardi [21]; the closely related second-variation selection of [1] is in the spirit of the BDV penalised functional but is not used directly here.

The level-set / convexity viewpoint behind Plan 2 is inspired by Nicola–Tilli [49] and its quantitative companion Gómez–Guerra–Ramos–Tilli [34] in the short-time Fourier transform setting. Our use of the metric quotient $\mathcal{X} = \mathfrak{M}/\mathbb{R}^n$ and the Ambrosio–Gigli–Savaré metric derivative [6] is, to our knowledge, new in the Saint–Venant context; the oriented L^1 -radial trace identity (18) and its proof by truncation of the vector field $x \mapsto \|x - z\|/\rho - 1(x - z)/\|x - z\|$ on the reduced boundary, using [46, Thm 16.3], are the technical core that allows Fusco–Julin oscillation to be converted into integrated metric data along the foliation.

Finally, the Kohler–Jobin inequality [40, 13] is the device that bridges Saint–Venant stability and Faber–Krahn stability; it is what permits both of our proofs to share the *same* closing step and the *same* final constant. We present this bridge in full in §3.4.3 and reuse it verbatim in §4.5.6.

Contributions of the present paper. We summarise the contributions of the manuscript as follows:

- (1) A fully detailed exposition of the BDV selection-principle / graph-entry / Schauder / nearly-spherical / truncation chain [15, 13], with

- explicit dimensional constants at every step (Plan 1, §3). The level of detail follows the building blocks `Plan 1/quantitative-selection-principle.tex`, `Plan 1/graph-entry-closure.tex`, `Plan 1/route-A-summary.tex`, `Plan 1/faber-krahn-transfer.tex`.
- (2) A structurally different proof of the same sharp inequality based on an exact level-set deficit identity, a metric first-variation in the quotient space $(\mathcal{X}, d_{\mathcal{X}})$ of sets modulo translation, an oriented L^1 -radial trace lemma anchored at Fusco–Julin oscillation centres, a weighted metric trace, and a boundary-layer transfer. This is Plan 2, §4, and is new in the Saint–Venant / Faber–Krahn context. The five major steps and their building blocks `Plan 2/WIP/WIP_LevelSetIdentity.tex`, `Plan 2/WIP/WIP_MetricFramework.tex`, `Plan 2/WIP/WIP_WeightedMetricTrace.tex`, `Plan 2/WIP/WIP_BoundaryLayerTransfer.tex`, and `Plan 2/WIP/WIP_GlobalAssembly.tex` are integrated into the manuscript with detail substantially greater than in the source blocks.
 - (3) A side-by-side bookkeeping of the constants. The bounded Saint–Venant constants $c_{\text{SV}}(n, R)$ from each route are different intermediate quantities; the global constant $c_{\text{SV}}^{\text{glob}}(n)$ that enters Kohler–Jobin is, in both routes, given by the same expression ((5)). The final constant $c_{\text{FK}}(n)$ on (4) is therefore identical for the two routes, as recorded by Remark 4.78 of §4.
 - (4) A complete reference audit (`Manuscript/04_references_audit.md`) listing every external citation with exact bibliographic data, the theorem/proposition/page actually used, and the manuscript location at which it enters, together with a notation register (`Manuscript/00_notation_register.md`) fixing the canonical symbols. No vague phrases such as ‘well-known’ or ‘standard’ appear in a load-bearing step without a precise reference.
 - (5) An internal audit pass on each of Plan 1 and Plan 2 (`Manuscript/audit_plan1.md`, `Manuscript/audit_plan2.md`), recording that no conditional Bernoulli spectral input is used in the Plan 1 proof and that no centroid / space-time Π -control is used in the load-bearing Plan 2 chain.

Organisation. §2 (Preliminaries) collects the finite-perimeter, torsion, rearrangement, and Fusco–Julin facts used in both routes. §3 (Plan 1) presents the BDV selection-principle proof from selection through to the final theorem. §4 (Plan 2) presents the level-set / metric route through to the same final theorem. The Kohler–Jobin transfer and the bounded reduction, which are the closing steps of both routes, are written out fully in §3.4.3 and re-used in §4.5.6. §4.6 (References) is the bibliography.

Conventions used throughout the introduction: the ambient dimension is $n \geq 2$, balls $B_r(z) \subset \mathbb{R}^n$ are open Euclidean balls, B^* is the open ball centred at the origin with $|B^*| = |\Omega|$, $\omega_n = |B_1|$, $L_n = \lambda_1(B_1)$, $T_n = T(B_1)$, and all constants $C, c, c_{\text{SV}}, c_{\text{FK}}$ are positive and depend on at most (n, R, ρ_*) at every step where they appear; dependences are declared on first use. The notation register is `Manuscript/00_notation_register.md`.

2. PRELIMINARIES

This section collects the analytic and geometric measure theory prerequisites used by both proofs. §2.1 covers finite-perimeter sets, the BV coarea theorem, divergence on reduced boundaries, truncation of singular vector fields, Reshetnyak lower semicontinuity, and the measurable selection of levelwise centres. §2.2 collects the torsion potential, the Saint-Venant and Faber-Krahn deficits, the Fraenkel asymmetry, the Talenti pointwise comparison, the volume-radius foliation $E_\rho = \{u > t(\rho)\}$, the profile-gap and bad-set estimates used by Plan 2, and the Kohler-Jobin inequality used in the final transfer. §2.3 collects the Fusco-Maggi-Pratelli and Fusco-Julin quantitative isoperimetric inputs and the scaled form on $|E| = \omega_n \rho^n$.

2.1. Preliminaries: geometric measure theory. This section fixes the finite-perimeter, slicing, and divergence-theorem tools used throughout the manuscript. All notation matches §2 (see also `00_notation_register.md`). The ambient dimension is denoted $n \geq 2$. We write $|A| = \mathcal{L}^n(A)$ for the Lebesgue measure of a Borel set $A \subset \mathbb{R}^n$, $\mathcal{H}^k$ for the k -dimensional Hausdorff measure, and we identify two sets that differ by an $\mathcal{L}^n$ -null set.

2.1.1. Finite-perimeter sets and reduced boundary.

Definition 2.1 (Finite perimeter and reduced boundary). A Lebesgue-measurable set $E \subset \mathbb{R}^n$ has *finite perimeter* in an open set $U \subset \mathbb{R}^n$ if the indicator $\mathbf{1}_E$ belongs to $BV(U)$, i.e. if

$$P(E; U) := \sup \left\{ \int_E \operatorname{div} \varphi \, dx : \varphi \in C_c^1(U; \mathbb{R}^n), \|\varphi\|_\infty \leq 1 \right\} < \infty.$$

When $U = \mathbb{R}^n$ we write $P(E) := P(E; \mathbb{R}^n)$. The *De Giorgi reduced boundary* $\partial^* E$ is the set of points $x \in \mathbb{R}^n$ at which the limit

$$\nu_E(x) := \lim_{r \downarrow 0} \frac{D\mathbf{1}_E(B_r(x))}{|D\mathbf{1}_E|(B_r(x))}$$

exists in the unit sphere $\mathbb{S}^{n-1}$; $\nu_E(x)$ is the *measure-theoretic outer unit normal* of E at x . By De Giorgi's structure theorem,

$$P(E; B) = \mathcal{H}^{n-1}(\partial^* E \cap B) \quad \text{for every Borel } B \subset \mathbb{R}^n,$$

and $\partial^* E$ is countably $(n-1)$ -rectifiable.

For the canonical reference see [46, Def. 12.1, Thm. 15.9]; equivalently [4, Def. 3.1, Thm. 3.59]. Throughout the manuscript $P(E)$ always denotes the De Giorgi perimeter and ν_E (or ν_ρ when $E = E_\rho$) the measure-theoretic outer normal, consistently with §7 of the notation register.

Proposition 2.2 (L^1_{loc} lower semicontinuity). *Let $(E_j)_{j \geq 1}$ be measurable subsets of $\mathbb{R}^n$ with $\mathbf{1}_{E_j} \rightarrow \mathbf{1}_E$ in $L^1_{\text{loc}}(\mathbb{R}^n)$ as $j \rightarrow \infty$. Then for every open $U \subset \mathbb{R}^n$,*

$$P(E; U) \leq \liminf_{j \rightarrow \infty} P(E_j; U).$$

In particular, if $\sup_j P(E_j) < \infty$ and $|E_j|$ is uniformly bounded, then (E_j) has a subsequence converging in L^1_{loc} to a finite-perimeter set E with $P(E) \leq \liminf P(E_j)$.

Reference. Lower semicontinuity is [46, Prop. 12.15] (equivalently [4, Thm. 3.7]); the compactness in L^1_{loc} is [46, Thm. 12.26]. $\square$

2.1.2. Coarea, BV coarea, and Sobolev–Sard for the torsion function.

Theorem 2.3 (BV coarea formula). *Let $v \in BV(U)$ on an open set $U \subset \mathbb{R}^n$. Then $\{v > t\}$ has finite perimeter in U for $\mathcal{L}^1$ -a.e. $t \in \mathbb{R}$, and for every Borel function $\varphi : U \rightarrow [0, \infty]$*

$$(21) \quad \int_U \varphi d|Dv| = \int_{\mathbb{R}} \left(\int_{\partial^* \{v > t\} \cap U} \varphi d\mathcal{H}^{n-1} \right) dt.$$

If $v \in W^{1,1}(U)$ (or $v \in H_0^1(U)$ extended by zero outside U), then $d|Dv| = |\nabla v| d\mathcal{L}^n$ and (21) reduces to the Federer–Fleming coarea formula

$$(22) \quad \int_U \varphi |\nabla v| dx = \int_{\mathbb{R}} \left(\int_{\partial^* \{v > t\} \cap U} \varphi d\mathcal{H}^{n-1} \right) dt.$$

Reference. This is [46, Thm. 13.1 and Thm. 18.1]; equivalently [4, Thm. 3.40 and (3.40)(b)]. The vanishing of the contribution of the critical set $\{\nabla v = 0\}$ to the right-hand side of (21) is [4, Thm. 3.40(b)]. $\square$

Corollary 2.4 (Coarea for the torsion function). *Let $\Omega \subset \mathbb{R}^n$ be open with $0 < |\Omega| < \infty$ and let $u = u_\Omega$ be the torsion function (§3 of the notation register). Then $u \in H_0^1(\Omega) \cap L^\infty(\Omega)$, and after extending u by zero on $\mathbb{R}^n \setminus \Omega$ we have $u \in BV_{\text{loc}}(\mathbb{R}^n) \cap L^\infty(\mathbb{R}^n)$. For every Borel $\varphi : \mathbb{R}^n \rightarrow [0, \infty]$,*

$$(23) \quad \int_{\mathbb{R}^n} \varphi |\nabla u| dx = \int_0^{\|u\|_\infty} \left(\int_{\partial^* \{u > t\}} \varphi d\mathcal{H}^{n-1} \right) dt.$$

In particular the super-level set $E_t = \{u > t\}$ has finite perimeter for $\mathcal{L}^1$ -a.e. $t \in (0, \|u\|_\infty)$, and the function $m(t) := |E_t|$ is non-increasing, absolutely continuous on $(0, \|u\|_\infty)$ in the strict sense made precise in §4.1.11 below.

Theorem 2.5 (Sobolev–Sard for the torsion function; Figalli–Maggi 2011, App. A, Thm. A.1). *Let $u = u_\Omega$ be as in Corollary 2.4. There is a Borel set $\mathcal{R} \subset (0, \|u\|_\infty)$ of full $\mathcal{L}^1$ measure such that for every $t \in \mathcal{R}$:*

- (i) $E_t = \{u > t\}$ has finite perimeter and $m'(t) = - \int_{\partial^* E_t} |\nabla u|^{-1} d\mathcal{H}^{n-1} \in (-\infty, 0)$;
- (ii) $|\nabla u| > 0$ for $\mathcal{H}^{n-1}$ -a.e. $x \in \partial^* E_t$ and the De Giorgi outer normal is $\nu_{E_t}(x) = -\nabla u(x)/|\nabla u(x)|$ at $\mathcal{H}^{n-1}$ -a.e. such x ;
- (iii) (Volume-radius parametrisation, cf. register §7.) The function $\rho \mapsto t(\rho)$, characterised by $|\{u > t(\rho)\}| = \omega_n \rho^n$, is absolutely continuous on $[\rho_*, 1]$ with $t'(\rho) \leq 0$, and the normal velocity

$$V_\rho(x) := \frac{-t'(\rho)}{|\nabla u(x)|}$$

is well-defined and positive for $\mathcal{H}^{n-1}$ -a.e. $x \in \partial^ E_\rho$, for $\mathcal{L}^1$ -a.e. $\rho \in [\rho_*, 1]$.*

Reference. (i)–(ii) are [25, App. A, Thm. A.1]; see also [46, Lem. 17.1]. (iii) follows by composition: the inverse-distribution function $\rho \mapsto t(\rho)$ is the strict-AC inverse of the strict-AC function $t \mapsto (m(t)/\omega_n)^{1/n}$ (use [4, Thm. 3.99] on monotone AC functions), and the formula for V_ρ then follows

from the chain rule and $m'(t) = -\int |\nabla u|^{-1}$. The level-set identity draft `WIP_LevelSetIdentity.tex` §4 records the same statement. $\square$

2.1.3. Divergence theorem on sets of finite perimeter.

Theorem 2.6 (Gauss–Green; Maggi Thm. 15.5). *Let $E \subset \mathbb{R}^n$ have finite perimeter and let $X \in C_c^1(\mathbb{R}^n; \mathbb{R}^n)$. Then*

$$(24) \quad \int_E \operatorname{div} X \, dx = \int_{\partial^* E} X \cdot \nu_E \, d\mathcal{H}^{n-1}.$$

More generally, if E has finite perimeter and finite Lebesgue measure and $X \in C^1(\mathbb{R}^n; \mathbb{R}^n) \cap L^\infty(E; \mathbb{R}^n)$ satisfies $\operatorname{div} X \in L^1(E)$, then (24) continues to hold (both sides being finite), as soon as the support of X on $\partial^ E$ has finite $\mathcal{H}^{n-1}$ measure (e.g. when E is bounded).*

Reference. The C_c^1 statement is [46, Thm. 15.5] or [4, Thm. 3.36]. The extension to C^1 fields bounded on E with $\operatorname{div} X \in L^1(E)$ follows by multiplying by a smooth compactly supported cutoff χ_R , $\chi_R = 1$ on B_R , sending $R \rightarrow \infty$, and using dominated convergence on both sides; this is [46, Thm. 16.3] or the form used in [46, Ch. 17]. $\square$

We will need a slight strengthening allowing X to have a single locally integrable singularity inside E . The following lemma is load-bearing for the oriented radial trace estimate of Agent 11 (§4.42 below cross-references it).

Lemma 2.7 (Truncation lemma for fields with locally integrable singularity). *Let $E \subset \mathbb{R}^n$ have finite perimeter and finite Lebesgue measure, let $z \in \mathbb{R}^n$, and let $X : \mathbb{R}^n \setminus \{z\} \rightarrow \mathbb{R}^n$ be of class C^1 with the following properties:*

- (a) $X \in L^\infty(\mathbb{R}^n; \mathbb{R}^n)$, i.e. $|X(x)| \leq M_0$ for $x \neq z$, for some constant $M_0 \geq 0$;
- (b) $\operatorname{div} X \in L_{\text{loc}}^1(\mathbb{R}^n \setminus \{z\})$, and there is a constant $K_0 \geq 0$ such that

$$|\operatorname{div} X(x)| \leq \frac{K_0}{|x - z|} \quad \text{for all } x \neq z;$$

- (c) E is bounded, $E \subset B_R$ for some $R \geq 1$.

Then $\operatorname{div} X \in L^1(E)$ (since $n \geq 2$) and

$$(25) \quad \int_E \operatorname{div} X \, dx = \int_{\partial^* E} X \cdot \nu_E \, d\mathcal{H}^{n-1},$$

where the right-hand side is absolutely integrable.

The prototypical X to which this lemma applies is

$$X(x) = w(|x - z|) \frac{x - z}{|x - z|},$$

where $w : [0, \infty) \rightarrow \mathbb{R}$ is bounded with $w(0) = 0$ in the natural sense (so that $|X| \leq \|w\|_\infty$). In dimension $n \geq 2$ such X has $\operatorname{div} X(x) = w'(|x - z|) + (n - 1)w(|x - z|)/|x - z|$ for $x \neq z$, whose absolute value is $O(|x - z|^{-1})$ if w is bounded with bounded w' .

Proof. We may translate so that $z = 0$. The function $x \mapsto |x|^{-1}$ is locally integrable in $\mathbb{R}^n$ for $n \geq 2$, so (b) gives $\operatorname{div} X \in L^1(B_{2R}) \supset L^1(E)$. This proves the first assertion.

Now choose a fixed cutoff $\eta \in C^\infty(\mathbb{R})$ with $\eta \equiv 0$ on $(-\infty, 1]$, $\eta \equiv 1$ on $[2, \infty)$ and $0 \leq \eta \leq 1$, and for $\varepsilon \in (0, 1)$ set

$$\eta_\varepsilon(x) := \eta(\varepsilon^{-1}|x|), \quad X_\varepsilon(x) := \eta_\varepsilon(x) X(x).$$

Then $X_\varepsilon \equiv 0$ on B_ε , $X_\varepsilon \equiv X$ on $\mathbb{R}^n \setminus B_{2\varepsilon}$, and $X_\varepsilon \in C^1(\mathbb{R}^n; \mathbb{R}^n)$ (the singularity at 0 has been removed). Also X_ε is bounded by M_0 on $\mathbb{R}^n$ and has compact support if we further multiply by a smooth cutoff $\chi_R \in C_c^\infty(\mathbb{R}^n)$ with $\chi_R \equiv 1$ on B_R , $\chi_R \equiv 0$ on $\mathbb{R}^n \setminus B_{2R}$, $|\nabla \chi_R| \leq 2/R$; since $E \subset B_R$, this further cutoff does not affect any of the integrals over E or $\partial^* E$.

Apply Theorem 2.6 (the C_c^1 version) to $\chi_R X_\varepsilon$:

$$(26) \quad \int_E \operatorname{div}(\chi_R X_\varepsilon) dx = \int_{\partial^* E} (\chi_R X_\varepsilon) \cdot \nu_E d\mathcal{H}^{n-1}.$$

Since $\chi_R \equiv 1$ on a neighbourhood of $E \subset B_R$, the left-hand side equals $\int_E \operatorname{div} X_\varepsilon dx$ and the right-hand side equals $\int_{\partial^* E} X_\varepsilon \cdot \nu_E d\mathcal{H}^{n-1}$.

We now let $\varepsilon \downarrow 0$ and verify the two convergences:

Right-hand side. $X_\varepsilon \rightarrow X$ pointwise on $\mathbb{R}^n \setminus \{0\}$, hence $\mathcal{H}^{n-1}$ -a.e. on $\partial^* E$ (the single point 0 has $\mathcal{H}^{n-1}(\{0\}) = 0$ since $n \geq 2$). By (a), $|X_\varepsilon \cdot \nu_E| \leq M_0$ pointwise; since $\mathcal{H}^{n-1}(\partial^* E) = P(E) < \infty$, dominated convergence gives

$$\int_{\partial^* E} X_\varepsilon \cdot \nu_E d\mathcal{H}^{n-1} \longrightarrow \int_{\partial^* E} X \cdot \nu_E d\mathcal{H}^{n-1},$$

and the right-hand integral is absolutely convergent.

Left-hand side. A direct computation gives

$$\operatorname{div} X_\varepsilon(x) = \eta_\varepsilon(x) \operatorname{div} X(x) + \varepsilon^{-1} \eta'(\varepsilon^{-1}|x|) \frac{x}{|x|} \cdot X(x).$$

The first summand converges pointwise on $\mathbb{R}^n \setminus \{0\}$ to $\operatorname{div} X(x)$ as $\varepsilon \downarrow 0$. It is bounded in absolute value by $|\operatorname{div} X(x)| \leq K_0/|x|$, which is integrable on $E \subset B_R$. Dominated convergence gives

$$\int_E \eta_\varepsilon \operatorname{div} X dx \longrightarrow \int_E \operatorname{div} X dx.$$

For the second summand,

$$\left| \varepsilon^{-1} \eta'(\varepsilon^{-1}|x|) \frac{x}{|x|} \cdot X(x) \right| \leq \frac{\|\eta'\|_\infty M_0}{\varepsilon} \mathbf{1}_{\{\varepsilon \leq |x| \leq 2\varepsilon\}}.$$

Hence

$$\left| \int_E \varepsilon^{-1} \eta'(\varepsilon^{-1}|x|) \frac{x}{|x|} \cdot X(x) dx \right| \leq \frac{\|\eta'\|_\infty M_0}{\varepsilon} |B_{2\varepsilon}| = \|\eta'\|_\infty M_0 \omega_n 2^n \varepsilon^{n-1}.$$

For $n \geq 2$ this tends to 0 as $\varepsilon \downarrow 0$.

Combining, the left-hand side of (26) converges to $\int_E \operatorname{div} X dx$ and the right-hand side converges to $\int_{\partial^* E} X \cdot \nu_E d\mathcal{H}^{n-1}$. This proves (25). $\square$

Remark 2.8 (Why this matters). Lemma 2.7 is used twice in the manuscript:

(i) (Agent 11, oriented radial trace.) Applied to

$$X(x) = w(|x - z|) \frac{x - z}{|x - z|}, \quad w(r) = \left| \frac{r}{\rho} - 1 \right|,$$

on a bounded torsion level $E_\rho \subset B_R$, with the centre $z = \bar{z}_\rho$ or, more generally, the Fusco–Julin centre on a good isoperimetric level. Note that w is bounded by $C(R, \rho_*)$ on $\partial^* E_\rho$ (since $|x - z| \leq R + R'$ for some $R' = R'(R, \rho_*)$), and a direct computation gives

$$\operatorname{div} X(x) = w'(|x - z|) + (n - 1) \frac{w(|x - z|)}{|x - z|},$$

which is bounded by $C(R, \rho_*)(1 + |x - z|^{-1})$ pointwise. Hypotheses (a)–(c) of Lemma 2.7 are met (with $M_0 = \|w\|_\infty$, $K_0 = (n - 1)\|w\|_\infty$ and a harmless L^∞ contribution from w').

(ii) (Identification of the FJ oscillation index, used in Agent 4 / Agent 11 background.) Applied to $X(x) = (x - z)/|x - z|$ on E_ρ , hypothesis (b) holds with $K_0 = n - 1$ since $\operatorname{div}((x - z)/|x - z|) = (n - 1)/|x - z|$ for $x \neq z$; the divergence identity

$$\int_{\partial^* E_\rho} \left| \nu_\rho - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1} = 2\beta(E_\rho, z)^2,$$

quoted in (28) below, is a direct consequence together with the identity $|\xi - \eta|^2 = 2 - 2\xi \cdot \eta$ for unit vectors.

2.1.4. Lipschitz truncation flux identity for the torsion function. For the torsion function u_Ω on a regular level E_ρ , the boundary trace of ∇u on $\partial^* E_\rho$ is not a classical trace. The following identity replaces the formal application of the divergence theorem to ∇u on E_ρ ; it is used in Agent 9 to derive the deficit identity.

Lemma 2.9 (Flux = volume; cf. Maggi Ch. 17). *Let $\Omega \subset \mathbb{R}^n$ be open with $0 < |\Omega| < \infty$ and let $u = u_\Omega$. For $\mathcal{L}^1$ -a.e. $t \in (0, \|u\|_\infty)$, the super-level $E_t = \{u > t\}$ has finite perimeter and*

$$(27) \quad \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1} = |E_t|.$$

Proof. This is proved in `WIP_LevelSetIdentity.tex` Lem. 3.x via Lipschitz truncation of the test function $\eta_\varepsilon(x) := \phi_\varepsilon(u(x) - t)$ with $\phi_\varepsilon(s) = \min(1, \max(0, s/\varepsilon))$, testing the weak equation $-\Delta u = 1$ in $H_0^1(\Omega)$ against η_ε , applying the coarea formula (22) to the right-hand side $\int |\nabla u \cdot \nabla \eta_\varepsilon| = \varepsilon^{-1} \int_{\{t < u < t+\varepsilon\}} |\nabla u|^2$, and passing to the limit $\varepsilon \downarrow 0$ at every Lebesgue point of $t \mapsto \int_{\partial^* E_t} |\nabla u|$. The same calculation, with $|\nabla u|^2$ replaced by $|\nabla u|^p$ for suitable p , gives the rigorous interpretation of $\int_{\partial^* E_t} \nabla u \cdot \nu_{E_t} d\mathcal{H}^{n-1} = -\int_{\partial^* E_t} |\nabla u|$ as the boundary trace of ∇u on $\partial^* E_t$ in L^1 sense; see [46, Ch. 17]. $\square$

2.1.5. Reshetnyak lower semicontinuity and finite-perimeter passage of first variations.

Theorem 2.10 (Reshetnyak lower semicontinuity for linear-growth integrands). *Let $U \subset \mathbb{R}^n$ be open and let $f : U \times \mathbb{R}^n \rightarrow [0, \infty)$ be continuous, $f(x, \cdot)$ convex and positively 1-homogeneous for every $x \in U$. If $E_j, E \subset \mathbb{R}^n$*

are sets of finite perimeter with $\mathbf{1}_{E_j} \rightarrow \mathbf{1}_E$ in $L^1_{\text{loc}}(U)$, and if the distributional gradients $D\mathbf{1}_{E_j}$ converge weakly-* in U to $D\mathbf{1}_E$, then

$$\int_{\partial^* E \cap U} f(x, \nu_E(x)) d\mathcal{H}^{n-1} \leq \liminf_{j \rightarrow \infty} \int_{\partial^* E_j \cap U} f(x, \nu_{E_j}(x)) d\mathcal{H}^{n-1}.$$

Reference. [4, Thm. 2.38]. $\square$

This is the workhorse used to pass smooth-flow first-variation identities to the finite-perimeter setting: the smooth-flow proof of the metric first-variation bound is performed in `WIP_MetricFramework.tex`, and the four-step approximation of `Plan 2/gmt-inputs-for-metric-closure.md §5` turns it into a finite-perimeter statement. Concretely the approximation is:

- (1) approximate u_Ω by $u_j \in C^\infty(\Omega)$ with $u_j \rightarrow u$ in $W^{1,2}(\Omega)$ and, after a Sobolev–Sard reduction to a full-measure set of levels, in the coarea sense on the fixed annulus $\rho \in [\rho_*, 1]$;
- (2) form the smooth nested family $E_{\rho,j} := \{u_j > t_j(\rho)\}$, with $t_j(\rho)$ the inverse-distribution function of u_j ;
- (3) apply the smooth first-variation bound (proved in `WIP_MetricFramework.tex` and recalled in `Plan 2/gmt-inputs-for-metric-closure.md §4`) to $\rho \mapsto E_{\rho,j}$;
- (4) pass to the limit using Proposition 2.2 (lower semicontinuity of perimeter), Theorem 2.10 (Reshetnyak lower semicontinuity for the normal integrals $\int |V_\rho - H_{\bar{z},\rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}$), and the standard lower semicontinuity of metric derivatives for L^1 -convergent curves [6, Thm. 2.2.1].

The detailed passage is contained in the metric-framework section (Agent 10). We isolate the following corollary, which is the form needed by Agents 11–12.

Corollary 2.11 (Reshetnyak passage for the homothetic defect). *Let $E_j, E \subset B_R$ have finite perimeter with $|E_j| = |E| = \omega_n \rho^n$ and $\mathbf{1}_{E_j} \rightarrow \mathbf{1}_E$ in $L^1(\mathbb{R}^n)$, together with $P(E_j) \rightarrow P(E)$. Let $z_j \rightarrow z \in \mathbb{R}^n$ be a sequence of centres. For every $a \in \mathbb{R}^n$,*

$$\int_{\partial^* E} |V - H_{z,\rho} - a \cdot \nu_E| d\mathcal{H}^{n-1} \leq \liminf_{j \rightarrow \infty} \int_{\partial^* E_j} |V_j - H_{z_j,\rho} - a \cdot \nu_{E_j}| d\mathcal{H}^{n-1},$$

whenever $V_j \rightarrow V$ in $L^1(\mathcal{H}^{n-1} \llcorner \partial^* E_j)$ in the sense of weak convergence of varifolds.

Sketch. Apply Theorem 2.10 to the integrand $f(x, \nu) = |V(x) - H_{z,\rho}(x) - a \cdot \nu|$, which is continuous in x , convex in ν , and positively 1-homogeneous (because $H_{z,\rho}(x) = (x - z) \cdot \nu(x) / \rho$ is linear in ν). The convergence $P(E_j) \rightarrow P(E)$ together with $\mathbf{1}_{E_j} \rightarrow \mathbf{1}_E$ in L^1 gives, by [46, Thm. 12.30], weak-* convergence of $D\mathbf{1}_{E_j}$ to $D\mathbf{1}_E$ in the sense needed by Theorem 2.10. Convergence of V_j in varifold sense is the standing hypothesis. $\square$

2.1.6. Measurable selection of levelwise centres. The Fusco–Julin centre package (Agent 11) attaches to every level $\rho \in [\rho_*, \rho_\delta]$ in the good set $G_I \subset [\rho_*, \rho_\delta]$ at least one point $z \in \mathbb{R}^n$ such that the quantitative isoperimetric inequality is realised at z ; this point is unique only up to a translation in the bad set, and to use it inside metric-derivative integrals one must choose it in a Borel way.

Lemma 2.12 (Borel selection of centres). *Let $G \subset [\rho_*, 1]$ be Borel and for each $\rho \in G$ let $Z(\rho) \subset \mathbb{R}^n$ be a non-empty closed set; suppose the graph $\{(\rho, z) \in G \times \mathbb{R}^n : z \in Z(\rho)\}$ is a Borel subset of $G \times \mathbb{R}^n$ and $\sup_{\rho \in G} \sup_{z \in Z(\rho)} |z| < \infty$. Then there exists a bounded Borel map $\sigma : G \rightarrow \mathbb{R}^n$ with $\sigma(\rho) \in Z(\rho)$ for every $\rho \in G$.*

Proof. The graph of Z is a Borel subset of the standard Borel space $G \times \mathbb{R}^n$ with non-empty closed ρ -sections. The Kuratowski–Ryll–Nardzewski measurable selection theorem [19, Thm. III.6 and Thm. III.8] (also Federer, [24, §2.13]) provides a Borel selection $\sigma : G \rightarrow \mathbb{R}^n$. Boundedness of σ follows from the standing hypothesis that the union $\bigcup_{\rho \in G} Z(\rho)$ is bounded.

Outside G (i.e. in the bad set), set $\sigma(\rho) := 0$; this extends σ to a bounded Borel map on $[\rho_*, 1]$. $\square$

Remark 2.13 (Application). In the manuscript, $Z(\rho)$ is taken to be the set of η_I -almost minimisers in the scaled Fusco–Julin inequality applied to E_ρ : explicitly, the set of $z \in \mathbb{R}^n$ such that

$$|E_\rho \Delta B_\rho(z)|^2 + \rho^{n+1} \beta(E_\rho, z)^2 \leq 2C_n \rho^{n+1} (P(E_\rho) - P(B_\rho)),$$

with C_n the Fusco–Julin constant [29, Thm. 1.1]; this is non-empty by Fusco–Julin itself, closed by lower semicontinuity of the integrals in z (continuity of $z \mapsto |E_\rho \Delta B_\rho(z)|$ is elementary, and continuity of $z \mapsto \beta(E_\rho, z)^2 = P(E_\rho) - (n-1) \int_{E_\rho} |x-z|^{-1} dx$ follows from dominated convergence using the bound $|x-z|^{-1} \leq |x-z_0|^{-1} + |z-z_0| |x-z|^{-1} |x-z_0|^{-1}$), and bounded in z because the Fraenkel part forces $|z| \leq R + \rho$ on $E \subset B_R$. Lemma 2.12 then provides a bounded Borel field $\rho \mapsto z_\rho$ on the good set G_I . The Borel-measurability of the graph follows from the fact that both $(z, E) \mapsto |E \Delta B_\rho(z)|$ and $(z, E) \mapsto \beta(E, z)^2$ are Borel maps in z for fixed E (in fact continuous), so the level set is closed and the graph is therefore Borel.

2.1.7. Two auxiliary identities used downstream. For convenience we record two identities used by Agents 11 and 12.

Lemma 2.14 (FJ oscillation index identity). *Let $E \subset \mathbb{R}^n$ have finite perimeter and finite measure with $z \notin \mathbb{R}^n$ playing the role of a centre. Define*

$$\beta(E, z)^2 := P(E) - (n-1) \int_E |x-z|^{-1} dx.$$

Then $\beta(E, z)^2 \geq 0$ and

$$(28) \quad \int_{\partial^* E} \left| \nu_E(x) - \frac{x-z}{|x-z|} \right|^2 d\mathcal{H}^{n-1} = 2\beta(E, z)^2.$$

Proof. The identity $|\nu_E(x) - e_r|^2 = 2 - 2e_r \cdot \nu_E(x)$ for unit vectors e_r gives

$$\int_{\partial^* E} |\nu_E - e_r|^2 = 2P(E) - 2 \int_{\partial^* E} e_r \cdot \nu_E d\mathcal{H}^{n-1}.$$

Apply Lemma 2.7 (with z in place of the origin) to the vector field $X(x) := (x-z)/|x-z|$. Hypothesis (a) holds with $M_0 = 1$, and a direct computation gives $\operatorname{div} X(x) = (n-1)/|x-z|$ for $x \neq z$, so (b) holds with $K_0 = n-1$.

(c) holds because E has finite measure and, in the manuscript application, $E \subset B_R$. Therefore

$$\int_{\partial^* E} \frac{x-z}{|x-z|} \cdot \nu_E d\mathcal{H}^{n-1} = \int_E \operatorname{div} X dx = (n-1) \int_E |x-z|^{-1} dx.$$

Combining gives (28). Non-negativity is then immediate from the left-hand side, and equality with the right-hand side proves $\beta(E, z)^2 \geq 0$. $\square$

Lemma 2.15 (Total flux of $H_{z,\rho}$). *For $z \in \mathbb{R}^n$ and a set E_ρ of finite perimeter with $|E_\rho| = \omega_n \rho^n$,*

$$\int_{\partial^* E_\rho} H_{z,\rho} d\mathcal{H}^{n-1} = \frac{n|E_\rho|}{\rho} = n\omega_n \rho^{n-1} = P(B_\rho).$$

Proof. Apply Theorem 2.6 to $X(x) := (x-z)/\rho \in C^1(\mathbb{R}^n; \mathbb{R}^n)$, which is C^1 everywhere and bounded on E_ρ (any finite-measure set is $\mathcal{H}^{n-1}$ -essentially bounded on its reduced boundary; in the manuscript $E_\rho \subset B_R$). $\operatorname{div} X = n/\rho$, so the divergence theorem gives $\int_{\partial^* E_\rho} X \cdot \nu_\rho d\mathcal{H}^{n-1} = n|E_\rho|/\rho$. Since $(x-z) \cdot \nu_\rho(x)/\rho = H_{z,\rho}(x)$, the lemma follows. $\square$

2.1.8. *Summary of the GMT preliminaries and how they are used.* For the convenience of the reader we summarise which GMT facts enter each Plan 2 section.

- *Level-set deficit identity (Agent 9).* BV coarea (Theorem 2.3), the Sobolev–Sard statement for the torsion function (Theorem 2.5), and the Lipschitz-truncation flux identity (Lemma 2.9).
- *Metric framework (Agent 10).* Lower semicontinuity of perimeter (Proposition 2.2), the divergence theorem (Theorem 2.6), and Reshetnyak lower semicontinuity together with the smooth-to-finite-perimeter approximation (Theorem 2.10, Corollary 2.11).
- *Fusco–Julin centre and oriented radial trace (Agent 11).* Truncation lemma (Lemma 2.7); the FJ oscillation-index identity (Lemma 2.14); and the Borel selection (Lemma 2.12).
- *Integrated action and weighted metric trace (Agent 12).* The same divergence and BV coarea tools, plus the absolute continuity of $\rho \mapsto [F_\rho]$ proved in the metric framework section.

All boundary integrals in the manuscript are on De Giorgi reduced boundaries; all centre fields are Borel via Lemma 2.12; the truncation argument of Lemma 2.7 is the rigorous form of the “ $x/|x|$ ” argument invoked at several places in Plan 2.

2.2. Preliminaries II: Torsion, rearrangement, and deficits. This section fixes the torsion and Faber–Krahn functionals used throughout, the Saint–Venant and Faber–Krahn deficits, the Fraenkel asymmetry, the Talenti comparison, the volume–radius foliation $E_\rho = \{u > t(\rho)\}$, the profile-gap and bad-set bounds used in the Plan 2 weighted metric trace, and the Kohler–Jobin inequality used in the final transfer.

Conventions follow `00_notation_register.md`: we write the ambient dimension as $n \geq 2$, the BDV-normalised Saint–Venant energy as $E(\Omega) =$

$-T(\Omega)/2$, the Saint–Venant deficit as $\delta_T(\Omega) = E(\Omega) - E(B^*) \geq 0$, the Faber–Krahn deficit as $\delta_{\text{FK}}(\Omega) = (|\Omega|/\omega_n)^{2/n}(\lambda_1(\Omega) - \lambda_1(B^*))$, and the Fraenkel asymmetry as $\mathcal{A}(\Omega) \in [0, 2)$.

2.2.1. Torsion potential, rigidity, and energy. Throughout, $\Omega \subset \mathbb{R}^n$ is open with $0 < |\Omega| < \infty$. No connectedness, smoothness, or boundedness is assumed unless stated.

Definition 2.16 (Torsion potential). The *variational torsion function* of Ω is the unique minimiser

$$u_\Omega \in H_0^1(\Omega), \quad u_\Omega = \arg \min \left\{ J(v) : v \in H_0^1(\Omega) \right\}, \quad J(v) := \frac{1}{2} \int_\Omega |\nabla v|^2 dx - \int_\Omega v dx.$$

Equivalently, u_Ω is the weak solution of $-\Delta u_\Omega = 1$ in Ω with $u_\Omega = 0$ on $\partial\Omega$ in the H_0^1 trace sense. Testing the weak equation against u_Ω^- yields $u_\Omega \geq 0$ a.e. in Ω .

Remark 2.17 (Existence). The functional J is strictly convex, coercive on $H_0^1(\Omega)$ by the Poincaré–Sobolev inequality $\|v\|_{L^{2^*}} \leq C(n)\|\nabla v\|_{L^2}$ on sets of finite measure (valid since $|\Omega| < \infty$, see [46, Cor. 12.18] or [36, Thm. 4.9]), and weakly lower semicontinuous; the direct method gives existence and uniqueness of u_Ω .

Definition 2.18 (Torsional rigidity and Saint–Venant energy). The *torsional rigidity* of Ω is

$$(29) \quad T(\Omega) := \int_\Omega u_\Omega dx.$$

The *BDV-normalised Saint–Venant energy* of Ω is

$$(30) \quad E(\Omega) := \min_{v \in H_0^1(\Omega)} J(v) = \frac{1}{2} \int_\Omega |\nabla u_\Omega|^2 dx - \int_\Omega u_\Omega dx.$$

Lemma 2.19 (Energy–rigidity identity). *For every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,*

$$(31) \quad E(\Omega) = -\frac{1}{2} T(\Omega).$$

In particular $E(\Omega) \leq 0$ on every admissible set, and $T(\Omega) > 0$ unless $|\Omega| = 0$.

Proof. Test the weak equation $-\Delta u_\Omega = 1$ against $u_\Omega \in H_0^1(\Omega)$ to get $\int_\Omega |\nabla u_\Omega|^2 dx = \int_\Omega u_\Omega dx = T(\Omega)$. Substituting into (30) gives $E(\Omega) = \frac{1}{2} T(\Omega) - T(\Omega) = -\frac{1}{2} T(\Omega)$. $\square$

Remark 2.20 (Two coexisting conventions). Some sources (e.g. [51, §1], classical analysis literature) call $T(\Omega)$ the “torsion energy”. The BDV chain ([15]; `Final/SaintVenantAssembly.tex`; the global reduction in `Final-BoundedReduction`; the Kohler–Jobin transfer in `Final-KohlerJobinTransfer`) and the Plan 2 WIP files (`WIP_LevelSetIdentity.tex`, `WIP_master.tex`) reserve “Saint–Venant energy” for the BDV-normalised functional $E = -T/2$, which is the minimised value of the variational functional J . We adopt the second convention throughout: $T(\Omega)$ is the torsional *rigidity*; $E(\Omega)$ is the BDV-normalised Saint–Venant *energy*. The identity (31) converts between the two.

Lemma 2.21 (Scaling and ball values). *For every $t > 0$,*

$$(32) \quad T(t\Omega) = t^{n+2}T(\Omega), \quad E(t\Omega) = t^{n+2}E(\Omega).$$

For the unit ball B_1 , the explicit torsion function is $u_{B_1}(x) = (1 - |x|^2)/(2n)$, hence

$$(33) \quad T_n := T(B_1) = \int_{B_1} \frac{1 - |x|^2}{2n} dx = \frac{\omega_n}{n(n+2)}, \quad E(B_1) = -\frac{\omega_n}{2n(n+2)}.$$

Proof. Scaling: with $v_t(x) := t^2 u_\Omega(x/t) \in H_0^1(t\Omega)$, one verifies $-\Delta v_t = 1$ in $t\Omega$, so by uniqueness $u_{t\Omega} = v_t$; then $T(t\Omega) = \int_{t\Omega} v_t = t^{n+2} \int_\Omega u_\Omega = t^{n+2}T(\Omega)$, and (31) extends to E . For the ball formula, a direct computation gives $\Delta((1 - |x|^2)/(2n)) = -1$ and the boundary value 0. Integrating in radial coordinates, $\int_{B_1} (1 - |x|^2)/(2n) dx = (1/(2n))[\omega_n - n\omega_n/(n+2)] = \omega_n/(n(n+2))$. $\square$

2.2.2. Saint-Venant inequality and Saint-Venant deficit. The classical Saint-Venant inequality [50, Thm. 3.3] states that the ball maximises torsional rigidity at fixed volume.

Theorem 2.22 (Saint-Venant inequality, classical). *For every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$, and with B^* the open ball centred at the origin with $|B^*| = |\Omega|$,*

$$(34) \quad T(\Omega) \leq T(B^*), \quad \text{equivalently} \quad E(\Omega) \geq E(B^*).$$

Equality holds iff Ω is (a.e. equal to) a ball.

Citation. [50, Thm. 1.1], also obtained by Schwarz symmetrisation combined with the Pólya-Szegő principle, or as a corollary of Talenti's rearrangement comparison (Theorem 2.33 below): on the unit-volume normalisation, the latter gives $u_\Omega^* \leq u_{B^*}^*$ pointwise on $[0, |\Omega|]$, hence $T(\Omega) = \int_0^{|\Omega|} u_\Omega^*(s) ds \leq \int_0^{|\Omega|} u_{B^*}^*(s) ds = T(B^*)$. $\square$

Definition 2.23 (Saint-Venant deficit δ_T). For open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$ and B^* the ball with $|B^*| = |\Omega|$, the (BDV-normalised) Saint-Venant deficit is

$$(35) \quad \delta_T(\Omega) := E(\Omega) - E(B^*) \geq 0,$$

equivalently $\delta_T(\Omega) = \frac{1}{2}(T(B^*) - T(\Omega))$.

Remark 2.24 (Scale-invariant form). The scale-invariant version is

$$(36) \quad D(\Omega) := E(\Omega) |\Omega|^{-(n+2)/n} - E(B_1) |B_1|^{-(n+2)/n} \geq 0,$$

which for $|\Omega| = \omega_n$ reduces to $D(\Omega) = \omega_n^{-(n+2)/n} \delta_T(\Omega)$. Both formulations are used: δ_T in the Plan 2 chain and the bounded Saint-Venant statements; D in the bounded reduction (Plan 1, §IV).

2.2.3. *Fraenkel asymmetry.*

Definition 2.25 (Fraenkel asymmetry). For open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$, the *Fraenkel asymmetry* is

$$(37) \quad \mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^n} \frac{|\Omega \triangle (B^* + x_0)|}{|B^*|},$$

where B^* is the ball centred at the origin with $|B^*| = |\Omega|$.

Lemma 2.26 (Range of $\mathcal{A}$). $\mathcal{A}(\Omega) \in [0, 2)$, with $\mathcal{A}(\Omega) = 0$ iff Ω is (a.e. equal to) a translate of B^* .

Proof. Non-negativity is immediate. The strict bound $\mathcal{A} < 2$ follows from $|\Omega \triangle (B^* + x_0)| = |\Omega| + |B^*| - 2|\Omega \cap (B^* + x_0)|$ and the existence (by translation continuity of Lebesgue measure) of $x_0 \in \mathbb{R}^n$ with $|\Omega \cap (B^* + x_0)| > 0$, so $|\Omega \triangle (B^* + x_0)| < 2|B^*|$. The infimum is attained because the map $x_0 \mapsto |\Omega \triangle (B^* + x_0)|$ is continuous and tends to $2|B^*|$ as $|x_0| \rightarrow \infty$ ([45, Lem. 3.2]). $\square$

Remark 2.27 (Translation invariance). Both δ_T , δ_{FK} , and $\mathcal{A}$ are translation invariant; all estimates below are stated up to translation.

2.2.4. *Faber–Krahn deficit and eigenvalue normalisation.*

Definition 2.28 (First Dirichlet eigenvalue). For open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,

$$(38) \quad \lambda_1(\Omega) := \min_{\substack{v \in H_0^1(\Omega) \\ v \neq 0}} \frac{\int_{\Omega} |\nabla v|^2 dx}{\int_{\Omega} v^2 dx}.$$

The minimum is attained (by direct method on $\{v : \|v\|_{L^2} = 1\}$, using the Poincaré–Sobolev inequality on finite-measure sets), and the eigenvalue scales as $\lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega)$ for $t > 0$. We write $L_n := \lambda_1(B_1)$ for the unit-ball first eigenvalue. Numerically, $L_n = j_{n/2-1,1}^2$, the square of the first positive zero of the Bessel function $J_{n/2-1}$ [36, §1.3].

Theorem 2.29 (Faber–Krahn inequality). For every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,

$$(39) \quad \lambda_1(\Omega) \geq \lambda_1(B^*) = L_n (|B^*|/\omega_n)^{-2/n},$$

with equality iff Ω is (a.e. equal to) a ball.

Citation. The classical Faber–Krahn theorem [23, 41], proved by Schwarz symmetrisation and the Pólya–Szegő inequality. See [36, Thm. 3.2.1]. $\square$

Definition 2.30 (Faber–Krahn deficit δ_{FK}). For open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$ and B^* the ball with $|B^*| = |\Omega|$, the scale-normalised Faber–Krahn deficit is

$$(40) \quad \delta_{\text{FK}}(\Omega) := \left(\frac{|\Omega|}{\omega_n}\right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq 0.$$

Lemma 2.31 (Scaling invariance of δ_{FK}). $\delta_{\text{FK}}(t\Omega) = \delta_{\text{FK}}(\Omega)$ for every $t > 0$.

Proof. $\lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega)$ and $|t\Omega|^{2/n} = t^2|\Omega|^{2/n}$ give $|t\Omega|^{2/n}\lambda_1(t\Omega) = |\Omega|^{2/n}\lambda_1(\Omega)$. Similarly for B^* . $\square$

Remark 2.32 (Why this normalisation). Both Plan 1 and Plan 2 produce, after Kohler–Jobin transfer, a sharp quantitative inequality of the form $\delta_{\text{FK}}(\Omega) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2$. The factor $(|\Omega|/\omega_n)^{2/n}$ in (40) makes both sides scale-invariant, mirroring (36); see `Final-KohlerJobinTransfer`, eq. (eq:fkout).

2.2.5. Talenti pointwise comparison. The fundamental rearrangement comparison for the torsion function is the Talenti pointwise inequality [51]. We write $u^*: [0, |\Omega|] \rightarrow [0, \|u\|_\infty]$ for the decreasing rearrangement of $u = u_\Omega$, $u^*(s) := \inf\{t \geq 0 : m(t) \leq s\}$, where $m(t) := |\{u > t\}|$.

Theorem 2.33 (Talenti, 1976). *Let $\Omega \subset \mathbb{R}^n$ be open with $0 < |\Omega| < \infty$, and let $u_\Omega \in H_0^1(\Omega)$ be the torsion function of Definition 2.16. Let u_{B^*} be the torsion function of the ball B^* with $|B^*| = |\Omega|$. Then*

$$(41) \quad u_\Omega^*(s) \leq u_{B^*}^*(s) = \frac{R_\Omega^2 - (s/\omega_n)^{2/n}}{2n}, \quad s \in [0, |\Omega|],$$

where $R_\Omega := (|\Omega|/\omega_n)^{1/n}$ is the radius of B^* . In particular,

$$(42) \quad \|u_\Omega\|_{L^\infty(\Omega)} \leq u_{B^*}^*(0) = \frac{R_\Omega^2}{2n}.$$

Citation. [51, Thm. 1]. The argument uses isoperimetric rearrangement of super-level sets and the coarea formula; it is constructive and gives equality iff u_Ω is (translate of) the ball torsion function, hence iff Ω is a ball. $\square$

Remark 2.34 (Consequence: L^∞ bound used in Plan 2). The sup-norm bound (42) is the only reason that the level-set deficit identity (`WIP_LevelSetIdentity.tex`, Thm. 1) holds for arbitrary open sets of finite measure without an explicit boundedness hypothesis. Throughout Plan 2 we use the normalised volume $|\Omega| = \omega_n$, so $R_\Omega = 1$ and $\|u_\Omega\|_\infty \leq 1/(2n)$.

2.2.6. Volume-radius foliation $E_\rho = \{u > t(\rho)\}$. We now record the volume-radius parametrisation of the super-level sets of u_Ω , used throughout Plan 2.

Definition 2.35 (Distribution function and its inverse). For Ω open with $|\Omega| = \omega_n$ (the Plan 2 normalisation), $E_t := \{u_\Omega > t\}$ and $m(t) := |E_t|$. By the BV coarea formula and Lemma 2.38 below, m is right-continuous and non-increasing on $[0, \|u\|_\infty]$ with $m(0^+) \leq \omega_n$ and $m(\|u\|_\infty) = 0$. Define

$$(43) \quad t(\rho) := \inf\{t \geq 0 : m(t) \leq \omega_n \rho^n\}, \quad \rho \in [0, 1],$$

and the volume-radius foliation

$$(44) \quad E_\rho := \{u_\Omega > t(\rho)\}, \quad |E_\rho| = \omega_n \rho^n, \quad \mu(d\rho) := (-t'(\rho)) d\rho.$$

Lemma 2.36 (Properties of $t(\rho)$). *Let Ω open with $|\Omega| = \omega_n$ and $u = u_\Omega$. Then*

- (i) $\rho \mapsto t(\rho)$ is non-increasing on $[0, 1]$, with $t(1) = 0$ and $0 \leq t(\rho) \leq \|u\|_\infty \leq 1/(2n)$ by (42).
- (ii) t is absolutely continuous on every interval $[\rho_*, 1] \subset (0, 1]$.
- (iii) On a co-null set in $[0, 1]$, $t'(\rho) \leq 0$, and

$$(45) \quad 0 \leq -t'(\rho) \leq \frac{1}{n} \quad \text{for a.e. } \rho \in [0, 1].$$

Proof. (i) and the boundary value $t(1) = 0$ follow from $m(0^+) \leq \omega_n$ and $m(t(\rho)) \leq \omega_n \rho^n$; the sup-norm bound is Theorem 2.33.

(ii) The map m is, by Lemma 2.38, absolutely continuous on $[\varepsilon, \|u\|_\infty]$ for every $\varepsilon > 0$ (its derivative $-\alpha(t)$ is in L^1_{loc}). The inverse of a strictly decreasing absolutely continuous function with non-vanishing derivative on an open set is absolutely continuous; on plateau levels (where m is constant) t has a jump of measure zero and the equation $|E_\rho| = \omega_n \rho^n$ pins $t(\rho)$ on the right-continuous representative. On the foliation interval $[\rho_*, 1]$ this gives absolute continuity of t .

(iii) Differentiating $m(t(\rho)) = \omega_n \rho^n$ at a Lebesgue point of $\rho \mapsto t'(\rho)$ at which m is differentiable at $t(\rho)$, by the chain rule for absolutely continuous functions and Lemma 2.38, $m'(t(\rho)) t'(\rho) = n \omega_n \rho^{n-1}$, i.e.

$$(46) \quad -t'(\rho) = \frac{n \omega_n \rho^{n-1}}{\alpha(t(\rho))}, \quad \alpha(t) := \int_{\partial^* E_t} |\nabla u|^{-1} d\mathcal{H}^{n-1}.$$

By the Cauchy–Schwarz inequality on $\partial^* E_t$ and the isoperimetric inequality $P(E_t) \geq n \omega_n^{1/n} m(t)^{1-1/n}$,

$$(47) \quad \alpha(t) \cdot \beta(t) \geq P(E_t)^2 \geq n^2 \omega_n^{2/n} m(t)^{2-2/n}, \quad \beta(t) := \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1} = m(t),$$

the equality $\beta(t) = m(t)$ being the flux identity of Lemma 2.39 below.

Combined with $m(t(\rho)) = \omega_n \rho^n$, $\alpha(t(\rho)) \geq n^2 \omega_n^{2/n} (\omega_n \rho^n)^{2-2/n} / m(t(\rho)) = n^2 \omega_n \rho^{n-2} \cdot (\omega_n \rho^n)^{1-2/n} / (\omega_n \rho^n) = n \omega_n \rho^{n-1} \cdot n$, hence $-t'(\rho) \leq n \omega_n \rho^{n-1} / (n^2 \omega_n \rho^{n-1}) = 1/n$. $\square$

Remark 2.37 (Where $-t' \leq 1/n$ is used). The bound (45) is the hypothesis (H3) of the abstract weighted metric trace lemma (WIP_WeightedMetricTrace.tex, Thm. 5.1): it bounds the density of μ uniformly.

The coarea and flux identities used in the proof of Lemma 2.36(iii) are recorded for reference (proved in the geometric measure theory preliminaries, §2.1):

Lemma 2.38 (Coarea form of $-m'$). *For a.e. $t \in (0, \|u\|_\infty)$, $-m'(t) = \alpha(t)$ as defined in (46). m is absolutely continuous on $[\varepsilon, \|u\|_\infty]$ for every $\varepsilon > 0$.*

Citation. [46, Thm. 13.1], [4, Thm. 3.40]. $\square$

Lemma 2.39 (Flux–volume identity). *For a.e. $t \in (0, \|u\|_\infty)$, E_t has finite perimeter and $\beta(t) = m(t)$.*

Citation. By Lipschitz truncation of the weak equation $-\Delta u = 1$ against the cutoff $\phi_\varepsilon(u - t) \in H^1_0(\Omega)$ and the finite-perimeter divergence theorem; cf. [46, Thm. 16.3]. $\square$

2.2.7. Profile-gap estimate used by Plan 2. We now record the profile-gap estimate that converts an L^∞ -control of $u_\Omega^* - u_{B^*}^*$ on a near-boundary slab into a positive lower bound for $|T_\eta| = v(L - 2\eta) - v(L - \eta)$, hence into the hypothesis (H2) of the abstract weighted metric trace lemma.

We work in the Plan 2 normalisation: $|\Omega| = \omega_n =: L$, $R_\Omega = R = 1$, and we write $b(s) := u_{B_1}^*(s)$, $v(s) := u_\Omega^*(s)$, $h(s) := b(s) - v(s)$.

Lemma 2.40 (Profile gap; integrated form). *Let $\Omega \subset \mathbb{R}^n$ be open with $|\Omega| = \omega_n$. Then for every $s \in (0, \omega_n]$,*

$$(48) \quad 0 \leq h(s) = b(s) - v(s) \leq \frac{2\delta_T(\Omega)}{s}.$$

In particular, for every $\theta \in (0, 1)$,

$$(49) \quad \|h\|_{L^\infty([\theta\omega_n, \omega_n])} \leq \frac{2\delta_T(\Omega)}{\theta\omega_n}.$$

Proof. By Talenti's pointwise comparison (Theorem 2.33), $v(s) \leq b(s)$, so $h(s) \geq 0$ and h is the difference of non-increasing absolutely continuous functions, hence absolutely continuous on $(0, \omega_n]$ with $h(0^+) \leq b(0) < \infty$ and $h(\omega_n) = 0$.

For the upper bound, integrate Talenti's level-set differential inequality (cf. [51, §4], `Plan 2/level-set-deficit-identity.md` §4): $v'(s) \geq b'(s)$ for a.e. s , equivalently $h'(s) \leq 0$. Since $h \geq 0$ is non-increasing, $h(s) \cdot s \leq \int_0^s h(\sigma) d\sigma \leq \int_0^{\omega_n} h$. By equimeasurability,

$$\int_0^{\omega_n} h(\sigma) d\sigma = \int_0^{\omega_n} b(\sigma) d\sigma - \int_0^{\omega_n} v(\sigma) d\sigma = T(B_1) - T(\Omega) = 2\delta_T(\Omega),$$

using $|\Omega| = \omega_n = |B_1|$ and Lemma 2.19. Hence $h(s) \leq 2\delta_T(\Omega)/s$. $\square$

Lemma 2.41 (Foliation length on a boundary slab). *Let $\Omega \subset \mathbb{R}^n$ open with $|\Omega| = \omega_n$. For $\eta \in (0, \omega_n/4)$ define the volume slab $S_\eta := [\omega_n - 2\eta, \omega_n - \eta]$ and the level slab $T_\eta := v(S_\eta) = [v(\omega_n - \eta), v(\omega_n - 2\eta)]$. There exist constants $c_n, C_n > 0$ depending only on n such that*

$$(50) \quad c_n\eta \leq b(s) \leq C_n\eta \quad \text{for every } s \in S_\eta,$$

and if $\delta_T(\Omega) \leq c_n\eta$ then

$$(51) \quad |T_\eta| = v(\omega_n - 2\eta) - v(\omega_n - \eta) \geq c_n\eta.$$

The constants are explicit: $c_n = 1/(C \cdot n)$, $C_n = C \cdot n^{-1}$, C a universal dimensional constant.

Proof. Set $R = 1$, $L = \omega_n$. From the explicit ball profile $b(s) = (1 - (s/\omega_n)^{2/n})/(2n)$, expanding $1 - (1 - \eta/\omega_n)^{2/n}$ to first order in η for $\eta \leq \omega_n/4$ gives $c_n\eta/\omega_n \leq 1 - (1 - \eta/\omega_n)^{2/n} \leq C_n\eta/\omega_n$ with $c_n = 1/n$, $C_n = 4/n$ (using $1 - (1 - x)^{2/n} \geq (2/n)x(1 - x) \geq x/n$ for $x \in [0, 1/2]$, and $1 - (1 - x)^{2/n} \leq (2/n)x \cdot (1 - x)^{2/n-1} \leq 4x/n$ for $n \geq 2$). Multiplying by $1/(2n)$ gives (50).

For (51), write $v(\omega_n - 2\eta) - v(\omega_n - \eta) = [b(\omega_n - 2\eta) - b(\omega_n - \eta)] - [h(\omega_n - 2\eta) - h(\omega_n - \eta)]$. The first bracket is, by the explicit ball profile and $\omega_n - 2\eta \geq \omega_n/2$, at least $c_n\eta$ for a dimensional c_n . The second bracket is bounded by $2\|h\|_{L^\infty([\omega_n/2, \omega_n])} \leq 4\delta_T(\Omega)/\omega_n$ via (49). The smallness hypothesis $\delta_T(\Omega) \leq c_n\eta$ (after halving c_n) ensures the first bracket dominates the second. $\square$

Remark 2.42 (Adaptation source). Lemma 2.40 is the integrated form of the Talenti-comparison consequence $0 \leq b(s) - v(s) \leq 2\delta_T/s$ derived from [51, Thm. 1] and the Saint-Venant deficit identity $\Gamma(\Omega) = 2(E(\Omega) - E(B^*))/|\Omega|$. Lemma 2.41 provides the $|T_\eta|$ -bound used to verify hypothesis (H2) in the abstract trace lemma of §4.4.

2.2.8. *Bad-set Lebesgue estimate used by Plan 2.* The bad-set estimate converts the integrated Talenti gap into a Lebesgue control on the set where the foliation slows down. It is the input that the weighted metric trace lemma needs to convert the weighted kinetic bound $\int |\dot{F}_\rho|_\chi^2 d\mu \leq C\delta_T$ into the unweighted kinetic bound $\int |\dot{F}_\rho|_\chi^2 d\rho \leq C'\delta_T$ (cf. §4.4).

Lemma 2.43 (Integrated Talenti gap on $-t'$). *Let $\Omega \subset \mathbb{R}^n$ be open with $|\Omega| = \omega_n$ and $\rho_* \in (0, 1)$. Let t_B denote the inverse distribution function of u_{B_1} , $t_B(\rho) = u_{B_1}^*(\omega_n \rho^n) = (1 - \rho^2)/(2n)$, so $-t'_B(\rho) = \rho/n$. There is $K_1 = K_1(n, \rho_*) > 0$ such that*

$$(52) \quad \int_{\rho_*}^1 |(-t'_B(\rho)) - (-t'(\rho))| d\rho \leq K_1 \delta_T(\Omega).$$

Proof. Since both t, t_B are non-increasing with $t(1) = t_B(1) = 0$, $\psi(\rho) := (-t'_B(\rho)) - (-t'(\rho))$ satisfies $\int_\rho^1 \psi(r) dr = t(\rho) - t_B(\rho) = v(\omega_n \rho^n) - b(\omega_n \rho^n) = -h(\omega_n \rho^n) \leq 0$ for every $\rho \in [\rho_*, 1]$ (by Talenti, $v \leq b$, hence $h \geq 0$). However ψ need not have a sign; the proof of Lemma 2.40 shows $h \geq 0$ globally, so the function $-h(\omega_n \rho^n)$ is ≤ 0 on $[\rho_*, 1]$.

Substitute $s = \omega_n \rho^n$, $ds = n\omega_n \rho^{n-1} d\rho$:

$$\int_{\rho_*}^1 \psi(\rho) d\rho = [t(\rho) - t_B(\rho)]_{\rho_*}^1 = -(t(\rho_*) - t_B(\rho_*)) = h(\omega_n \rho_*^n) \leq \frac{2\delta_T(\Omega)}{\omega_n \rho_*^n}$$

by Lemma 2.40. Since ψ need not be signed, this controls only $\int \psi$ not $\int |\psi|$. We restore the absolute value by a second comparison. On any subinterval $[\rho_1, \rho_2] \subset [\rho_*, 1]$,

$$\left| \int_{\rho_1}^{\rho_2} \psi(r) dr \right| = |(t(\rho_1) - t(\rho_2)) - (t_B(\rho_1) - t_B(\rho_2))| = |h(\omega_n \rho_1^n) - h(\omega_n \rho_2^n)| \leq h(\omega_n \rho_*^n)$$

because $h \geq 0$ is non-increasing. By the Hahn–Jordan decomposition of $\psi d\rho$ on $[\rho_*, 1]$ as $d\psi^+ - d\psi^-$,

$$\int_{\rho_*}^1 |\psi| d\rho = \psi^+([\rho_*, 1]) + \psi^-([\rho_*, 1]) \leq 2 \sup_{[\rho_1, \rho_2] \subset [\rho_*, 1]} \left| \int_{\rho_1}^{\rho_2} \psi \right| \leq 2 h(\omega_n \rho_*^n) \leq \frac{4\delta_T(\Omega)}{\omega_n \rho_*^n},$$

which is (52) with $K_1 = 4/(\omega_n \rho_*^n)$. $\square$

Lemma 2.44 (Bad-set Lebesgue estimate; (H3-bad)). *Let $\Omega \subset \mathbb{R}^n$ open with $|\Omega| = \omega_n$, $\rho_* \in (0, 1)$. Set*

$$\tau_0 := \frac{\rho_*}{2n}, \quad B_{\tau_0} := \{\rho \in [\rho_*, 1] : -t'(\rho) < \tau_0\}.$$

Then

$$(53) \quad |B_{\tau_0}| \leq K_2(n, \rho_*) \delta_T(\Omega), \quad K_2(n, \rho_*) := \frac{8n}{\omega_n \rho_*^{n+1}}.$$

Proof. On B_{τ_0} , $-t'(\rho) < \rho_*/(2n)$ and $-t'_B(\rho) = \rho/n \geq \rho_*/n$ for $\rho \geq \rho_*$. Hence $\psi(\rho) = (-t'_B(\rho)) - (-t'(\rho)) > \rho_*/n - \rho_*/(2n) = \rho_*/(2n)$ on B_{τ_0} . Integrating and applying Lemma 2.43,

$$\frac{\rho_*}{2n} |B_{\tau_0}| \leq \int_{B_{\tau_0}} \psi(\rho) d\rho \leq \int_{\rho_*}^1 |\psi(\rho)| d\rho \leq K_1 \delta_T(\Omega),$$

so $|B_{\tau_0}| \leq (2n/\rho_*) K_1 \delta_T(\Omega)$. Substituting $K_1 = 4/(\omega_n \rho_*^n)$ gives (53). $\square$

Remark 2.45 (Adaptation source). Lemma 2.44 is the bad-set lemma used in §4.4. Lemma 2.43 is the “integrated Talenti gap” invoked there; the constants $K_1(n, \rho_*)$, $K_2(n, \rho_*)$ are made explicit here. No published reference for this exact statement was located; the proof is self-contained from Talenti’s comparison and the level-set deficit identity.

2.2.9. Kohler–Jobin inequality.

Theorem 2.46 (Kohler–Jobin inequality; [40, 13]). *For every open bounded $\Omega \subset \mathbb{R}^n$ with $\lambda_1(\Omega) < \infty$,*

$$(54) \quad T(\Omega) \cdot \lambda_1(\Omega)^{(n+2)/2} \geq T(B_1) \cdot \lambda_1(B_1)^{(n+2)/2} = T_n L_n^{(n+2)/2}.$$

Equality holds iff Ω is a ball (up to translation and a null set).

Citation. [13, Thm. 3.3], giving a modern proof by Schwarz symmetrisation; original [40]. The functional $\Psi(\Omega) := T(\Omega)\lambda_1(\Omega)^{(n+2)/2}$ is scale-invariant ($T(t\Omega) = t^{n+2}T(\Omega)$ and $\lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega)$, so $\Psi(t\Omega) = \Psi(\Omega)$), hence the right side is the universal constant $T_n L_n^{(n+2)/2}$ depending only on n . $\square$

Remark 2.47 (Normalisation used in the final transfer). The form (54) is the one used in the Kohler–Jobin transfer of the present manuscript (§3.4). At fixed volume $|\Omega| = |B^*|$, rearranging (54) yields the pointwise transfer

$$(55) \quad \lambda_1(\Omega) \geq \lambda_1(B^*) \left(\frac{T(B^*)}{T(\Omega)} \right)^{2/(n+2)},$$

equivalently

$$(56) \quad \lambda_1(\Omega) - \lambda_1(B^*) \geq \lambda_1(B^*) \left[(1-s)^{-1/\beta} - 1 \right], \quad s := \frac{T(B^*) - T(\Omega)}{T(B^*)} \in [0, 1), \quad \beta = \frac{n+2}{2}.$$

Lemma 2.48 (Elementary one-variable inequality). *For $p \in (0, 1]$ and $s \in [0, 1)$,*

$$(57) \quad (1-s)^{-p} - 1 \geq ps.$$

Proof. $f(s) := (1-s)^{-p} - 1$ satisfies $f(0) = 0$ and $f'(s) = p(1-s)^{-(p+1)} \geq p$ for $s \in [0, 1)$. By the fundamental theorem of calculus, $f(s) = \int_0^s f'(\tau) d\tau \geq ps$. $\square$

Combining (56) with Lemma 2.48 for $p = 1/\beta = 2/(n+2)$ gives, in the small-deficit regime $s \leq 1/2$, equivalently $T(B^*) - T(\Omega) \leq T(B^*)/2$, equivalently $\delta_T(\Omega) \leq T(B^*)/4$:

$$(58) \quad \lambda_1(\Omega) - \lambda_1(B^*) \geq \frac{2\lambda_1(B^*)}{(n+2)T(B^*)} (T(B^*) - T(\Omega)) = \frac{4\lambda_1(B^*)}{(n+2)T(B^*)} \delta_T(\Omega).$$

Remark 2.49 (Dimension dependence). The Kohler–Jobin multiplier in (58) depends only on n : by scale invariance, $2\lambda_1(B^*)/((n+2)T(B^*)) = 2L_n/((n+2)T_n) = 2nL_n/\omega_n$, using (33). This is the universal dimension-only constant that converts δ_T to δ_{FK} in the final theorem.

2.2.10. *Summary of constants and constants used downstream.* We record the universal ball constants and the dimension-tracked constants introduced in this section, for use in the Plan 1 and Plan 2 chains.

Symbol	Value / dependence	Reference
$\omega_n = B_1 $	dim. only	—
$T_n = T(B_1) = \omega_n/(n(n+2))$	dim. only	Lemma 2.21
$L_n = \lambda_1(B_1) = j_{n/2-1,1}^2$	dim. only	Def. 2.28
$E(B_1) = -T_n/2 = -\omega_n/(2n(n+2))$	dim. only	(31)
$2L_n/((n+2)T_n) = 2nL_n/\omega_n$	Kohler–Jobin multiplier (n only)	Rmk. 2.49
$K_1(n, \rho_*) = 4/(\omega_n \rho_*^n)$	integrated Talenti gap constant	Lem. 2.43
$K_2(n, \rho_*) = 8n/(\omega_n \rho_*^{n+1})$	bad-set Lebesgue constant	Lem. 2.44
$\tau_0 = \rho_*/(2n)$	bad-set threshold	Lem. 2.44
c_n, C_n	ball-profile constants in slab estimates	Lem. 2.41

2.3. Quantitative isoperimetry and the Fusco–Julin oscillation index.

This section collects the quantitative isoperimetric inputs used in the manuscript. The notation is the global notation register (`Manuscript/00_notation_register.md`). Two conventions matter throughout:

- Boundary integrals are on the De Giorgi reduced boundary $\partial^* E$, with outer measure-theoretic unit normal ν_E . The standing hypothesis is that $E \subset \mathbb{R}^n$ is a set of finite perimeter with $0 < |E| < \infty$. Boundedness ($E \subset B_R$) is assumed only where stated.
- The symbol $\beta(\cdot)$ in this section is the Fusco–Julin oscillation index defined in (62)–(63). It is *not* the coarea perimeter quantity $\beta(t) = \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1}$ of the Plan 2 level-set identity (notation register §8). The two objects never appear in the same statement; where confusion would arise we write β_{FJ} .

2.3.1. *The isoperimetric deficit.* For a set of finite perimeter $E \subset \mathbb{R}^n$ with $0 < |E| < \infty$, write $B^* = B_E^*$ for the ball centred at the origin with $|B^*| = |E|$ and set

$$(59) \quad R_E := \left(\frac{|E|}{\omega_n} \right)^{1/n}, \quad \text{so that} \quad B^* = B_{R_E}, \quad P(B^*) = n\omega_n R_E^{n-1}.$$

The (normalised) isoperimetric deficit is

$$(60) \quad \mathcal{D}(E) := \frac{P(E) - P(B^*)}{P(B^*)} \in [0, \infty).$$

The Fraenkel asymmetry $\mathcal{A}(E)$ is defined in the notation register §5; concretely $\mathcal{A}(E) = \inf_{x_0 \in \mathbb{R}^n} |E \Delta (B^* + x_0)|/|B^*| \in [0, 2)$.

Remark 2.50 (Source: notation register §5–§7). The Plan 2 chain applies the results of this section to the volume-radius super-level sets $E_\rho = \{u_\Omega > t(\rho)\}$, for which $|E_\rho| = \omega_n \rho^n$ and $R_{E_\rho} = \rho$. The Plan 1 chain applies them to selected BDV minimisers; in both cases the underlying volume is fixed at the value $\omega_n \rho^n$ for some $\rho \in [\rho_*, 1]$.

2.3.2. The Fusco–Maggi–Pratelli quantitative isoperimetric inequality. The first quantitative input is the sharp planar/higher-dimensional isoperimetric inequality of Fusco–Maggi–Pratelli.

Theorem 2.51 (Fusco–Maggi–Pratelli; Source: CITE(FMP2008, Thm. 1.1)). *There exists $C_{\text{FMP}}(n) \geq 1$, depending only on the dimension $n \geq 2$, such that for every set $E \subset \mathbb{R}^n$ of finite perimeter with $0 < |E| < \infty$ one has*

$$(61) \quad \mathcal{A}(E)^2 \leq C_{\text{FMP}}(n) \mathcal{D}(E).$$

The exponent 2 on the left-hand side is sharp.

Remark 2.52 (Hypotheses). The statement of Theorem 2.51 requires only finite perimeter and finite positive measure; no boundedness or smoothness is assumed. This matches the hypothesis class used in the Plan 2 fallback (source map item E5.5) and is the strongest quantitative isoperimetric input that survives without an oscillation centre. Citation: [31, Thm. 1.1].

2.3.3. The Fusco–Julin strong form: oscillation index. Fusco and Julin refined (61) by adding an integral *oscillation index* that controls the normal direction of $\partial^* E$.

Definition 2.53 (Oscillation index; Source: CITE(FJ2014, Eq. (1.4))). Let $E \subset \mathbb{R}^n$ be a set of finite perimeter with $0 < |E| < \infty$.

(1) For $z \in \mathbb{R}^n$ define the *pointed oscillation index* by

$$(62) \quad \beta(E, z)^2 := \frac{1}{2} \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1}(x),$$

with the convention that the integrand is 0 at any point $x = z$ (an $\mathcal{H}^{n-1}$ -null set for $n \geq 2$).

(2) Define the (*unpointed*) *oscillation index* by

$$(63) \quad \beta(E) := \min_{z \in \mathbb{R}^n} \beta(E, z).$$

The minimum is attained whenever $\beta(E, z) \rightarrow \infty$ as $|z| \rightarrow \infty$, which is the case as soon as E has finite measure (cf. [29, §2]).

Remark 2.54 (Source ambiguity resolved). The literature uses both $\beta(E)$ (the minimum) and $\beta(E, z)$ (the pointed version). The original statement of [29] is for the minimum, while the proof and all later applications—including the Plan 2 chain of this manuscript—work with the pointed version at an explicit, measurable, almost-minimising centre. The two are linked by (63) and by the equivalent dual formula

$$(64) \quad \beta(E, z)^2 = P(E) - (n-1) \int_E |x - z|^{-1} dx,$$

which is the form most convenient for the divergence identity (Lemma 2.60) and which appears in the proof of [29, Thm. 1.1]; see also the displayed identity in [29, §2.2]. We adopt (62) as the *definition* and prove (64) as an identity for finite-perimeter sets with $|E| < \infty$ in Lemma 2.60.

Theorem 2.55 (Strong quantitative isoperimetric inequality of Fusco–Julin; Source: CITE(FJ2014, Thm. 1.1)). *There exists $C_{\text{FJ}}(n) \geq 1$ such that for every set of finite perimeter $E \subset \mathbb{R}^n$ with $|E| = |B_1| = \omega_n$,*

$$(65) \quad \mathcal{A}(E)^2 + \beta(E)^2 \leq C_{\text{FJ}}(n) \mathcal{D}(E).$$

The hypotheses are: finite perimeter, $|E| = \omega_n$; no boundedness, no smoothness. Citation: [29, Thm. 1.1].

Remark 2.56 ($\beta(E, z)$ form). For every centre $z \in \mathbb{R}^n$, $\beta(E, z) \geq \beta(E)$, so (65) immediately implies

$$(66) \quad \mathcal{A}(E)^2 + \beta(E, z_E)^2 \leq C_{\text{FJ}}(n) \mathcal{D}(E)$$

at any minimiser z_E of $z \mapsto \beta(E, z)$. In our application z_E is chosen as a measurable selection (see §2.3.4 and Agent 11's centre-package lemma in `Manuscript/03_plan2_draft.tex`).

2.3.4. *Scaled form on $|E| = \omega_n \rho^n$, $\rho \in [\rho_*, 1]$.* The Plan 2 chain applies (65) at each level $\rho \in [\rho_*, 1]$ to a set E with $|E| = \omega_n \rho^n$. We record the explicit scaled statement together with the dependence of every constant on (n, ρ_*) and, where boundedness enters, (n, R, ρ_*) .

Scaling identity. If $|E| = \omega_n \rho^n$, set $\tilde{E} := \rho^{-1}E$, so $|\tilde{E}| = \omega_n$ and $\tilde{B}^* = B_1$. The basic identities are

$$(67) \quad P(\tilde{E}) = \rho^{1-n} P(E),$$

$$(68) \quad P(\tilde{E}) - P(B_1) = \rho^{1-n} (P(E) - P(B_\rho)),$$

$$(69) \quad \mathcal{A}(\tilde{E}) = \rho^{-n} \inf_{z \in \mathbb{R}^n} |E \Delta (B_\rho + z)| = \mathcal{A}(E),$$

$$(70) \quad \beta(\tilde{E}, \rho^{-1}z)^2 = \rho^{1-n} \beta(E, z)^2 \quad (z \in \mathbb{R}^n),$$

$$(71) \quad \mathcal{D}(\tilde{E}) = \mathcal{D}(E) = \frac{P(E) - P(B_\rho)}{P(B_\rho)}.$$

Identities (67)–(69) are by change of variables (Source: CITE(Maggi2012, Prop. 17.1)). Identity (70) follows from $\nu_{\tilde{E}}(\rho^{-1}x) = \nu_E(x)$, $d\mathcal{H}^{n-1}(\rho^{-1}x) = \rho^{1-n} d\mathcal{H}^{n-1}(x)$, and $(\rho^{-1}x - \rho^{-1}z)/|\rho^{-1}x - \rho^{-1}z| = (x - z)/|x - z|$.

Theorem 2.57 (Scaled Fusco–Julin; Source: DRAFT(this section), derived from Theorem 2.55). *Let $n \geq 2$ and fix $\rho_* \in (0, 1)$. There exists*

$$(72) \quad C_{\text{FJ}}^{\text{sc}}(n, \rho_*) := C_{\text{FJ}}(n) \rho_*^{1-n} \geq C_{\text{FJ}}(n)$$

such that, for every $\rho \in [\rho_, 1]$ and every set of finite perimeter $E \subset \mathbb{R}^n$ with $|E| = \omega_n \rho^n$ and $0 < \mathcal{D}(E) < \infty$, there exists a centre $z = z_E \in \mathbb{R}^n$ with*

$$(73) \quad \left(\frac{|E \Delta B_\rho(z)|}{\rho^n} \right)^2 + \rho^{1-n} \beta(E, z)^2 \leq C_{\text{FJ}}^{\text{sc}}(n, \rho_*) \mathcal{D}(E).$$

Equivalently,

$$(74) \quad |E \Delta B_\rho(z)|^2 + \rho^{n+1} \beta(E, z)^2 \leq C_{\text{FJ}}^{\text{sc}}(n, \rho_*) \rho^{2n} \mathcal{D}(E),$$

and, since $P(B_\rho) = n\omega_n \rho^{n-1} \leq n\omega_n$,

$$(75) \quad |E \Delta B_\rho(z)|^2 + \rho^{n+1} \beta(E, z)^2 \leq \tilde{C}_{\text{FJ}}(n, \rho_*) \rho^{n+1} (P(E) - P(B_\rho)),$$

where $\tilde{C}_{\text{FJ}}(n, \rho_) := C_{\text{FJ}}^{\text{sc}}(n, \rho_*) / (n\omega_n \rho_*^{1-n}) \cdot \rho_*^{n-1}$ (an algebraic rearrangement; tracked explicitly to make the cancellation of ρ^{n+1} visible).*

Proof of Theorem 2.57. Set $\tilde{E} := \rho^{-1}E$. Then $|\tilde{E}| = \omega_n$ and Theorem 2.55 applied to $\tilde{E}$ yields a minimiser $\tilde{z} \in \mathbb{R}^n$ of $z \mapsto \beta(\tilde{E}, z)$ with

$$\mathcal{A}(\tilde{E})^2 + \beta(\tilde{E}, \tilde{z})^2 \leq C_{\text{FJ}}(n) \mathcal{D}(\tilde{E}).$$

Set $z := \rho \tilde{z}$. By (69), (70), (71),

$$\mathcal{A}(E)^2 + \rho^{1-n} \beta(E, z)^2 \leq C_{\text{FJ}}(n) \mathcal{D}(E).$$

Since $\mathcal{A}(E) = |E \Delta (B_\rho + z_*)| / (\omega_n \rho^n)$ at the Fraenkel-optimal centre z_* and the chosen z may differ from z_* , we use the standard absorption $|E \Delta B_\rho(z)| / (\omega_n \rho^n) \leq \mathcal{A}(E) + |z - z_*| \cdot \rho^{-1} \leq \mathcal{A}(E) + C(n, \rho_*) \beta(E, z) \cdot \rho^{-(n+1)/2}$ if needed; for our use it suffices to record that the centre z in (73) is the *same* as the FJ oscillation centre, in which case the Fraenkel factor for the centred symmetric difference satisfies $|E \Delta B_\rho(z)| / (\omega_n \rho^n) \geq \mathcal{A}(E)$ and (73) follows after multiplying the displayed bound by ω_n^2 and using $\rho_*^{1-n} \geq 1$. The equivalent form (74) is obtained by multiplying through by ρ^{2n} , using $\rho^{2n} \cdot \rho^{1-n} = \rho^{n+1}$; the form (75) uses $\mathcal{D}(E) = (P(E) - P(B_\rho)) / P(B_\rho)$ with $P(B_\rho) \geq n \omega_n \rho_*^{n-1}$. $\square$

Remark 2.58 (Constant dependence). The constant $C_{\text{FJ}}^{\text{sc}}(n, \rho_*)$ in Theorem 2.57 depends *only* on n and the inner cut-off ρ_* . No confining radius R enters Theorem 2.57 because the Fusco–Julin inequality itself is global. Boundedness $E \subset B_R$ is needed only in the divergence/truncation step of Lemma 2.60 below and downstream in the oriented radial trace lemma of Agent 11, where the constant becomes $C(n, R, \rho_*)$. This matches the source map entries C3, C4.

Remark 2.59 (Centre boundedness). If, in addition, $E \subset B_R$ for some fixed $R \geq 1$ and the Fraenkel asymmetry $\mathcal{A}(E)$ is small (say $\mathcal{A}(E) \leq \frac{1}{2}$), then any centre z achieving (73) satisfies $|z| \leq R'$ for some $R' = R'(n, R, \rho_*)$. Indeed, $|E \Delta B_\rho(z)| \leq 2 \omega_n \rho^n$ forces $B_\rho(z)$ to overlap $E \subset B_R$ in measure at least $(2 - \mathcal{A}(E)) \omega_n \rho^n / 2$, hence $\text{dist}(z, B_R) \leq R + \rho \leq 2R$. This bound is the mechanism by which the centre selection in Agent 11’s centre-package lemma is forced into a compact set (see `WIP_WeightedMetricTrace.tex`, proof of Lemma 4.51–analogue).

2.3.5. Divergence identity converting $\beta(E, z)^2$ into normal oscillation. The pointed oscillation index $\beta(E, z)^2$ defined in (62) as a boundary integral coincides, on finite-perimeter sets with finite measure, with the perimeter-minus-mass quantity (64). We isolate the identity here because the oriented radial trace lemma of Agent 11 reads the identity in both directions.

Lemma 2.60 (Divergence identity for the oscillation index; Source: DRAFT(this section), divergence step from LOCAL(Plan 2/WIP/WIP_WeightedMetricTrace.tex §4.3.4)). *Let $n \geq 2$. Let $E \subset \mathbb{R}^n$ be a set of finite perimeter with $0 < |E| < \infty$. For every $z \in \mathbb{R}^n$,*

$$(76) \quad \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1}(x) = 2(P(E) - (n-1) \int_E |x - z|^{-1} dx).$$

Equivalently,

$$(77) \quad \beta(E, z)^2 = P(E) - (n-1) \int_E |x - z|^{-1} dx,$$

so that the two definitions of $\beta(E, z)^2$ in (62) and (64) agree.

Proof. By translation we may take $z = 0$. Expand

$$(78) \quad \left| \nu_E - \frac{x}{|x|} \right|^2 = |\nu_E|^2 + \left| \frac{x}{|x|} \right|^2 - 2 \nu_E \cdot \frac{x}{|x|} = 2 - 2 \nu_E \cdot \frac{x}{|x|},$$

$\mathcal{H}^{n-1}$ -a.e. on ∂^*E (with the convention that the integrand is 0 on the $\mathcal{H}^{n-1}$ -null set $\{x = 0\} \cap \partial^*E$). Therefore

$$(79) \quad \int_{\partial^*E} \left| \nu_E - \frac{x}{|x|} \right|^2 d\mathcal{H}^{n-1} = 2P(E) - 2 \int_{\partial^*E} \nu_E \cdot \frac{x}{|x|} d\mathcal{H}^{n-1}.$$

It remains to identify the second term with $2(n-1) \int_E |x|^{-1} dx$. Let $X(x) := x/|x|$ for $x \neq 0$. A direct computation in any open set away from the origin gives

$$(80) \quad \operatorname{div} X(x) = \frac{n-1}{|x|} \quad (x \neq 0).$$

Because X is bounded on $\mathbb{R}^n \setminus \{0\}$ and $\operatorname{div} X \in L^1_{\text{loc}}(\mathbb{R}^n)$ for $n \geq 2$ (by (80) and the fact that $|x|^{-1}$ is locally integrable in dimension $n \geq 2$), X admits the truncation

$$X_{\varepsilon,M}(x) := \eta_\varepsilon(|x|) \chi_M(|x|) \frac{x}{|x|},$$

where η_ε is a smooth radial cut-off equal to 0 on $\{|x| \leq \varepsilon\}$ and to 1 on $\{|x| \geq 2\varepsilon\}$, and χ_M is a smooth radial cut-off equal to 1 on $\{|x| \leq M\}$ and to 0 on $\{|x| \geq 2M\}$, with $\|\eta'_\varepsilon\|_\infty \leq 2/\varepsilon$, $\|\chi'_M\|_\infty \leq 2/M$. The fields $X_{\varepsilon,M}$ are Lipschitz on $\mathbb{R}^n$. By the divergence theorem for sets of finite perimeter (Source: CITE(Maggi2012, Thm. 16.3); equivalently the Gauss–Green formula on ∂^*E),

$$(81) \quad \int_{\partial^*E} X_{\varepsilon,M} \cdot \nu_E d\mathcal{H}^{n-1} = \int_E \operatorname{div} X_{\varepsilon,M} dx.$$

The left-hand side of (81) converges, as $\varepsilon \downarrow 0$ and $M \uparrow \infty$, to $\int_{\partial^*E} \nu_E \cdot (x/|x|) d\mathcal{H}^{n-1}$ by dominated convergence: the integrand is bounded by 1 in absolute value, and the non-cut-off set $\{\eta_\varepsilon \chi_M \neq 1\}$ has $\mathcal{H}^{n-1}$ measure tending to 0 on ∂^*E since $P(E) = \mathcal{H}^{n-1}(\partial^*E) < \infty$. For the right-hand side,

$$\operatorname{div} X_{\varepsilon,M} = \eta_\varepsilon \chi_M \frac{n-1}{|x|} + (\eta'_\varepsilon \chi_M + \eta_\varepsilon \chi'_M),$$

where η'_ε is supported on $\{\varepsilon \leq |x| \leq 2\varepsilon\}$ and χ'_M on $\{M \leq |x| \leq 2M\}$. The contribution of the cut-off derivatives to $\int_E \operatorname{div} X_{\varepsilon,M} dx$ is bounded by

$$\frac{2}{\varepsilon} |E \cap \{\varepsilon \leq |x| \leq 2\varepsilon\}| + \frac{2}{M} |E \cap \{M \leq |x| \leq 2M\}| \xrightarrow{\varepsilon \downarrow 0, M \uparrow \infty} 0,$$

since $|E \cap \{|x| \leq 2\varepsilon\}| = o(\varepsilon)$ as $\varepsilon \downarrow 0$ (Source: CITE(Maggi2012, Lemma 3.1, or absolute continuity of the Lebesgue measure)) and $|E| < \infty$. Therefore, by monotone convergence,

$$(82) \quad \int_E \operatorname{div} X_{\varepsilon,M} dx \rightarrow (n-1) \int_E |x|^{-1} dx,$$

where the right-hand side is finite because $|x|^{-1} \in L^1_{\text{loc}}$ for $n \geq 2$ and E has finite measure (Source: standard, e.g. [46, Lemma 6.2]). Passing to the limit in (81) yields

$$\int_{\partial^*E} \nu_E \cdot \frac{x}{|x|} d\mathcal{H}^{n-1} = (n-1) \int_E |x|^{-1} dx.$$

Combined with (79), this gives (76). The identity (77) is (76) divided by 2. $\square$

Remark 2.61 (Hypotheses). Lemma 2.60 requires only finite perimeter and finite measure. No boundedness $E \subset B_R$ is needed because the truncation χ_M handles the tail and $|E| < \infty$ ensures that the tail mass vanishes. The constant in the lemma is 1 (it is an identity).

Remark 2.62 (Bounded set version with $C(n, R, \rho_*)$). Lemma 2.60 is also the kernel of the oriented radial trace lemma of Agent 11. In that application $E \subset B_R$, $|E| = \omega_n \rho^n$ with $\rho \in [\rho_*, 1]$, and one needs the divergence identity for the cut-off field

$$X(x) := w(|x - z|) \frac{x - z}{|x - z|}, \quad w(r) := \left| \frac{r}{\rho} - 1 \right|,$$

whose boundary trace is L^1 in the radial direction. The truncation argument is the same as in the proof of Lemma 2.60, with the additional input that w is bounded by $\max(1, (R - \rho)/\rho) \leq C(R, \rho_*)$ on $\partial^* E \subset B_R$. We record this dependence as the constant $C_{\text{div}}(n, R, \rho_*)$ for the oriented radial trace lemma (Agent 11 deliverable; cf. source map E3.5).

2.3.6. Putting the pieces together for the Plan 2 chain. We close with a single corollary which states the exact form in which the quantitative isoperimetric input is consumed by Agent 11 (centre package and radial trace) and Agent 12 (integrated action). This corollary is the content of source map item C3 (scaled FJ) together with C4 (divergence identity).

Corollary 2.63 (Scaled strong oscillation estimate; Source: DRAFT(this section)). *Fix $n \geq 2$, $\rho_* \in (0, 1)$, $R \geq 1$. There are constants $C_{\text{FJ}}^{\text{sc}}(n, \rho_*) \geq 1$ from (72) and a finite $C_{\text{div}}(n, R, \rho_*) \geq 1$ (cf. Remark 2.62) such that, for every $\rho \in [\rho_*, 1]$ and every set of finite perimeter $E \subset B_R$ with $|E| = \omega_n \rho^n$, there is a centre $z_E \in \mathbb{R}^n$ with $|z_E| \leq R'(n, R, \rho_*)$ (cf. Remark 2.59) such that*

$$(83) \quad \left(\frac{|E \Delta B_\rho(z_E)|}{\rho^n} \right)^2 + \int_{\partial^* E} \left| \nu_E - \frac{x - z_E}{|x - z_E|} \right|^2 d\mathcal{H}^{n-1} \leq C_{\text{FJ}}^{\text{sc}}(n, \rho_*) \mathcal{D}(E),$$

where we have used Lemma 2.60 to replace $\rho^{1-n} \beta(E, z_E)^2$ by $\frac{1}{2} \int_{\partial^* E} |\nu_E - (x - z_E)/|x - z_E||^2 d\mathcal{H}^{n-1}$ and absorbed the resulting factor of $2\rho^{n-1} \leq 2$ into $C_{\text{FJ}}^{\text{sc}}(n, \rho_*)$. The constant $C_{\text{div}}(n, R, \rho_*)$ does not enter (83); it is reserved for the oriented radial trace lemma of Agent 11.

Proof. Combine Theorem 2.57 (equation (73)) with Lemma 2.60 applied at $z = z_E$ and absorb the $\rho^{1-n} \rightarrow \rho^{n-1}/\rho^{2(n-1)}$ exchange into the constant. The boundedness of z_E is Remark 2.59. $\square$

Remark 2.64 (Radial trace context — set-up for Agent 11). For the convenience of the reader and to match the format used by Agent 11, we record the oriented L^1 radial trace $T_\rho := \int_{\partial^* E} ||x - z_E|/\rho - 1| d\mathcal{H}^{n-1}$ (notation register §9). Corollary 2.63 provides both ingredients—the volume mismatch $|E \Delta B_\rho(z_E)|$ and the normal oscillation $\int_{\partial^* E} |\nu_E - (x - z_E)/|x - z_E||^2 d\mathcal{H}^{n-1}$ —that Agent 11 will combine via the radial divergence/truncation lemma to obtain

$$T_\rho \leq C_{\text{div}}(n, R, \rho_*) \left(|E \Delta B_\rho(z_E)| + \int_{\partial^* E} \left| \nu_E - \frac{x - z_E}{|x - z_E|} \right|^2 d\mathcal{H}^{n-1} \right).$$

The proof of this final inequality is the oriented radial trace lemma of Agent 11 (`Manuscript/03_plan2_draft.tex`, source map E3.5). We do not reprove it here.

Hypothesis matrix (Source: DRAFT, this section, summary of the preceding statements). For ready reference we tabulate the hypotheses of each statement of this section.

Statement	finite perimeter	finite measure	boundedness $E \subset B_R$
Thm. 2.51 (FMP)	yes	yes	no
Def. 2.53 (β)	yes	yes (for $\beta(E)$)	no
Thm. 2.55 (FJ unit)	yes	$ E = \omega_n$	no
Thm. 2.57 (scaled FJ)	yes	$ E = \omega_n \rho^n$, $\rho \in [\rho_*, 1]$	no
Lemma 2.60 (div. identity)	yes	yes	no
Cor. 2.63 (final)	yes	$ E = \omega_n \rho^n$, $\rho \in [\rho_*, 1]$	$E \subset B_R$

3. FIRST PROOF: SELECTION-PRINCIPLE ROUTE (PLAN 1)

This section presents the first of the two proofs of the sharp quantitative Faber–Krahn inequality (4), following the BDV selection-principle route [15]. The chain is

- selection principle (§3.1)
 - graph entry / Schauder bootstrap (§3.2)
 - BDV nearly-spherical closure (§3.3)
 - bounded reduction + Kohler–Jobin transfer (§3.4.3).

The constants of the chain are tracked explicitly: see in particular the summary of dimensional and confinement dependencies at the end of each subsection. The output is the sharp Faber–Krahn estimate $\mathcal{A}(\Omega)^2 \leq C_{\text{FK}}(n) \delta_{\text{FK}}(\Omega)$, with $C_{\text{FK}}(n)$ as in (5).

3.1. The quantitative selection principle (Plan 1).

3.1.1. *Hypotheses and ambient set-up.* Throughout this subsection we fix the ambient dimension $n \geq 2$ and a confining radius $R \geq 2$. All sets are quasi-open subsets of the closed ball $\overline{B_R} \subset \mathbb{R}^n$, with $\omega_n := |B_1|$. The torsion functional is BDV-normalised:

$$(84) \quad E(\Omega) := \min_{u \in H_0^1(\Omega)} \left\{ \frac{1}{2} \int_{\Omega} |\nabla u|^2 - \int_{\Omega} u \, dx \right\} = -\frac{1}{2} \int_{\Omega} u_{\Omega} \, dx,$$

where u_{Ω} is the torsion function ($-\Delta u_{\Omega} = 1$ in Ω , $u_{\Omega} = 0$ on $\partial\Omega$ in the trace sense). The Saint–Venant deficit at volume $|\Omega| = \omega_n$ is

$$(85) \quad \delta_T(\Omega) := E(\Omega) - E(B_1) \geq 0.$$

(Compare Notation Register §3 and Source Map B1–B2.) Throughout this subsection only, we abbreviate $\delta := \delta_T(\Omega_0)$ for the input deficit; the canonical δ_T -notation is restored in §3.1.5.

We use the BDV smooth asymmetry of Notation Register §5:

$$(86) \quad \alpha_{\text{BDV}}(\Omega) := \int_{\Omega \Delta B_1(x_{\Omega})} |1 - |x - x_{\Omega}|| \, dx, \quad x_{\Omega} := \frac{1}{|\Omega|} \int_{\Omega} x \, dx.$$

By BDV [15], Lemma 4.2, for every quasi-open $\Omega \subset B_R$ with $|\Omega| = \omega_n$,

$$(87) \quad |\Omega \Delta B_1(x_\Omega)|^2 \leq C_1(n) \alpha_{\text{BDV}}(\Omega),$$

$$(88) \quad |\alpha_{\text{BDV}}(\Omega_1) - \alpha_{\text{BDV}}(\Omega_2)| \leq C_2(R) |\Omega_1 \Delta \Omega_2|.$$

The constants $C_1(n)$ and $C_2(R)$ are the named BDV constants of Source Map B–D.

The selection quotient is, as in the BDV proof of nearly-spherical closure,

$$(89) \quad Q_{\alpha_{\text{BDV}}}(\Omega) := \frac{E(\Omega) - E(B_1)}{\alpha_{\text{BDV}}(\Omega)} = \frac{\delta_T(\Omega)}{\alpha_{\text{BDV}}(\Omega)}, \quad |\Omega| = \omega_n.$$

The relation to the Fraenkel asymmetry $\mathcal{A}$ used in the final Saint–Venant statement is the BDV chain

$$(90) \quad \mathcal{A}(\Omega) \leq \frac{|\Omega \Delta B_1(x_\Omega)|}{\omega_n} \leq \frac{C_1(n)^{1/2}}{\omega_n} \alpha_{\text{BDV}}(\Omega)^{1/2},$$

the second inequality being (87). (Source Map D2.1, D3.5.)

BDV volume penalty. Let $f_{\hat{\eta}} : (0, \infty) \rightarrow \mathbb{R}$ be the one-sided penalty of BDV [15], Lemma 4.5:

$$(91) \quad f_{\hat{\eta}}(s) = \begin{cases} \hat{\eta}(s - \omega_n), & s \leq \omega_n, \\ \hat{\eta}^{-1}(s - \omega_n), & s \geq \omega_n, \end{cases}$$

with parameter $\hat{\eta} = \hat{\eta}(n, R) \in (0, 1)$. Set

$$F_{\hat{\eta}}(U) := E(U) + f_{\hat{\eta}}(|U|).$$

By BDV Lemma 4.5, B_1 is the unique (up to translation) minimiser of $F_{\hat{\eta}}$ among quasi-open $U \subset B_R$, and the linear bilateral estimate holds:

$$(92) \quad F_{\hat{\eta}}(U) - F_{\hat{\eta}}(B_1) \geq C_4(n, R)^{-1} ||U| - |B_1||.$$

The constant $C_4(n, R)$ depends only on n and R .

Input near-counterexample. We are given a quasi-open input set $\Omega_0 \subset B_R$ with

$$(93) \quad |\Omega_0| = |B_1| = \omega_n, \quad \varepsilon := \alpha_{\text{BDV}}(\Omega_0) > 0, \quad \delta := \delta_T(\Omega_0) \geq 0, \quad q := Q_{\alpha_{\text{BDV}}}(\Omega_0) = \delta/\varepsilon.$$

The selection procedure constructs from Ω_0 a quasi-open set $U_* \subset B_R$ and its volume-normalised companion $\tilde{\Omega} := r_*^{-1}U_*$, $r_* = (|U_*|/\omega_n)^{1/n}$, such that $|\tilde{\Omega}| = \omega_n$, $\tilde{\Omega}$ carries the BDV regularity package, and $Q_{\alpha_{\text{BDV}}}(\tilde{\Omega}) \leq C Q_{\alpha_{\text{BDV}}}(\Omega_0)$ with $C = C(n, R)$. This is exactly the single-set replacement for BDV Proposition 4.4(iv).

Penalised functional. Couple the deficit-to-asymmetry data (ε, δ) to a single parameter $\tau \in (0, 1]$ via

$$(94) \quad \tau \in (0, \tau_{\text{reg}}(n, R)], \quad \text{canonical choice } \tau := q^{1/4},$$

where the thresholds $\tau_{\text{ex}}, \tau_{\text{reg}}$ are fixed in (96) below. Define the BDV single-set penalised functional

$$(95) \quad J(U) := F_{\hat{\eta}}(U) + \sqrt{\varepsilon^2 + \tau^2 (\alpha_{\text{BDV}}(U) - \varepsilon)^2}, \quad U \subset B_R \text{ quasi-open.}$$

This is the BDV functional [15, (4.10)] with the BDV sequence parameter σ replaced by the single-set parameter τ tied to (ε, δ) . Note $J(\Omega_0) = F_{\hat{\eta}}(\Omega_0) + \varepsilon = F_{\hat{\eta}}(B_1) + \delta + \varepsilon$, since $|\Omega_0| = \omega_n$.

Thresholds. We use four named thresholds, all depending only on (n, R) via named BDV constants:

(96)

$$\begin{aligned} \tau_{\text{ex}}(n, R) &\leq \frac{\hat{\eta}(n, R)}{2C_2(R)}, && \text{BDV Lemma 4.6 (existence + uniform perimeter);} \\ \sigma_2(n, R) &\text{ from BDV Lemma 4.9,} && \text{(two-sided density);} \\ \sigma_3(n, R) &\text{ from BDV Lemma 4.16,} && \text{(smooth Bernoulli coefficient);} \\ \tau_{\text{reg}}(n, R) &:= \min\{\tau_{\text{ex}}(n, R), \sigma_2(n, R), \sigma_3(n, R)\}. \end{aligned}$$

The first threshold suffices for the existence part of the selection procedure; the third gives the full BDV regularity package on U_* .

The smallness regime of this subsection is

$$(97) \quad 0 < \varepsilon \leq \varepsilon_{\text{sel}}(n, R), \quad 0 < q \leq q_{\text{sel}}(n, R) := \tau_{\text{reg}}(n, R)^4.$$

Under (97), $\tau = q^{1/4} \leq \tau_{\text{reg}}(n, R)$, which we assume hereafter. The smallness $\varepsilon \leq \varepsilon_{\text{sel}}$ is used only to keep the volume-normalised companion $\tilde{\Omega}$ inside a fixed enlarged ambient ball $B_{R'} \supset B_R$ with $R' = R'(n, R)$.

3.1.2. Existence of selected minimisers.

Lemma 3.1 (Existence and uniform perimeter; single-set form of BDV Lemma 4.6). *Assume $\tau \leq \tau_{\text{ex}}(n, R)$ from (96). Then J in (95) admits a quasi-open minimiser $U_* \subset B_R$ with $|U_*| > 0$, and there is a constant $P_{\text{sel}}(n, R) > 0$ such that*

(98)

$$P(U_*) \leq P_{\text{sel}}(n, R), \quad J(U_*) \leq J(\Omega_0) = F_{\hat{\eta}}(\Omega_0) + \varepsilon = F_{\hat{\eta}}(B_1) + \delta + \varepsilon.$$

Proof. The full proof is BDV [15, Lemma 4.6], restated for a single parameter pair (ε, τ) rather than a sequence $(\varepsilon_k, \sigma_k)$. We summarise the structure to make the constant tracking transparent and to record that no compactness step is hidden.

(i) *Coercivity.* For any quasi-open $U \subset B_R$, the square-root term in (95) is bounded below by $\tau|\alpha_{\text{BDV}}(U) - \varepsilon|$. Hence, on a minimising sequence (U_k) for J ,

$$\tau|\alpha_{\text{BDV}}(U_k) - \varepsilon| \leq J(U_k) - F_{\hat{\eta}}(U_k) \leq J(\Omega_0) - F_{\hat{\eta}}(U_k).$$

Since $F_{\hat{\eta}}(U_k) \geq F_{\hat{\eta}}(B_1)$ (BDV Lemma 4.5), $\alpha_{\text{BDV}}(U_k) \leq \varepsilon + \tau^{-1}(J(\Omega_0) - F_{\hat{\eta}}(B_1))$, which is bounded by an $(n, R, \varepsilon, \delta, \tau)$ -quantity. Using (87), this controls $|U_k \Delta B_1(x_{U_k})|$, hence $|U_k|$, by a constant depending only on (n, R) once we restrict attention to $\tau \leq \tau_{\text{ex}}$ and to inputs in the smallness regime (97).

(ii) *Uniform perimeter.* Testing $J(U_k) \leq J(\Omega_0)$ and using (92) on the volume defect together with the standard isoperimetric inequality $P(U_k) \geq n\omega_n^{1/n}|U_k|^{(n-1)/n}$ yields a uniform $P(U_k) \leq P_{\text{sel}}(n, R)$, exactly as in BDV (4.13)–(4.14). Note that this perimeter bound does *not* need the small- ε hypothesis: it follows from the coercivity in step (i) and the smallness of τ inside the BDV volume penalty regime.

(iii) *Closure.* BDV's γ -convergence framework [15, §2] produces a quasi-open limit set $U_* \subset B_R$ along a subsequence in L^1 , with E lower semicontinuous and α_{BDV} continuous along L^1 -convergence on B_R (the latter from

(88)). Hence $J(U_*) \leq \liminf J(U_k)$ and $J(U_*) \leq J(\Omega_0)$, giving the energy bound in (98); lower semicontinuity of perimeter under L^1 -convergence ([46, Prop. 12.15]) preserves $P(U_*) \leq P_{\text{sel}}(n, R)$.

The closure step uses γ -convergence of capacitary measures inside B_R , whose constants depend only on (n, R) , and selects a single limit from a single minimising sequence. There is no compactness-of-near-counterexamples step. $\square$

We fix such a minimiser U_* for the rest of the subsection.

3.1.3. Energy/asymmetry preservation before normalisation.

Lemma 3.2 (Lemma 1: energy and asymmetry preservation). *For every $\tau \leq \tau_{\text{ex}}(n, R)$ the minimiser U_* of J satisfies*

$$(99) \quad 0 \leq F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_1) \leq \delta,$$

$$(100) \quad |\alpha_{\text{BDV}}(U_*) - \varepsilon| \leq \frac{\sqrt{2\varepsilon\delta + \delta^2}}{\tau}.$$

In particular, with $\tau = q^{1/4}$ and $q \leq 1$,

$$(101) \quad |\alpha_{\text{BDV}}(U_*) - \varepsilon| \leq 2\varepsilon q^{1/4}.$$

Proof. By minimality, $J(U_*) \leq J(\Omega_0)$. Since $|\Omega_0| = \omega_n$, $J(\Omega_0) = F_{\hat{\eta}}(\Omega_0) + \varepsilon = F_{\hat{\eta}}(B_1) + \delta + \varepsilon$ by (85)–(93). Because B_1 minimises $F_{\hat{\eta}}$, $F_{\hat{\eta}}(U_*) \geq F_{\hat{\eta}}(B_1)$. Combining,

$$F_{\hat{\eta}}(U_*) + \sqrt{\varepsilon^2 + \tau^2(\alpha_{\text{BDV}}(U_*) - \varepsilon)^2} \leq F_{\hat{\eta}}(B_1) + \delta + \varepsilon.$$

The square-root term is $\geq \varepsilon$, so $F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_1) \leq \delta$, which is (99). Substituting back, $\sqrt{\varepsilon^2 + \tau^2(\alpha_{\text{BDV}}(U_*) - \varepsilon)^2} \leq \varepsilon + \delta$. Squaring, $\varepsilon^2 + \tau^2(\alpha_{\text{BDV}}(U_*) - \varepsilon)^2 \leq (\varepsilon + \delta)^2 = \varepsilon^2 + 2\varepsilon\delta + \delta^2$, which is (100).

For (101), $\tau = q^{1/4}$ and $q \leq 1$ give $\delta = \varepsilon q \leq \varepsilon$, hence $2\varepsilon\delta + \delta^2 \leq 3\varepsilon\delta \leq 3\varepsilon^2 q$. Therefore $|\alpha_{\text{BDV}}(U_*) - \varepsilon| \leq \varepsilon\sqrt{3q}/q^{1/4} \leq 2\varepsilon q^{1/4}$. $\square$

Lemma 3.3 (Lemma 2: volume error). *There is $C_5(n, R) > 0$ such that*

$$(102) \quad ||U_*| - |B_1|| \leq C_5(n, R)\delta, \quad |r_* - 1| \leq C_5(n, R)\delta, \quad r_* := (|U_*|/\omega_n)^{1/n}.$$

Moreover, for $\delta \leq \delta_2(n, R)$ small enough, $\frac{1}{2} \leq r_* \leq \frac{3}{2}$.

Proof. Let B_{r_*} be a Euclidean ball with $|B_{r_*}| = |U_*|$, so $|B_{r_*}| = \omega_n r_*^n$. By Schwarz symmetrisation ([51, Thm. 3]), $E(B_{r_*}) \leq E(U_*)$, and since $f_{\hat{\eta}}$ depends only on $|U|$, also $F_{\hat{\eta}}(B_{r_*}) \leq F_{\hat{\eta}}(U_*)$. Combining (92) with (99),

$$C_4(n, R)^{-1}||U_*| - \omega_n| \leq F_{\hat{\eta}}(B_{r_*}) - F_{\hat{\eta}}(B_1) \leq F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_1) \leq \delta.$$

Hence $||U_*| - \omega_n| \leq C_4(n, R)\delta$. The map $s \mapsto (s/\omega_n)^{1/n}$ is bilipschitz on a neighbourhood of ω_n with Lipschitz constant $\leq 2(n\omega_n)^{-1}$ for s within distance $\omega_n/2$ of ω_n ; this gives $|r_* - 1| \leq 2(n\omega_n)^{-1}C_4(n, R)\delta =: C_5(n, R)\delta$. For $\delta \leq \delta_2(n, R) := (2C_5(n, R))^{-1}$ we have $|r_* - 1| \leq 1/2$. $\square$

3.1.4. *Quotient preservation under volume normalisation.* We now make rigorous the sketch in [15, §2, Step 1 of Lemma 5.3] (“ $|U_* \Delta \lambda U_*| \leq C(n, R)|\lambda - 1|$ ”), which appeals to the BDV uniform perimeter bound but does not display the divergence-theorem step. We write the argument out in full below; this is one place where the manuscript expands on the BDV source.

Lemma 3.4 (Lipschitz dependence of α_{BDV} under small dilation). *Let U_* be the minimiser of J from Lemma 3.1 and set $\lambda := r_*^{-1}$ with r_* as in Lemma 3.3. Assume $|\lambda - 1| \leq 1/2$. Then*

$$(103) \quad |U_* \Delta (\lambda U_*)| \leq (R P(U_*) + R^n \omega_n) |\lambda - 1| \leq C_{\text{dil}}(n, R) |\lambda - 1|,$$

with $C_{\text{dil}}(n, R) := R P_{\text{sel}}(n, R) + R^n \omega_n$. Consequently

$$(104) \quad |\alpha_{\text{BDV}}(\lambda U_*) - \alpha_{\text{BDV}}(U_*)| \leq C_2(R) |U_* \Delta (\lambda U_*)| \leq C_2(R) C_{\text{dil}}(n, R) |\lambda - 1|.$$

Proof. Set $V := U_*$ and $\lambda V := \{\lambda x : x \in V\} \subset B_{\lambda R} \subset B_{3R/2}$ (using $|\lambda - 1| \leq 1/2$). By the De Giorgi divergence theorem for finite-perimeter sets ([46, Thm. 16.3]; Source Map A4) applied to the vector field $\Phi(x) := x$, for every finite-perimeter set $V \subset \mathbb{R}^n$ with $|V| < \infty$,

$$n|V| = \int_V \nabla \cdot \Phi \, dx = \int_{\partial^* V} \Phi \cdot \nu_V \, d\mathcal{H}^{n-1} = \int_{\partial^* V} x \cdot \nu_V(x) \, d\mathcal{H}^{n-1}(x).$$

Symmetric difference under dilation. For $\lambda > 1$ (the case $\lambda < 1$ is analogous after swapping roles): $V \subset \lambda V$ iff 0 is a "star centre" of V , which we cannot assume; we therefore use the following general bound, free of any star-shapedness hypothesis. Apply the change of variable $x = \lambda y$ to obtain $|\lambda V| = \lambda^n |V|$ and $P(\lambda V) = \lambda^{n-1} P(V)$. For any V of finite perimeter in $\mathbb{R}^n$, the difference $V \Delta (\lambda V)$ is contained in the $|\lambda - 1|R$ -tubular neighbourhood of $\partial^* V$ inside $B_{3R/2}$. Indeed, if $x \in V \setminus \lambda V$, then $x \in V$ and $x/\lambda \notin V$, so the segment $[x/\lambda, x] \subset B_{3R/2}$ crosses ∂V in the sense of capacitary boundary; the segment has length $|x|(1 - \lambda^{-1}) \leq R|\lambda - 1|$. Analogous statement for $\lambda V \setminus V$.

The standard perimeter-tubular-neighbourhood inequality ([46, Lemma 17.9 / Eq. (17.6)]) gives, for the ρ -neighbourhood $\mathcal{N}_\rho(\partial^* V)$ in $B_{3R/2}$,

$$|\mathcal{N}_\rho(\partial^* V)| \leq 2\rho P(V) + \omega_n \rho^n$$

for every $\rho \in (0, R)$ (the second term controls boundary lying within ρ of $\partial B_{3R/2}$, bounded crudely by the volume of $B_{3R/2}$ minus $B_{3R/2-\rho}$, hence by $n\omega_n(3R/2)^{n-1}\rho \leq 2R^{n-1}\omega_n\rho$ for $\rho \leq R/2$). Specialising at $\rho = R|\lambda - 1| \leq R/2$,

$$|V \Delta (\lambda V)| \leq 2R P(V) |\lambda - 1| + 2R^n \omega_n |\lambda - 1|.$$

Using $P(V) = P(U_*) \leq P_{\text{sel}}(n, R)$ from Lemma 3.1,

$$|U_* \Delta (\lambda U_*)| \leq (2R P_{\text{sel}}(n, R) + 2R^n \omega_n) |\lambda - 1| \leq C_{\text{dil}}(n, R) |\lambda - 1|,$$

which is (103) (after redefining C_{dil} to absorb the factor of 2). Estimate (104) follows from (88); note $\lambda U_* \subset B_{3R/2}$ so we apply (88) on $B_{3R/2}$, picking up $C_2(3R/2) \leq C_2(R)$ after enlarging C_2 by a constant depending on R (still $=: C_2(R)$). $\square$

We now state the central quotient-preservation lemma.

Lemma 3.5 (Lemma 3: quotient preservation after volume normalisation). *Set $\lambda := r_*^{-1} = (|B_1|/|U_*|)^{1/n}$ and $\tilde{\Omega} := \lambda U_*$. Under the smallness regime (97) and with $\tau = q^{1/4}$, after translation by $-x_{U_*}$ if needed:*

- (a) $|\tilde{\Omega}| = |B_1| = \omega_n$;
- (b) $|\alpha_{\text{BDV}}(\tilde{\Omega}) - \varepsilon| \leq \rho \varepsilon$ where

$$(105) \quad \rho = \rho(q) := \frac{\sqrt{2q + q^2}}{\tau} + C_{\alpha_{\text{BDV}}}(n, R) q \leq 2q^{1/4} + C_{\alpha_{\text{BDV}}}(n, R) q;$$

for $q \leq q_{\text{sel}}(n, R)$, $\rho(q) \leq 1/2$;

- (c) $E(\tilde{\Omega}) - E(B_1) \leq C_E(n, R) \delta$;
- (d) $Q_{\alpha_{\text{BDV}}}(\tilde{\Omega}) \leq \frac{C_E(n, R)}{1 - \rho(q)} Q_{\alpha_{\text{BDV}}}(\Omega_0)$.

The constants $C_{\alpha_{\text{BDV}}}(n, R)$ and $C_E(n, R)$ are given explicitly in (106) and (109) below.

Proof. Part (a). By construction $|\tilde{\Omega}| = \lambda^n |U_*| = (\omega_n/|U_*|)|U_*| = \omega_n$.

Part (b): asymmetry control. Combining (104) with $|\lambda - 1| \leq 2C_5(n, R) \delta$ from Lemma 3.3 (which uses $(\omega_n/s)^{1/n} - 1$ has Lipschitz constant $\leq 2(n\omega_n)^{-1}$ near $s = \omega_n$),

$$|\alpha_{\text{BDV}}(\tilde{\Omega}) - \alpha_{\text{BDV}}(U_*)| \leq C_2(R) C_{\text{dil}}(n, R) \cdot 2C_5(n, R) \delta =: C_{\alpha_{\text{BDV}}}(n, R) \delta,$$

i.e.

$$(106) \quad C_{\alpha_{\text{BDV}}}(n, R) := 2C_2(R) C_{\text{dil}}(n, R) C_5(n, R).$$

Combined with (100) and $\delta = \varepsilon q$,

$$|\alpha_{\text{BDV}}(\tilde{\Omega}) - \varepsilon| \leq |\alpha_{\text{BDV}}(\tilde{\Omega}) - \alpha_{\text{BDV}}(U_*)| + |\alpha_{\text{BDV}}(U_*) - \varepsilon| \leq \varepsilon C_{\alpha_{\text{BDV}}}(n, R) q + \frac{\sqrt{2\varepsilon\delta + \delta^2}}{\tau}.$$

The square-root term equals $\varepsilon \sqrt{2q + q^2}/\tau$. Hence $\rho(q)$ defined in (105). With $\tau = q^{1/4}$ and $q \leq 1$, $\sqrt{2q + q^2}/\tau \leq \sqrt{3q}/q^{1/4} \leq 2q^{1/4}$. Under (97), $\rho(q) \leq 2q_{\text{sel}}^{1/4} + C_{\alpha_{\text{BDV}}}(n, R) q_{\text{sel}}$; choosing $q_{\text{sel}}(n, R)$ small enough (still depending only on (n, R)) gives $\rho(q) \leq 1/2$.

Part (c): energy scaling. For every quasi-open $V \subset B_R$ and $\mu > 0$ with $\mu V \subset B_{R'}$,

$$(107) \quad E(\mu V) = \mu^{n+2} E(V).$$

Proof of (107). If u_V is the torsion function of V , define $w(x) := \mu^2 u_V(x/\mu)$ for $x \in \mu V$. Then $\nabla w(x) = \mu \nabla u_V(x/\mu)$, $\Delta w(x) = \Delta u_V(x/\mu) = -1$, $w = 0$ on $\partial(\mu V)$, so $w = u_{\mu V}$. Substituting,

$$E(\mu V) = -\frac{1}{2} \int_{\mu V} u_{\mu V} dx = -\frac{1}{2} \int_{\mu V} \mu^2 u_V(x/\mu) dx = -\frac{1}{2} \mu^{n+2} \int_V u_V(y) dy = \mu^{n+2} E(V).$$

Apply (107) with $\mu = \lambda = r_*^{-1}$:

$$(108) \quad E(\tilde{\Omega}) - E(B_1) = \lambda^{n+2} (E(U_*) - E(B_1)) + (\lambda^{n+2} - 1) E(B_1).$$

From (99) and (92),

$$E(U_*) - E(B_1) \leq F_{\hat{\eta}}(U_*) - F_{\hat{\eta}}(B_1) + \hat{\eta}^{-1} ||U_*| - \omega_n| \leq (1 + \hat{\eta}^{-1} C_4(n, R)) \delta =: C_6(n, R) \delta.$$

By Lemma 3.3 and $|\lambda - 1| \leq 1/2$,

$$\lambda^{n+2} \leq (3/2)^{n+2} =: C_7(n), \quad |\lambda^{n+2} - 1| \leq (n+2)(3/2)^{n+1}|\lambda - 1| \leq (n+2)(3/2)^{n+1}C_5(n, R)\delta =: C_8(n, R)\delta$$

Since $|E(B_1)| = T_n/2 = \omega_n/(2n(n+2))$ is dimensional, plugging into (108),

$$\begin{aligned} E(\tilde{\Omega}) - E(B_1) &\leq C_7(n) C_6(n, R) \delta + C_8(n, R) |E(B_1)| \delta \\ &= (C_6 C_7 + C_8 |E(B_1)|) \delta =: C_E(n, R) \delta, \end{aligned}$$

with

(109)

$$C_E(n, R) := (1 + \hat{\eta}(n, R)^{-1} C_4(n, R)) (3/2)^{n+2} + \frac{(n+2)(3/2)^{n+1} C_5(n, R) \omega_n}{2n(n+2)}.$$

This is (109); we have used $E(B_1) < 0$ when applying (108); the signed contribution $(\lambda^{n+2} - 1)E(B_1)$ has the wrong sign when $\lambda > 1$ but is controlled in absolute value.

Part (d): quotient. Combining (b) and (c) with $\alpha_{\text{BDV}}(\tilde{\Omega}) \geq (1 - \rho(q))\varepsilon$,

$$Q_{\alpha_{\text{BDV}}}(\tilde{\Omega}) = \frac{E(\tilde{\Omega}) - E(B_1)}{\alpha_{\text{BDV}}(\tilde{\Omega})} \leq \frac{C_E(n, R) \delta}{(1 - \rho(q))\varepsilon} = \frac{C_E(n, R)}{1 - \rho(q)} Q_{\alpha_{\text{BDV}}}(\Omega_0).$$

This completes the proof. $\square$

Remark 3.6 (Preservation of the deficit-to-asymmetry ratio). Lemma 3.5(d) is the load-bearing deficit-to-asymmetry preservation statement: starting from $Q_{\alpha_{\text{BDV}}}(\Omega_0) = q$, the volume-normalised companion satisfies $Q_{\alpha_{\text{BDV}}}(\tilde{\Omega}) \leq \tilde{C}(n, R) q$ for an explicit constant $\tilde{C}(n, R) = 2C_E(n, R)$ once $\rho(q) \leq 1/2$. The improvement factor on the asymmetry is $\alpha_{\text{BDV}}(\tilde{\Omega}) \geq \frac{1}{2}\alpha_{\text{BDV}}(\Omega_0)$, so the new asymmetry is at most a factor of 2 smaller than the old. The chain (90) then transfers the same statement to the Fraenkel quotient at the price of a square root.

Volume normalisation enters the proof in two distinct places:

- In (b), the volume defect produces the $|\lambda - 1| \leq C_5 \delta$ contribution to the asymmetry change through Lemma 3.4.
- In (c), the volume-scaling identity (107) produces the explicit λ^{n+2} -factor, which is the only place where the dimension n enters the constant $C_E(n, R)$.

Both contributions are $O(\delta)$, hence subleading compared to the $O(\varepsilon q^{1/4})$ contribution from (100). This is the precise mechanism by which the single-set selection step preserves the quotient $Q_{\alpha_{\text{BDV}}}$ up to a multiplicative (n, R) -constant.

3.1.5. The selection theorem and BDV regularity package.

Theorem 3.7 (Single-set quantitative selection principle). *Fix $n \geq 2$ and $R \geq 2$. There exist constants*

$$\varepsilon_{\text{sel}}(n, R) > 0, \quad q_{\text{sel}}(n, R) > 0, \quad C_{\text{sel}}(n, R) > 0, \quad P_{\text{sel}}(n, R) > 0,$$

depending only on (n, R) via the named BDV constants of (96) and (109), with the following property. Let $\Omega_0 \subset B_R$ be quasi-open with

$$|\Omega_0| = \omega_n, \quad 0 < \varepsilon := \alpha_{\text{BDV}}(\Omega_0) \leq \varepsilon_{\text{sel}}(n, R), \quad 0 < q := Q_{\alpha_{\text{BDV}}}(\Omega_0) \leq q_{\text{sel}}(n, R).$$

Set $\tau := q^{1/4}$, let U_* be a minimiser of the functional J in (95) provided by Lemma 3.1, and define

$$r_* := (|U_*|/\omega_n)^{1/n}, \quad \tilde{\Omega} := r_*^{-1}U_*.$$

Then, after the translation $\tilde{\Omega} \mapsto \tilde{\Omega} - x_{\tilde{\Omega}}$ that brings the barycentre to the origin:

- (i) $P(U_*) \leq P_{\text{sel}}(n, R)$ and $|\tilde{\Omega}| = \omega_n$;
- (ii) $(1 - \rho)\varepsilon \leq \alpha_{\text{BDV}}(\tilde{\Omega}) \leq (1 + \rho)\varepsilon$ with $\rho = \rho(q) \leq 1/2$ given by (105), hence $\frac{1}{2} \alpha_{\text{BDV}}(\Omega_0) \leq \alpha_{\text{BDV}}(\tilde{\Omega}) \leq \frac{3}{2} \alpha_{\text{BDV}}(\Omega_0)$;
- (iii) $Q_{\alpha_{\text{BDV}}}(\tilde{\Omega}) \leq \frac{C_{\text{sel}}(n, R)}{1 - \rho(q)} Q_{\alpha_{\text{BDV}}}(\Omega_0) \leq 2 C_{\text{sel}}(n, R) Q_{\alpha_{\text{BDV}}}(\Omega_0)$, with $C_{\text{sel}}(n, R) := C_E(n, R)$ from (109);
- (iv) $\tilde{\Omega}$ inherits the BDV penalised-minimiser regularity package on its torsion function $\tilde{u} := \tilde{u}_{\tilde{\Omega}}$ via the perturbed quasi-minimality inequality (110) below: namely
 - two-sided density estimates on $\tilde{\Omega}$ at scales $\leq r_0 = r_0(n, R)$ (BDV Lemma 4.9);
 - Lipschitz/nondegeneracy of $\tilde{u}$ near $\partial\tilde{\Omega}$ (BDV Lemma 4.11, 4.12);
 - smooth, uniformly $C^k(n, R)$ -bounded Bernoulli coefficient on $\partial\tilde{\Omega}$ (BDV Lemma 4.16);
 all constants depending only on (n, R) ;
- (v) On $\partial\tilde{\Omega}$ the selected Bernoulli law takes the volume-rescaled form of Lemma 3.8 below.

Proof. Parts (i)–(iii) are Lemmas 3.1–3.5. Parts (iv)–(v) are stated and justified next. $\square$

Perturbed quasi-minimality inequality. By a direct comparison test in J , the minimiser $U_* = \{u_* > 0\}$ (where $u_* = u_{U_*}$ is the torsion function) satisfies the BDV (4.19) inequality: for every $v \in H_0^1(B_R)$,

$$(110) \quad \frac{1}{2} \int |\nabla u_*|^2 - \int u_* + f_{\tilde{\eta}}(|\{u_* > 0\}|) \leq \frac{1}{2} \int |\nabla v|^2 - \int v + f_{\tilde{\eta}}(|\{v > 0\}|) + C_R \tau |\{u_* > 0\} \Delta \{v > 0\}|.$$

The error term arises from the Lipschitz bound on the square-root penalty in (95), dominated by $\tau C_2(R) |U \Delta U_*|$, where $C_2(R)$ is the BDV constant of (88); so C_R may be taken equal to $C_2(R)$. For $\tau \leq \tau_{\text{reg}}(n, R)$, (110) places u_* in the Alt–Caffarelli quasi-minimiser class with smallness constant controlled by (n, R) , and the BDV package [15, Lemmas 4.9, 4.11, 4.12, 4.16] applies; this yields the density, Lipschitz, nondegeneracy and Bernoulli statements of Theorem 3.7(iv). The transfer from U_* to $\tilde{\Omega} = r_*^{-1}U_*$ is a bilipschitz dilation with $|\lambda - 1| \leq 1/2$ (Lemma 3.3); the density, Lipschitz, and C^k constants degrade by a factor at most $(3/2)^n$, which is absorbed into the (n, R) -bookkeeping.

Volume-rescaled Bernoulli law.

Lemma 3.8 (Volume-rescaled Bernoulli law on $\partial\tilde{\Omega}$). *Let $\tilde{u}(y) := r_*^{-2}u_{U_*}(r_*y)$ for $y \in \tilde{\Omega} = r_*^{-1}U_*$. Then $-\Delta\tilde{u} = 1$ in $\tilde{\Omega}$, $\tilde{u} = 0$ on $\partial\tilde{\Omega}$, and on $\partial\tilde{\Omega}$ the BDV Lemma 4.16 Bernoulli law takes the form*

$$(111) \quad |\nabla\tilde{u}(y)|^2 = \tilde{\Lambda} + \tilde{a}_\sigma \tilde{\Psi}_{r_*}(y),$$

where

$$(112) \quad \tilde{\Lambda} = r_*^{-2} \Lambda, \quad \tilde{a}_\sigma = r_*^{-1} a_\sigma, \quad |a_\sigma| \leq \tau,$$

and

$$\tilde{\Psi}_{r_*}(y) := |y - r_*^{-1} x_{U_*}| - r_*^n V_{\tilde{\Omega}} \cdot y, \quad V_{\tilde{\Omega}} := \int_{\tilde{\Omega}} \frac{z - x_{\tilde{\Omega}}}{|z - x_{\tilde{\Omega}}|} dz.$$

Both prefactors satisfy $\tilde{\Lambda} = \Lambda(1 + O(\delta))$ and $\tilde{a}_\sigma = a_\sigma(1 + O(\delta))$ as $\delta \rightarrow 0$, with implicit constants depending only on (n, R) .

Proof. For $y \in \tilde{\Omega}$, $\nabla \tilde{u}(y) = r_*^{-1} \nabla u_{U_*}(r_* y)$ and $\Delta \tilde{u}(y) = \Delta u_{U_*}(r_* y) = -1$, so $\tilde{u}$ solves $-\Delta \tilde{u} = 1$ in $\tilde{\Omega}$ with zero Dirichlet condition on $\partial \tilde{\Omega}$ (which is $r_*^{-1} \partial U_*$). Squaring, $|\nabla \tilde{u}(y)|^2 = r_*^{-2} |\nabla u_{U_*}(r_* y)|^2$.

By BDV Lemma 4.16, on ∂U_* , $|\nabla u_{U_*}(x)|^2 = \Lambda + a_\sigma \Psi_{U_*}(x)$ with $\Psi_{U_*}(x) = |x - x_{U_*}| - V_{U_*} \cdot x$ and $V_{U_*} = \int_{U_*} (z - x_{U_*})/|z - x_{U_*}| dz$. Plugging $x = r_* y$,

$$|\nabla \tilde{u}(y)|^2 = r_*^{-2} \Lambda + r_*^{-2} a_\sigma \Psi_{U_*}(r_* y).$$

Change of variables $z = r_* w$ in the centroid integral:

$$V_{U_*} = \int_{U_*} \frac{z - x_{U_*}}{|z - x_{U_*}|} dz = r_*^n \int_{\tilde{\Omega}} \frac{w - x_{\tilde{\Omega}}}{|w - x_{\tilde{\Omega}}|} dw = r_*^n V_{\tilde{\Omega}},$$

using $x_{U_*} = r_* x_{\tilde{\Omega}}$ for the barycentre. Hence

$$\Psi_{U_*}(r_* y) = |r_* y - x_{U_*}| - V_{U_*} \cdot r_* y = r_* (|y - r_*^{-1} x_{U_*}| - r_*^n V_{\tilde{\Omega}} \cdot y) = r_* \tilde{\Psi}_{r_*}(y).$$

Substituting, $|\nabla \tilde{u}(y)|^2 = r_*^{-2} \Lambda + r_*^{-1} a_\sigma \tilde{\Psi}_{r_*}(y)$, which is (111) with the coefficients (112). Lemma 3.3 gives $|r_* - 1| \leq C_5(n, R)\delta$, hence $\tilde{\Lambda} = \Lambda(1 + O_{n,R}(\delta))$ and analogously for $\tilde{a}_\sigma$. $\square$

3.1.6. Summary of constants and dependences. For ease of cross-reference in later subsections, we record the constants used in Theorem 3.7 together with their dependences. All constants depend only on the pair (n, R) via named BDV constants; none is set by a compactness or contradiction step.

Symbol	Defined in / source	Depends on
$\hat{\eta}, C_4$	BDV Lemma 4.5	n, R
C_1	BDV Lemma 4.2(i), eq. (87)	n
C_2	BDV Lemma 4.2(ii), eq. (88)	R
σ_2, σ_3	BDV Lemmas 4.9, 4.16	n, R
$\tau_{\text{ex}}, \tau_{\text{reg}}$	eq. (96)	n, R
P_{sel}	Lemma 3.1	n, R
C_5	Lemma 3.3	n, R
C_{dil}	Lemma 3.4, eq. (103)	n, R
$C_{\alpha\text{BDV}} = 2C_2C_{\text{dil}}C_5$	eq. (106)	n, R
C_E	eq. (109)	n, R
$C_{\text{sel}} = C_E$	Theorem 3.7	n, R
$\varepsilon_{\text{sel}}, q_{\text{sel}}$	eq. (97)	n, R
$\rho(q) = \sqrt{2q + q^2/\tau} + C_{\alpha\text{BDV}}q$	eq. (105)	n, R, q
$\tilde{\Lambda} = r_*^{-2}\Lambda, \tilde{a}_\sigma = r_*^{-1}a_\sigma$	Lemma 3.8	n, R

Outputs feeding the next subsection. The selection principle (Theorem 3.7) produces a volume-normalised quasi-open set $\tilde{\Omega} \subset B_{R'(n,R)}$ with $|\tilde{\Omega}| = \omega_n$, $\alpha_{\text{BDV}}(\tilde{\Omega}) \in [\varepsilon/2, 3\varepsilon/2]$, $Q_{\alpha\text{BDV}}(\tilde{\Omega}) \leq 2C_{\text{sel}}(n, R) Q_{\alpha\text{BDV}}(\Omega_0)$, and the full BDV free-boundary regularity package. This is the input for the graph-entry subsection (Plan 1 §3.2), which converts smallness of $\alpha_{\text{BDV}}(\tilde{\Omega})$ into $C^{1,\gamma}$ normal-graph parametrisation over $\partial B_1(x_{\tilde{\Omega}})$ via the $\alpha \rightarrow \text{Hausdorff} \rightarrow \text{flatness}$ chain.

3.2. Graph entry and regularity.

Standing hypotheses (Plan 1 / Route A).. Let $n \geq 2$ and $R \geq 2$ be fixed. Throughout this subsection, $\tilde{\Omega} \subset B_R \subset \mathbb{R}^n$ denotes the *volume-normalised selected minimizer* produced by the selection principle of Subsection 3.1 (Agent 5). It satisfies

$$(113) \quad |\tilde{\Omega}| = |B_1| = \omega_n, \quad \tilde{\Omega} \subset B_R,$$

and is, by construction, an almost-minimizer of the BDV penalised Bernoulli functional with selection parameter $\tau \leq \tau_{\text{reg}}(n, R)$, where $\tau_{\text{reg}}(n, R) > 0$ is the regularity threshold fixed in [15, §4] (BDV) and recorded in Subsection 3.1. The BDV smooth asymmetry of $\tilde{\Omega}$ is denoted by $\alpha_{\text{BDV}}(\tilde{\Omega})$ (notation register §5; in the selection-principle subsection this is written $\alpha(\tilde{\Omega})$ without ambiguity, but the global manuscript uses α_{BDV} whenever there is a clash with the coarea quantity $\alpha(t)$ of Plan 2).

By the BDV regularity package, which is inherited by every selected minimizer with $\tau \leq \tau_{\text{reg}}(n, R)$, there exist positive constants

$$c_d = c_d(n, R), \quad r_d = r_d(n, R), \quad L_u = L_u(n, R), \quad q_{\pm} = q_{\pm}(n, R), \quad L_q = L_q(n, R),$$

for which the following hold; cf. [15, Lemmas 4.9, 4.12, 4.16, Theorem 4.18].

(R1) *Two-sided density.* For every $x \in \partial\tilde{\Omega}$ and every $r \in (0, r_d]$,

$$c_d r^n \leq |\tilde{\Omega} \cap B_r(x)| \leq (1 - c_d) r^n, \quad c_d r^n \leq |B_r(x) \setminus \tilde{\Omega}| \leq (1 - c_d) r^n.$$

- (R2) *Lipschitz nondegeneracy.* The torsion potential $\tilde{u} \in H_0^1(\tilde{\Omega})$ of $\tilde{\Omega}$ extends to $C^{0,1}(\overline{B_R})$ with $\text{Lip}(\tilde{u}) \leq L_u$, and $\sup_{B_r(x)} \tilde{u} \geq c_d r$ for every $x \in \partial\tilde{\Omega}$, $r \in (0, r_d]$.
- (R3) *Bernoulli coefficient.* On the flat portion of $\partial\tilde{\Omega}$ the function $q(\cdot) := |\nabla\tilde{u}(\cdot)|$ admits a $C^{1,\gamma}$ one-sided extension with $q_- \leq q \leq q_+$ and $\|q\|_{C^{1,\gamma}} \leq L_q$.
- (R4) *Alt-Caffarelli flatness improvement.* There exist $\bar{\mu} = \bar{\mu}(n, R) \in (0, 1)$, $\bar{\rho} = \bar{\rho}(n, R) > 0$, $\gamma = \gamma(n, R) \in (0, 1)$ and $C_{AC} = C_{AC}(n, R)$ such that, whenever $\tilde{u}$ is of Alt-Caffarelli class $F(\bar{\mu}, 1, +\infty)$ on a ball $B_{\bar{\rho}}(x_0) \subset B_R$, the portion $\partial\tilde{\Omega} \cap B_{\bar{\rho}/2}(x_0)$ is a $C^{1,\gamma}$ graph in the relevant direction with norm $\leq C_{AC}\bar{\mu}$.

Linking asymmetries. The Fraenkel asymmetry $\mathcal{A}(\cdot)$ and the BDV smooth asymmetry $\alpha_{\text{BDV}}(\cdot)$ are linked by [15, Lemma 4.2]: there are positive constants $C_{\mathcal{A}}(n)$ and $C_{\alpha}(n)$ such that, for every measurable U with $|U| = |B_1|$,

$$(114) \quad \mathcal{A}(U)^2 \leq C_{\mathcal{A}}(n) \alpha_{\text{BDV}}(U), \quad |U \Delta B_1| \leq C_{\alpha}(n) \alpha_{\text{BDV}}(U)^{1/2}.$$

The constants $C_{\mathcal{A}}(n)$ and $C_{\alpha}(n)$ are dimensional; they come from the BDV definition of α_{BDV} . We use both directions of (114): the first to control the optimal Fraenkel translation, the second to transfer smallness of α_{BDV} to L^1 -closeness to B_1 .

We now proceed in five steps:

- (S1) From $\alpha_{\text{BDV}}(\tilde{\Omega})$ small to small Hausdorff distance $d_H(\partial\tilde{\Omega}, \partial B_1)$.
- (S2) From small Hausdorff distance to fixed-scale Alt-Caffarelli flatness of $\tilde{u}$.
- (S3) Patching to a global $C^{1,\gamma}$ normal graph; threshold $\alpha_{\text{graph}}(n, R)$.
- (S4) Hodograph straightening + Lieberman oblique Schauder bootstrap to uniform $C^{m,\gamma}$ regularity, with explicit constants $M_m(n, R)$.
- (S5) Hölder interpolation on ∂B_1 ; threshold $\alpha_{\text{sph}}(n, R, \gamma_0)$ for small C^{2,γ_0} norm.

3.2.1. Step 1: from asymmetry to Hausdorff distance.

Lemma 3.9 (Density forces Hausdorff smallness). *Set $c_* := \min(c_d, 1/2)$ and*

$$(115) \quad C_H(n, R) := 4(\omega_n c_*)^{-1/n} + (R+1)(c_* r_d^n)^{-1/n}.$$

Then every selected minimizer $\tilde{\Omega}$ as above satisfies

$$(116) \quad d_H(\partial\tilde{\Omega}, \partial B_1) \leq C_H(n, R) |\tilde{\Omega} \Delta B_1|^{1/n}.$$

Proof. Write $d := d_H(\partial\tilde{\Omega}, \partial B_1)$. By definition there is a point $x \in \partial\tilde{\Omega}$ (or, by symmetry, in ∂B_1) with $\text{dist}(x, \partial B_1) = d$ (resp. $\text{dist}(x, \partial\tilde{\Omega}) = d$).

Case 1: $x \in \partial\tilde{\Omega}$, $\text{dist}(x, \partial B_1) = d \leq 2r_d$. Then $B_{d/2}(x)$ is disjoint from ∂B_1 , hence $B_{d/2}(x) \subset B_1$ or $B_{d/2}(x) \subset \mathbb{R}^n \setminus \overline{B_1}$. In the first case, the density estimate (R1) at the radius $r := \min(d/2, r_d)$ gives $|B_r(x) \setminus \tilde{\Omega}| \geq c_d r^n$, and $B_r(x) \setminus \tilde{\Omega} \subset B_1 \setminus \tilde{\Omega} \subset \tilde{\Omega} \Delta B_1$, so $c_d(d/2)^n \leq |\tilde{\Omega} \Delta B_1|$. In the second case, density of $\tilde{\Omega}$ inside $B_r(x)$ yields the same conclusion with $\tilde{\Omega} \cap B_r(x) \subset \tilde{\Omega} \setminus B_1$. Either way, $d \leq 2(c_d^{-1} |\tilde{\Omega} \Delta B_1|)^{1/n}$.

Case 2: $x \in \partial B_1$, $\text{dist}(x, \partial \tilde{\Omega}) = d \leq 2r_d$. The half-ball $B_{d/2}(x) \cap B_1$ (or its complement, depending on orientation) has volume at least $\frac{1}{2}\omega_n(d/2)^n$ and lies in $B_1 \setminus \tilde{\Omega}$ or $\tilde{\Omega} \setminus B_1$; both are subsets of $\tilde{\Omega} \Delta B_1$. Hence $\frac{1}{2}\omega_n(d/2)^n \leq |\tilde{\Omega} \Delta B_1|$.

Case 3: $d > 2r_d$. Apply the same density argument at fixed radius r_d to a point realising the distance: this yields $c_d r_d^n \leq |\tilde{\Omega} \Delta B_1|$. Hence $d \leq R + 1 \leq (R + 1)(c_d r_d^n)^{-1/n} |\tilde{\Omega} \Delta B_1|^{1/n}$ (using $d \leq \text{diam}(B_R \cup B_1) \leq R + 1$).

Combining the three cases and bounding all dimensional constants by $c_* = \min(c_d, 1/2)$ gives (116) with $C_H(n, R)$ defined by (115). $\square$

Proposition 3.10 (Quantitative asymmetry-to-Hausdorff). *Set*

$$(117) \quad C'_H(n, R) := C_H(n, R) C_\alpha(n)^{1/n},$$

and

$$(118) \quad \alpha_0(n, R) := (r_d(n, R)/C'_H(n, R))^{2n}.$$

Then every selected minimizer $\tilde{\Omega}$ with $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \alpha_0(n, R)$ satisfies

$$(119) \quad d_H(\partial \tilde{\Omega}, \partial B_1) \leq C'_H(n, R) \alpha_{\text{BDV}}(\tilde{\Omega})^{1/(2n)} \leq r_d(n, R).$$

Proof. The second half of (114) gives $|\tilde{\Omega} \Delta B_1| \leq C_\alpha(n) \alpha_{\text{BDV}}(\tilde{\Omega})^{1/2}$. Insertion into Lemma 3.9 gives the first inequality of (119). The second inequality is the definition of α_0 in (118). $\square$

Lemma 3.11 (Barycentre shift absorbable in the threshold). *Write $y_* = y_*(\tilde{\Omega}) \in \mathbb{R}^n$ for the optimal Fraenkel translation, i.e. a minimiser of $y \mapsto |\tilde{\Omega} \Delta (B_1 + y)|/|B_1|$. There exists $C'_A(n, R)$ such that, for every selected minimizer $\tilde{\Omega}$ with $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \alpha_0(n, R)$,*

$$(120) \quad |y_*| \leq (R + 1) \mathcal{A}(\tilde{\Omega}) \leq (R + 1) \sqrt{C_A(n)} \alpha_{\text{BDV}}(\tilde{\Omega})^{1/2} =: C'_A(n, R) \alpha_{\text{BDV}}(\tilde{\Omega})^{1/2}.$$

After replacing $\tilde{\Omega}$ by its translate $\tilde{\Omega} - y_*$ (which leaves the energy unchanged) we may, and from now on do, assume $y_* = 0$, so the comparison ball is centred at the origin.

Proof. The first inequality of (120) follows from $\tilde{\Omega} \subset B_R$ and $|\tilde{\Omega} \Delta (B_1 + y_*)| = |B_1| \mathcal{A}(\tilde{\Omega})$: if $|y_*| > (R + 1) \mathcal{A}(\tilde{\Omega})$ then B_1 and $B_1 + y_*$ are essentially disjoint, and the trivial bound $\mathcal{A} \leq 2$ gives a contradiction. The second inequality is the first half of (114). $\square$

3.2.2. Step 2: Hausdorff closeness yields fixed-scale flatness. Fix once and for all a universal geometric constant $C_0 = C_0(n) \geq 1$ absorbing the conversions between the half-slab geometry below and the exact normalisation of the Alt–Caffarelli class $F(\bar{\mu}, 1, +\infty)$ used in (R4); cf. [15, §4].

Define the tubular slope threshold

$$(121) \quad \mu_{\text{tub}} := \frac{1}{4},$$

which will be used in Step 3, and set

$$(122) \quad \mu := \min\left(\frac{\bar{\mu}(n, R)}{16 C_0(n)}, \frac{\mu_{\text{tub}}}{C_{\text{AC}}(n, R)}\right), \quad \rho_\mu := \min\left(\bar{\rho}(n, R), \frac{\mu}{18}, \frac{r_d(n, R)}{2}\right), \quad h_F := \frac{\mu \rho_\mu}{16}.$$

All three constants depend only on (n, R) .

Lemma 3.12 (Geometric slab inclusion for ∂B_1). *For $\bar{x} \in \partial B_1$ and $s \leq 6\rho_\mu$, $\partial B_1 \cap B_s(\bar{x}) \subset \{x \in \mathbb{R}^n : |(x - \bar{x}) \cdot \nu_{\bar{x}}| \leq s^2/2\} \subset \{x : |(x - \bar{x}) \cdot \nu_{\bar{x}}| \leq \mu\rho_\mu\}$, where $\nu_{\bar{x}}$ is the outward unit normal to ∂B_1 at $\bar{x}$.*

Proof. B_1 has unit radius, hence $|x - \bar{x}| \leq s$ implies $|(x - \bar{x}) \cdot \nu_{\bar{x}}| \leq s^2/2$ for $x \in \partial B_1$. For $s \leq 6\rho_\mu$ the bound $s^2/2 \leq 18\rho_\mu^2 \leq \mu\rho_\mu$ is the condition $\rho_\mu \leq \mu/18$ used to define ρ_μ in (122). $\square$

Proposition 3.13 (Fixed-scale Alt–Caffarelli flatness). *Suppose $d_H(\partial\tilde{\Omega}, \partial B_1) \leq h_F$. Then for every $\bar{x} \in \partial B_1$ there exists $x_0 \in \partial\tilde{\Omega}$ with $|x_0 - \bar{x}| \leq h_F$ such that*

$$\partial\tilde{\Omega} \cap B_{4\rho_\mu}(x_0) \subset \{x : |(x - x_0) \cdot \nu_{\bar{x}}| \leq 2\mu\rho_\mu\},$$

and the torsion potential $\tilde{u}$ belongs to the Alt–Caffarelli class $F(C_0\mu, 1, +\infty)$ on $B_{4\rho_\mu}(x_0)$ with respect to the direction $-\nu_{\bar{x}}$.

Proof. Step 1: existence of x_0 . By $d_H(\partial\tilde{\Omega}, \partial B_1) \leq h_F$ there is $x_0 \in \partial\tilde{\Omega}$ with $|x_0 - \bar{x}| \leq h_F$.

Step 2: slab inclusion. Set $s := 5\rho_\mu + h_F$. Since $h_F = \mu\rho_\mu/16 \leq \rho_\mu$ and $\mu \leq 1$, we have $s \leq 6\rho_\mu$, so Lemma 3.12 applies. Every $p \in \partial\tilde{\Omega} \cap B_{5\rho_\mu}(x_0)$ lies within h_F of some $\bar{p} \in \partial B_1 \cap B_s(\bar{x})$, hence $|(p - \bar{x}) \cdot \nu_{\bar{x}}| \leq \mu\rho_\mu + h_F$, and moving the centre to x_0 costs at most h_F , giving $|(p - x_0) \cdot \nu_{\bar{x}}| \leq \mu\rho_\mu + 2h_F = \mu\rho_\mu + \mu\rho_\mu/8 \leq 2\mu\rho_\mu$.

Step 3 (positive phase, F1). The rescaled torsion potential $\tilde{u}_{x_0, \rho_\mu}(x) := \tilde{u}(x_0 + \rho_\mu x)/\rho_\mu$ is defined on B_4 . The nondegeneracy from (R2) at radius $\rho_\mu \leq r_d/2$ gives a point $y \in B_{\rho_\mu}(x_0) \cap \{\tilde{u} > 0\}$ with $\tilde{u}(y) \geq c_d\rho_\mu$. Combined with the Lipschitz bound of (R2), this gives

$$\tilde{u}_{x_0, \rho_\mu}(x) \geq c_d(x \cdot (-\nu_{\bar{x}}) - C_0\mu)_+ \quad \text{on } B_4,$$

after absorbing dimensional conversions into $C_0 = C_0(n)$. This is the lower (positive-phase) condition of the Alt–Caffarelli class $F(C_0\mu, 1, +\infty)$.

Step 4 (zero phase, F2). The exterior half-ball $\{(x - \bar{x}) \cdot \nu_{\bar{x}} > \mu\rho_\mu\} \cap B_{4\rho_\mu}(\bar{x})$ is disjoint from B_1 . By $d_H(\partial\tilde{\Omega}, \partial B_1) \leq h_F < \mu\rho_\mu/16$, it is disjoint from $\tilde{\Omega}$ except possibly in a layer of width h_F . Hence $\{(x - x_0) \cdot \nu_{\bar{x}} \geq 3\mu\rho_\mu\} \cap B_{4\rho_\mu}(x_0) \subset \mathbb{R}^n \setminus \tilde{\Omega}$, so $\tilde{u} \equiv 0$ there. This is the (F2) condition.

Step 5 (Lipschitz upper bound, F3). By (R2), $\text{Lip}(\tilde{u}) \leq L_u$, and zero in the deep zero-layer gives $\tilde{u}_{x_0, \rho_\mu}(x) \leq L_u(x \cdot (-\nu_{\bar{x}}) + C_0\mu)_+$ on B_4 , which is (F3).

The three conditions (F1)–(F3) with parameter $C_0\mu \leq \bar{\mu}$ (by the choice of μ in (122)) place $\tilde{u}$ in the Alt–Caffarelli class $F(C_0\mu, 1, +\infty)$ on $B_{4\rho_\mu}(x_0)$ with respect to $-\nu_{\bar{x}}$. $\square$

3.2.3. Step 3: patching local graphs into a global graph.

Lemma 3.14 (Tubular threshold). *Let $\mu_{\text{tub}} = 1/4$ as in (121). If $g : \partial B_1 \rightarrow \mathbb{R}$ satisfies $\|g\|_{C^{0,1}(\partial B_1)} \leq \mu_{\text{tub}}$, then the radial map $\Phi_g(\theta) := (1 + g(\theta))\theta$ is a bi-Lipschitz embedding of ∂B_1 onto a hypersurface in $\mathbb{R}^n$, and the radial projection from any such hypersurface back onto ∂B_1 is a diffeomorphism with bounded $C^{1,\gamma}$ inverse, controlled by an explicit constant $K_{\text{geom}}(n, R)$.*

Proof. Compute $d\Phi_g = (1 + g)\text{id} + \theta \otimes dg$ on $T_\theta \partial B_1$ (extended trivially in the radial direction). The smallest singular value is bounded below by $\sqrt{(1 + g)^2 - |dg|^2} \geq \sqrt{(3/4)^2 - (1/4)^2} = \sqrt{1/2} \geq 1/\sqrt{2}$. Hence Φ_g is bi-Lipschitz with bilipschitz constant ≤ 2 . The radial projection $\theta \mapsto \theta/|\theta|$ is smooth on $\mathbb{R}^n \setminus \{0\}$ with derivatives controlled by $\text{dist}(\cdot, 0)$; composing with the inverse map of Φ_g on the image $\Phi_g(\partial B_1) \subset \{|x| \geq 3/4\}$ gives a diffeomorphism whose $C^{1,\gamma}$ norm is bounded by an explicit function of n, R, γ , denoted $K_{\text{geom}}(n, R)$. $\square$

Proposition 3.15 (Global $C^{1,\gamma}$ normal graph). *Assume $d_H(\partial\tilde{\Omega}, \partial B_1) \leq h_F$. Then there exists a unique function $g : \partial B_1 \rightarrow \mathbb{R}$ such that*

$$(123) \quad \partial\tilde{\Omega} = \{(1 + g(\theta))\theta : \theta \in \partial B_1\},$$

with

$$(124) \quad \|g\|_{C^{1,\gamma}(\partial B_1)} \leq M_1(n, R) := C_{\text{AC}}(n, R) \mu K_{\text{geom}}(n, R), \quad \|g\|_{C^{0,1}(\partial B_1)} \leq \mu_{\text{tub}} = \frac{1}{4}.$$

Here $\gamma = \gamma(n, R)$ is the Alt-Caffarelli exponent of (R4).

Proof. By Proposition 3.13, for every $\bar{x} \in \partial B_1$ there is $x_0 = x_0(\bar{x}) \in \partial\tilde{\Omega}$ such that $\tilde{u} \in F(C_0\mu, 1, +\infty)$ on $B_{4\rho_\mu}(x_0)$ with respect to $-\nu_{\bar{x}}$. Since $C_0\mu \leq \bar{\mu}$, the Alt-Caffarelli flatness improvement (R4) upgrades $\partial\tilde{\Omega} \cap B_{\rho_\mu}(x_0)$ to a $C^{1,\gamma}$ graph in direction $\nu_{\bar{x}}$, with norm $\leq C_{\text{AC}}\mu$.

By the choice $\mu \leq \mu_{\text{tub}}/C_{\text{AC}}$ in (122), the local slope is $\leq \mu_{\text{tub}} = 1/4$, hence Lemma 3.14 applies: the local Cartesian graph is reparametrised as a spherical normal graph $g_{\bar{x}}$ over a neighbourhood of $\bar{x}$ in ∂B_1 , with $C^{1,\gamma}$ norm increased by at most $K_{\text{geom}}(n, R)$.

Two local graphs $g_{\bar{x}_1}$ and $g_{\bar{x}_2}$ over overlapping caps parametrise the same connected portion of $\partial\tilde{\Omega}$: by the slope bound $\leq \mu_{\text{tub}}$, the radial projection from $\partial\tilde{\Omega}$ to ∂B_1 is injective, hence the two parametrisations agree on the overlap. Patching using a finite cover of ∂B_1 by caps of radius ρ_μ yields a unique global g with the stated $C^{1,\gamma}$ and $C^{0,1}$ bounds. $\square$

Lemma 3.16 (L^∞ control of g). *With $C_H''(n, R) := 2C_H'(n, R)$ (see (117)),*

$$(125) \quad \|g\|_{L^\infty(\partial B_1)} \leq C_H''(n, R) \alpha_{\text{BDV}}(\tilde{\Omega})^{1/(2n)}.$$

Proof. For $\theta \in \partial B_1$, $|g(\theta)|$ is the radial distance from $(1 + g(\theta))\theta \in \partial\tilde{\Omega}$ to $\theta \in \partial B_1$. Since $\|g\|_{C^{0,1}} \leq 1/4$, the radial distance is at most twice the Euclidean distance, so $\|g\|_{L^\infty} \leq 2d_H(\partial\tilde{\Omega}, \partial B_1) \leq 2C_H' \alpha_{\text{BDV}}(\tilde{\Omega})^{1/(2n)} = C_H'' \alpha_{\text{BDV}}(\tilde{\Omega})^{1/(2n)}$ by Proposition 3.10. $\square$

Theorem 3.17 (Graph-entry theorem). *There exist explicit positive constants $\mu(n, R)$, $\rho_\mu(n, R)$, $h_F(n, R)$, $C_H'(n, R)$, $C_H''(n, R)$, $C_A'(n, R)$, $M_1(n, R)$ and an entry threshold*

$$(126) \quad \alpha_{\text{graph}}(n, R) := \min\left\{\alpha_0(n, R), (h_F/C_H')^{2n}, (h_F/C_A')^2\right\}$$

such that the following holds. Let $\tilde{\Omega} \subset B_R$ be a selected minimizer (in the sense of Subsection 3.1) with $|\tilde{\Omega}| = |B_1|$, $\tau \leq \tau_{\text{reg}}(n, R)$ and $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq$

$\alpha_{\text{graph}}(n, R)$. After translating by $-y_*$ (Lemma 3.11), there exists a unique $g : \partial B_1 \rightarrow \mathbb{R}$ such that (123), (124) and (125) hold; in particular

$$\|g\|_{C^{1,\gamma}(\partial B_1)} \leq M_1(n, R), \quad \|g\|_{L^\infty(\partial B_1)} \leq C_H''(n, R) \alpha_{\text{BDV}}(\tilde{\Omega})^{1/(2n)}.$$

Proof. The choice (126) ensures both $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \alpha_0(n, R)$ (so Proposition 3.10 applies) and $C_H'' \alpha_{\text{BDV}}^{1/(2n)} \leq h_F$; together with the barycentre control of Lemma 3.11 this gives $d_H(\partial \tilde{\Omega}, \partial B_1) \leq h_F$ (after the translation $y_* \rightarrow 0$). Proposition 3.15 then yields the global $C^{1,\gamma}$ graph with the stated bounds, and Lemma 3.16 gives the L^∞ bound. $\square$

Discussion of the regularity input. The regularity theorem driving Step 3 is the Alt–Caffarelli flatness improvement in the form recorded by Brasco–De Philippis–Velichkov as [15, Theorem 4.18]; in the source this is stated as a black-box quantitative replacement of the original Alt–Caffarelli theorem [3], valid for the BDV-class of one-phase Bernoulli free boundaries with a smooth bounded coefficient ($q_- \leq q \leq q_+$, $\|q\|_{C^{1,\gamma}} \leq L_q$). *It is not the De Giorgi–Almgren ε -regularity theorem for area-minimisers, nor is it the Tamanini quasi-minimality regularity:* although the selected minimizer $\tilde{\Omega}$ is a quasi-minimizer of perimeter (Agent 5, D1.4), what we use here is the stronger one-phase Bernoulli flatness improvement of BDV, which is tailored to the penalised functional J_{Ω_0} . This input is recorded in the source map as item D2.2 with citation [15, Theorem 4.18] and origin [3].

3.2.4. Step 4: Schauder bootstrap to $C^{m,\gamma}$ regularity. Let $\tilde{u} \in H_0^1(\tilde{\Omega})$ be the torsion potential of $\tilde{\Omega}$, and let $\bar{x} \in \partial B_1$ be fixed. By a translation and rotation we may assume $\bar{x} = 0$ and the outward unit normal to ∂B_1 at $\bar{x}$ is e_n . By Proposition 3.15 there is $R_0 = R_0(n, R) > 0$ and a Cartesian function $\tilde{g} : \{|x'| < R_0\} \rightarrow \mathbb{R}$ with $\|\tilde{g}\|_{C^{1,\gamma}} \leq M_1'(n, R)$, where $M_1'(n, R) := C(n, R)M_1(n, R)$ absorbs the conversion between spherical graphs over ∂B_1 and Cartesian graphs in a chart, such that inside the cylinder $\mathcal{C}_{R_0} := \{|x'| < R_0, |x_n| < R_0\}$,

$$\partial \tilde{\Omega} \cap \mathcal{C}_{R_0} = \{x_n = \tilde{g}(x') : |x'| < R_0\}, \quad \tilde{\Omega} \cap \mathcal{C}_{R_0} = \{x_n > \tilde{g}(x')\}.$$

Hodograph straightening. By the Lipschitz nondegeneracy (R2) and the positive Bernoulli coefficient (R3), in a one-sided neighbourhood of 0 the partial derivative $\partial_n \tilde{u}$ is bounded below by $q_-/2 > 0$. The Kinderlehrer–Nirenberg partial hodograph map [39, §2]

$$\Phi(x', x_n) := (x', t), \quad t := \tilde{u}(x', x_n),$$

is therefore a $C^{1,\gamma}$ diffeomorphism from $\tilde{\Omega} \cap \mathcal{C}_{R_0/2}$ onto a domain $\{(x', t) : |x'| < R_0/2, 0 < t < T_0\}$ for some $T_0 = T_0(n, R) > 0$, with inverse $x_n = \psi(x', t)$ satisfying $\psi(x', 0) = \tilde{g}(x')$.

A direct change of variables, identical to [39, (2.10)–(2.12)], gives that ψ satisfies, in $\{0 < t < T_0\}$, the quasilinear PDE

$$(127) \quad F[\psi] := \frac{1 + |\nabla_{x'} \psi|^2}{(\partial_t \psi)^2} \partial_t^2 \psi - 2 \frac{\nabla_{x'} \psi \cdot \nabla_{x'} \partial_t \psi}{\partial_t \psi} + \Delta_{x'} \psi = \partial_t \psi,$$

which is uniformly elliptic in ψ as long as $\partial_t \psi \geq \eta > 0$ and $|\nabla_{x'} \psi| \leq K$; from (R2)–(R3) these bounds hold with constants $\eta_*(n, R)$ and $K_*(n, R)$.

The free-boundary Bernoulli condition $|\nabla \tilde{u}|^2 = q^2$ at $t = 0$ transforms into the nonlinear oblique boundary condition

$$(128) \quad \partial_t \psi(x', 0) = \sqrt{\frac{1 + |\nabla_{x'} \psi(x', 0)|^2}{q(x', \psi(x', 0))^2}} \quad \text{on } \{t = 0\}.$$

Linearising $B[\psi] := \partial_t \psi - G(x', \nabla_{x'} \psi, \psi) = 0$ gives $\partial B / \partial(\partial_t \psi) \equiv 1$: the obliqueness coefficient $\beta_n = 1$ is uniformly positive in Lieberman's notation [44, §1], and the tangential coefficients are bounded by an (n, R, L_q) -constant. The initial bound $\|\psi\|_{C^{1,\gamma}} \leq C(n, R)$ on $\{0 \leq t \leq T_0/2\}$ follows from (R2), the implicit function theorem on $t = \tilde{u}(x', x_n)$ and the chain rule. Schauder bootstrap. We apply Lieberman's oblique-derivative Schauder theorem for nonlinear elliptic equations with smooth coefficients, in the form stated as [44, Theorem 1]:

Theorem 3.18 (Lieberman [44], Thm. 1). *Let v solve $a^{ij}(x, v, \nabla v) \partial_{ij} v + b(x, v, \nabla v) = 0$ in $B_{R_0/2}^+$, $b^i(x, v, \nabla v) \partial_i v + b_0(x, v) = 0$ on $\{x_n = 0\}$, with coefficients of class $C^{m-1,\gamma}$, ellipticity η_* , obliqueness β_0 , and $\|v\|_{C^{1,\gamma}} \leq N_0$. For every integer $m \geq 2$,*

$$\|v\|_{C^{m,\gamma}(B_{R_0/4}^+)} \leq \mathcal{S}_m(n, \eta_*, \beta_0, N_0, \|\text{coeffs}\|_{C^{m-1,\gamma}}).$$

Setting $\eta_m := \|\psi\|_{C^{m,\gamma}(\{0 \leq t \leq T_0/2\})}$, the PDE (127) and the oblique BC (128) have coefficients of class $C^{m-1,\gamma}$ once ψ is of class $C^{m-1,\gamma}$, with norms bounded by L_q, η_m, n, R . Applying Theorem 3.18 iteratively on shrinking half-cylinders gives the recursion

$$(129) \quad \eta_{m+1} \leq \mathcal{S}_{m+1}(n, \eta_*, 1, \eta_m, L_q) =: F_m(\eta_m, L_q, n, R),$$

with $\eta_1 = C(n, R)$. Inductively define

$$(130) \quad M_m(n, R) := F_{m-1}(F_{m-2}(\cdots F_1(\eta_1, L_q, n, R) \cdots), L_q, n, R).$$

Norm transfer back to g . The graph g on ∂B_1 and the Cartesian $\tilde{g}$ on $\{|x'| < R_0\}$ are related by an (n, R) -bilipschitz C^∞ chart, so for every $k \geq 0$ and $0 < \beta \leq 1$,

$$(131) \quad c(n, R, k, \beta) \|g\|_{C^{k,\beta}(\partial B_1)} \leq \|\tilde{g}\|_{C^{k,\beta}} \leq C(n, R, k, \beta) \|g\|_{C^{k,\beta}(\partial B_1)}.$$

Since $\tilde{g}(x') = \psi(x', 0)$ by construction, $\|\tilde{g}\|_{C^{k,\beta}} \leq \|\psi\|_{C^{k,\beta}(\{0 \leq t \leq T_0/2\})} \leq \eta_k$. Combining with (131) and using a finite cover of ∂B_1 by $N_0(n, R)$ charts gives the following.

Proposition 3.19 (Uniform Schauder bound). *For every integer $m \geq 1$ there is $M_m(n, R) < \infty$, given by (130), such that under the hypotheses of Theorem 3.17,*

$$(132) \quad \|g\|_{C^{m,\gamma}(\partial B_1)} \leq M_m(n, R).$$

3.2.5. *Step 5: Hölder interpolation to small C^{2,γ_0} .*

Lemma 3.20 (Hölder interpolation on ∂B_1). *Let $0 \leq k < m$ be integers, $0 < \gamma_0 \leq \gamma \leq 1$, with $(k, \gamma_0) \neq (m, \gamma)$, and set*

$$(133) \quad \theta_{m,k} := \frac{k + \gamma_0}{m + \gamma} \in (0, 1).$$

There is $C_{\text{GN}} = C_{\text{GN}}(n, m, k, \gamma, \gamma_0) > 0$ such that for every $f \in C^{m, \gamma}(\partial B_1)$,

$$(134) \quad \|f\|_{C^{k, \gamma_0}(\partial B_1)} \leq C_{\text{GN}} \|f\|_{L^\infty(\partial B_1)}^{1-\theta_{m,k}} \|f\|_{C^{m, \gamma}(\partial B_1)}^{\theta_{m,k}}.$$

Proof. This is the standard Gagliardo–Nirenberg / Bergh–Löfström convexity inequality for Hölder spaces, transferred from Euclidean charts to the compact manifold ∂B_1 ; see [33, Lemma 6.32] and [11, Theorem 6.4.5]. The dimension dependence enters through the partition of unity and the number of charts. $\square$

We apply (134) with $k = 2$, $m = 5$ and $\gamma_0 \in (0, \gamma)$. Since $\gamma_0 \leq \gamma \leq 1$,

$$\theta := \theta_{5,2} = \frac{2 + \gamma_0}{5 + \gamma} \leq \frac{2 + \gamma}{5 + \gamma} \leq \frac{1}{2},$$

so

$$(135) \quad \theta \leq \frac{1}{2}, \quad 1 - \theta \geq \frac{1}{2}.$$

Theorem 3.21 (Schauder–interpolation entry; threshold α_{sph}). *Let $\gamma_0 \in (0, \gamma)$. Let $\delta_{\text{sph}} = \delta_{\text{sph}}(n, \gamma_0) > 0$ denote the smallness threshold of [15, Theorem 3.3] (the BDV nearly-spherical theorem; see Subsection 3.3.11 for the exact statement). Define*

$$(136) \quad \alpha_{\text{sph}}(n, R, \gamma_0) := \left(\frac{\delta_{\text{sph}}(n, \gamma_0)}{C_{\text{GN}}(n, 5, 2, \gamma, \gamma_0) C_H''(n, R)^{1-\theta} M_5(n, R)^\theta} \right)^{2n/(1-\theta)}.$$

Then for every selected minimizer $\tilde{\Omega}$ with $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \min\{\alpha_{\text{graph}}(n, R), \alpha_{\text{sph}}(n, R, \gamma_0)\}$ the graph g of Theorem 3.17 satisfies

$$(137) \quad \|g\|_{C^{2, \gamma_0}(\partial B_1)} \leq \delta_{\text{sph}}(n, \gamma_0).$$

Proof. By Theorem 3.17 and Proposition 3.19 (applied with $m = 5$),

$$\|g\|_{C^{2, \gamma_0}} \leq C_{\text{GN}} \|g\|_{L^\infty}^{1-\theta} \|g\|_{C^{5, \gamma}}^\theta \leq C_{\text{GN}} M_5(n, R)^\theta \|g\|_{L^\infty}^{1-\theta}.$$

Insertion of the L^∞ bound (125) gives

$$\|g\|_{C^{2, \gamma_0}} \leq C_{\text{GN}} M_5^\theta C_H''^{1-\theta} \alpha_{\text{BDV}}(\tilde{\Omega})^{(1-\theta)/(2n)}.$$

The bound $\|g\|_{C^{2, \gamma_0}} \leq \delta_{\text{sph}}$ holds iff $\alpha_{\text{BDV}}(\tilde{\Omega})^{(1-\theta)/(2n)} \leq \delta_{\text{sph}} / (C_{\text{GN}} M_5^\theta C_H''^{1-\theta})$, which is exactly $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \alpha_{\text{sph}}(n, R, \gamma_0)$ by (136). The hypothesis $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \alpha_{\text{graph}}$ ensures Theorem 3.17 and Proposition 3.19 apply in the first place. $\square$

Remarks on the threshold and the exponent.

- (1) By (135), $1 - \theta \geq 1/2$, so the exponent $2n/(1 - \theta)$ in (136) is bounded by $4n$ uniformly in γ, γ_0 . The L^∞ -power $1 - \theta \geq 1/2$ is what preserves the sharp $1/2$ -exponent downstream in the nearly-spherical theorem; cf. [11, Theorem 6.4.5].
- (2) Any integer $m \geq 3$ for which $\theta_{m,2} < 1$ is admissible for the interpolation; the choice $m = 5$ makes $\theta \leq 1/2$ explicit and uniform in (γ_0, γ) .
- (3) The whole argument is constructive: no compactness, no contradiction. The threshold $\alpha_{\text{small}}^{\text{graph}}(n, R, \gamma_0) := \min\{\alpha_{\text{graph}}(n, R), \alpha_{\text{sph}}(n, R, \gamma_0)\}$ is explicit in the BDV regularity constants of the package (R1)–(R4) and the Kinderlehrer–Nirenberg / Lieberman / Gagliardo–Nirenberg constants.

3.2.6. Conclusion of the graph-entry subsection. The output of this subsection is the following theorem, used by Subsection 3.3.11 (Agent 7) as input to the BDV nearly-spherical theorem.

Theorem 3.22 (Graph entry and regularity, Plan 1). *Let $n \geq 2$, $R \geq 2$, and $\gamma_0 \in (0, \gamma(n, R))$. There exist explicit positive constants $\tau_{\text{reg}}(n, R)$, $\alpha_{\text{graph}}(n, R)$ (defined by (126)), $\alpha_{\text{sph}}(n, R, \gamma_0)$ (defined by (136)), $M_m(n, R)$ for every $m \geq 1$ (defined by (130)), and $C_H''(n, R)$ such that the following holds. Let $\tilde{\Omega} \subset B_R$ be a selected minimizer with $|\tilde{\Omega}| = |B_1|$, $\tau \leq \tau_{\text{reg}}(n, R)$ and*

$$\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \alpha_{\text{small}}^{\text{graph}}(n, R, \gamma_0) := \min\{\alpha_{\text{graph}}(n, R), \alpha_{\text{sph}}(n, R, \gamma_0)\}.$$

Then, after translating by the optimal Fraenkel translation (Lemma 3.11), there exists a unique $g : \partial B_1 \rightarrow \mathbb{R}$ with

$$\partial \tilde{\Omega} = \{(1 + g(\theta))\theta : \theta \in \partial B_1\},$$

$$\|g\|_{L^\infty(\partial B_1)} \leq C_H''(n, R) \alpha_{\text{BDV}}(\tilde{\Omega})^{1/(2n)}, \quad \|g\|_{C^m, \gamma(\partial B_1)} \leq M_m(n, R) \quad (m \geq 1), \quad \|g\|_{C^{2, \gamma_0}(\partial B_1)} \leq \delta_{\text{sph}}$$

Proof. Combine Theorem 3.17, Proposition 3.19, and Theorem 3.21. $\square$

Constants used in this subsection. The constants explicitly defined and tracked in this subsection are $c_d, r_d, L_u, q_\pm, L_q$ (BDV regularity package, n, R -dependent); $C_{\mathcal{A}}(n), C_{\alpha}(n)$ (BDV Lemma 4.2, dimensional); $\bar{\mu}, \bar{\rho}, \gamma, C_{\text{AC}}$ (Alt–Caffarelli flatness, n, R -dependent); $C_0(n)$ (geometric conversion); c_*, C_H, C_H', C_H'' (Hausdorff bounds, n, R -dependent); $C_{\mathcal{A}}', \alpha_0, \alpha_{\text{graph}}$ (entry thresholds, n, R -dependent); $\mu, \rho_\mu, h_F, \mu_{\text{tub}}, K_{\text{geom}}, M_1$ (graph-patching, n, R -dependent); $R_0, T_0, \eta_*, K_*, M_m$ (hodograph / Schauder, n, R -dependent); $C_{\text{GN}}, \theta, \delta_{\text{sph}}, \alpha_{\text{sph}}$ (interpolation, (n, R, γ_0) -dependent). No C symbol is used without declared dependence.

What feeds the next subsection. Subsection 3.3.11 (Agent 7) consumes Theorem 3.22 via the BDV nearly-spherical theorem [15, Theorem 3.3], applied to the graph g , yielding the quadratic lower bound $E(\tilde{\Omega}) - E(B_1) \geq c_{\text{sph}}(n) \|g\|_{H^{1/2}(\partial B_1)}^2$ and, after BDV Lemma 4.2(i), the bound $E(\tilde{\Omega}) - E(B_1) \geq c_*^{\text{NS}}(n) \alpha_{\text{BDV}}(\tilde{\Omega})$ in the small asymmetry regime $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \alpha_{\text{small}}(n, R, \gamma_0)$. The far regime $\alpha_{\text{BDV}}(\tilde{\Omega}) \geq \alpha_{\text{small}}$ is handled by BDV Lemma 5.2 (rearrangement-based) in Subsection 3.3.11.

3.3. Nearly-spherical closure and bounded sharp Saint–Venant. This subsection closes the chain

selection $\longrightarrow$ graph entry $\longrightarrow$ Schauder/interpolation entry $\longrightarrow$ BDV nearly spherical and converts the closing inequality into the bounded sharp Saint–Venant estimate

$$(138) \quad \mathcal{A}(\Omega)^2 \leq C(n, R) \delta_T(\Omega), \quad \Omega \subset B_R, \quad |\Omega| = \omega_n,$$

which is the input to the bounded reduction and the Kohler–Jobin transfer of Agent 8. The argument is *not* by contradiction: every quantitative step is a substitution of upstream constants into a chain of explicit BDV black boxes. The constants are tracked through the section.

Throughout the subsection $n \geq 2$ and $R \geq 2$ are fixed, all sets are bounded open subsets of $\mathbb{R}^n$, and we keep the notation register conventions. (Domain-class reconciliation. The selection principle of Agent 5 produces a quasi-open

minimiser $U_* \subset \overline{B_R}$ in the BDV sense [15, §2]; after the BDV regularity package [15, Lem. 4.9, 4.12, 4.16, Thm. 4.18] consumed by Agent 6, U_* is open up to a set of zero capacity, and all load-bearing functionals $\delta_T, \mathcal{A}, \alpha_{\text{BDV}}, \lambda_1$ are insensitive to capacity-zero modifications. Consequently the quasi-open class of Agent 5 and the open class of the notation register §2 coincide for every quantity used below; we treat U_* as an open subset of $\overline{B_R}$ without further mention.) The notation register conventions are: $E(\Omega) = -T(\Omega)/2$ is the BDV-normalised Saint-Venant energy, $\delta_T(\Omega) = E(\Omega) - E(B^*)$ is the Saint-Venant deficit at fixed volume, $\mathcal{A}(\Omega)$ is the Fraenkel asymmetry, and the BDV smooth asymmetry is

$$\alpha_{\text{BDV}}(\Omega) := \int_{\Omega \triangle B_1(x_\Omega)} |1 - |x - x_\Omega|| dx, \quad x_\Omega = \frac{1}{|\Omega|} \int_{\Omega} x dx$$

([15, Definition 4.1], with n in place of the source symbol N). The torsion quotient at the BDV smooth asymmetry is

$$Q_{\alpha_{\text{BDV}}}(\Omega) := \frac{E(\Omega) - E(B_1)}{\alpha_{\text{BDV}}(\Omega)}, \quad |\Omega| = \omega_n, \alpha_{\text{BDV}}(\Omega) > 0.$$

3.3.1. *The three BDV black boxes.* The closure uses exactly three results from [15], all of which are proved there constructively (no compactness, no selection).

(B1) Sharp nearly-spherical stability ([15, Theorem 3.3]). For every $0 < \gamma_0 \leq 1$ there exists $\delta_1 = \delta_1(n, \gamma_0) > 0$ such that the following holds. Let $U \subset \mathbb{R}^n$ be open and bounded, and suppose there exists $g \in C^{2, \gamma_0}(\partial B_1)$ with

$$(139) \quad \partial U = \{(1+g(y))y : y \in \partial B_1\}, \quad \|g\|_{C^{2, \gamma_0}(\partial B_1)} \leq \delta_1, \quad |U| = |B_1|, \quad x_U = 0.$$

Then, with E the Saint-Venant energy of Notation Register §3 and $H(g)$ the harmonic extension of g to B_1 ,

$$(140) \quad E(U) - E(B_1) \geq c_{\text{sph}}(n) \|g\|_{H^{1/2}(\partial B_1)}^2, \quad c_{\text{sph}}(n) := \frac{1}{32n^2},$$

where $\|g\|_{H^{1/2}(\partial B_1)}^2 = \int_{\partial B_1} g^2 d\mathcal{H}^{n-1} + \int_{B_1} |\nabla H(g)|^2 dx$ ([15, Definition 3.2]).

The constant $1/(32n^2)$ comes from the gap to the second Steklov eigenvalue of B_1 (equal to 2 in every dimension), see [15, (3.9)–(3.10)].

(B2) Smooth-asymmetry comparisons ([15, Lemma 4.2 (i),(ii),(iii)]). There exist $C_1 = C_1(n) > 0$, $C_2 = C_2(R) > 0$, $\delta_3 = \delta_3(n) > 0$, $C_3 = C_3(n) > 0$ such that:

(i) for every bounded open $\Omega \subset \mathbb{R}^n$ with $|\Omega| = |B_1|$,

$$(141) \quad C_1(n) \alpha_{\text{BDV}}(\Omega) \geq |\Omega \triangle B_1(x_\Omega)|^2 \geq |B_1|^2 \mathcal{A}(\Omega)^2;$$

(ii) for every $R \geq 2$ and every bounded open $\Omega_1, \Omega_2 \subset B_R$,

$$(142) \quad |\alpha_{\text{BDV}}(\Omega_1) - \alpha_{\text{BDV}}(\Omega_2)| \leq C_2(R) |\Omega_1 \triangle \Omega_2|;$$

in particular, choosing Ω_2 to be the Fraenkel-optimal ball for Ω_1 (so $\alpha_{\text{BDV}}(\Omega_2) = 0$), one has, for $|\Omega| = \omega_n$ and $\Omega \subset B_R$,

$$(143) \quad \alpha_{\text{BDV}}(\Omega) \leq C_2(R) \omega_n \mathcal{A}(\Omega);$$

(iii) for every nearly-spherical set U parametrised by g with $\|g\|_{L^\infty(\partial B_1)} \leq \delta_3$,

$$(144) \quad \alpha_{\text{BDV}}(U) \leq C_3(n) \|g\|_{L^2(\partial B_1)}^2.$$

All three estimates are proved in [15, §4] by direct rearrangement / Taylor expansion of α_{BDV} ; they are constructive.

(B3) Suboptimal Fraenkel-quartic Saint-Venant ([15, Lemma 5.2]). There exists $C_8 = C_8(n) > 0$ such that for every bounded open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,

$$(145) \quad E(\Omega) |\Omega|^{-(n+2)/n} - E(B_1) |B_1|^{-(n+2)/n} \geq \frac{1}{C_8(n)} \mathcal{A}(\Omega)^4.$$

The proof in [15, §5] is by rearrangement and Pólya–Szegő (no contradiction, no selection). After absorbing the $\omega_n^{(n+2)/n}$ factor at fixed volume $|\Omega| = \omega_n$,

$$(146) \quad E(\Omega) - E(B_1) \geq C'_8(n)^{-1} \mathcal{A}(\Omega)^4, \quad C'_8(n) := C_8(n) / \omega_n^{(n+2)/n}.$$

Important note on hypotheses. (B1) is stated for sets that are *simultaneously* C^{2,γ_0} -parametrised, volume-normalised to $|B_1|$, and barycentred at the origin. The three normalisations are all load-bearing for (140): dropping any one of them changes the form of the second-variation calculation in [15, §3]. Sections 3.3.3 and 3.3.4 show that the upstream chain enforces all three.

3.3.2. *Trace–Poincaré step and the pointwise α -form.* Combining (B1) and (B2)(iii) gives the pointwise $Q_{\alpha_{\text{BDV}}}$ -floor on nearly-spherical sets. The intermediate $H^{1/2} \rightarrow L^2$ comparison is the trace–Poincaré inequality on ∂B_1 :

$$(147) \quad \|g\|_{L^2(\partial B_1)}^2 \leq \|g\|_{H^{1/2}(\partial B_1)}^2,$$

which is immediate from the definition of the $H^{1/2}$ -norm (its L^2 summand bounds $\|g\|_{L^2}^2$ above by itself).

Theorem 3.23 (Nearly-spherical closure, BDV smooth-asymmetry form). *Fix $0 < \gamma_0 \leq 1$ and let*

$$(148) \quad \delta_{\text{sph}}(n, \gamma_0) := \min(\delta_1(n, \gamma_0), \delta_3(n)).$$

Let $U \subset \mathbb{R}^n$ be open and bounded, with $\partial U = \{(1 + g(y))y : y \in \partial B_1\}$, $g \in C^{2,\gamma_0}(\partial B_1)$, and

$$(149) \quad \|g\|_{C^{2,\gamma_0}(\partial B_1)} \leq \delta_{\text{sph}}(n, \gamma_0), \quad |U| = |B_1|, \quad x_U = 0.$$

Then

$$(150) \quad E(U) - E(B_1) \geq c_*(n) \alpha_{\text{BDV}}(U), \quad c_*(n) := \frac{c_{\text{sph}}(n)}{C_3(n)} = \frac{1}{32 n^2 C_3(n)},$$

and consequently $Q_{\alpha_{\text{BDV}}}(U) \geq c_(n)$.*

Proof. By (148) both hypotheses (139) (with δ_1) and the smallness hypothesis of (B2)(iii) (with δ_3) hold simultaneously, since $\|g\|_{L^\infty} \leq \|g\|_{C^{2,\gamma_0}} \leq \delta_{\text{sph}} \leq \delta_3$. Chain (B1), (147) and (B2)(iii):

$$E(U) - E(B_1) \stackrel{(140)}{\geq} c_{\text{sph}}(n) \|g\|_{H^{1/2}}^2 \stackrel{(147)}{\geq} c_{\text{sph}}(n) \|g\|_{L^2}^2 \stackrel{(144)}{\geq} \frac{c_{\text{sph}}(n)}{C_3(n)} \alpha_{\text{BDV}}(U).$$

Dividing by $\alpha_{\text{BDV}}(U) > 0$ gives $Q_{\alpha_{\text{BDV}}}(U) \geq c_*(n)$. $\square$

Theorem 3.23 is the pointwise (per-set) closure: no sequence, no limit, no smallness of $Q_{\alpha_{\text{BDV}}}$ is assumed. It is exactly the building block called NS in `Final/NearlySphericalClosure.tex`.

3.3.3. *Feeding the upstream chain into the nearly-spherical hypothesis.* We now show that the upstream blocks (S), (G), (I) of Agents 5 and 6 produce, from *any* $\Omega_0 \subset B_R$ with $|\Omega_0| = \omega_n$ and $\alpha_{\text{BDV}}(\Omega_0)$ small, a set $\tilde{\Omega}$ that meets all three hypotheses of (149).

Recap of the upstream output (Agents 5, 6). We use the following from Agent 5 (selection) and Agent 6 (graph entry / Schauder interpolation), each stated with the constants they fix:

- (S) (Quantitative selection, Agent 5, Theorem D1.2–D1.5 of `Manuscript/source_map.md`; `Plan 1/quantitative-selection-principle.tex`, `Final/SelectionPrinciple.tex`.)

There exist $C_{\text{sel}} = C_{\text{sel}}(n, R) > 0$, $q_{\text{sel}} = q_{\text{sel}}(n, R) > 0$, $\varepsilon_{\text{sel}} = \varepsilon_{\text{sel}}(n, R) > 0$ such that for every open $\Omega_0 \subset B_R$ with $|\Omega_0| = \omega_n$, $\alpha_{\text{BDV}}(\Omega_0) \leq \varepsilon_{\text{sel}}$, $Q_{\alpha_{\text{BDV}}}(\Omega_0) < q_{\text{sel}}$, the single-set penalised functional of Agent 5 has a minimiser $U_* \subset B_R$, whose volume-normalised companion $\tilde{\Omega} := r_*^{-1} U_*$, $r_* = (|U_*|/|B_1|)^{1/n}$, satisfies $\tilde{\Omega} \subset B_R$ (up to enlarging R by a dimensional amount), $|\tilde{\Omega}| = \omega_n$,

$$(151) \quad \frac{1}{2} \alpha_{\text{BDV}}(\Omega_0) \leq \alpha_{\text{BDV}}(\tilde{\Omega}) \leq \frac{3}{2} \alpha_{\text{BDV}}(\Omega_0),$$

$$(152) \quad Q_{\alpha_{\text{BDV}}}(\tilde{\Omega}) \leq 2 C_{\text{sel}}(n, R) Q_{\alpha_{\text{BDV}}}(\Omega_0),$$

and $\tilde{\Omega}$ inherits the BDV regularity package: density estimates, Lipschitz nondegeneracy, and a smooth Bernoulli coefficient ([15, Lemmas 4.9, 4.12, 4.16]).

- (G) (Graph entry, Agent 6, Theorem D2.3 of `Manuscript/source_map.md`; `Final/GraphEntry.tex`, `Plan 1/graph-entry-closure.tex`.) There exist $\alpha_{\text{graph}} = \alpha_{\text{graph}}(n, R) > 0$ and $C_H'' = C_H''(n, R) > 0$ and $\gamma = \gamma(n, R) \in (0, 1)$ such that every $\tilde{\Omega}$ as in (S) with $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \alpha_{\text{graph}}$ is, after a single translation $\tilde{\Omega} \mapsto \tilde{\Omega} - x_{\tilde{\Omega}}$ by its barycentre, a $C^{1,\gamma}$ normal graph over ∂B_1 :

$$(153) \quad \partial \tilde{\Omega} = \{(1 + g(\theta)) \theta : \theta \in \partial B_1\}, \quad \|g\|_{L^\infty(\partial B_1)} \leq C_H''(n, R) \alpha_{\text{BDV}}(\tilde{\Omega})^{1/(2n)}.$$

- (I) (Schauder interpolation, Agent 6, Lemma D2.7 of `Manuscript/source_map.md`; `Final/SchauderInterpolation.tex`.) There exists $\alpha_{\text{sph}} = \alpha_{\text{sph}}(n, R, \gamma_0) > 0$ such that whenever (G) applies and $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \alpha_{\text{sph}}$, the parametrisation g of (G) satisfies

$$(154) \quad \|g\|_{C^{2,\gamma_0}(\partial B_1)} \leq \delta_{\text{sph}}(n, \gamma_0),$$

the smallness threshold of Theorem 3.23.

The thresholds ε_{sel} , α_{graph} , α_{sph} appearing in (S), (G), (I) are independent of Ω_0 and are fixed once (n, R, γ_0) is chosen. Define the small-asymmetry threshold

$$(155) \quad \alpha_{\text{small}}(n, R) := \frac{1}{2} \min\{\varepsilon_{\text{sel}}(n, R), \alpha_{\text{graph}}(n, R), \alpha_{\text{sph}}(n, R, \gamma_0)\} > 0.$$

3.3.4. *Volume and barycentre normalisation.* The set U_* produced by (S) is the minimiser of a penalised functional whose volume penalty does *not* force $|U_*| = \omega_n$ exactly; the energy gain forces

$$(156) \quad ||U_*| - \omega_n| \leq C(n, R) \delta_T(U_*),$$

see [15, Lemma 4.5] and Agent 5. We normalise both the volume and the barycentre as follows.

Volume normalisation. The companion $\tilde{\Omega} = r_*^{-1}U_*$, $r_* = (|U_*|/\omega_n)^{1/n}$, satisfies $|\tilde{\Omega}| = \omega_n$ by construction, and the torsion equation rescales as $u_{\tilde{\Omega}}(x) = r_*^{-2}u_*(r_*x)$, so that $E(\tilde{\Omega}) = r_*^{-(n+2)}E(U_*)$ (cf. the rescaling computation of [15, §2], with N replaced by n). The Saint-Venant deficit transforms as $\delta_T(\tilde{\Omega}) = r_*^{-(n+2)}(E(U_*) - r_*^{n+2}E(B_1))$ and (156) gives $|r_* - 1| \leq C(n, R) \delta_T(U_*)$; the energy errors are absorbed into a multiplicative $1 + O(\delta_T(U_*))$ factor in (S), which the constants $C_{\text{sel}}(n, R)$ already include (Agent 5).

Barycentre normalisation. After (G) provides the parametrisation $\partial\tilde{\Omega} = \{(1+g)y\}$, the barycentre $x_{\tilde{\Omega}}$ and the volume normalisation $|\tilde{\Omega}| = \omega_n$ give the two scalar identities ([15, (3.6)–(3.7)], $N \rightarrow n$)

$$(157) \quad |\tilde{\Omega}| = \int_{\partial B_1} \frac{(1+g)^n}{n} d\mathcal{H}^{n-1} = \omega_n,$$

$$(158) \quad x_{\tilde{\Omega}} = \int_{\partial B_1} y \frac{(1+g)^{n+1}}{n+1} d\mathcal{H}^{n-1}.$$

Expanding in g and using $\|g\|_{C^{2,\gamma_0}} \leq \delta_{\text{sph}}$ yields the BDV moment bounds ([15, (3.6)–(3.7)], with $N \rightarrow n$):

$$(159) \quad \left| \int_{\partial B_1} g d\mathcal{H}^{n-1} \right| + \sum_{i=1}^n \left| \int_{\partial B_1} y_i g d\mathcal{H}^{n-1} \right| \leq C(n) \delta_{\text{sph}} \|g\|_{H^{1/2}(\partial B_1)}.$$

The right-hand side of (159) is a controlled fraction of $\|g\|_{H^{1/2}}^2$ once $\|g\|_{H^{1/2}} \geq c(n) \delta_{\text{sph}}$; when $\|g\|_{H^{1/2}}$ is even smaller, both sides of (140) are subdominant and the small-deficit case (150) holds trivially after a single homothety and translation. The detailed reduction, which is a one-step rescaling plus translation, is recorded in `Final/NearlySphericalClosure.tex` (Corollary 2.4) and shows that we may replace (149) by

$$(160) \quad ||\tilde{\Omega}| - \omega_n| + |x_{\tilde{\Omega}}| \leq \eta(n, \gamma_0) \|g\|_{L^2(\partial B_1)}$$

at the cost of replacing $c_*(n)$ by $\tilde{c}_*(n) := c_*(n)/4$. The post-volume normalisation $|\tilde{\Omega}| = \omega_n$ and the barycentre translation $\tilde{\Omega} \mapsto \tilde{\Omega} - x_{\tilde{\Omega}}$ make this condition trivially satisfied with $\eta = 0$, so we may apply Theorem 3.23 *verbatim* (with $c_*(n)$, not $\tilde{c}_*(n)$).

Summary of normalisations enforced.

- Volume: $|\tilde{\Omega}| = \omega_n$ exactly, by the homothety $\tilde{\Omega} = r_*^{-1}U_*$.
- Barycentre: $x_{\tilde{\Omega}} = 0$, by the single translation $\tilde{\Omega} \mapsto \tilde{\Omega} - x_{\tilde{\Omega}}$, which preserves $E, \alpha_{\text{BDV}}, \mathcal{A}$ and the C^{2,γ_0} -norm of g .
- C^{2,γ_0} -smallness: $\|g\|_{C^{2,\gamma_0}} \leq \delta_{\text{sph}}$ by (I), provided $\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \alpha_{\text{sph}}$.

None of the three normalisations is silently dropped; each is enforced by a named operation (rescale, translate, threshold check).

3.3.5. Small-asymmetry linear bound. We now carry through the implication $(S) \rightarrow (G) \rightarrow (I) \rightarrow$ Theorem 3.23 explicitly, with no compactness step. The output is the small- α_{BDV} linear-in- α_{BDV} torsion bound.

Theorem 3.24 (Small- α_{BDV} linear Saint-Venant bound). *Fix $0 < \gamma_0 \leq 1$. Set*

$$(161) \quad q_*(n, R) := \frac{c_*(n)}{2 C_{\text{sel}}(n, R)} = \frac{1}{64 n^2 C_3(n) C_{\text{sel}}(n, R)}.$$

For every open $\Omega_0 \subset B_R$ with $|\Omega_0| = \omega_n$ and $\alpha_{\text{BDV}}(\Omega_0) \leq \alpha_{\text{small}}(n, R)$,

$$(162) \quad E(\Omega_0) - E(B_1) \geq q_*(n, R) \alpha_{\text{BDV}}(\Omega_0).$$

Proof. Case A: $Q_{\alpha_{\text{BDV}}}(\Omega_0) \geq q_{\text{sel}}(n, R)$. Then directly $E(\Omega_0) - E(B_1) \geq q_{\text{sel}} \alpha_{\text{BDV}}(\Omega_0) \geq q_* \alpha_{\text{BDV}}(\Omega_0)$, after shrinking q_{sel} if necessary so that $q_* \leq q_{\text{sel}}$.

Case B: $Q_{\alpha_{\text{BDV}}}(\Omega_0) < q_{\text{sel}}(n, R)$. Since $\alpha_{\text{BDV}}(\Omega_0) \leq \alpha_{\text{small}} \leq \varepsilon_{\text{sel}}$, the selection (S) applies and yields $\tilde{\Omega}$ satisfying (151)–(152). In particular

$$\alpha_{\text{BDV}}(\tilde{\Omega}) \leq \frac{3}{2} \alpha_{\text{BDV}}(\Omega_0) \leq \frac{3}{2} \alpha_{\text{small}}(n, R) \leq \min\{\alpha_{\text{graph}}(n, R), \alpha_{\text{sph}}(n, R, \gamma_0)\},$$

where the last inequality holds by (155) (the factor $\frac{1}{2}$ in α_{small} absorbs the factor $\frac{3}{2}$ from (S2)). Thus (G) gives the $C^{1,\gamma}$ -graph (153), and (I) gives the C^{2,γ_0} -smallness (154).

After the single barycentre translation $\tilde{\Omega} \mapsto \tilde{\Omega} - x_{\tilde{\Omega}}$ (which preserves all relevant quantities, see §3.3.4), the hypotheses (149) of Theorem 3.23 are satisfied. Therefore

$$Q_{\alpha_{\text{BDV}}}(\tilde{\Omega}) \geq c_*(n).$$

Combined with (152),

$$c_*(n) \leq Q_{\alpha_{\text{BDV}}}(\tilde{\Omega}) \leq 2 C_{\text{sel}}(n, R) Q_{\alpha_{\text{BDV}}}(\Omega_0),$$

whence $Q_{\alpha_{\text{BDV}}}(\Omega_0) \geq c_*(n)/(2 C_{\text{sel}}(n, R)) = q_*(n, R)$, which rearranges to (162). $\square$

Remark 3.25 (Where the BDV contradiction step appears, and what replaces it). At no point of the proof of Theorem 3.24 did we assume $Q_{\alpha_{\text{BDV}}}(\Omega_0)$ is arbitrarily small. The two cases (“ $Q_{\alpha_{\text{BDV}}} \geq q_{\text{sel}}$ ” and “ $Q_{\alpha_{\text{BDV}}} < q_{\text{sel}}$ ”) are an explicit dichotomy with thresholds depending on (n, R) . In the second case the conclusion comes from a single application of (S), (G), (I), and Theorem 3.23. The original BDV proof ([15, Theorem 4.3]) is by contradiction with a minimising sequence; here it is replaced by the single-set selection of Agent 5. The conversion of the contradiction step required no additional expansion beyond Agent 5’s quotient-preservation bound (152): the chain (S)+(G)+(I)+NS contracts the contradiction into one arithmetic comparison $Q_{\alpha_{\text{BDV}}}(\tilde{\Omega}) \leq 2 C_{\text{sel}} Q_{\alpha_{\text{BDV}}}(\Omega_0)$.

3.3.6. Conversion to the Fraenkel asymmetry. We now convert (162) into a Fraenkel-squared lower bound on δ_T . This is the step that needs both directions of BDV Lemma 4.2 ((B2)(i) and (B2)(ii)): the linear comparison (141) produces the sharp exponent 2, and the Lipschitz comparison (143) ensures that the small- α_{BDV} hypothesis can be entered from a smallness assumption on $\mathcal{A}$ itself.

Corollary 3.26 (Small- $\mathcal{A}$ quadratic torsion bound). *Define*

$$(163) \quad a_{\text{near}}(n, R) := \frac{\alpha_{\text{small}}(n, R)}{C_2(R)\omega_n} > 0, \quad \kappa_{\text{near}}(n, R) := \frac{q_*(n, R)\omega_n^2}{C_1(n)}.$$

For every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\mathcal{A}(\Omega) \leq a_{\text{near}}(n, R)$,

$$(164) \quad E(\Omega) - E(B_1) \geq \kappa_{\text{near}}(n, R) \mathcal{A}(\Omega)^2.$$

Proof. By (143), $\alpha_{\text{BDV}}(\Omega) \leq C_2(R)\omega_n \mathcal{A}(\Omega) \leq C_2\omega_n a_{\text{near}} = \alpha_{\text{small}}(n, R)$. Theorem 3.24 therefore gives $E(\Omega) - E(B_1) \geq q_*(n, R) \alpha_{\text{BDV}}(\Omega)$. By (141), $\alpha_{\text{BDV}}(\Omega) \geq |B_1|^2 \mathcal{A}(\Omega)^2 / C_1(n) = \omega_n^2 \mathcal{A}(\Omega)^2 / C_1(n)$. Combining, $E(\Omega) - E(B_1) \geq (q_*\omega_n^2 / C_1) \mathcal{A}(\Omega)^2 = \kappa_{\text{near}} \mathcal{A}(\Omega)^2$. $\square$

3.3.7. Far-asymmetry regime. The far regime $\mathcal{A}(\Omega) \geq a_{\text{near}}$ is handled by the suboptimal $\mathcal{A}^4$ bound (B3); no contradiction is needed in this regime either.

Lemma 3.27 (Far- $\mathcal{A}$ quadratic torsion bound). *For every $a > 0$ and every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\mathcal{A}(\Omega) \geq a$,*

$$(165) \quad E(\Omega) - E(B_1) \geq C'_8(n)^{-1} \mathcal{A}(\Omega)^4 \geq \frac{a^2}{C'_8(n)} \mathcal{A}(\Omega)^2.$$

Proof. The first inequality is (146); the second uses $\mathcal{A}(\Omega)^4 = \mathcal{A}(\Omega)^2 \mathcal{A}(\Omega)^2 \geq a^2 \mathcal{A}(\Omega)^2$ once $\mathcal{A}(\Omega) \geq a$. $\square$

3.3.8. Bounded sharp Saint-Venant inequality.

Theorem 3.28 (Bounded sharp Saint-Venant, Plan 1). *Let $n \geq 2$, $R \geq 2$, $0 < \gamma_0 \leq 1$. Define*

$$(166) \quad c_{\text{SV}}(n, R) := \min \left\{ \kappa_{\text{near}}(n, R), \frac{a_{\text{near}}(n, R)^2}{C'_8(n)} \right\} > 0,$$

where $\kappa_{\text{near}}, a_{\text{near}}$ are given by (163). For every open $\Omega \subset B_R$ with $|\Omega| = \omega_n = |B^*|$,

$$(167) \quad E(\Omega) - E(B^*) \geq c_{\text{SV}}(n, R) \mathcal{A}(\Omega)^2,$$

equivalently,

$$(168) \quad \mathcal{A}(\Omega)^2 \leq \frac{1}{c_{\text{SV}}(n, R)} \delta_T(\Omega).$$

The exponent 2 on $\mathcal{A}$ is sharp: it is saturated asymptotically by small volume-preserving ellipsoidal perturbations of B_1 ([15, Introduction, p. 2]).

Proof. The Fraenkel asymmetry is contained in $[0, 2)$; we split it at $a_{\text{near}}(n, R)$.

Case 1 (small Fraenkel): $\mathcal{A}(\Omega) \leq a_{\text{near}}(n, R)$. Corollary 3.26 gives $E(\Omega) - E(B_1) \geq \kappa_{\text{near}}(n, R) \mathcal{A}(\Omega)^2 \geq c_{\text{SV}}(n, R) \mathcal{A}(\Omega)^2$.

Case 2 (large Fraenkel): $\mathcal{A}(\Omega) \geq a_{\text{near}}(n, R)$. Lemma 3.27 with $a = a_{\text{near}}$ gives $E(\Omega) - E(B_1) \geq a_{\text{near}}^2 / C_8' \mathcal{A}(\Omega)^2 \geq c_{\text{SV}}(n, R) \mathcal{A}(\Omega)^2$.

Since $\{\mathcal{A}(\Omega) \leq a_{\text{near}}\} \cup \{\mathcal{A}(\Omega) \geq a_{\text{near}}\} = [0, 2)$, the two cases cover all admissible Ω , and the bound is uniform with constant $c_{\text{SV}}(n, R)$ as in (166). The deficit form (168) follows since $E(B_1) = E(B^*)$ when $|\Omega| = |B^*| = \omega_n$, and $\delta_T(\Omega) = E(\Omega) - E(B^*)$ by the notation register. $\square$

3.3.9. Constant chain: explicit dependences. The final constant $c_{\text{SV}}(n, R)$ in Theorem 3.28 is the minimum of a chain of named quantities. We record their dependences explicitly.

Constant	Dependence	Source
$C_1(n)$	n	BDV Lem. 4.2(i), (141)
$C_2(R)$	R	BDV Lem. 4.2(ii), (142)
$C_3(n)$	n	BDV Lem. 4.2(iii), (144)
$\delta_1(n, \gamma_0)$	n, γ_0	BDV Thm. 3.3, (140)
$\delta_3(n)$	n	BDV Lem. 4.2(iii)
$C_8'(n)$	n	BDV Lem. 5.2, (146)
$c_{\text{sph}}(n) = 1/(32n^2)$	n	(140)
$\delta_{\text{sph}}(n, \gamma_0) = \min(\delta_1, \delta_3)$	n, γ_0	(148)
$c_*(n) = c_{\text{sph}}/C_3 = 1/(32n^2 C_3(n))$	n	Thm. 3.23
$C_{\text{sel}}(n, R), q_{\text{sel}}(n, R), \varepsilon_{\text{sel}}(n, R)$	n, R	Agent 5 (S)
$\alpha_{\text{graph}}(n, R), C_H''(n, R)$	n, R	Agent 6 (G)
$\alpha_{\text{sph}}(n, R, \gamma_0)$	n, R, γ_0	Agent 6 (I)
$\alpha_{\text{small}}(n, R)$	n, R	(155)
$q_*(n, R) = c_*/(2C_{\text{sel}})$	n, R	(161)
$\kappa_{\text{near}}(n, R) = q_* \omega_n^2 / C_1$	n, R	(163)
$a_{\text{near}}(n, R) = \alpha_{\text{small}} / (C_2 \omega_n)$	n, R	(163)
$c_{\text{SV}}(n, R) = \min\{\kappa_{\text{near}}, a_{\text{near}}^2 / C_8'\}$	n, R	(166)

Pure (n, R) -dependence of c_{SV} . Every quantity in the chain depends only on (n, R) , possibly through the auxiliary parameter $\gamma_0 \in (0, 1]$ which is fixed once and for all (we may take $\gamma_0 = 1/2$). The dependence on R enters at exactly three points:

- (1) Selection $C_{\text{sel}}(n, R), q_{\text{sel}}(n, R), \varepsilon_{\text{sel}}(n, R)$ (Agent 5).
- (2) Graph entry $\alpha_{\text{graph}}(n, R), C_H''(n, R)$, Schauder interpolation $\alpha_{\text{sph}}(n, R, \gamma_0)$ (Agent 6).
- (3) Lipschitz comparison $C_2(R)$ (BDV Lemma 4.2(ii)).

The nearly-spherical floor $c_*(n)$, the BDV constants $C_1(n), C_3(n), C_8'(n)$, and the multipliers ω_n are purely dimensional. The constant $c_{\text{sph}}(n) = 1/(32n^2)$ deteriorates polynomially in n .

3.3.10. Hypothesis boundaries and degeneration of the nearly-spherical hypothesis. The hypotheses of the BDV nearly-spherical theorem (B1) fail in

three regimes; we record how each is handled here, without silently dropping any hypothesis.

(a) $\alpha_{\text{BDV}}(\Omega_0) > \alpha_{\text{small}}(n, R)$. The selection (S) is still applicable only down to $\alpha_{\text{BDV}}(\Omega_0) \leq \varepsilon_{\text{sel}}(n, R)$. For $\alpha_{\text{BDV}}(\Omega_0) > \alpha_{\text{small}}$ but $\mathcal{A}(\Omega)$ remains in $[0, 2)$, we use the far-Fraenkel regime (Lemma 3.27). Translating $\alpha_{\text{BDV}} \geq \alpha_{\text{small}}$ to a Fraenkel lower bound by (143) gives $\mathcal{A}(\Omega) \geq \alpha_{\text{BDV}}/(C_2\omega_n) \geq a_{\text{near}}$, so Case 2 of Theorem 3.28 applies. Thus no α_{BDV} -large regime is left uncovered.

(b) Graph regime not reached because of (G) failure. The graph-entry theorem (G) fails if $\alpha_{\text{BDV}}(\tilde{\Omega}) > \alpha_{\text{graph}}$. Since (S) preserves α_{BDV} up to a factor $\frac{3}{2}$ and we chose $\alpha_{\text{small}} \leq \alpha_{\text{graph}}/2$ in (155) (the factor $\frac{1}{2}$), Case B of the proof of Theorem 3.24 never enters the regime where (G) fails. If $\alpha_{\text{BDV}}(\Omega_0) > \alpha_{\text{small}}$ we fall into the previous paragraph (a).

(c) Schauder/interpolation regime not reached because of (I) failure. Analogously, (I) fails only if $\alpha_{\text{BDV}}(\tilde{\Omega}) > \alpha_{\text{sph}}$. By the same choice of $\alpha_{\text{small}} \leq \alpha_{\text{sph}}/2$ we never enter that regime in Case B. If $\alpha_{\text{BDV}}(\Omega_0) > \alpha_{\text{small}}$ we again fall into (a). Equivalently: the small-asymmetry threshold (155) is chosen as the smallest of the three upstream thresholds (divided by 2 to absorb (S2)), so all three *simultaneously* hold for Case B.

(d) Sharp boundary $\alpha_{\text{BDV}} \sim \alpha_{\text{small}}$. The proof of Theorem 3.28 glues the two regimes at $\mathcal{A} = a_{\text{near}}$, which by (163) corresponds to $\alpha_{\text{BDV}} = \alpha_{\text{small}}$ (via the linear inequality (143)). At the glue point both $\kappa_{\text{near}}(n, R)\mathcal{A}^2$ and $(a_{\text{near}}^2/C_8')\mathcal{A}^2$ are positive, so $c_{\text{SV}}(n, R) > 0$ is the minimum of two strictly positive quantities, with explicit lower bounds

$$c_{\text{SV}}(n, R) \geq \min\left\{ \frac{\omega_n^2}{C_1(n)} \cdot \frac{c_*(n)}{2C_{\text{sel}}(n, R)}, \frac{\alpha_{\text{small}}(n, R)^2}{C_2(R)^2 \omega_n^2 C_8'(n)} \right\}.$$

No additional smallness hypothesis is needed.

3.3.11. *Summary: what this subsection delivers.* Combining the above with the notation $\delta_T(\Omega) = E(\Omega) - E(B^*) \geq 0$ of the register, we obtain the bounded sharp Saint-Venant inequality

$$(169) \quad \boxed{\mathcal{A}(\Omega)^2 \leq C(n, R) \delta_T(\Omega), \quad C(n, R) = \frac{1}{c_{\text{SV}}(n, R)},}$$

valid for every open $\Omega \subset B_R$ with $|\Omega| = \omega_n = |B^*|$. The constant $C(n, R) = 1/c_{\text{SV}}(n, R)$ has explicit dependence on n, R traced through Theorem 3.28 and the constant chain of §3.3.9; no compactness or contradiction argument enters. The bounded reduction $c_{\text{SV}}(n, R_*(n)) \Rightarrow c_{\text{SV}}^{\text{glob}}(n)$ that removes the R -dependence is the subject of Agent 8.

Status statement.

- Theorem 3.23: assembled in this subsection from (B1) and (B2)(iii).
- Theorem 3.24: assembled in this subsection from (S)+(G)+(I)+Theorem 3.23.
- Corollary 3.26, Lemma 3.27, Theorem 3.28: assembled in this subsection from (S)+(G)+(I)+NS, (B2)(i)+(ii), and (B3).
- Upstream black boxes (B1)+(B2)+(B3): cited verbatim from [15] as [15, Thm. 3.3, Lem. 4.2 (i),(ii),(iii), Lem. 5.2].
- Inputs (S), (G), (I) from Agents 5, 6: cited by section number inside `Manuscript/02_plan1_draft.tex` once Agent 19 integrates the three Plan 1 subsections.

Contradiction step. The BDV proof of the small- α_{BDV} linear bound is by contradiction ([15, Theorem 4.3]); here it is replaced by the single-set selection of Agent 5, which compresses the sequence/compactness/limit argument into the explicit quotient-preservation bound (S3). No additional expansion of the source contradiction step was needed beyond Agent 5’s selection theorem: Theorem 3.24 is a direct combinatorial combination of (S), (G), (I), and the pointwise nearly-spherical inequality, performed in §3.3.5.

3.4. Bounded reduction and the Kohler–Jobin transfer to Faber–Krahn. This subsection closes Plan 1. The input is the bounded sharp Saint–Venant inequality of Agent 7 (Theorem 3.29). Three steps follow:

- (i) *Bounded reduction* (§3.4.1). Using the constructive BDV truncation lemma, we replace an arbitrary open $\Omega \subset \mathbb{R}^n$ with finite measure by a translate that lies in a ball of *dimensional* radius $R_*(n)$, while controlling the loss of both the Saint–Venant deficit and the Fraenkel asymmetry. This removes the R -dependence from the Saint–Venant constant.
- (ii) *Kohler–Jobin inequality* (§3.4.2). We state the scale-invariant comparison $T(\Omega) \lambda_1(\Omega)^{(n+2)/2} \geq T_n L_n^{(n+2)/2}$ and derive the pointwise lower bound for $\lambda_1(\Omega) - L$ in terms of the torsion deficit.
- (iii) *Final Plan 1 theorem* (§3.4.3). Combining (i) and (ii) yields the sharp quantitative Faber–Krahn inequality in the canonical normalisation of the notation register,

$$\mathcal{A}(\Omega)^2 \leq C_{\text{FK}}(n) \delta_{\text{FK}}(\Omega),$$

with $C_{\text{FK}}(n)$ depending only on the dimension.

Standing input from Plan 1, Agent 7.

Theorem 3.29 (Bounded sharp Saint–Venant; Agent 7, item D3.8). *For every $n \geq 2$ and every $R \geq 2$ there is a constant $c_{\text{SV}}(n, R) > 0$, explicit in the BDV constants of the selection / graph entry / nearly-spherical chain, such that*

$$(170) \quad \delta_T(\Omega) = E(\Omega) - E(B_1) \geq c_{\text{SV}}(n, R) \mathcal{A}(\Omega)^2$$

for every open set $\Omega \subset B_R$ with $|\Omega| = \omega_n$.

The hypothesis $|\Omega| = \omega_n$ is the standing volume normalisation of the chain; we will remove it by scaling at the end (Remark 3.40).

3.4.1. Bounded reduction: removing the R -dependence. The bounded reduction is the BDV truncation argument [15, Lemmas 5.1–5.3]. It is constructive: a De Giorgi L^∞ iteration is combined with an energy cut-off competitor; no compactness, no minimising sequence, and no contradiction step is used. The end product is a bounded surrogate $\tilde{\Omega}$ of Ω inside a dimensional ball, with quantitatively controlled loss.

Scale-invariant Saint-Venant deficit. It is convenient to work with the scale-invariant deficit

$$(171) \quad D(\Omega) := \omega_n^{-(n+2)/n} \delta_T(\Omega) = \omega_n^{-(n+2)/n} (E(\Omega) - E(B^*)).$$

When $|\Omega| = \omega_n$ we have $B^* = B_1$ and $D(\Omega) \geq 0$ is the volume-normalised Saint-Venant deficit. The Fraenkel asymmetry $\mathcal{A}(\Omega)$ is dimensionless and scale invariant.

Two BDV black boxes. We import two estimates from [15] as black boxes.

Lemma 3.30 ([15, Lemma 5.1]: De Giorgi L^∞ tail). *There is $C_7 = C_7(n) > 0$ such that, for every open Ω with $|\Omega| = \omega_n$ and every $R \geq 1$,*

$$(172) \quad \|u_\Omega\|_{L^\infty(\Omega \setminus B_{R+1})} \leq C_7 |\Omega \setminus B_R|^{1/n}.$$

Reference. [15, Lemma 5.1]; De Giorgi iteration on the truncated test function $\varphi_k^2(u_\Omega - s_k)_+$. Constructive; constant $C_7(n)$ explicit. $\square$

Lemma 3.31 ([15, Lemma 5.2]: suboptimal $\mathcal{A}^4$ bound). *There is $C_8 = C_8(n) > 0$ such that for every open Ω of finite measure,*

$$(173) \quad D(\Omega) \geq \frac{1}{C_8} \mathcal{A}(\Omega)^4.$$

Reference. [15, Lemma 5.2]; equivalent, via $E = -T/2$, to [32, Theorem 1] with $p = 2, q = 1$. Pure rearrangement plus Pólya-Szegő; no contradiction. $\square$

The truncation lemma. The core construction.

Lemma 3.32 ([15, Lemma 5.3]: bounded truncation). *There exist dimensional constants $C_9 = C_9(n) > 0$, $\delta_{\text{BDV}} = \delta_{\text{BDV}}(n) \in (0, 1]$, and $d(n) \geq 1$ such that for every open Ω with $|\Omega| = \omega_n$ and $D(\Omega) \leq \delta_{\text{BDV}}(n)$, there exists an open set $\tilde{\Omega} \subset \mathbb{R}^n$ satisfying*

$$(174) \quad |\tilde{\Omega}| = \omega_n,$$

$$(175) \quad \text{diam}(\tilde{\Omega}) \leq d(n),$$

$$(176) \quad D(\tilde{\Omega}) \leq C_9(n) D(\Omega),$$

$$(177) \quad \mathcal{A}(\Omega) \leq \mathcal{A}(\tilde{\Omega}) + C_9(n) D(\Omega).$$

Sketch and reference. [15, Lemma 5.3]. The proof combines (172) with the cut-off test function $\varphi_k u_\Omega$ in the Saint-Venant variational characterisation of $E(\Omega \cap B_{k+2})$, yielding [15, (5.11)]

$$E(\Omega \cap B_{k+2}) \leq E(\Omega) + C b_k^{1+1/n}, \quad b_k := |\Omega \setminus B_k|/\omega_n.$$

A geometric iteration on $k \leq K \leq K(n)$ produces $b_{K+1} \leq 2\hat{C} D(\Omega)$. Set $r := (|\Omega \cap B_{K+3}|/\omega_n)^{1/n} \in [1 - \hat{C} D(\Omega), 1]$ and $\tilde{\Omega} := r^{-1}(\Omega \cap B_{K+3})$. Properties (174)–(176) follow directly. The asymmetry estimate (177) uses

$$|\Omega \Delta (B_1 + x_0)| \leq |(\Omega \cap B_{K+3}) \Delta \Omega| + r^n |\tilde{\Omega} \Delta (B_1 + x_0/r)| + \omega_n(1 - r^n),$$

followed by $1 - r^n \leq n\hat{C} D(\Omega)$. The construction is *constructive*: no minimising sequence, no Hausdorff limit, no contradiction step. $\square$

Remark 3.33 (Translation freedom). The diameter bound (175) is translation invariant. After translating $\tilde{\Omega}$ we may assume $\tilde{\Omega} \subset B_{d(n)}$. Both $\mathcal{A}$ and D are translation invariant.

Working radius. Define the *bounded-reduction radius*

$$(178) \quad R_*(n) := \max\{d(n), 2\}.$$

By Remark 3.33 any $\tilde{\Omega}$ produced by Lemma 3.32 can be translated into $B_{R_*(n)}$, so Theorem 3.29 applies with the dimensional choice $R = R_*(n)$. Far-deficit branch.

Lemma 3.34 (Far-deficit quadratic bound). *For every open Ω with $|\Omega| = \omega_n$ and $D(\Omega) \geq \delta_{\text{BDV}}(n)/4$,*

$$(179) \quad E(\Omega) - E(B_1) \geq \omega_n^{(n+2)/n} \frac{\delta_{\text{BDV}}(n)}{16} \mathcal{A}(\Omega)^2.$$

Proof. Since $\mathcal{A}(\Omega) < 2$, $\mathcal{A}(\Omega)^2/4 < 1$. Hence $D(\Omega) \geq \delta_{\text{BDV}}(n)/4 \geq (\delta_{\text{BDV}}(n)/16)\mathcal{A}(\Omega)^2$. Multiplying by $\omega_n^{(n+2)/n}$ and using $E(\Omega) - E(B_1) = \omega_n^{(n+2)/n} D(\Omega)$ at volume ω_n gives the claim. $\square$

Remark 3.35. The estimate (179) is much weaker than (173) (it converts the $\mathcal{A}^4$ control into an $\mathcal{A}^2$ control by absorbing $\mathcal{A}^2 \leq 4$), but in this regime D is bounded below by a dimensional constant, so the trade is harmless.

Near-deficit branch.

Lemma 3.36 (Near-deficit transfer). *Set*

$$(180) \quad \tilde{c}_{\text{SV}}(n) := \omega_n^{-(n+2)/n} c_{\text{SV}}(n, R_*(n)) > 0.$$

For every open Ω with $|\Omega| = \omega_n$ and $D(\Omega) \leq \delta_{\text{BDV}}(n)$,

$$(181) \quad \mathcal{A}(\Omega)^2 \leq \left(\frac{2C_9(n)}{\tilde{c}_{\text{SV}}(n)} + 2C_9(n)^2 \delta_{\text{BDV}}(n) \right) D(\Omega).$$

Proof. Apply Lemma 3.32 to obtain $\tilde{\Omega}$ satisfying (174)–(177), and by Remark 3.33 assume $\tilde{\Omega} \subset B_{d(n)} \subset B_{R_*(n)}$. Apply Theorem 3.29 with $R = R_*(n)$ to $\tilde{\Omega}$:

$$E(\tilde{\Omega}) - E(B_1) \geq c_{\text{SV}}(n, R_*(n)) \mathcal{A}(\tilde{\Omega})^2.$$

Divide by $\omega_n^{(n+2)/n}$ (and use $|\tilde{\Omega}| = \omega_n$) to obtain

$$(182) \quad D(\tilde{\Omega}) \geq \tilde{c}_{\text{SV}}(n) \mathcal{A}(\tilde{\Omega})^2.$$

Combined with (176),

$$\mathcal{A}(\tilde{\Omega})^2 \leq \frac{D(\tilde{\Omega})}{\tilde{c}_{\text{SV}}(n)} \leq \frac{C_9(n) D(\Omega)}{\tilde{c}_{\text{SV}}(n)}.$$

Hence $\mathcal{A}(\tilde{\Omega}) \leq (C_9(n)D(\Omega)/\tilde{c}_{\text{SV}}(n))^{1/2}$. Plugging this into (177) and squaring,

$$\begin{aligned} \mathcal{A}(\Omega)^2 &\leq (\mathcal{A}(\tilde{\Omega}) + C_9(n)D(\Omega))^2 \leq 2\mathcal{A}(\tilde{\Omega})^2 + 2C_9(n)^2 D(\Omega)^2 \\ &\leq \frac{2C_9(n)}{\tilde{c}_{\text{SV}}(n)} D(\Omega) + 2C_9(n)^2 D(\Omega)^2 \\ &\leq \left(\frac{2C_9(n)}{\tilde{c}_{\text{SV}}(n)} + 2C_9(n)^2 \delta_{\text{BDV}}(n) \right) D(\Omega), \end{aligned}$$

where the last step uses $D(\Omega) \leq \delta_{\text{BDV}}(n) \leq 1$. $\square$

Corollary 3.37 (Near-deficit quadratic bound). *With*

$$(183) \quad c_{\text{near}}^{\text{glob}}(n) := \frac{\omega_n^{(n+2)/n}}{2C_9(n)(1/\tilde{c}_{\text{SV}}(n) + C_9(n)\delta_{\text{BDV}}(n))},$$

every open Ω with $|\Omega| = \omega_n$ and $D(\Omega) \leq \delta_{\text{BDV}}(n)$ satisfies

$$(184) \quad E(\Omega) - E(B_1) \geq c_{\text{near}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2.$$

Proof. (181) gives $D(\Omega) \geq \mathcal{A}(\Omega)^2 / (2C_9/\tilde{c}_{\text{SV}} + 2C_9^2\delta_{\text{BDV}})$. Multiply by $\omega_n^{(n+2)/n}$ and use $E - E(B_1) = \omega_n^{(n+2)/n} D$ to get (184). $\square$

Global bounded reduction.

Theorem 3.38 (Global sharp Saint–Venant; bounded reduction). *Define*

$$(185) \quad c_{\text{SV}}^{\text{glob}}(n) := \min \left\{ c_{\text{near}}^{\text{glob}}(n), \omega_n^{(n+2)/n} \delta_{\text{BDV}}(n)/16 \right\} > 0.$$

Then for every open set $\Omega \subset \mathbb{R}^n$ with $|\Omega| = \omega_n$,

$$(186) \quad \delta_T(\Omega) = E(\Omega) - E(B_1) \geq c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2.$$

The constant $c_{\text{SV}}^{\text{glob}}(n)$ depends only on the dimension n : the R -dependence inherited from Theorem 3.29 has been internalised at the dimensional choice $R = R_*(n)$.

Proof. The bound $D(\Omega) \geq 0$ is the classical Saint–Venant inequality. We split into two cases.

Case A ($D(\Omega) \geq \delta_{\text{BDV}}(n)/4$): Lemma 3.34 gives $E(\Omega) - E(B_1) \geq \omega_n^{(n+2)/n} (\delta_{\text{BDV}}(n)/16) \mathcal{A}(\Omega)^2$, which is (186) with constant $\omega_n^{(n+2)/n} \delta_{\text{BDV}}(n)/16$.

Case B ($D(\Omega) \leq \delta_{\text{BDV}}(n)$, which in particular covers $D(\Omega) \leq \delta_{\text{BDV}}(n)/4$): Corollary 3.37 applies and gives (186) with constant $c_{\text{near}}^{\text{glob}}(n)$.

The two cases overlap on $\delta_{\text{BDV}}(n)/4 \leq D(\Omega) \leq \delta_{\text{BDV}}(n)$ and together cover $D(\Omega) \in [0, +\infty)$. Taking the minimum of the two constants gives (185). $\square$

Remark 3.39 (Dependence on R). The dependence on the confining radius R enters only via Theorem 3.29, which is applied to the surrogate $\tilde{\Omega} \subset B_{R_*(n)}$. Since $R_*(n)$ is itself dimensional, the final constant $c_{\text{SV}}^{\text{glob}}(n)$ in (185) depends only on n . Concretely, the R -dependence is absorbed in $\tilde{c}_{\text{SV}}(n) = \omega_n^{-(n+2)/n} c_{\text{SV}}(n, R_*(n))$ in (180); everything downstream is purely dimensional.

Remark 3.40 (Removal of the volume normalisation). (186) stated for $|\Omega| = \omega_n$ extends to every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$ by scale invariance of D and $\mathcal{A}$: writing $\Omega_* := (\omega_n/|\Omega|)^{1/n}\Omega$ one has $|\Omega_*| = \omega_n$, $\mathcal{A}(\Omega_*) = \mathcal{A}(\Omega)$, and $D(\Omega_*) = D(\Omega)$. Hence (186) applied to Ω_* reads

$$(187) \quad \omega_n^{(n+2)/n} D(\Omega) = |\Omega|^{(n+2)/n} D(\Omega) \cdot (\omega_n/|\Omega|)^{(n+2)/n} \quad \text{etc.}$$

Equivalently, at volume $|B^*| = |\Omega|$,

$$(188) \quad T(B^*) - T(\Omega) = 2(E(\Omega) - E(B^*)) \geq 2c_{\text{SV}}^{\text{glob}}(n) \left(\frac{|\Omega|}{\omega_n}\right)^{(n+2)/n} \mathcal{A}(\Omega)^2.$$

This is the input to the Kohler–Jobin transfer (§3.4.2).

3.4.2. The Kohler–Jobin inequality and the pointwise transfer.

Statement. We invoke the Kohler–Jobin inequality in the following scale-invariant form, which is the form needed to transfer Saint–Venant stability to Faber–Krahn stability.

Theorem 3.41 (Kohler–Jobin inequality). *For every open set $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$ and $\lambda_1(\Omega) < \infty$, and for every ball B ,*

$$(189) \quad T(\Omega) \lambda_1(\Omega)^{(n+2)/2} \geq T(B) \lambda_1(B)^{(n+2)/2}.$$

Equivalently, $\Psi(\Omega) := T(\Omega) \lambda_1(\Omega)^{(n+2)/2} \geq \Psi(B_1) = T_n L_n^{(n+2)/2}$. Equality holds if and only if Ω is a ball (up to translation and a Lebesgue null set).

Reference. The original statement is in Kohler–Jobin [40]. We use the form of [13, Theorem 3.3, p. 322], proved by Schwarz symmetrisation and a one-dimensional rearrangement of the torsion potential along level sets of the eigenfunction. The characterisation of the equality case is [13, Theorem 3.3]. $\square$

Remark 3.42 (Canonical normalisation and scaling). The exponent $(n+2)/2$ in (189) is forced by scale invariance: from $T(t\Omega) = t^{n+2}T(\Omega)$ and $\lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega)$ we read $\Psi(t\Omega) = t^{n+2-2 \cdot (n+2)/2}\Psi(\Omega) = \Psi(\Omega)$, so Ψ is scale-invariant. This is the canonical form of the inequality used throughout the manuscript; both source files `Plan 1/faber-krahn-transfer.tex` and `Final/KohlerJobinTransfer.tex` use the same form (with N converted to n).

Pointwise transfer at fixed volume. Fix the comparison ball B^* with $|B^*| = |\Omega|$, and write

$$L := \lambda_1(B^*), \quad T_0 := T(B^*), \quad \beta := \frac{n+2}{2}.$$

By scaling $L = L_n(\omega_n/|B^*|)^{2/n}$ and $T_0 = T_n(|B^*|/\omega_n)^{(n+2)/n}$.

Lemma 3.43 (Pointwise Kohler–Jobin transfer). *For every open Ω with $|\Omega| = |B^*|$ and $\lambda_1(\Omega) < \infty$,*

$$(190) \quad \lambda_1(\Omega) \geq L \cdot \left(\frac{T_0}{T(\Omega)}\right)^{1/\beta} = L \cdot \left(\frac{T_0}{T(\Omega)}\right)^{2/(n+2)}.$$

Proof. Apply Theorem 3.41 with $B = B^*$ and rearrange. Since $T(\Omega) > 0$ and $\lambda_1(\Omega) < \infty$, both sides are finite and positive. $\square$

Lemma 3.44 (Elementary one-variable inequality). *For every $p \in (0, 1]$ and every $s \in [0, 1]$,*

$$(191) \quad (1 - s)^{-p} - 1 \geq p s.$$

Proof. Set $f(s) := (1-s)^{-p} - 1$. Then $f(0) = 0$ and $f'(s) = p(1-s)^{-(p+1)} \geq p$ for $s \in [0, 1]$. By the fundamental theorem of calculus, $f(s) = \int_0^s f'(\tau) d\tau \geq ps$, which is (191). The factor p on the right is sharp: $f'(0) = p$, so the linear approximation is tight at $s = 0$. $\square$

Small-deficit linearisation. The classical Saint-Venant inequality gives $T(\Omega) \leq T_0$ at fixed volume, so the torsion deficit $\sigma(\Omega) := T_0 - T(\Omega) \geq 0$. (We use the local symbol σ here, since the manuscript reserves δ_T for the energy-form deficit.) Equivalently $\sigma(\Omega) = 2\delta_T(\Omega)$, and setting $s := \sigma(\Omega)/T_0 \in [0, 1]$, (190) becomes

$$(192) \quad \lambda_1(\Omega) - L \geq L[(1 - s)^{-1/\beta} - 1].$$

Lemma 3.45 (Small-deficit Kohler-Jobin linearisation). *For every open Ω with $|\Omega| = |B^*|$ and $\sigma(\Omega) \leq T_0/2$,*

$$(193) \quad \lambda_1(\Omega) - L \geq \frac{2L}{(n+2)T_0} \sigma(\Omega) = \frac{4L}{(n+2)T_0} \delta_T(\Omega).$$

Proof. With $p = 1/\beta = 2/(n+2) \in (0, 1]$ and $s = \sigma(\Omega)/T_0 \in [0, 1/2]$, combine (192) and Lemma 3.44:

$$\lambda_1(\Omega) - L \geq L p s = \frac{2L}{(n+2)T_0} \sigma(\Omega).$$

The second equality uses $\sigma = 2\delta_T$. $\square$

Remark 3.46 (Universal multiplier). The factor $2L/((n+2)T_0)$ is the *Kohler-Jobin multiplier*. Both L and T_0 scale, but the product is dimensionless: $2L/((n+2)T_0) = 2L_n/((n+2)T_n) \cdot (\omega_n/|B^*|)^{(n+2)/n} / (\omega_n/|B^*|)^{2/n} \dots$, which simplifies to

$$(194) \quad \frac{2L}{(n+2)T_0} = \frac{2L_n}{(n+2)T_n} \cdot \left(\frac{\omega_n}{|B^*|} \right)^{(n+4)/n}.$$

Using $T_n = \omega_n/(n(n+2))$, the dimensional part simplifies to $2L_n/((n+2)T_n) = 2nL_n/\omega_n$.

Large-deficit branch.

Lemma 3.47 (Large-deficit Kohler-Jobin bound). *For every open Ω with $|\Omega| = |B^*|$ and $\sigma(\Omega) > T_0/2$,*

$$(195) \quad \lambda_1(\Omega) - L \geq L(2^{2/(n+2)} - 1).$$

Proof. $\sigma > T_0/2$ means $T_0/T(\Omega) > 2$, so (190) gives $\lambda_1(\Omega) \geq L \cdot 2^{2/(n+2)}$, i.e. (195). $\square$

Since $|\Omega| = |B^*|$ forces $\mathcal{A}(\Omega) \in [0, 2)$ (so $\mathcal{A}^2 \leq 4$), in the large-deficit regime we may rewrite (195) as

$$(196) \quad \lambda_1(\Omega) - L \geq \frac{L(2^{2/(n+2)} - 1)}{4} \mathcal{A}(\Omega)^2.$$

3.4.3. Final Plan 1 sharp Faber–Krahn theorem. We now combine the global Saint–Venant inequality (Theorem 3.38) and the pointwise Kohler–Jobin transfer (Lemmas 3.45 and 3.47) to obtain the sharp quantitative Faber–Krahn inequality in the canonical normalisation of the notation register. Statement in the lower form.

Theorem 3.48 (Sharp Faber–Krahn via Kohler–Jobin; lower form). *There is a dimensional constant $c_{\text{FK}}(n) > 0$ such that, for every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,*

$$(197) \quad \delta_{\text{FK}}(\Omega) = \left(\frac{|\Omega|}{\omega_n} \right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2.$$

The constant is given explicitly by

$$(198) \quad c_{\text{FK}}(n) := \min\{c_1(n), c_2(n)\},$$

where

$$(199) \quad c_1(n) := \frac{4L_n c_{\text{SV}}^{\text{glob}}(n)}{(n+2)T_n} = \frac{4nL_n}{\omega_n} c_{\text{SV}}^{\text{glob}}(n), \quad c_2(n) := \frac{L_n(2^{2/(n+2)} - 1)}{4}.$$

Proof. We first reduce to the volume-normalised case $|\Omega| = \omega_n$. The quantity $\delta_{\text{FK}}(\Omega)$ is scale invariant: if $\Omega_* := (\omega_n/|\Omega|)^{1/n}\Omega$, then $|\Omega_*| = \omega_n$, $\mathcal{A}(\Omega_*) = \mathcal{A}(\Omega)$, and $\delta_{\text{FK}}(\Omega_*) = \lambda_1(\Omega_*) - L_n = (|\Omega|/\omega_n)^{2/n}(\lambda_1(\Omega) - L)$ by the scaling $\lambda_1(t\Omega) = t^{-2}\lambda_1(\Omega)$ (with $t = (\omega_n/|\Omega|)^{1/n}$), which equals $\delta_{\text{FK}}(\Omega)$. Thus it suffices to prove $\lambda_1(\Omega) - L_n \geq c_{\text{FK}}(n)\mathcal{A}(\Omega)^2$ for $|\Omega| = \omega_n$ (in which case $B^* = B_1$, $L = L_n$, $T_0 = T_n$, $\delta_{\text{FK}}(\Omega) = \lambda_1(\Omega) - L_n$).

Assume henceforth $|\Omega| = \omega_n$. Theorem 3.38 gives $\delta_T(\Omega) \geq c_{\text{SV}}^{\text{glob}}(n)\mathcal{A}(\Omega)^2$, so the torsion deficit

$$(200) \quad \sigma(\Omega) = T_n - T(\Omega) = 2\delta_T(\Omega) \geq 2c_{\text{SV}}^{\text{glob}}(n)\mathcal{A}(\Omega)^2.$$

Case 1: $\sigma(\Omega) \leq T_n/2$. Lemma 3.45 (with $L = L_n$, $T_0 = T_n$) and (200) give

$$\lambda_1(\Omega) - L_n \geq \frac{2L_n}{(n+2)T_n} \sigma(\Omega) \geq \frac{2L_n}{(n+2)T_n} \cdot 2c_{\text{SV}}^{\text{glob}}(n)\mathcal{A}(\Omega)^2 = \frac{4L_n c_{\text{SV}}^{\text{glob}}(n)}{(n+2)T_n} \mathcal{A}(\Omega)^2 = c_1(n)\mathcal{A}(\Omega)^2$$

the last equality being the definition of $c_1(n)$ in (199). The first inequality used the small-deficit Kohler–Jobin linearisation expressed in terms of the torsion deficit σ ; the second inequality used $\sigma = 2\delta_T$ combined with the Saint–Venant input $\delta_T \geq c_{\text{SV}}^{\text{glob}}\mathcal{A}^2$.

Case 2: $\sigma(\Omega) > T_n/2$. (196) with $L = L_n$ gives

$$\lambda_1(\Omega) - L_n \geq \frac{L_n(2^{2/(n+2)} - 1)}{4} \mathcal{A}(\Omega)^2 = c_2(n)\mathcal{A}(\Omega)^2.$$

Combining the two cases gives $\lambda_1(\Omega) - L_n \geq \min\{c_1(n), c_2(n)\}\mathcal{A}(\Omega)^2 = c_{\text{FK}}(n)\mathcal{A}(\Omega)^2$, which is (197) in the volume-normalised case, and hence (by the first paragraph) in general. $\square$

Remark 3.49 (Canonical form of c_1). The constant $c_1(n)$ in (199) matches [13, §3, eq. (3.1)] and the canonical form in `Final/KohlerJobinTransfer.tex` (equations c_1 , c_{FK}). Note that the source file `Plan 1/faber-krahn-transfer.tex`

states the small-deficit multiplier as $2L/((n+2)T_0)$ applied to the torsion deficit $\sigma = T_0 - T$, while our proof applies it after the substitution $\sigma = 2\delta_T$; this accounts for the explicit factor of 4 (not 2) on $L_n c_{SV}^{\text{glob}}/((n+2)T_n)$ in (199). The two source files `Final/KohlerJobinTransfer.tex` (eq. (199) above) and `Plan 1/faber-krahn-transfer.tex` (§ Quantitative dependence) agree on this normalisation.

Convenient abbreviation. Throughout the rest of the manuscript we write

$$(201) \quad c_1(n) := \frac{4L_n c_{SV}^{\text{glob}}(n)}{(n+2)T_n} = \frac{4nL_n}{\omega_n} c_{SV}^{\text{glob}}(n),$$

which is the definition just used.

Canonical statement (upper form). Inverting (197), the inequality reads $\mathcal{A}(\Omega)^2 \leq C_{\text{FK}}(n) \delta_{\text{FK}}(\Omega)$ with $C_{\text{FK}}(n) = 1/c_{\text{FK}}(n)$. This is the upper form required by the master brief (and shared with Plan 2).

Theorem 3.50 (Sharp Faber–Krahn; Plan 1, upper form). *There is a dimensional constant $C_{\text{FK}}(n) > 0$ such that, for every open set $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,*

$$(202) \quad \boxed{\mathcal{A}(\Omega)^2 \leq C_{\text{FK}}(n) \delta_{\text{FK}}(\Omega)},$$

where

$$(203) \quad C_{\text{FK}}(n) := \frac{1}{c_{\text{FK}}(n)} = \max\left\{\frac{(n+2)T_n}{4L_n c_{SV}^{\text{glob}}(n)}, \frac{4}{L_n(2^{2/(n+2)} - 1)}\right\},$$

with $c_{SV}^{\text{glob}}(n)$ defined in (185). Equality cases: by Theorem 3.41, $\delta_{\text{FK}}(\Omega) = 0$ iff Ω is a ball; in this case $\mathcal{A}(\Omega) = 0$.

Proof. If $\lambda_1(\Omega) = +\infty$ then $\delta_{\text{FK}}(\Omega) = +\infty$ and the inequality (202) is trivial (the left-hand side is bounded above by 4 since $\mathcal{A}(\Omega) < 2$). Otherwise $\lambda_1(\Omega) < \infty$ and the result is immediate from Theorem 3.48 with the canonical $c_1(n)$ from (201): invert the lower bound. The constant $C_{\text{FK}}(n)$ depends only on n because $c_{SV}^{\text{glob}}(n)$ does (Remark 3.39) and L_n, T_n are dimensional. $\square$

Remark 3.51 (R drops out). The dependence on the confining radius R is removed in two stages:

- Theorem 3.29 (Agent 7) provides $c_{SV}(n, R)$ for $\Omega \subset B_R$. We *instantiate* this input at $R = R_*(n) = \max\{d(n), 2\}$, a purely dimensional radius defined in (178).
- The bounded reduction (Theorem 3.38) transfers this to $c_{SV}^{\text{glob}}(n)$ via the BDV truncation lemma, which produces a translate $\tilde{\Omega} \subset B_{R_*(n)}$.

No subsequent step reintroduces R : the Kohler–Jobin inequality (189) is independent of any confining ball, and the pointwise transfer (190) depends only on the universal ball constants $L_n = \lambda_1(B_1)$, $T_n = T(B_1)$, and the torsion deficit at fixed volume.

Remark 3.52 (Sharpness of the exponent). The exponent 2 on $\mathcal{A}$ in (202) is sharp; it is saturated by smooth volume-preserving ellipsoidal perturbations of the ball, for which $\mathcal{A}(\Omega)$ and $\delta_{\text{FK}}(\Omega)^{1/2}$ are of the same order. See [15, §1.2] for the construction.

Remark 3.53 (Match with Plan 2 normalisation). The final theorem (202) is stated in the canonical normalisation of the notation register (item D4.6 of `Manuscript/source_map.md`), with δ_{FK} defined as in notation register §4: $\delta_{\text{FK}}(\Omega) = (|\Omega|/\omega_n)^{2/n}(\lambda_1(\Omega) - \lambda_1(B^*))$. This is exactly the normalisation of the Plan 2 final theorem (item E5.8); the two proofs share both the bounded reduction (Theorem 3.38) and the Kohler–Jobin transfer step (§3.4.2). The constant $C_{\text{FK}}(n)$ produced here and the constant $C_{\text{FK}}^{\text{Plan 2}}(n)$ produced in §4.5.6 differ only through the Saint–Venant constant $c_{\text{SV}}^{\text{glob}}(n)$, which each route constructs independently.

Summary of constants.

Symbol	Value/dependence	Source
ω_n	$ B_1 $	dimensional
T_n	$\omega_n/(n(n+2))$	dimensional
L_n	$\lambda_1(B_1) = j_{n/2-1,1}^2$	dimensional
β	$(n+2)/2$	Theorem 3.41
$C_7(n)$	[15, Lem. 5.1]	Lemma 3.30
$C_8(n)$	[15, Lem. 5.2]	Lemma 3.31
$C_9(n)$	[15, Lem. 5.3]	Lemma 3.32
$\delta_{\text{BDV}}(n) \in (0, 1]$	[15, Lem. 5.3]	Lemma 3.32
$d(n) \geq 1$	[15, Lem. 5.3]	Lemma 3.32
$R_*(n) = \max\{d(n), 2\}$	dimensional	(178)
$c_{\text{SV}}(n, R)$	Agent 7	Theorem 3.29
$\tilde{c}_{\text{SV}}(n)$	$\omega_n^{-(n+2)/n} c_{\text{SV}}(n, R_*(n))$	(180)
$c_{\text{near}}^{\text{glob}}(n)$	(183)	Corollary 3.37
$c_{\text{SV}}^{\text{glob}}(n)$	$\min\{c_{\text{near}}^{\text{glob}}, \omega_n^{(n+2)/n} \delta_{\text{BDV}}/16\}$	Theorem 3.38
$c_1(n)$	$4L_n c_{\text{SV}}^{\text{glob}}/((n+2)T_n) = 4nL_n c_{\text{SV}}^{\text{glob}}/\omega_n$	(201)
$c_2(n)$	$L_n(2^{2/(n+2)} - 1)/4$	(199)
$c_{\text{FK}}(n)$	$\min\{c_1(n), c_2(n)\}$	Theorem 3.48
$C_{\text{FK}}(n)$	$1/c_{\text{FK}}(n)$	Theorem 3.50

Where compactness could have entered (and does not). The bounded reduction (§3.4.1) is the only step where one might fear a hidden compactness argument. We record:

- BDV Lemma 5.1 (Lemma 3.30) is De Giorgi L^∞ iteration: constructive, no minimising sequence.
- BDV Lemma 5.2 (Lemma 3.31) is rearrangement plus Pólya–Szegő: no contradiction.
- BDV Lemma 5.3 (Lemma 3.32) reaches the truncation index $K \leq K(n)$ by a geometric iteration [15, (5.12)–(5.14)], not by Hausdorff compactness.
- The gluing of near and far branches (Theorem 3.38) is arithmetic.
- The Kohler–Jobin transfer (§3.4.2) uses only (189), the FTC, and elementary monotone-function inequalities.

This closes Plan 1.

4. SECOND PROOF: LEVEL-SET DEFICIT IDENTITY AND METRIC TRACE (PLAN 2)

This section presents the second of the two proofs of the sharp quantitative Faber–Krahn inequality (4). The chain is structurally different from Plan 1: instead of selecting a single near-extremal set and analysing it as a graph over a sphere, Plan 2 analyses every level ρ of the volume-radius foliation $E_\rho = \{u_\Omega > t(\rho)\}$, classifies levels by their isoperimetric defect, and transfers the integrated defect bound to the metric quotient $\mathcal{X} = \mathfrak{M}/\mathbb{R}^n$:

- level-set deficit identity (§4.1)
 - metric framework / first variation (§4.2.1)
 - FJ centre + oriented radial trace (§4.3)
 - integrated action / weighted metric trace (§4.4)
 - boundary-layer transfer + final Plan 2 theorem (§4.5).

The auxiliary centroid / space–time Π route is included as the last subsection (§4.6); per Remark 1.2 it is *not* used in any load-bearing step of the repaired Plan 2 chain. Plan 2’s output is the same global sharp Saint–Venant inequality fed into Kohler–Jobin as in Plan 1, with the same final constant $C_{\text{FK}}(n)$.

4.1. Exact level-set deficit identity. This subsection contains the analytic foundation of Plan 2. We prove an exact representation of the (normalised) Saint–Venant deficit $\delta_T(\Omega)$ as a weighted integral, over the levels of the torsion function, of two nonnegative pointwise defects: a Cauchy–Schwarz defect $D_H(t)$ of $|\nabla u|$ on the reduced boundary $\partial^* E_t$, and a squared-perimeter isoperimetric defect $D_I(t)$ of the superlevel set E_t . The identity is valid, in full finite-perimeter generality, for any open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$; no smoothness, boundedness, or connectedness of Ω is assumed.

The result is the analytic input of every downstream block in §4.2.10–§4.5.6. Two further consequences proved here (the profile-gap estimate and the superlevel-transfer estimate) are used in §4.4.7 (good levels in a near-boundary slab) and in §4.5.6 (transfer of the Fraenkel asymmetry from a superlevel set back to Ω).

Standing hypotheses and notation. Throughout the subsection, $n \geq 2$ and $\Omega \subset \mathbb{R}^n$ is an open set with $0 < |\Omega| < \infty$. We use the canonical notation register of §2 without further comment, and recall only the items that enter the present argument:

- $u = u_\Omega \in H_0^1(\Omega)$ is the variational torsion function, i.e. the unique minimiser of $J(v) := \frac{1}{2} \int_\Omega |\nabla v|^2 - \int_\Omega v$ on $H_0^1(\Omega)$, equivalently the weak solution of $-\Delta u = 1$ in Ω with zero H_0^1 -trace; testing $-\Delta u = 1$ against $u^- \in H_0^1(\Omega)$ gives $u \geq 0$ a.e.
- $E(\Omega) := -\frac{1}{2} \int_\Omega u \, dx = -\frac{1}{2} T(\Omega)$ is the BDV-normalised Saint–Venant energy (notation register §3), and $B^* \subset \mathbb{R}^n$ is the open ball with

$|B^*| = |\Omega|$, centred at the origin, of (volume) radius

$$R_\Omega := \left(\frac{|\Omega|}{\omega_n} \right)^{1/n}.$$

(We write R_Ω rather than R to avoid the conflict with the confining radius of §2.)

- For $t \geq 0$, $E_t := \{u > t\}$ (precise representative [46, §15]), $m(t) := |E_t|$ is right-continuous and nonincreasing, with $m(0^+) \leq |\Omega|$ and $m(\|u\|_\infty) = 0$. The decreasing rearrangement is $u^*(s) := \inf\{t \geq 0 : m(t) \leq s\}$ for $s \in [0, |\Omega|]$.
- $\partial^* E_t$ is the De Giorgi reduced boundary [22], and ν_{E_t} is the measure-theoretic outer unit normal. We set

$$(204) \quad \alpha(t) := \int_{\partial^* E_t} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1}, \quad \beta(t) := \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1},$$

defined for a.e. t (justified by Lemma 4.1–4.3 below), and

$$c_n := n^2 \omega_n^{2/n}.$$

Talenti L^∞ bound. By Talenti's pointwise rearrangement comparison [51, Theorem 1] (see also [38, Theorem 1.3.2]),

$$(205) \quad u_\Omega^*(s) \leq u_{B^*}^*(s) = \frac{R_\Omega^2 - (s/\omega_n)^{2/n}}{2n}, \quad s \in [0, |\Omega|],$$

so

$$(206) \quad \|u\|_{L^\infty(\Omega)} \leq \frac{R_\Omega^2}{2n} < \infty.$$

This is the sole reason boundedness of Ω is unnecessary at the analytic level. Combined with $u \in H_0^1(\Omega)$ extended by zero, (206) gives $u \in BV_{\text{loc}}(\mathbb{R}^n) \cap L^\infty$; the BV coarea formula [4, Theorem 3.40], [46, Theorem 13.1] then applies, and for every Borel $\varphi \geq 0$,

$$(207) \quad \int_\Omega |\nabla u| \varphi dx = \int_{-\infty}^\infty \left(\int_{\partial^* E_t} \varphi d\mathcal{H}^{n-1} \right) dt.$$

The set

$$\mathcal{R} := \{t \in (0, \|u\|_\infty) : P(E_t) < \infty \text{ and } \partial^* E_t \subset \Omega\}$$

has full Lebesgue measure in $(0, \|u\|_\infty)$, and all statements “for a.e. t ” below mean for a.e. $t \in \mathcal{R}$. The countable set of plateau levels $\{t : |\{u = t\}| > 0\}$ is, in particular, Lebesgue-null and is automatically discarded by the dt -integral.

4.1.1. *Step 1: coarea identity for $-m'(t)$.*

Lemma 4.1 (Coarea form of $-m'$). *For a.e. $t \in (0, \|u\|_\infty)$,*

$$(208) \quad -m'(t) = \alpha(t) = \int_{\partial^* E_t} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1}.$$

In particular m is absolutely continuous on $[\varepsilon, \|u\|_\infty]$ for every $\varepsilon > 0$, and $\alpha \in L_{\text{loc}}^1((0, \|u\|_\infty))$.

Proof. Fix $0 < \alpha_0 < \beta_0 < \|u\|_\infty$ and apply (207) to the Borel function $\varphi(x) = |\nabla u(x)|^{-1} \mathbf{1}_{\{|\nabla u| > 0\}}(x) \mathbf{1}_{[\alpha_0, \beta_0]}(u(x))$. The product $|\nabla u| \varphi$ equals $\mathbf{1}_{\{\alpha_0 < u < \beta_0\} \cap \{|\nabla u| > 0\}}$ almost everywhere; the contribution of $\{|\nabla u| = 0\}$ to the BV coarea slicing vanishes by [4, 3.40(b)], so

$$(209) \quad |E_{\alpha_0} \setminus E_{\beta_0}| = \int_{\Omega} \mathbf{1}_{\{\alpha_0 < u < \beta_0\}} dx = \int_{\alpha_0}^{\beta_0} \left(\int_{\partial^* E_t} \frac{1}{|\nabla u|} d\mathcal{H}^{n-1} \right) dt = \int_{\alpha_0}^{\beta_0} \alpha(t) dt.$$

Outside the at most countable set of plateau levels, $|E_{\alpha_0} \setminus E_{\beta_0}| = m(\alpha_0) - m(\beta_0)$, so $m(\alpha_0) - m(\beta_0) = \int_{\alpha_0}^{\beta_0} \alpha(t) dt$. Differentiating in β_0 at Lebesgue points of α yields (208). Local integrability of α on $(0, \|u\|_\infty)$ follows because (209) bounds $\int_{\alpha_0}^{\beta_0} \alpha(t) dt \leq |\Omega| < \infty$, and absolute continuity of m on $[\varepsilon, \|u\|_\infty]$ follows directly from (209). $\square$

Remark 4.2. The countable set of plateau levels $\{t : |\{u = t\}| > 0\}$ contributes jumps to m that are absorbed by the dt -integral in Theorem 4.13 below. Equation (208) is not asserted on these levels.

4.1.2. Step 2: weak divergence/flux identity. The next lemma is the rigorous BV/Lipschitz-truncation replacement for the formal divergence identity $\int_{\partial^* E_t} \nabla u \cdot \nu_{E_t} d\mathcal{H}^{n-1} = \int_{E_t} \Delta u dx$ on the finite-perimeter superlevel E_t . The point is that the trace of ∇u on the reduced boundary $\partial^* E_t$ does not exist in general; the Lipschitz truncation of $u - t$ avoids invoking that trace.

Lemma 4.3 (Flux equals volume). *For a.e. $t \in (0, \|u\|_\infty)$, E_t has finite perimeter and*

$$(210) \quad \beta(t) = \int_{\partial^* E_t} |\nabla u| d\mathcal{H}^{n-1} = m(t).$$

Proof. Fix $t \in \mathcal{R}$ with $|E_t| < \infty$, $P(E_t) < \infty$, and u not having a plateau at t ; this holds for a.e. t .

Step 1 (Lipschitz truncation). For each $\varepsilon > 0$ define $\phi_\varepsilon: \mathbb{R} \rightarrow [0, 1]$ by $\phi_\varepsilon(s) := \min(1, \max(0, s/\varepsilon))$ and set $\eta_\varepsilon(x) := \phi_\varepsilon(u(x) - t)$. Since ϕ_ε is Lipschitz with $\phi_\varepsilon(0) = 0$ and $u \in H_0^1(\Omega)$, the chain rule for Sobolev compositions [4, Theorem 3.99] yields $\eta_\varepsilon \in H_0^1(\Omega)$ with

$$(211) \quad \nabla \eta_\varepsilon(x) = \varepsilon^{-1} \nabla u(x) \mathbf{1}_{\{t < u < t+\varepsilon\}}(x).$$

Test the weak equation $-\Delta u = 1$ on $H_0^1(\Omega)$ against η_ε :

$$(212) \quad \int_{\Omega} \nabla u \cdot \nabla \eta_\varepsilon dx = \int_{\Omega} \eta_\varepsilon dx.$$

By (211), the left-hand side of (212) equals

$$(213) \quad \int_{\Omega} \nabla u \cdot \nabla \eta_\varepsilon dx = \frac{1}{\varepsilon} \int_{\{t < u < t+\varepsilon\}} |\nabla u|^2 dx.$$

Step 2 (Coarea applied to the LHS). Applying (207) with $\varphi(x) = |\nabla u(x)| \mathbf{1}_{(t, t+\varepsilon)}(u(x))$,

$$(214) \quad \int_{\{t < u < t+\varepsilon\}} |\nabla u|^2 dx = \int_t^{t+\varepsilon} \left(\int_{\partial^* E_\tau} |\nabla u| d\mathcal{H}^{n-1} \right) d\tau = \int_t^{t+\varepsilon} \beta(\tau) d\tau.$$

Step 3 (Limit of the right-hand side of (212)). Since t is not a plateau level, $|\{u = t\}| = 0$, so $\eta_\varepsilon \rightarrow \mathbf{1}_{E_t}$ pointwise a.e. as $\varepsilon \downarrow 0$. Because $0 \leq \eta_\varepsilon \leq$

$\mathbf{1}_{\{u>t\}} \leq \mathbf{1}_{E_t} + \mathbf{1}_{\{u=t\}}$ and $|E_t| < \infty$, the dominated convergence theorem gives

$$\int_{\Omega} \eta_{\varepsilon} dx \xrightarrow{\varepsilon \downarrow 0} |E_t| = m(t).$$

Step 4 (Limit of (214)). The function $\tau \mapsto \beta(\tau)$ is locally integrable on $(0, \|u\|_{\infty})$: by (207) with $\varphi(x) = |\nabla u(x)| \mathbf{1}_{(0, \|u\|_{\infty})}(u(x))$, and using the Dirichlet identity $\int_{\Omega} |\nabla u|^2 dx = \int_{\Omega} u dx < \infty$ (test $-\Delta u = 1$ against $u \in H_0^1(\Omega)$),

$$\int_0^{\|u\|_{\infty}} \beta(\tau) d\tau = \int_{\Omega} |\nabla u|^2 dx = \int_{\Omega} u dx < \infty.$$

Therefore, at every Lebesgue point of β (i.e. for a.e. t),

$$\frac{1}{\varepsilon} \int_t^{t+\varepsilon} \beta(\tau) d\tau \xrightarrow{\varepsilon \downarrow 0} \beta(t).$$

Step 5 (Conclusion). Combining (212)–(214) with Steps 3 and 4 gives $\beta(t) = m(t)$ for a.e. $t \in \mathcal{R}$. $\square$

Remark 4.4 (Formal divergence-theorem reading). If one had a one-sided trace of ∇u on $\partial^* E_t$ (i.e. classical smoothness near the level surface), the relation $\nabla u \cdot \nu_{E_t} = -|\nabla u|$ on $\partial^* E_t$ would follow from $u > t$ inside E_t , and Lemma 4.3 would reduce to the standard divergence theorem [46, Theorem 16.3] applied to ∇u :

$$m(t) = \int_{E_t} 1 dx = - \int_{E_t} \Delta u dx = - \int_{\partial^* E_t} \nabla u \cdot \nu_{E_t} d\mathcal{H}^{n-1} = \beta(t).$$

The Lipschitz-truncation step above is the rigorous replacement when no such trace is available; see [46, Chapter 17] for the general theory of fluxes on finite-perimeter sets without traces.

4.1.3. *Step 3: distributional derivatives of the rearranged primitive.* Define

$$(215) \quad I(s) := \int_{\{u>u^*(s)\}} u dx, \quad s \in [0, |\Omega|].$$

Lemma 4.5 (Layer-cake form of I and first derivative). *For every $s \in [0, |\Omega|]$,*

$$(216) \quad I(s) = \int_0^s u^*(\sigma) d\sigma,$$

I is absolutely continuous on $[0, |\Omega|]$, and $I'(s) = u^(s)$ for a.e. $s \in (0, |\Omega|)$.*

Proof. By the equimeasurability of u and u^* on $(0, |\Omega|)$ [37, §2], [51], one has $\int_{\Omega} g(u) dx = \int_0^{|\Omega|} g(u^*(\sigma)) d\sigma$ for every Borel $g \geq 0$. Choose $g(t) := t \mathbf{1}_{\{t>u^*(s)\}}$. By the definition of u^* , the set $\{\sigma \in (0, |\Omega|) : u^*(\sigma) > u^*(s)\}$ equals $(0, s)$ up to Lebesgue-null sets, so

$$I(s) = \int_{\Omega} g(u) dx = \int_0^{|\Omega|} g(u^*(\sigma)) d\sigma = \int_0^s u^*(\sigma) d\sigma,$$

which is (216). By (205), $u^* \in L^{\infty}(0, |\Omega|) \subset L^1(0, |\Omega|)$, so (216) expresses I as the indefinite integral of a bounded Borel function; the Lebesgue differentiation theorem then gives the absolute continuity of I on $[0, |\Omega|]$ and $I'(s) = u^*(s)$ a.e. $\square$

The next lemma is the *corrected* form of the second derivative of I . A naive derivation gives $I''(s) = -1/a(u^*(s))$ with $a(t) = \int_{\partial^* E_t} |\nabla u|$, i.e. with the *flux* $\beta(t)$ in place of the *coarea* $\alpha(t)$. By Lemma 4.3, $\beta(t) = m(t)$, so that expression would give $I''(s) = -1/m(u^*(s)) = -1/s$, which is correct only when $|\nabla u|$ is $\mathcal{H}^{n-1}$ -a.e. constant on $\partial^* E_t$ (i.e. on the ball). The general identity uses α , and the difference $\alpha\beta - P(E_t)^2$ between the two is precisely the Cauchy–Schwarz defect D_H of §4.1.5 below.

Lemma 4.6 (Corrected second derivative of I). *The rearrangement u^* is absolutely continuous on $[s_0, |\Omega|]$ for every $0 < s_0 < |\Omega|$, and*

$$(217) \quad I''(s) = (u^*)'(s) = \frac{1}{m'(u^*(s))} = -\frac{1}{\alpha(u^*(s))} \quad \text{for a.e. } s \in (0, |\Omega|),$$

where I'' denotes the absolutely continuous representative of the distributional derivative of $I' = u^*$ on $[s_0, |\Omega|]$.

Proof. By Lemma 4.1, m is absolutely continuous on $[\varepsilon, \|u\|_\infty]$ for every $\varepsilon > 0$, with $m'(t) = -\alpha(t) \leq 0$ a.e. The set of plateau values $\{t : |\{u = t\}| > 0\}$ is at most countable (monotonicity of m); outside it, m is strictly decreasing on its domain of strict monotonicity. Consequently the right-continuous generalised inverse u^* of m is also strictly decreasing and absolutely continuous on $[s_0, |\Omega|]$ for every $s_0 \in (0, |\Omega|)$. Furthermore, $\alpha(t) > 0$ for a.e. t (otherwise m would not be strictly decreasing), so $1/\alpha(t)$ is well-defined a.e. Applying the inverse-function theorem for monotone absolutely continuous functions [4, Theorem 3.99], [10, Lemma 1.4] to $m(u^*(s)) = s$ yields

$$(u^*)'(s) = \frac{1}{m'(u^*(s))} = -\frac{1}{\alpha(u^*(s))}$$

for a.e. s . Since $I'(s) = u^*(s)$ a.e. by Lemma 4.5, the distributional derivative of I' on $[s_0, |\Omega|]$ coincides with the classical a.e. derivative of u^* , giving (217). $\square$

Cauchy–Schwarz on the reduced boundary. Applying the Cauchy–Schwarz inequality on $\partial^* E_t$ to the pair of functions $(|\nabla u|^{1/2}, |\nabla u|^{-1/2})$ gives

$$(218) \quad P(E_t)^2 = \left(\int_{\partial^* E_t} 1 d\mathcal{H}^{n-1} \right)^2 \leq \alpha(t) \beta(t),$$

with equality if and only if $|\nabla u|$ is $\mathcal{H}^{n-1}$ -a.e. constant on $\partial^* E_t$.

4.1.4. *Step 4: pointwise convexity formula.* Following the Nicola–Tilli convexity-gap idea adapted to the torsion problem, introduce the auxiliary function

$$(219) \quad G(s) := I(s) + \frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}}, \quad s \in [0, |\Omega|].$$

A direct computation gives

$$\frac{d^2}{ds^2} \frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}} = \frac{(1+2/n)(2/n)}{(4+2n)\omega_n^{2/n}} s^{2/n-1} = \frac{1}{n^2 \omega_n^{2/n} s^{1-2/n}} = \frac{1}{c_n s^{1-2/n}}.$$

Lemma 4.7 (Pointwise convexity formula). *For a.e. $s \in (0, |\Omega|)$, with $t = u^*(s)$ (equivalently $s = m(t)$ on the strictly decreasing branch of m),*

$$(220) \quad G''(s) = \frac{1}{c_n s^{1-2/n}} - \frac{1}{\alpha(t)}.$$

Proof. Combine Lemma 4.6 ($I''(s) = -1/\alpha(u^*(s))$) with the explicit second derivative of the polynomial corrector $\frac{s^{1+2/n}}{(4+2n)\omega_n^{2/n}}$ computed above. $\square$

4.1.5. *Step 5: the two pointwise defects.*

Definition 4.8 (Cauchy–Schwarz and isoperimetric defects). For a.e. $t \in (0, \|u\|_\infty)$, set

$$(221) \quad D_H(t) := \alpha(t) \beta(t) - P(E_t)^2 = \left(\int_{\partial^* E_t} \frac{1}{|\nabla u|} \right) \left(\int_{\partial^* E_t} |\nabla u| \right) - P(E_t)^2,$$

$$(222) \quad D_I(t) := P(E_t)^2 - c_n m(t)^{2-2/n}.$$

Lemma 4.9 (Pointwise nonnegativity). *For a.e. $t \in (0, \|u\|_\infty)$, $D_H(t) \geq 0$ with equality if and only if $|\nabla u|$ is $\mathcal{H}^{n-1}$ -a.e. constant on $\partial^* E_t$; and $D_I(t) \geq 0$ with equality if and only if E_t is (equivalent to) a ball.*

Proof. $D_H \geq 0$ is the Cauchy–Schwarz inequality (218). The equality case is the standard equality case in Cauchy–Schwarz.

$D_I \geq 0$ is the squared form of the De Giorgi isoperimetric inequality $P(E_t) \geq n\omega_n^{1/n} |E_t|^{1-1/n}$ valid for every set of finite perimeter [22, Theorem II], [46, Theorem 14.1]; equality holds iff E_t is $\mathcal{H}^n$ -equivalent to a ball. $\square$

4.1.6. *Step 6: pointwise-to-integrated convexity identity.*

Lemma 4.10 (Pointwise-to-integrated convexity identity). *With $L := |\Omega|$,*

$$(223) \quad \int_0^L s G''(s) ds = \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{c_n m(t)^{1-2/n}} dt.$$

Proof. By Lemma 4.7,

$$(224) \quad \int_0^L s G''(s) ds = \int_0^L s \left(\frac{1}{c_n s^{1-2/n}} - \frac{1}{\alpha(u^*(s))} \right) ds.$$

Perform the change of variable $s = m(t)$, $ds = m'(t) dt = -\alpha(t) dt$ on the strictly decreasing branch of m . The countable set of plateau levels of u is a Lebesgue-null set in t , and contributes nothing to the integral. As s runs from 0 to L , t runs from $\|u\|_\infty$ to 0; reversing orientation,

$$(225) \quad \int_0^L s G''(s) ds = \int_0^{\|u\|_\infty} m(t) \left(\frac{1}{c_n m(t)^{1-2/n}} - \frac{1}{\alpha(t)} \right) \alpha(t) dt = \int_0^{\|u\|_\infty} \left(\frac{m(t) \alpha(t)}{c_n m(t)^{1-2/n}} - m(t) \right) dt.$$

By Lemma 4.3, $m(t) = \beta(t)$ a.e., so $m(t)\alpha(t) = \alpha(t)\beta(t)$. Adding and subtracting $P(E_t)^2/(c_n m(t)^{1-2/n})$ in the integrand of (225),

$$\frac{m(t) \alpha(t)}{c_n m(t)^{1-2/n}} - m(t) = \frac{\alpha(t) \beta(t) - c_n m(t)^{2-2/n}}{c_n m(t)^{1-2/n}} = \frac{D_H(t) + D_I(t)}{c_n m(t)^{1-2/n}}.$$

Substituting back into (225) gives (223). $\square$

4.1.7. *Step 7: endpoint convexity gap.* With $L := |\Omega|$, define

$$(226) \quad \Gamma(\Omega) := G'(L) - \frac{G(L)}{L}.$$

Lemma 4.11 (Endpoint convexity-gap formula). $\Gamma(\Omega) = \frac{1}{L} \int_0^L s G''(s) ds$.

Proof. $I \in W^{1,\infty}(0, L) \subset W_{\text{loc}}^{2,1}((0, L])$ by Lemmas 4.5–4.6, and the polynomial corrector in (219) is smooth on $(0, L]$ and Lipschitz on $[0, L]$; so G is absolutely continuous on $[0, L]$ and in $W_{\text{loc}}^{2,1}((0, L])$, with $G(0) = I(0) = 0$. By Lemma 4.5,

$$G'(s) = u^*(s) + \frac{(1 + 2/n)s^{2/n}}{(4 + 2n)\omega_n^{2/n}} = u^*(s) + \frac{s^{2/n}}{2n\omega_n^{2/n}},$$

using $\frac{1+2/n}{4+2n} = \frac{1}{2n}$; G' is bounded on $[0, L]$ by (206). Integration by parts on (ε, L) gives

$$\int_{\varepsilon}^L s G''(s) ds = L G'(L) - \varepsilon G'(\varepsilon) - G(L) + G(\varepsilon).$$

As $\varepsilon \downarrow 0$, $\varepsilon G'(\varepsilon) \rightarrow 0$ and $G(\varepsilon) \rightarrow G(0) = 0$ by the boundedness of G' and the absolute continuity of G . Therefore $\int_0^L s G''(s) ds = L G'(L) - G(L)$; dividing by L yields the claim. $\square$

4.1.8. *Step 8: endpoint identification with the Saint-Venant deficit.*

Lemma 4.12 (Endpoint value).

$$(227) \quad \Gamma(\Omega) = \frac{2}{|\Omega|} (E(\Omega) - E(B^*)).$$

Proof. At $s = L$, $u^*(L) = 0$ by definition of u^* (the lowest level is the zero level). By Lemma 4.5, $I(L) = \int_0^L u^*(\sigma) d\sigma = \int_{\Omega} u dx$ (using equimeasurability), so

$$\frac{G(L)}{L} = \frac{1}{L} \int_{\Omega} u dx + \frac{L^{2/n}}{(4 + 2n)\omega_n^{2/n}}.$$

Also $I'(L) = u^*(L) = 0$, so

$$G'(L) = \frac{L^{2/n}}{2n\omega_n^{2/n}}.$$

Combining,

$$(228) \quad \Gamma(\Omega) = \frac{L^{2/n}}{2n\omega_n^{2/n}} - \frac{L^{2/n}}{(4 + 2n)\omega_n^{2/n}} - \frac{1}{L} \int_{\Omega} u dx = \frac{L^{2/n}}{2n(n + 2)\omega_n^{2/n}} - \frac{1}{L} \int_{\Omega} u dx,$$

where we used $4 + 2n = 2(n + 2)$ and $\frac{1}{2n} - \frac{1}{2(n+2)} = \frac{1}{2n(n+2)}$.

For the ball B^* of volume $L = \omega_n R_{\Omega}^n$, the torsion function is $u_{B^*}(x) = (R_{\Omega}^2 - |x|^2)/(2n)$ (the unique solution of $-\Delta u_{B^*} = 1$ in B^* , $u_{B^*} = 0$ on ∂B^*). Direct integration in spherical coordinates,

$$\int_{B^*} u_{B^*} dx = \frac{1}{2n} \int_{B^*} (R_{\Omega}^2 - |x|^2) dx = \frac{n\omega_n}{2n} \int_0^{R_{\Omega}} (R_{\Omega}^2 - r^2) r^{n-1} dr = \frac{\omega_n R_{\Omega}^{n+2}}{2n(n + 2)}.$$

Since $L = \omega_n R_\Omega^n$, $L^{2/n} = \omega_n^{2/n} R_\Omega^2$, hence

$$\frac{L^{2/n}}{2n(n+2)\omega_n^{2/n}} = \frac{R_\Omega^2}{2n(n+2)} = \frac{1}{L} \cdot \frac{\omega_n R_\Omega^{n+2}}{2n(n+2)} = \frac{1}{L} \int_{B^*} u_{B^*} dx.$$

Inserting back in (228) and using $E(\Omega) = -\frac{1}{2} \int_\Omega u$ (and the analogous identity for B^*),

$$\Gamma(\Omega) = \frac{1}{L} \left(\int_{B^*} u_{B^*} dx - \int_\Omega u dx \right) = \frac{2}{L} (E(\Omega) - E(B^*)).$$

This identification of $\Gamma(B^*)$ via the explicit ball torsion function is the classical Pólya–Szegő content [50], [9, §II.5]. $\square$

4.1.9. The exact deficit identity.

Theorem 4.13 (Exact level-set deficit identity). *Let $n \geq 2$ and let $\Omega \subset \mathbb{R}^n$ be open with $0 < |\Omega| < \infty$. Let $u = u_\Omega \in H_0^1(\Omega)$ be the variational torsion function. Then for a.e. $t \in (0, \|u\|_\infty)$, the superlevel set $E_t = \{u > t\}$ has finite perimeter, the coarea identity (208), the flux identity (210), and the corrected second-derivative identity (217) hold, and*

$$(229) \quad \boxed{\frac{2}{|\Omega|} (E(\Omega) - E(B^*)) = \frac{1}{|\Omega|} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{n^2 \omega_n^{2/n} m(t)^{1-2/n}} dt,}$$

where D_H, D_I are the nonnegative pointwise defects of (221)–(222) and B^* is the ball with $|B^*| = |\Omega|$.

Proof. Lemmas 4.10, 4.11, 4.12 together give

$$\frac{2}{|\Omega|} (E(\Omega) - E(B^*)) = \Gamma(\Omega) = \frac{1}{L} \int_0^L s G''(s) ds = \frac{1}{L} \int_0^{\|u\|_\infty} \frac{D_H(t) + D_I(t)}{c_n m(t)^{1-2/n}} dt.$$

The pointwise statements (208), (210), (217) are proved in Lemmas 4.1, 4.3, 4.6. $\square$

The deficit identity (229) thus expresses $\delta_T(\Omega)$ in the form

$$\delta_T(\Omega) = E(\Omega) - E(B^*) = \frac{1}{2} \int_0^{\|u\|_\infty} (D_H(t) + D_I(t)) w(t) dt,$$

with the *explicit weight*

$$(230) \quad w(t) := \frac{1}{c_n m(t)^{1-2/n}} = \frac{1}{n^2 \omega_n^{2/n} m(t)^{1-2/n}}.$$

This is the form quoted in the master brief and in §2; the dimensional constant $c_n = n^2 \omega_n^{2/n}$ is the only constant appearing in the identity itself.

Cauchy–Schwarz/Serrin variance form of D_H . The following corollary recasts D_H as a level-set Serrin variance defect, in the form used in §4.4.7–§4.5.6 when combining D_H with normal-oscillation bounds.

Corollary 4.14 (Weighted variance form of D_H). *For a.e. $t \in (0, \|u\|_\infty)$, with $f := |\nabla u|$ on $\partial^* E_t$ and $\bar{f} := m(t)/P(E_t)$,*

$$(231) \quad \int_{\partial^* E_t} \frac{(f - \bar{f})^2}{f} d\mathcal{H}^{n-1} = \frac{m(t)}{P(E_t)^2} D_H(t).$$

Proof. Direct computation, using Lemma 4.3 ($\int f d\mathcal{H}^{n-1} = \beta(t) = m(t)$) and $\int 1 d\mathcal{H}^{n-1} = P(E_t)$:

$$\int_{\partial^* E_t} \frac{(f - \bar{f})^2}{f} d\mathcal{H}^{n-1} = \int_{\partial^* E_t} f d\mathcal{H}^{n-1} - 2\bar{f} P(E_t) + \bar{f}^2 \int_{\partial^* E_t} \frac{1}{f} d\mathcal{H}^{n-1} = m(t) - \frac{2m(t)}{1} + \frac{m(t)^2}{P(E_t)^2} \alpha(t).$$

The first two terms collapse to $-m(t)$, and using $\alpha(t)\beta(t) = \alpha(t)m(t)$,

$$\int_{\partial^* E_t} \frac{(f - \bar{f})^2}{f} d\mathcal{H}^{n-1} = \frac{m(t)^2}{P(E_t)^2} \alpha(t) - m(t) = \frac{m(t)(\alpha(t)m(t) - P(E_t)^2)}{P(E_t)^2} = \frac{m(t)}{P(E_t)^2} D_H(t),$$

since $\alpha m = \alpha \beta$. This is (231). $\square$

4.1.10. *Profile-gap consequences (used in §4.4.7–§4.5.6).* The deficit identity (229) controls weighted averages of $D_H + D_I$ over levels; the two consequences below extract *pointwise* good levels in a controlled near-boundary slab, together with a transfer estimate from a superlevel set back to Ω . These are the only consequences of the deficit identity used in the rest of Plan 2. Setup. Write

$$L := |\Omega|, \quad R_\Omega := (L/\omega_n)^{1/n}, \quad \delta_T(\Omega) := E(\Omega) - E(B^*), \quad v(s) := u_\Omega^*(s), \quad b(s) := u_{B^*}^*(s).$$

By Talenti (205), $b(s) - v(s) \geq 0$ on $[0, L]$, and the explicit ball profile gives $b(s) = (R_\Omega^2 - (s/\omega_n)^{2/n})/(2n)$.

Lemma 4.15 (Profile-gap estimate). *The function $h(s) := b(s) - v(s)$ is nonnegative and nonincreasing on $[0, L]$, $h(L) = 0$, and*

$$(232) \quad \int_0^L h(s) ds = \int_{B^*} u_{B^*} dx - \int_\Omega u dx = 2\delta_T(\Omega).$$

Consequently, for every $s \in (0, L]$,

$$(233) \quad 0 \leq b(s) - v(s) \leq \frac{2\delta_T(\Omega)}{s}.$$

Proof. By Talenti's level-set differential inequality $(u_\Omega^*)'(s) \geq (u_{B^*}^*)'(s)$ for a.e. $s \in (0, L)$ [51, Theorem 1, (3.6)], $h = b - v$ has $h'(s) \leq 0$ a.e., i.e. h is nonincreasing. Also $h(L) = b(L) - v(L) = 0 - 0 = 0$. The identity (232) follows from $\int_0^L b = \int_{B^*} u_{B^*}$ and $\int_0^L v = \int_\Omega u$ (equimeasurability), combined with $E(\Omega) - E(B^*) = \frac{1}{2}(\int_{B^*} u_{B^*} - \int_\Omega u)$. Since h is nonincreasing and nonnegative on $[0, L]$, for any $s \in (0, L]$,

$$s h(s) \leq \int_0^s h(\sigma) d\sigma \leq \int_0^L h(\sigma) d\sigma = 2\delta_T(\Omega),$$

which is (233). $\square$

Good-level slab lemma. Define the weighted level measure

$$(234) \quad d\nu(t) := \frac{dt}{c_n m(t)^{1-2/n}}.$$

Theorem 4.13 reads $\int_0^{\|u\|^\infty} (D_H(t) + D_I(t)) d\nu(t) = 2\delta_T(\Omega)$.

Lemma 4.16 (Good level in a near-boundary slab). *Let $0 < \eta < L/4$, and set*

$$S_\eta := [L - 2\eta, L - \eta], \quad T_\eta := v(S_\eta) = [v(L - \eta), v(L - 2\eta)].$$

There exists a dimensional constant $c = c(n) > 0$ such that the following holds. Assume the smallness condition

$$(235) \quad \delta_T(\Omega) \leq c R_\Omega^2 \eta.$$

Then for every $s \in S_\eta$ the level $t_s := v(s)$ satisfies

$$(236) \quad c \frac{R_\Omega^2}{L} \eta \leq t_s \leq C \frac{R_\Omega^2}{L} \eta,$$

for some dimensional $C = C(n)$, and

$$(237) \quad |T_\eta| \geq c \frac{R_\Omega^2}{L} \eta.$$

Moreover there exists $t_\eta^ \in T_\eta$ with*

$$(238) \quad D_H(t_\eta^*) + D_I(t_\eta^*) \leq \frac{2 \delta_T(\Omega)}{\nu(T_\eta)} \leq C' \frac{L^{2-2/n} \delta_T(\Omega)}{\eta R_\Omega^2}$$

for a dimensional constant $C' = C'(n)$, and $\eta \leq |\Omega \setminus E_{t_\eta^}| \leq 2\eta$.*

Proof. Profile-height estimates (236). By Lemma 4.15, on $S_\eta \subset [L/2, L]$,

$$0 \leq b(s) - v(s) \leq \frac{2 \delta_T(\Omega)}{s} \leq \frac{4 \delta_T(\Omega)}{L}.$$

For $s = L - \sigma$, $\sigma \in [\eta, 2\eta]$, the explicit ball profile gives

$$b(s) = \frac{R_\Omega^2}{2n} \left(1 - (1 - \sigma/L)^{2/n} \right).$$

The elementary one-dimensional inequalities $\frac{2}{n} \frac{\sigma}{L} (1 - \frac{\sigma}{L})^{2/n-1} \leq 1 - (1 - \sigma/L)^{2/n} \leq \frac{2}{n} \frac{\sigma}{L}$ for $0 < \sigma < L/4$ give, with new dimensional constants c_1, C_1 ,

$$c_1 \frac{R_\Omega^2}{L} \sigma \leq b(s) \leq C_1 \frac{R_\Omega^2}{L} \sigma.$$

Combined with the profile gap and (235) chosen so that $4\delta_T/L \leq c_1 R_\Omega^2 \eta / (2L)$, this gives (236) after relabelling c, C .

Lower bound (237). $|T_\eta| = v(L - 2\eta) - v(L - \eta)$. Using $v = b - h$ and the monotonicity of h ,

$$v(L - 2\eta) - v(L - \eta) = [b(L - 2\eta) - b(L - \eta)] - [h(L - 2\eta) - h(L - \eta)] \geq [b(L - 2\eta) - b(L - \eta)] - \frac{4\delta_T(\Omega)}{L},$$

since $0 \leq h(L - 2\eta) - h(L - \eta) \leq h(L - 2\eta) \leq 4\delta_T/L$ (using $h(L - 2\eta) \leq 2\delta_T/(L - 2\eta) \leq 4\delta_T/L$). On the other hand, $b(L - 2\eta) - b(L - \eta) \geq c_2 R_\Omega^2 \eta / L$ for a dimensional c_2 , by the same elementary one-dimensional estimate above. Under (235) (with $c \leq c_2/8$), $4\delta_T/L \leq c_2 R_\Omega^2 \eta / (2L)$, giving (237).

Good-level existence (238). By the area formula (equivalently the coarea identity (208) integrated),

$$\nu(T_\eta) = \int_{T_\eta} \frac{dt}{c_n m(t)^{1-2/n}} \geq \frac{1}{c_n L^{1-2/n}} |T_\eta| \geq \frac{c R_\Omega^2 \eta}{c_n L^{2-2/n}}$$

(using $m(t) \leq L$ on T_η and (237)). The deficit identity (229) integrated over T_η and the mean-value inequality give the existence of $t_\eta^* \in T_\eta$ with

$$D_H(t_\eta^*) + D_I(t_\eta^*) \leq \frac{1}{\nu(T_\eta)} \int_{T_\eta} (D_H + D_I) d\nu \leq \frac{2\delta_T(\Omega)}{\nu(T_\eta)}.$$

The second inequality in (238) now follows from the lower bound on $\nu(T_\eta)$. Finally, $t_\eta^* \in T_\eta$ means $m(t_\eta^*) \in [L - 2\eta, L - \eta]$, i.e. $\eta \leq |\Omega \setminus E_{t_\eta^*}| \leq 2\eta$. $\square$

Superlevel transfer of Fraenkel asymmetry. This is the elementary boundary-layer transfer used in §4.5.6. Let $\mathcal{A}(\Omega) := \inf_{x_0 \in \mathbb{R}^n} |\Omega \Delta (B^* + x_0)|/|B^*|$ be the Fraenkel asymmetry (notation register §5).

Lemma 4.17 (Superlevel asymmetry transfer). *Let $E_t = \{u > t\}$ with $E_t \subset \Omega$, and let $s := |E_t| = L - \eta$ with $0 < \eta < L$. Let $B^{(s)}$ be a ball with $|B^{(s)}| = s$ realising the Fraenkel infimum for E_t , and let $B^{(L)}$ be the concentric ball of volume L . Then*

$$(239) \quad \mathcal{A}(\Omega) \leq \mathcal{A}(E_t) + \frac{4\eta}{L}.$$

Proof. By the triangle inequality for symmetric differences,

$$|\Omega \Delta B^{(L)}| \leq |\Omega \setminus E_t| + |E_t \Delta B^{(s)}| + |B^{(s)} \Delta B^{(L)}|.$$

Now $|\Omega \setminus E_t| = \eta$, $|E_t \Delta B^{(s)}| = s \mathcal{A}(E_t) \leq L \mathcal{A}(E_t)$, and $B^{(s)} \subset B^{(L)}$ so $|B^{(s)} \Delta B^{(L)}| = L - s = \eta$. Hence

$$\mathcal{A}(\Omega) \leq \frac{|\Omega \Delta B^{(L)}|}{L} \leq \frac{|E_t \Delta B^{(s)}|}{L} + \frac{2\eta}{L} \leq \mathcal{A}(E_t) + \frac{2\eta}{L}.$$

Allowing for the asymmetry to be computed at a different centre than the optimal one (which can cost at most $|\Omega \setminus E_t|/L = \eta/L$) gives (239) after enlarging the constant from 2 to 4 (any centre admissible for $\mathcal{A}(E_t)$ is admissible for $\mathcal{A}(\Omega)$ up to the boundary-layer correction). $\square$

4.1.11. *Failure modes and what is not assumed.* We close this subsection by recording the regularity assumptions that are *not* used and the ones that are. Both lists are required inputs of Agent 16's Plan 2 audit (§4.6).

- (1) *Ω need not be bounded.* The only consequence of $|\Omega| < \infty$ is the Talenti L^∞ bound (206). Boundedness $\Omega \subset B_R$ enters only later in §4.2.10–§4.5.6, where strong isoperimetric inequalities and radial-trace estimates are used.
- (2) *Ω need not be smooth.* Lemmas 4.1 and 4.3 use only BV coarea [46, Theorem 13.1] and the Lipschitz truncation of $u - t$; neither requires $\partial\Omega$ to be a manifold. There is no use of a one-sided trace of ∇u on $\partial^* E_t$ (cf. Remark 4.4).
- (3) *Ω need not be connected.* The torsion function decouples across connected components and the identity is additive.
- (4) *The identity is a.e., not pointwise.* Lemmas 4.1, 4.3, 4.6 hold for t outside an $\mathcal{L}^1$ -null set of bad levels: at most countably many plateau levels of u , the Sobolev–Sard exceptional set, and BV-coarea exceptional levels. Levels in this null set contribute zero to the dt -integral.

- (5) *All integrals are on reduced boundaries.* For a.e. t , $\mathcal{H}^{n-1}(\partial E_t \setminus \partial^* E_t) = 0$ [22], [46, Theorem 16.2], so cusps, self-tangencies, and the unreduced part of ∂E_t contribute nothing. No graph parametrisation of $\partial^* E_t$ is used or available in general.
- (6) *The corrected I'' .* The use of α (and not β) in Lemma 4.6 is essential. The naive guess $I'' = -1/\beta = -1/m$ holds only on the ball, where $|\nabla u|$ is level-constant and Cauchy–Schwarz is sharp; in general it would silently delete D_H from (229).

Where the identity is used downstream. The metric framework (§4.2.10) extracts a first-variation/velocity defect from the integrated combination $D_H + D_I$ via the homothetic radial vector field. The weighted metric trace (§4.4.7) uses Lemma 4.16 to extract a good isoperimetric-defect level in a near-boundary slab, then upgrades the integrated action bound to a point-wise endpoint at $\hat{\rho}$. The boundary-layer transfer (§4.5.6) uses Lemma 4.17 to transfer the asymmetry from $E_{\hat{\rho}}$ back to Ω , squaring the resulting $O(\sqrt{\delta_T})$ boundary layer to preserve the sharp exponent.

4.2. The metric framework: quotient space, absolute continuity of the foliation, and finite-perimeter first variation. In this subsection we construct the metric ambient space $\mathcal{X}$ in which the rescaled volume–radius foliation $\rho \mapsto [F_\rho]$ lives, prove that this curve is absolutely continuous in $\mathcal{X}$, and establish the *finite-perimeter metric first-variation bound* (T) on the metric derivative $|F_\rho|_{\mathcal{X}}$ in terms of the normal velocity V_ρ and the homothetic velocity $H_{z_\rho, \rho}$. The latter is used downstream by the weighted metric trace argument (Agent 12) to convert the level-set deficit identity (Agent 9) into a metric-energy bound. No global Lipschitz boundary regularity is assumed on Ω or on the level sets E_ρ ; the only regularity used is finite perimeter, which holds for a.e. level by the BV coarea theorem.

4.2.1. Standing hypotheses. Throughout this subsection, $n \geq 2$, $R \geq 2$, and $0 < \rho_* < 1$ are fixed. We assume that

$$(240) \quad \Omega \subset \mathbb{R}^n \text{ is open, bounded, with } \Omega \subset B_R \text{ and } |\Omega| = \omega_n.$$

Let $u \in H_0^1(\Omega) \cap L^\infty(\Omega)$ be the variational torsion potential (Sec. 2.2.10, item B1). Standard Stampacchia regularity gives $u \in W_{\text{loc}}^{2,p}(\Omega)$ for every $p \in (1, \infty)$, and the BV coarea theorem (item A2; [46, Thm. 13.1], [4, Thm. 3.40]) implies that for a.e. $t \in (0, \|u\|_\infty)$ the super-level set $E_t = \{u > t\}$ has finite perimeter, with reduced boundary $\partial^* E_t$, outer measure-theoretic normal ν_t , and $P(E_t) = \mathcal{H}^{n-1}(\partial^* E_t)$. Reparametrising by volume (item B6),

$$(241) \quad E_\rho := \{u > t(\rho)\}, \quad |E_\rho| = \omega_n \rho^n, \quad \rho \in [\rho_*, 1],$$

yields a function $\rho \mapsto t(\rho)$ that is absolutely continuous on $[\rho_*, 1]$ with $t'(\rho) \leq 0$ a.e. The corresponding measure on $[\rho_*, 1]$ is $\mu(d\rho) := (-t'(\rho)) d\rho$ (notation register §7).

For a Borel-measurable map $\rho \mapsto z_\rho \in \mathbb{R}^n$ with $\sup_{\rho \in [\rho_*, 1]} |z_\rho| \leq R' < \infty$ we set

$$(242) \quad H_{z_\rho, \rho}(x) := \frac{(x - z_\rho) \cdot \nu_\rho(x)}{\rho}, \quad x \in \partial^* E_\rho.$$

The Borel measurability of $\rho \mapsto z_\rho$ is the only hypothesis imposed in this subsection; the precise centre choice (centroid $\bar{z}_\rho$ or Fusco–Julin oscillation centre) is fixed downstream (Agents 11–12).

4.2.2. The metric quotient space $\mathcal{X}$. We denote by $\mathfrak{M}$ the set of $\mathcal{L}^n$ -equivalence classes of characteristic functions $\mathbf{1}_A$ with $|A| < \infty$, equipped with the L^1 metric

$$d_{L^1}(\mathbf{1}_A, \mathbf{1}_C) := \|\mathbf{1}_A - \mathbf{1}_C\|_{L^1(\mathbb{R}^n)} = |A \Delta C|.$$

The translation group $\mathbb{R}^n$ acts on $\mathfrak{M}$ by $a \cdot \mathbf{1}_A := \mathbf{1}_{A+a}$.

Definition 4.18 (Metric quotient). Define the quotient space

$$(243) \quad \mathcal{X} := \mathfrak{M}/\mathbb{R}^n, \quad d_{\mathcal{X}}([A], [C]) := \inf_{a \in \mathbb{R}^n} |A \Delta (C + a)|.$$

Lemma 4.19 (Attainment of the translation infimum). *If $A, C \subset \mathbb{R}^n$ have finite measure and are both contained in a common ball B_M , then the infimum in (243) is attained at some $a_* \in \overline{B_{2M}}$.*

Proof. For $|a| \geq 2M$ the translate $C + a$ is disjoint from A , so $|A \Delta (C + a)| = |A| + |C| \geq |A \Delta C|$. Hence the infimum is unchanged when restricted to $\overline{B_{2M}}$. The function $a \mapsto |A \Delta (C + a)|$ is continuous (dominated convergence applied to $|\mathbf{1}_A - \mathbf{1}_{C+a}|$), and therefore attains its minimum on the compact set $\overline{B_{2M}}$. $\square$

Proposition 4.20 (Metric properties of $(\mathcal{X}, d_{\mathcal{X}})$). *The pair $(\mathcal{X}, d_{\mathcal{X}})$ is a complete separable metric space. Restricted to classes of sets contained in a fixed ball B_M , it is a Polish space, and the restriction is sequentially compact in the following weak sense: any sequence $(\mathbf{1}_{A_k}) \subset \mathfrak{M}$ with $A_k \subset B_M$ admits, after extracting a subsequence and translating by suitable $a_k \in \overline{B_{2M}}$, an L^1 -limit $\mathbf{1}_{A_\infty}$ with $A_\infty \subset B_{3M}$.*

Proof. That $d_{\mathcal{X}}$ is a metric is immediate from the symmetry $|A \Delta C| = |C \Delta A|$ and the triangle inequality $|A \Delta (C + a + b)| \leq |A \Delta (C + a)| + |D \Delta ((C + a) + b)|$; the explicit verification reduces to translation-invariance of L^1 and is recorded in [5, Def. 2.1, Lem. 2.2] (and is essentially [6, Def. 5.5.1]).

Completeness. Let $([A_k])$ be Cauchy in $\mathcal{X}$. By passing to a subsequence we may assume $d_{\mathcal{X}}([A_{k+1}], [A_k]) \leq 2^{-k}$, and by Lemma 4.19 (applied after a uniform translation to bring A_1 into a ball B_M and using $|A_k| = |A_1|$ if we further restrict to a single mass class) we may pick $a_k \in \mathbb{R}^n$ with $|A_{k+1} \Delta (A_k + a_k)| \leq 2^{-k}$. Setting $\tilde{A}_k := A_k - (a_1 + \dots + a_{k-1})$ produces an L^1 -Cauchy sequence of characteristic functions in $\mathbb{R}^n$, which converges to some $\mathbf{1}_{A_\infty}$ in L^1 by completeness of $L^1(\mathbb{R}^n; [0, 1])$ ([46, Prop. 12.10]). The limit class $[A_\infty]$ is the required limit in $\mathcal{X}$.

Separability and Polish structure on $\{A \subset B_M\}$. On the restricted subset, finite unions of dyadic cubes contained in B_M form a countable L^1 -dense subset of $\{\mathbf{1}_A : A \subset B_M\}$ ([46, Thm. 1.18]), and projection to $\mathcal{X}$ preserves density.

Sequential compactness. For $A_k \subset B_M$, the sets are uniformly bounded with uniform measure; by BV compactness ([46, Thm. 12.26], equivalently [4, Thm. 3.23], with the perimeter bound $P(A_k) \leq P(B_M)$ only required when one wants limit-of-perimeter information, which we do not need here),

the sequence $(\mathbf{1}_{A_k})$ is precompact in $L^1_{\text{loc}}(\mathbb{R}^n)$ after suitable translations. Tightness is enforced by Lemma 4.19: there is $a_k \in \overline{B_{2M}}$ with $A_k + a_k \subset B_{3M}$ achieving the infimum, and we extract along this normalised sequence. $\square$

Remark 4.21 (Why we do not use perimeter in the metric). We deliberately do *not* equip $\mathcal{X}$ with a perimeter-augmented distance: the absolute-continuity statement we need (Theorem 4.25 below) and the metric first-variation bound (T) only require the L^1 metric. The perimeter enters separately through the velocity integrals on $\partial^* E_\rho$, see (251).

4.2.3. *The rescaled curve $\rho \mapsto [F_\rho]$ and metric derivative.*

Definition 4.22 (Rescaled curve). For $\rho \in [\rho_*, 1]$ put

$$(244) \quad F_\rho := \rho^{-1} E_\rho \subset \mathbb{R}^n, \quad [F_\rho] \in \mathcal{X}.$$

By construction $|F_\rho| = \omega_n = |B_1|$ for all $\rho \in [\rho_*, 1]$. Since $\Omega \subset B_R$ and $\rho \geq \rho_*$, $F_\rho \subset B_{R/\rho_*}$ uniformly, and Lemma 4.19 applies to all pairs $([F_{\rho_1}], [F_{\rho_2}])$ with $a_* \in \overline{B_{2R/\rho_*}}$.

Definition 4.23 (Metric derivative; [6, Thm. 1.1.2]). A curve $\gamma : I \rightarrow \mathcal{X}$ is *absolutely continuous* if there is $m \in L^1(I)$ such that

$$d_{\mathcal{X}}(\gamma(s), \gamma(\sigma)) \leq \int_\sigma^s m(\tau) d\tau \quad \text{for } \sigma \leq s \text{ in } I.$$

The *metric derivative*

$$|\dot{\gamma}|_{\mathcal{X}}(\rho) := \lim_{h \rightarrow 0} \frac{d_{\mathcal{X}}(\gamma(\rho+h), \gamma(\rho))}{|h|}$$

then exists for a.e. $\rho \in I$ and is the smallest admissible m .

4.2.4. *Pointwise definition of the normal velocity.* The normal velocity of the flow $\rho \mapsto E_\rho$ is, formally,

$$(245) \quad V_\rho(x) = \frac{-t'(\rho)}{|\nabla u(x)|},$$

defined where $|\nabla u(x)| > 0$. The next proposition makes this pointwise on a.e. reduced boundary.

Proposition 4.24 (Sobolev–Sard for V_ρ ; item A3). *Let $u \in W^{2,p}_{\text{loc}}(\Omega)$ for some $p > n$. Then for a.e. $\rho \in [\rho_*, 1]$,*

$$(246) \quad \mathcal{H}^{n-1}(\partial^* E_\rho \cap \{|\nabla u| = 0\}) = 0,$$

and consequently $V_\rho : \partial^ E_\rho \rightarrow (0, \infty)$ is $\mathcal{H}^{n-1}$ -a.e. finite and strictly positive. The flux identity holds:*

$$(247) \quad \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = \frac{d}{d\rho} |E_\rho| = n\omega_n \rho^{n-1} =: A_\rho.$$

Proof. The Stampacchia regularity $-\Delta u = 1$ in Ω gives $u \in W^{2,p}_{\text{loc}}(\Omega)$ for every $p < \infty$. Fix any $p > n$. By the Figalli–Maggi Sobolev–Sard theorem ([25, Appendix A, Theorem A.1]), for every $u \in W^{2,p}_{\text{loc}}$ with $p > n$,

$$\mathcal{H}^{n-1}(\{u = t\} \cap \{|\nabla u| = 0\}) = 0 \quad \text{for a.e. } t \in \mathbb{R}.$$

The De Giorgi structure theorem ([46, § 15]) implies $\partial^*\{u > t\} \subset \{u = t\}$ up to an $\mathcal{H}^{n-1}$ -null set for a.e. t ; reparametrising $t = t(\rho)$ via (241) yields (246). The flux identity (247) is the coarea formula ([46, Thm. 13.1]) applied to the BV function $\rho \mapsto |E_\rho|$ followed by Lebesgue differentiation. $\square$

In particular, $V_\rho \in L^1(\partial^* E_\rho, \mathcal{H}^{n-1})$ for a.e. $\rho \in [\rho_*, 1]$, with L^1 -norm equal to $A_\rho = n\omega_n \rho^{n-1}$ uniformly in ρ .

4.2.5. *Absolute continuity of $\rho \mapsto [F_\rho]$.*

Theorem 4.25 (Absolute continuity in $\mathcal{X}$). *Under the standing hypotheses, the map*

$$[\rho_*, 1] \ni \rho \longmapsto [F_\rho] \in \mathcal{X}$$

is absolutely continuous. Indeed, there exists a function $m \in L^1([\rho_, 1])$ with*

$$(248) \quad d_{\mathcal{X}}([F_{\rho_2}], [F_{\rho_1}]) \leq \int_{\rho_1}^{\rho_2} m(\rho) d\rho, \quad \rho_* \leq \rho_1 \leq \rho_2 \leq 1,$$

where one may take

$$(249) \quad m(\rho) = \rho_*^{-n} \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} + n\rho_*^{-n-1} \cdot R/\rho_* \leq C(n, R, \rho_*).$$

Proof. For $\rho_* \leq \rho < \sigma \leq 1$,

$$F_\sigma \Delta F_\rho = (\sigma^{-1} E_\sigma) \Delta (\rho^{-1} E_\rho).$$

By Lemma 4.19, $d_{\mathcal{X}}([F_\sigma], [F_\rho]) \leq |F_\sigma \Delta F_\rho|$. We estimate this last symmetric difference. Write $F_\sigma \Delta F_\rho = (F_\sigma \Delta (\sigma^{-1} E_\rho)) \Delta ((\sigma^{-1} E_\rho) \Delta F_\rho)$, so

$$(250) \quad |F_\sigma \Delta F_\rho| \leq |\sigma^{-1}(E_\sigma \Delta E_\rho)| + |\sigma^{-1} E_\rho \Delta \rho^{-1} E_\rho|.$$

For the first term, $E_\rho \subset E_\sigma$ when $\rho \leq \sigma$ (the foliation is nested), so $|E_\sigma \Delta E_\rho| = |E_\sigma| - |E_\rho| = \omega_n(\sigma^n - \rho^n)$, and the coarea flux (247) integrated on $[\rho, \sigma]$ gives

$$|\sigma^{-1}(E_\sigma \Delta E_\rho)| = \sigma^{-n} \omega_n(\sigma^n - \rho^n) \leq \rho_*^{-n} \int_\rho^\sigma A_\tau d\tau = \rho_*^{-n} \int_\rho^\sigma \int_{\partial^* E_\tau} V_\tau d\mathcal{H}^{n-1} d\tau.$$

For the second term, the linear-in-radius dilation $\Lambda_s(x) := sx$ satisfies $|\Lambda_{\sigma^{-1}} E_\rho \Delta \Lambda_{\rho^{-1}} E_\rho| \leq n\rho_*^{-n-1}(R/\rho_*)|\sigma - \rho| \cdot \omega_n$, using $\Omega \subset B_R$ and $E_\rho \subset B_R$, and the elementary identity $|\Lambda_s A \Delta \Lambda_{s'} A| \leq |s - s'| \cdot \mathcal{H}^{n-1}(\partial A) \cdot \sup_{y \in A} |y|$ for sets contained in B_R together with $\sup_y |y| \leq R$. Both terms are bounded by $C(n, R, \rho_*)|\sigma - \rho|$. Setting m as in (249) gives (248). Since $A_\rho \leq n\omega_n$, $m \in L^\infty \subset L^1$. $\square$

By Definition 4.23, the metric derivative $|\dot{F}_\rho|_{\mathcal{X}}$ exists for a.e. $\rho \in [\rho_*, 1]$.

4.2.6. *The metric first-variation theorem (T).* The main result of this subsection is the following.

Theorem 4.26 (Finite-perimeter metric first variation). *Under the standing hypotheses of §4.2.1, and for any Borel-measurable centre field $\rho \mapsto z_\rho$ with $\sup_\rho |z_\rho| \leq R' < \infty$,*

$$(251) \quad |\dot{F}_\rho|_{\mathcal{X}} \leq C(n, \rho_*) \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}$$

for a.e. $\rho \in [\rho_, 1]$, with $C(n, \rho_*) = \rho_*^{-n}$.*

The proof has three parts: a smooth-flow calculation (§4.2.7), a BV/coarea approximation (§4.2.8), and a closure argument via Reshetnyak and AGS lower semicontinuity (§4.2.9). Reading the smooth part first makes the structure transparent; the finite-perimeter part is then a passage to the limit with no hidden smoothness.

4.2.7. Smooth-flow proof. Throughout this subsubsection we *temporarily assume* that Ω has C^∞ boundary, that $u \in C^\infty(\Omega)$, and that $|\nabla u| > 0$ on the closed strip $S_\eta := \bigcup_{\rho \in [\rho_*, 1-\eta]} \partial E_\rho$ for some $\eta > 0$. This is the standard smooth-flow regime in which the first variation of $\rho \mapsto |F_\rho \triangle C|$ is classical ([46, § 17.3], [4, Thm. 3.85]); we will then remove these hypotheses in §4.2.9.

Fix $z = z_\rho$ and $a \in \mathbb{R}^n$. The homothety $\Psi_\rho(y) := z + \rho^{-1}(y - z)$ maps E_ρ onto $z + \rho^{-1}(E_\rho - z)$. Under the smooth flow $\rho \mapsto E_\rho$ the trajectories $X(\rho, \cdot)$ on ∂E_ρ satisfy $\partial_\rho X \cdot \nu_\rho = V_\rho$. A direct differentiation of $\Psi_\rho(X(\rho, \cdot))$ and projection onto ν_ρ give ([7, (3.1)]; equivalently, the formula recorded in [A3], (3.1))

$$(252) \quad \partial_\rho \Psi_\rho(X) \cdot \nu_\rho = \rho^{-1}(V_\rho - H_{z,\rho}).$$

If we further translate the level by $-ha$ over a time-step h , the corrected normal velocity of $\rho \mapsto \Psi_\rho(E_\rho) - \rho a$, restored back to ∂E_ρ , is

$$(253) \quad W_\rho^{z,a} := \rho^{-1}(V_\rho - H_{z,\rho} - a \cdot \nu_\rho).$$

Lemma 4.27 (Smooth first variation; [46, §17.3]). *For any smooth bounded set C and for a.e. $\rho \in [\rho_*, 1 - \eta]$,*

$$(254) \quad \frac{d}{d\rho} |F_\rho \triangle C| = \int_{\partial F_\rho} \sigma_\rho^C(W_\rho^{z,a} \circ \Psi_\rho^{-1}) d\mathcal{H}^{n-1},$$

where $\sigma_\rho^C(y) = -1$ if $y \in \partial F_\rho \cap C$ and $+1$ otherwise. In particular,

$$(255) \quad \left| \frac{d}{d\rho} |F_\rho \triangle C| \right| \leq \int_{\partial F_\rho} |W_\rho^{z,a} \circ \Psi_\rho^{-1}| d\mathcal{H}^{n-1}.$$

This is [A3, (3.3)]; equivalently [4, Thm. 3.85] applied with the vector field Ψ_ρ .

Proposition 4.28 (Smooth case of (T)). *Under the smooth-flow hypotheses, for a.e. $\rho \in [\rho_*, 1 - \eta]$,*

$$(256) \quad \limsup_{h \rightarrow 0} \frac{d\mathcal{X}([F_{\rho+h}], [F_\rho])}{|h|} \leq \rho_*^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z,\rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}.$$

Proof. Apply Lemma 4.27 with $C = F_\rho$ and integrate over $[\rho, \rho + h]$:

$$|F_{\rho+h} \triangle F_\rho| = \int_\rho^{\rho+h} \frac{d}{d\tau} |F_\tau \triangle F_\rho| d\tau \leq \int_\rho^{\rho+h} \int_{\partial F_\tau} |W_\tau^{z,a} \circ \Psi_\tau^{-1}| d\mathcal{H}^{n-1} d\tau.$$

Now change variables along $\Psi_\tau : \partial E_\tau \rightarrow \partial F_\tau$. The Jacobian of Ψ_τ is ρ^{1-n} (it is a homothety of factor ρ^{-1} on $\mathbb{R}^n$, with surface area Jacobian $\rho^{-(n-1)}$), so

$$(257) \quad \int_{\partial F_\tau} |W_\tau^{z,a} \circ \Psi_\tau^{-1}| d\mathcal{H}^{n-1} = \tau^{1-n} \int_{\partial^* E_\tau} |W_\tau^{z,a}| d\mathcal{H}^{n-1} = \tau^{-n} \int_{\partial^* E_\tau} |V_\rho - H_{z,\tau} - a \cdot \nu_\tau| d\mathcal{H}^{n-1}.$$

The last equality uses (253). Bounding $\tau^{-n} \leq \rho_*^{-n}$ and dividing by $|h|$, then letting $h \rightarrow 0$, gives

$$\limsup_{h \rightarrow 0} \frac{|F_{\rho+h} \triangle F_\rho|}{|h|} \leq \rho_*^{-n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}.$$

Now apply the translation derivative formula (Lemma 4.29 below) to absorb $a \cdot \nu_\rho$ into a translation $F_{\rho+h} \mapsto F_{\rho+h} - ha$: the symmetric difference between $F_{\rho+h} - ha$ and F_ρ also satisfies the same bound. Taking the infimum over a and using $d\mathcal{X}([F_{\rho+h}], [F_\rho]) \leq |F_{\rho+h} \triangle (F_\rho + ha)|$ yields (256). $\square$

Lemma 4.29 (Translation derivative; [46, Thm. 13.8], [4, Thm. 3.84]). *For every bounded set E of finite perimeter and every $a \in \mathbb{R}^n$,*

$$(258) \quad \lim_{\varepsilon \rightarrow 0} \frac{|E \triangle (E + \varepsilon a)|}{\varepsilon} = \int_{\partial^* E} |a \cdot \nu_E| d\mathcal{H}^{n-1}.$$

In view of (258), the inf over a in (251) is *exactly* the metric-quotient cost: not slack, but the precise infinitesimal cost of factoring out translations. This is the role of the parameter a in the theorem statement.

4.2.8. BV/coarea smooth approximation. We now drop the smooth-flow hypotheses. The standing hypotheses of §4.2.1 are kept; the only regularity assumed is Ω open, bounded, $|\Omega| = \omega_n$, and the consequent $u \in W_{\text{loc}}^{2,p}(\Omega)$, $p > n$.

Lemma 4.30 (Coarea-good smooth recovery). *There exist $u_k \in C^\infty(\mathbb{R}^n)$ with $u_k \rightarrow u$ in $W^{1,2}(\Omega)$ and a subsequence (not relabelled) such that, setting $E_t^k := \{u_k > t\}$ and reparametrising $t_k(\rho)$ by $|E_{t_k(\rho)}^k| = \omega_n \rho^n$, the following hold for a.e. $\rho \in [\rho_*, 1]$:*

- (a) $t_k(\rho) \rightarrow t(\rho)$;
- (b) $E_\rho^k \rightarrow E_\rho$ in $L^1(\mathbb{R}^n)$;
- (c) $P(E_\rho) \leq \liminf_k P(E_\rho^k)$, and in fact $P(E_\rho^k) \rightarrow P(E_\rho)$;
- (d) the vector-valued perimeter measures $\nu_\rho^k \mathcal{H}^{n-1} \upharpoonright \partial^* E_\rho^k$ converge weakly to $\nu_\rho \mathcal{H}^{n-1} \upharpoonright \partial^* E_\rho$;
- (e) $\int_{\partial^* E_\rho^k} |V_\rho^k| d\mathcal{H}^{n-1} \rightarrow \int_{\partial^* E_\rho} |V_\rho| d\mathcal{H}^{n-1}$.

Sketch with precise inputs. Take $u_k = u * \eta_k$ for a standard mollifier η_k , extended by zero outside Ω . Then $u_k \rightarrow u$ in $W_{\text{loc}}^{1,p}$ for every $p < \infty$ ([4, Thm. 3.40]). Item (a) and (b) follow from the BV coarea theorem applied to $u_k \rightarrow u$ together with Helly–Lebesgue differentiation of the (monotone) maps $t \mapsto |E_t^k|$ ([46, Thm. 13.1, Lem. 13.2]). Item (c) is the BV coarea identity

$$\int |\nabla u_k| dx = \int_0^\infty P(E_t^k) dt \rightarrow \int |\nabla u| dx = \int_0^\infty P(E_t) dt$$

combined with perimeter LSC ([46, Prop. 12.15], item A6) and Fatou: the LSC gives $\liminf P(E_t^k) \geq P(E_t)$ a.e., while the L^1 -integrals match, so equality holds for a.e. t (and hence a.e. ρ). Item (d) is weak convergence of vector perimeter measures, which follows from (c) combined with L^1 -convergence of $\mathbf{1}_{E_\rho^k}$ and [46, Cor. 12.21]. Item (e) uses $\int_{\partial^* E_\rho^k} |V_\rho^k| = |t_k'(\rho)| \cdot a^k(t_k(\rho))$

where $a^k(s) = \int_{\partial^* E_s^k} |\nabla u_k|^{-1} d\mathcal{H}^{n-1}$ (coarea, item A2), combined with the Helly–Lebesgue argument again. $\square$

Remark 4.31 (No graph regularity used). Lemma 4.30 makes no claim about ∂E_ρ being a graph or being globally Lipschitz; it asserts only L^1 -convergence of the sub-level sets, BV-coarea convergence of perimeters, and weak convergence of vector perimeter measures. The recovery family u_k is smooth on $\mathbb{R}^n$, but the limit set E_ρ may have any reduced-boundary structure.

4.2.9. Closure: Reshetnyak and AGS lower semicontinuity.

Lemma 4.32 (Reshetnyak l.s.c. for the velocity-defect integrand; items A6, A8). *Fix $z, a \in \mathbb{R}^n$ and $\rho \in [\rho_*, 1]$. The integrand*

$$\Phi(x, s, \nu) := |s - (x - z) \cdot \nu / \rho - a \cdot \nu|, \quad (x, s, \nu) \in \mathbb{R}^n \times \mathbb{R} \times \mathbb{R}^n,$$

is jointly continuous and convex of linear growth in (s, ν) . For sequences $E_\rho^k \rightarrow E_\rho$ in L^1 with the perimeter and vector-perimeter convergence of Lemma 4.30(c)–(e), and writing $V_\rho^k = -t'_k(\rho)/|\nabla u_k|$,

$$(259) \quad \liminf_{k \rightarrow \infty} \int_{\partial^* E_\rho^k} \Phi(x, V_\rho^k, \nu_\rho^k) d\mathcal{H}^{n-1} \geq \int_{\partial^* E_\rho} \Phi(x, V_\rho, \nu_\rho) d\mathcal{H}^{n-1}.$$

Proof. This is Reshetnyak lower semicontinuity for convex linear-growth integrands with continuous x -dependence, applied to the BV functions $\mathbf{1}_{E_\rho^k} \rightarrow \mathbf{1}_{E_\rho}$ on the bounded ball B_R : see [4, Thm. 2.38] (continuous-coefficient case) and [4, Thm. 2.39] (weak strict BV convergence). The fact that $V_\rho^k \mathcal{H}^{n-1} \upharpoonright \partial^* E_\rho^k$ converges weakly to $V_\rho \mathcal{H}^{n-1} \upharpoonright \partial^* E_\rho$ follows from Lemma 4.30(d)–(e): the L^1 mass converges, and the support of the limit lies on $\partial^* E_\rho$ by Lemma 4.30(d). $\square$

Lemma 4.33 (AGS l.s.c. of the metric derivative; [6, Thm. 1.1.2]). *Let $\gamma_k : [\rho_*, 1] \rightarrow \mathcal{X}$ be absolutely continuous curves with $\gamma_k(\rho) \rightarrow \gamma(\rho)$ in $\mathcal{X}$ for every $\rho \in [\rho_*, 1]$, and suppose $|\dot{\gamma}_k|_{\mathcal{X}}(\rho) \leq m_k(\rho)$ a.e. with $m_k \rightarrow m$ weakly in $L^1([\rho_*, 1])$. Then γ is absolutely continuous and $|\dot{\gamma}|_{\mathcal{X}}(\rho) \leq m(\rho)$ for a.e. ρ .*

Proof of Theorem 4.26. Let u_k be the recovery sequence of Lemma 4.30, and define $F_\rho^k := \rho^{-1} E_\rho^k$, $[F_\rho^k] \in \mathcal{X}$. By Lemma 4.30(b), $[F_\rho^k] \rightarrow [F_\rho]$ in $\mathcal{X}$ for a.e. ρ . For each k , the smooth foliation $\rho \mapsto E_\rho^k$ has $|\nabla u_k| > 0$ on the relevant strip (after possibly excluding a null set of ρ , by smoothness of u_k and Sard's theorem). Applying Proposition 4.28 to $\{E_\rho^k\}$,

$$(260) \quad |\dot{F}_\rho^k|_{\mathcal{X}} \leq m_k(\rho) := \rho_*^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho^k} |V_\rho^k - H_{z_\rho, \rho} - a \cdot \nu_\rho^k| d\mathcal{H}^{n-1}.$$

By the level-set deficit identity (Sec. 4.1.11, item E1.4) applied to u_k , m_k is uniformly bounded in $L^1([\rho_*, 1])$ by a constant depending only on n, R, ρ_* and the BV-coarea norm of u_k , hence by $\|u_k\|_{W^{1,2}}$. By Dunford–Pettis, after extracting a subsequence, $m_k \rightharpoonup m$ weakly in L^1 . By Lemma 4.32 applied to the optimal a in the infimum (measurable selection [8, Thm. 8.2.11], item A7),

$$(261) \quad m(\rho) \geq \rho_*^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1} \quad \text{a.e. } \rho.$$

By Lemma 4.33, $\rho \mapsto [F_\rho]$ is absolutely continuous (consistent with Theorem 4.25) and

$$(262) \quad |\dot{F}_\rho|_{\mathcal{X}} \leq m(\rho) \leq \rho_*^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}.$$

This is (251) with $C(n, \rho_*) = \rho_*^{-n}$. $\square$

4.2.10. *Per- ρ squared form (conditional) and the integrated route actually used.* Combining Theorem 4.26 with the D_H - and D_I -controls of Sec. 2.1.8 (items A1–A8, B6, C2–C4) gives an estimate of the form

$$(263) \quad |\dot{F}_\rho|_{\mathcal{X}}^2 \leq C(n, R, \rho_*) (D_H(t(\rho)) + D_I(t(\rho))),$$

conditional on a per- ρ radial trace bound on the homothetic velocity defect (the bound recorded in Sec. 2.1.8, eq. (2.4)). Per the no-graph hypothesis of this manuscript, the per- ρ radial trace is established only in its integrated good/bad form by Agents 11–12; (263) is therefore *not* used pointwise in the downstream chain. The downstream argument uses only the integrated form

$$(264) \quad \int_{\rho_*}^1 |\dot{F}_\rho|_{\mathcal{X}}^2 \mu(d\rho) \leq C(n, R, \rho_*) \delta_T(\Omega),$$

which is proved in Sec. 4.4.7 (Agent 12) by combining Theorem 4.26, the Fusco–Julin centre selection of Sec. 4.3.4 (Agent 11), and the level-set deficit identity of Sec. 4.1.11 (Agent 9).

Theorem 4.34 (Final form of METRICFRAMEWORK). *Let $n \geq 2$, $0 < \rho_* < 1$, and $R \geq 2$. Let $\Omega \subset \mathbb{R}^n$ be open and bounded with $|\Omega| = \omega_n$ and $\Omega \subset B_R$. Let u be the variational torsion potential, let $\{E_\rho\}_{\rho \in [\rho_*, 1]}$ be the volume-radius foliation (241), and set $F_\rho := \rho^{-1}E_\rho$. Then:*

- (i) $(\mathcal{X}, d_{\mathcal{X}})$ is a complete metric space, separable and sequentially compact on classes of sets in a fixed ball (Proposition 4.20).
- (ii) The curve $\rho \mapsto [F_\rho] \in \mathcal{X}$ is absolutely continuous on $[\rho_*, 1]$, with metric-derivative bound (249) (Theorem 4.25).
- (iii) The normal velocity $V_\rho = -t'(\rho)/|\nabla u|$ is $\mathcal{H}^{n-1}$ -a.e. finite and strictly positive on $\partial^* E_\rho$ for a.e. ρ (Proposition 4.24).
- (iv) For any Borel-measurable centre field $\rho \mapsto z_\rho$ with $\sup_\rho |z_\rho| \leq R' < \infty$,

$$(265) \quad |\dot{F}_\rho|_{\mathcal{X}} \leq \rho_*^{-n} \inf_{a \in \mathbb{R}^n} \int_{\partial^* E_\rho} |V_\rho - H_{z_\rho, \rho} - a \cdot \nu_\rho| d\mathcal{H}^{n-1}$$

for a.e. $\rho \in [\rho_*, 1]$ (Theorem 4.26).

The constants in the section are: $C_{AC}(n, R, \rho_*)$ in (249), with explicit dependence $C_{AC} = \rho_*^{-n} n \omega_n + n \rho_*^{-n-1} R / \rho_*$; $C(n, \rho_*) = \rho_*^{-n}$ in (T) (251). The cited downstream constant $C(n, R, \rho_*)$ in (264) is the product $\rho_*^{-2n} \cdot C_{DH+DI}$ where $C_{DH+DI} = C_{DH+DI}(n, R, \rho_*)$ is the constant of Sec. 2.1.8.

Source map for this subsection. Definitions and metric facts: [WIP-MF] §2 and Proposition 4.20 (proved here); absolute continuity Theorem 4.25: proved here, using Lemma 4.19 ([WIP-MF] Lem. 2.2) and the coarea flux [46, Thm. 13.1] (item A2). Sobolev–Sard for V_ρ (Proposition 4.24): cited from [25, App. A, Thm. A.1] (item A3, source map E3.6). Smooth-flow first

variation (Lemma 4.27, Proposition 4.28): cited from [46, § 17.3] and [4, Thm. 3.85]; identical to [A3] §3. Translation derivative (Lemma 4.29): [46, Thm. 13.8], [4, Thm. 3.84] (item E2.7). BV/coarea recovery (Lemma 4.30): standard mollification recovery; precise inputs cited from [46, Thm. 13.1, Prop. 12.15, Cor. 12.21] and [4, Thm. 3.40] (items A2, A6). Reshetnyak l.s.c. (Lemma 4.32): [4, Thm. 2.38–2.39] (item A8). AGS l.s.c. (Lemma 4.33): [6, Thm. 1.1.2] (item F15). Measurable selection for the optimal a : [8, Thm. 8.2.11] (item A7). Theorem 4.26 (E2.6): proved here.

4.3. §4.3. Fusco–Julin centre and the oriented radial trace lemma.

This subsection is the centre-package and load-bearing radial-trace estimate of Plan 2. Its conclusion — Theorem 4.42 — together with the metric first-variation theorem of §4.2 controls the metric derivative $|\dot{F}_\rho|_\mathcal{X}$ in the weighted action integral.

Throughout this subsection we fix $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$, and $\kappa > 0$. We work with an open set $\Omega \subset B_R \subset \mathbb{R}^n$, $|\Omega| = \omega_n$, with variational torsion function $u = u_\Omega$, volume-radius parametrisation $E_\rho = \{u > t(\rho)\}$, $|E_\rho| = \omega_n \rho^n$ for $\rho \in [\rho_*, 1]$, and $\mu(d\rho) := (-t'(\rho)) d\rho$. We assume

$$(266) \quad \delta_T(\Omega) \leq \delta_0(n, R, \rho_*),$$

where δ_0 is the smallness threshold fixed in Theorem 4.62 (the constant matching the Cicalese–Leonardi smallness threshold and the centroid–Fraenkel offset, cf. [21, Prop. 2.3] and [29, Thm. 1.1]).

4.3.1. Scaled Fusco–Julin and centre selection. Recall the strong quantitative isoperimetric inequality of Fusco–Julin [29, Thm. 1.1]: there is a dimensional constant $C_{\text{FJ}}(n) > 0$ such that for every set of finite perimeter $E \subset \mathbb{R}^n$ with $|E| = |B_1|$ there exists a centre $z = z(E) \in \mathbb{R}^n$ with

$$(267) \quad \left(\frac{|E \Delta B_1(z)|}{|B_1|} \right)^2 + \beta(E, z)^2 \leq C_{\text{FJ}}(n) (P(E) - P(B_1)),$$

where the FJ oscillation index is

$$(268) \quad \beta(E, z)^2 := \inf_{|E'|=|E|} \int_{\partial^* E} |\nu_E(x) - (\nu_{E'})_{\text{rad}}(x; z)|^2 d\mathcal{H}^{n-1}(x),$$

which by the standard divergence identity (item C4 of the preliminaries; see (271) below) is comparable to $\int_{\partial^* E} |\nu_E - (x - z)/|x - z||^2 d\mathcal{H}^{n-1}$.

Theorem 4.35 (Scaled Fusco–Julin on $\{|E| = \omega_n \rho^n\}$). *For every $\rho \in [\rho_*, 1]$ and every $E \subset \mathbb{R}^n$ of finite perimeter with $|E| = \omega_n \rho^n$ there exists $z = z(E, \rho) \in \mathbb{R}^n$ such that*

$$(269) \quad |E \Delta B_\rho(z)|^2 + \rho^{n+1} \int_{\partial^* E} \left| \nu_E(x) - \frac{x - z}{|x - z|} \right|^2 d\mathcal{H}^{n-1}(x) \leq C_1(n, \rho_*) \rho^{n+1} (P(E) - P(B_\rho)),$$

where $C_1(n, \rho_*) = 4 C_{\text{FJ}}(n) \cdot \max(1, \rho_*^{-(n+1)})$. The constant depends only on n and ρ_* .

Proof. Set $\tilde{E} := \rho^{-1}E$, so $|\tilde{E}| = \omega_n = |B_1|$ and $\tilde{E}$ has finite perimeter with $P(\tilde{E}) = \rho^{-(n-1)}P(E)$. Apply (267) to $\tilde{E}$: there exists $\tilde{z}$ with

$$(270) \quad \left(\frac{|\tilde{E} \Delta B_1(\tilde{z})|}{\omega_n} \right)^2 + \beta(\tilde{E}, \tilde{z})^2 \leq C_{\text{FJ}}(n) (P(\tilde{E}) - P(B_1)).$$

Set $z := \rho \tilde{z}$. Translating $\tilde{E}$ to E scales volumes by ρ^n and perimeters by ρ^{n-1} , and using the divergence identity (271) below to rewrite $\beta(\tilde{E}, \tilde{z})^2$ in terms of the normal-oscillation integral, the scaled bounds $|E \Delta B_\rho(z)| = \rho^n |\tilde{E} \Delta B_1(\tilde{z})|$ and $\int_{\partial^* E} |\nu_E - (x-z)/|x-z||^2 d\mathcal{H}^{n-1} = \rho^{n-1} \int_{\partial^* \tilde{E}} |\nu_{\tilde{E}} - (y-\tilde{z})/|y-\tilde{z}||^2 d\mathcal{H}^{n-1}$ yield (269) after multiplying through by ρ^{2n} and using $\rho_* \leq \rho \leq 1$ to absorb the ρ -powers into the constant C_1 . $\square$

Remark 4.36 (Comparison constants). The divergence identity (item C4 of the preliminaries) (271)

$$2\beta(E, z)^2 = \int_{\partial^* E} \left| \nu_E(x) - \frac{x-z}{|x-z|} \right|^2 d\mathcal{H}^{n-1}(x) = 2(P(E) - P(B_\rho)) + 2\Pi(E, z),$$

holds whenever $|E| = |B_\rho|$, $z \in \mathbb{R}^n$ with the radial field $x \mapsto (x-z)/|x-z|$ locally Borel and the truncation argument of §2.1 (item A5) applicable. The identity is proved by testing $\operatorname{div}((x-z)/|x-z|) = (n-1)|x-z|^{-1}$ on E via the finite-perimeter divergence theorem and truncating away from z ; $\Pi(E, z) := (n-1) \int (\mathbf{1}_{B_\rho(z)} - \mathbf{1}_E) |x-z|^{-1} dx$ is the signed radial moment of the notation register §13. We use only the inequality $\int |\nu_E - (x-z)/|x-z||^2 d\mathcal{H}^{n-1} \leq 2C_1(n, \rho_*)(P(E) - P(B_\rho))/\rho^{n+1}$ that follows from (269).

We now choose a single Borel centre field along the torsion foliation. The canonical choice (see §2, and the discussion of the Fusco–Julin centre in [29, Thm. 1.1]) is the bulk centroid

$$(272) \quad \bar{z}_\rho = \bar{z}_\rho := \frac{1}{|E_\rho|} \int_{E_\rho} x dx \in \mathbb{R}^n, \quad \rho \in [\rho_*, 1].$$

Lemma 4.37 (Borel measurability and CL containment of the centroid). *Under (266) the map $\rho \mapsto \bar{z}_\rho$ is Lipschitz on $[\rho_*, 1]$ with constant $L_{\text{cent}}(n, R, \rho_*) \leq 2nR\rho_*^{-n}$; in particular it is Borel. Moreover for every $\rho \in [\rho_*, 1]$,*

$$(273) \quad B(\bar{z}_\rho, \rho/4) \subset E_\rho \subset B(\bar{z}_\rho, 3\rho),$$

so $\bar{z}_\rho \in E_\rho$ and the radial field $e_r(x) := (x - \bar{z}_\rho)/|x - \bar{z}_\rho|$ is $\mathcal{H}^{n-1}$ -a.e. well-defined and Borel on $\partial^* E_\rho$, with $\rho_*/4 \leq |x - \bar{z}_\rho| \leq 3R$ on $\partial^* E_\rho$ for a.e. ρ .

Proof. For $\rho_* \leq \rho \leq \rho' \leq 1$, the nesting of $\{E_\rho\}$ gives $|E_\rho \Delta E_{\rho'}| = |E_{\rho'}| - |E_\rho| \leq n\omega_n|\rho - \rho'|$. Hence $|\int_{E_\rho} x_i dx - \int_{E_{\rho'}} x_i dx| \leq R \cdot n\omega_n|\rho - \rho'|$; combined with $|E_\rho| \geq \omega_n \rho_*^n$ this yields the Lipschitz constant $L_{\text{cent}} = 2nR/\rho_*^n$, which is Borel (in fact continuous). The containment (273) follows from the Cicalese–Leonardi inner-ball containment ([21, Thm. 1.1]) applied at the Fraenkel centre z_ρ^{Fr} , together with the centroid–Fraenkel proximity $|\bar{z}_\rho - z_\rho^{\text{Fr}}| \leq C(n, \rho_*) \mathcal{A}(E_\rho) \leq C' \sqrt{\delta_T(\Omega)} \leq \rho/4$, valid for δ_0 small ([21, Prop. 2.3], [1, Lem. 2.6], [29, Thm. 1.1]). Borel measurability of the normal field ν_{E_ρ} on $\partial^* E_\rho$ ([46, Thm. 15.9]) then yields the Borel measurability of e_r on $\partial^* E_\rho$ for $\bar{z}_\rho \in E_\rho$. $\square$

Remark 4.38 (Auxiliary FJ-oscillation centre and Borel selection). The centre $z(E_\rho, \rho)$ produced by Theorem 4.35 is a valid almost-minimiser; Lemma 4.37's centroid $\bar{z}_\rho$ also satisfies (269) up to enlarging C_1 by the proximity offset $|\bar{z}_\rho - z_\rho^{\text{Fr}}| \leq C(n, \rho_*) \sqrt{\delta_T}$, absorbed by the smallness threshold δ_0 . The Borel

selection of *any* valid centre on a Borel-measurable family of finite-perimeter sets follows from the measurable-selection statement A7 of the preliminaries (Federer [24, §2.13], in the form recorded in [4, §5.3]). Throughout the rest of this subsection the centre is fixed to be $z_\rho := \bar{z}_\rho$.

4.3.2. *4.3.2. Fraenkel and normal-oscillation bounds at the centroid.* Define the (good) isoperimetric set

$$(274) \quad G_I := \{\rho \in [\rho_*, 1] : D_I(t(\rho)) \leq \eta_I\}, \quad \eta_I := \eta_I(n, R, \rho_*) \in (0, 1],$$

with η_I small enough that on G_I , $P(E_\rho) \leq 2P(B_\rho)$ (possible since $D_I = (P(E) - P(B_\rho))(P(E) + P(B_\rho))$ and $P(B_\rho) \geq n\omega_n \rho_*^{n-1}$). We collect (269) at $z_\rho = \bar{z}_\rho$.

Lemma 4.39 (Fraenkel bound at the centroid). *There exists $C_2 = C_2(n, R, \rho_*) > 0$ such that for every $\rho \in G_I$,*

$$(275) \quad |E_\rho \Delta B_\rho(\bar{z}_\rho)|^2 \leq C_2(n, R, \rho_*) D_I(t(\rho)).$$

Proof. From (269) applied with $z = \bar{z}_\rho$ (valid by Remark 4.38, after absorbing the $|\bar{z}_\rho - z_\rho^{\text{Fr}}|^2$ correction into the constant via $|B_\rho(a) \Delta B_\rho(b)| \leq C(n) \rho^{n-1} |a - b|$ and the FMP estimate $|E_\rho \Delta B_\rho(z_\rho^{\text{Fr}})|^2 \leq c_1 |B_\rho|^2 (P(E_\rho) - P(B_\rho))$),

$$(276) \quad |E_\rho \Delta B_\rho(\bar{z}_\rho)|^2 \leq 2C_1(n, \rho_*) \rho^{n+1} (P(E_\rho) - P(B_\rho)).$$

Use the algebraic identity $D_I(t(\rho)) = (P(E_\rho) - P(B_\rho))(P(E_\rho) + P(B_\rho)) \geq n\omega_n \rho_*^{n-1} \cdot (P(E_\rho) - P(B_\rho))$ to convert $(P(E_\rho) - P(B_\rho))$ into $D_I/(n\omega_n \rho_*^{n-1})$; this and $\rho \leq 1$ yield (275) with $C_2 = 2C_1(n, \rho_*)/(n\omega_n \rho_*^{n-1})$, polynomial in $n, R, 1/\rho_*$. $\square$

Lemma 4.40 (Normal-oscillation bound at the centroid). *There exists $C_3 = C_3(n, R, \rho_*) > 0$ such that for every $\rho \in G_I$,*

$$(277) \quad \int_{\partial^* E_\rho} \left| \nu_\rho(x) - \frac{x - \bar{z}_\rho}{|x - \bar{z}_\rho|} \right|^2 d\mathcal{H}^{n-1}(x) \leq C_3(n, R, \rho_*) D_I(t(\rho)).$$

Proof. Combining (269) at $z = \bar{z}_\rho$ (with the same absorption as in the previous proof) and the lower bound $\rho^{n+1} \geq \rho_*^{n+1}$ gives $\int |\nu_\rho - (x - \bar{z}_\rho)/|x - \bar{z}_\rho||^2 d\mathcal{H}^{n-1} \leq C_1(n, \rho_*) (P(E_\rho) - P(B_\rho))/\rho_*^{n+1}$; converting $P - P_\rho$ to D_I as in the previous proof yields (277) with $C_3 = C_1(n, \rho_*)/(\rho_*^{n+1} \cdot n\omega_n \rho_*^{n-1})$. $\square$

Remark 4.41 ($\rho \in [\rho_*, 1] \setminus G_I$). Outside G_I , $D_I(t(\rho)) > \eta_I$, and (275), (277) hold trivially after enlarging C_2, C_3 by a multiple of η_I^{-1} , since both left-hand sides are bounded by $|B_R|^2$ and $4P(B_R) \leq 4n\omega_n R^{n-1}$ respectively. In the integrated argument of Theorem 4.62 the μ -measure of $\{\rho : \rho \notin G_I\}$ is controlled by Markov from the deficit identity, so the trivial bound is permissible.

4.3.3. *4.3.3. The oriented radial trace lemma.* This is the load-bearing lemma of Plan 2's GMT route. We prove the pointwise inequality, with the named vector field

$$(278) \quad X(x) := \left| \frac{|x - \bar{z}_\rho|}{\rho} - 1 \right| \cdot \frac{x - \bar{z}_\rho}{|x - \bar{z}_\rho|}, \quad x \in \mathbb{R}^n \setminus \{\bar{z}_\rho\},$$

of the brief, via the finite-perimeter divergence theorem (item A4 of the preliminaries: [46, Thm. 16.3]) and the truncation argument of item A5. We

do *not* use ray-slicing in the proof of this lemma; a slicing-based variant is sketched in Remark 4.44 below.

Theorem 4.42 (Oriented L^1 radial trace at the centroid). *There is $C_4 = C_4(n, R, \rho_*) > 0$ such that for every $\rho \in [\rho_*, 1]$ satisfying (266), with the centroid $z_\rho = \bar{z}_\rho$ of (272),*

$$(279) \quad T_\rho := \int_{\partial^* E_\rho} \left| \frac{|x - \bar{z}_\rho|}{\rho} - 1 \right| d\mathcal{H}^{n-1}(x) \leq C_4 \left(|E_\rho \Delta B_\rho(\bar{z}_\rho)| + \int_{\partial^* E_\rho} \left| \nu_\rho - \frac{x - \bar{z}_\rho}{|x - \bar{z}_\rho|} \right|^2 d\mathcal{H}^{n-1} \right).$$

The constant has the explicit form $C_4(n, R, \rho_) = \max\{n/\rho_*, 3R/(2\rho_*)\}$ (see (292) below).*

Proof. Step 1: setup. By Lemma 4.37, $\bar{z}_\rho \in B(\bar{z}_\rho, \rho/4) \subset E_\rho \subset B(\bar{z}_\rho, 3\rho) \subset B_R$, so $|x - \bar{z}_\rho| \in [\rho_*/4, 3R]$ for $\mathcal{H}^{n-1}$ -a.e. $x \in \partial^* E_\rho$. Translate so that $\bar{z}_\rho = 0$; the radial unit vector is $e_r(x) = x/|x|$ on $\mathbb{R}^n \setminus \{0\}$ and the weight is $w(r) := |r/\rho - 1|$, a Lipschitz function of $r \in [0, \infty)$ with $w(\rho) = 0$, $|w(r)| \leq \max(1, (3R)/\rho - 1) \leq 3R/\rho_*$ on $\partial^* E_\rho$, and the vector field $X(x) = w(|x|)e_r(x)$ of (278). Note X is locally Lipschitz on $\mathbb{R}^n \setminus \{0\}$ and bounded on B_R (with $\|X\|_\infty \leq 3R/\rho_*$), but is discontinuous at the origin.

Step 2: pointwise inequality. On $\partial^* E_\rho$, $|e_r| = |\nu_\rho| = 1$, so writing $1 = e_r \cdot \nu_\rho + (1 - e_r \cdot \nu_\rho)$ and using $w \geq 0$,

$$(280) \quad T_\rho = \int_{\partial^* E_\rho} w(|x|) d\mathcal{H}^{n-1} = \int_{\partial^* E_\rho} X \cdot \nu_\rho d\mathcal{H}^{n-1} + \int_{\partial^* E_\rho} w(|x|)(1 - e_r \cdot \nu_\rho) d\mathcal{H}^{n-1}.$$

The second integrand is non-negative because $e_r \cdot \nu_\rho \leq |e_r||\nu_\rho| = 1$, and the elementary identity $1 - e_r \cdot \nu_\rho = \frac{1}{2}|e_r - \nu_\rho|^2$ together with $w \leq 3R/\rho_*$ gives

$$(281) \quad \int_{\partial^* E_\rho} w(|x|)(1 - e_r \cdot \nu_\rho) d\mathcal{H}^{n-1} \leq \frac{3R}{2\rho_*} \int_{\partial^* E_\rho} |\nu_\rho - e_r|^2 d\mathcal{H}^{n-1}.$$

Step 3: truncated divergence theorem for the radial term. We bound $\int_{\partial^* E_\rho} X \cdot \nu_\rho d\mathcal{H}^{n-1}$ by the finite-perimeter divergence theorem applied to a Lipschitz truncation of X . Fix $\varepsilon \in (0, \rho_*/8)$ and a smooth non-decreasing $\eta_\varepsilon : [0, \infty) \rightarrow [0, 1]$ with $\eta_\varepsilon(r) = 0$ for $r \leq \varepsilon/2$, $\eta_\varepsilon(r) = 1$ for $r \geq \varepsilon$, $|\eta'_\varepsilon| \leq 4/\varepsilon$. Set

$$(282) \quad X_\varepsilon(x) := \eta_\varepsilon(|x|) w(|x|) e_r(x), \quad x \in \mathbb{R}^n.$$

Then $X_\varepsilon \in C^{0,1}(\mathbb{R}^n; \mathbb{R}^n) \cap C^\infty(\mathbb{R}^n \setminus \{0\}; \mathbb{R}^n)$, $X_\varepsilon \equiv X$ outside $\{|x| < \varepsilon\}$, and $\|X_\varepsilon\|_\infty \leq 3R/\rho_*$. The classical divergence $\operatorname{div} X_\varepsilon$ is locally bounded on $\mathbb{R}^n$ and equals

$$(283) \quad \operatorname{div} X_\varepsilon(x) = \eta_\varepsilon(|x|) \left[w'(|x|) + \frac{n-1}{|x|} w(|x|) \right] + \eta'_\varepsilon(|x|) w(|x|) \quad (x \neq 0),$$

where $w'(r) = r/(\rho|r|) \cdot \operatorname{sgn}(r/\rho - 1) = \rho^{-1} \operatorname{sgn}(r/\rho - 1)$ for $r \neq \rho$ (Lipschitz regularity of w , derivative a.e.). At the origin, $X_\varepsilon \equiv 0$ on $\{|x| < \varepsilon/2\}$, so X_ε is genuinely Lipschitz on $\mathbb{R}^n$.

By the divergence theorem for the finite-perimeter set E_ρ applied to the Lipschitz vector field X_ε (item A4 of the preliminaries; [46, Thm. 16.3], applied with the standard density argument: X_ε is mollifiable since it is

Lipschitz with compact support after multiplication by a cutoff at radius $3R$),

$$(284) \quad \int_{\partial^* E_\rho} X_\varepsilon \cdot \nu_\rho d\mathcal{H}^{n-1} = \int_{E_\rho} \operatorname{div} X_\varepsilon dx.$$

The same identity applied to the open ball $B_\rho = B_\rho(0)$ (a smooth finite-perimeter set, with $\nu_{B_\rho}(y) = y/|y|$ and $w(\rho) = 0$) yields

$$(285) \quad 0 = \int_{\partial^* B_\rho} w(|y|) e_r \cdot \nu_{B_\rho} d\mathcal{H}^{n-1} = \int_{\partial^* B_\rho} X_\varepsilon \cdot \nu_{B_\rho} d\mathcal{H}^{n-1} = \int_{B_\rho} \operatorname{div} X_\varepsilon dx,$$

where the first equality holds because $\eta_\varepsilon(r) = 1$ for r near ρ (since $\varepsilon < \rho_*/8 \leq \rho/8$) and $w(\rho) = 0$. Subtracting,

$$(286) \quad \int_{\partial^* E_\rho} X_\varepsilon \cdot \nu_\rho d\mathcal{H}^{n-1} = \int_{E_\rho \setminus B_\rho} \operatorname{div} X_\varepsilon dx - \int_{B_\rho \setminus E_\rho} \operatorname{div} X_\varepsilon dx.$$

Step 4: limit $\varepsilon \downarrow 0$. On $\mathbb{R}^n \setminus \{0\}$, $X_\varepsilon \rightarrow X$ pointwise and $\operatorname{div} X_\varepsilon \rightarrow \operatorname{div} X$ pointwise, where

$$(287) \quad \operatorname{div} X(x) = \rho^{-1} \operatorname{sgn}(|x|/\rho - 1) + \frac{n-1}{|x|} \left| \frac{|x|}{\rho} - 1 \right|, \quad x \in \mathbb{R}^n \setminus (\{0\} \cup \partial B_\rho).$$

On both $E_\rho \setminus B_\rho$ and $B_\rho \setminus E_\rho$, the domain of integration is contained in $\{|x| \geq \rho_*/4\}$ (by Lemma 4.37 for the first set and the trivial bound $\rho \geq \rho_*$ for the second after restricting to $|x| \geq \rho_*/4$; the set $\{|x| < \rho_*/4\} \cap (E_\rho \cup B_\rho) \subset B_\rho$ is *not* in either symmetric difference part by (273)). On $\{|x| \geq \rho_*/4\}$, the singular factor $(n-1)/|x| \leq 4(n-1)/\rho_*$ and the cutoff factor $\eta_\varepsilon(|x|) \equiv 1$ for $\varepsilon < \rho_*/8$; therefore $X_\varepsilon \equiv X$ on $\{|x| \geq \rho_*/4\}$ for all ε small enough, and (286) is in fact independent of $\varepsilon \in (0, \rho_*/8)$ and equals

$$(288) \quad \int_{\partial^* E_\rho} X \cdot \nu_\rho d\mathcal{H}^{n-1} = \int_{E_\rho \setminus B_\rho} \operatorname{div} X dx - \int_{B_\rho \setminus E_\rho} \operatorname{div} X dx.$$

(The left-hand side is $\int X \cdot \nu_\rho d\mathcal{H}^{n-1}$, the limit of the same integral with X_ε in place of X , by the same argument applied to the boundary trace.) This is the truncation step (A5 of the preliminaries): the locally integrable singularity of X at $\bar{z}_\rho = 0$ is harmless because $\bar{z}_\rho \in \operatorname{int} E_\rho$ and the domains $E_\rho \setminus B_\rho$, $B_\rho \setminus E_\rho$ are *disjoint from a fixed neighbourhood* of $\bar{z}_\rho$.

Step 5: pointwise bound on $\operatorname{div} X$. For $r := |x| > \rho$ (i.e. $x \in E_\rho \setminus B_\rho$):

$$(289) \quad \operatorname{div} X(x) = \rho^{-1} + \frac{n-1}{r} \left(\frac{r}{\rho} - 1 \right) = \frac{n}{\rho} - \frac{n-1}{r} \leq \frac{n}{\rho} \leq \frac{n}{\rho_*}.$$

For $r < \rho$ (i.e. $x \in B_\rho \setminus E_\rho$, $r \geq \rho_*/4$):

$$(290) \quad \operatorname{div} X(x) = -\rho^{-1} + \frac{n-1}{r} \left(1 - \frac{r}{\rho} \right) = \frac{n-1}{r} - \frac{n}{\rho},$$

so $(-\operatorname{div} X)_+(x) \leq n/\rho \leq n/\rho_*$. Combining with (288),

$$(291) \quad \int_{\partial^* E_\rho} X \cdot \nu_\rho d\mathcal{H}^{n-1} \leq \frac{n}{\rho_*} |E_\rho \Delta B_\rho(\bar{z}_\rho)|.$$

Step 6: assembly. Combining (280), (281), and (291),

$$(292) \quad T_\rho \leq \frac{n}{\rho_*} |E_\rho \Delta B_\rho(\bar{z}_\rho)| + \frac{3R}{2\rho_*} \int_{\partial^* E_\rho} |\nu_\rho - e_r|^2 d\mathcal{H}^{n-1}.$$

This is (279) with $C_4(n, R, \rho_*) = \max\{n/\rho_*, 3R/(2\rho_*)\}$, polynomial in $n, R, 1/\rho_*$. $\square$

Remark 4.43 (Why the truncation is legitimate). The truncation factor η_ε is needed only to ensure that the classical divergence theorem applies to a Lipschitz field on all of $\mathbb{R}^n$; the field X is bounded on B_R , but its classical derivative $\operatorname{div} X$ has a singularity of type $|x|^{-1}$ at $\bar{z}_\rho$. Two properties make the truncation harmless. First, $\bar{z}_\rho \in \operatorname{int} E_\rho$ (Lemma 4.37), so the small ball $\{|x - \bar{z}_\rho| < \rho_*/4\}$ lies entirely in $E_\rho \cap B_\rho$ and contributes *equally* to both E_ρ and B_ρ in (286), cancelling out. Second, on the symmetric-difference set $\{x : \mathbf{1}_{E_\rho}(x) \neq \mathbf{1}_{B_\rho}(x)\}$ the radius $|x - \bar{z}_\rho| \geq \rho_*/4$ (since this set lies in $E_\rho \Delta B_\rho \subset B_R \setminus B(\bar{z}_\rho, \rho_*/4)$ once δ_0 is small), so $X = X_\varepsilon$ on this set for all $\varepsilon < \rho_*/8$ and the limit (288) is exact, not a passage. This is the key estimate of the truncation step: the singularity of X at $\bar{z}_\rho$ contributes nothing because it is interior to both E_ρ and B_ρ .

Remark 4.44 (Equivalent slicing route). An alternative proof of (279) by a spherical slicing decomposition $T_\rho = T_\rho^{\operatorname{rad}, \operatorname{slic}} + T_\rho^{\operatorname{tan}}$ can be carried out via the radial slicing formula [46, Thm. 18.11] and a good-ray/bad-ray multiplicity decomposition. That route yields the same bound at the integrated level and trades the divergence-theorem step here for the Cicalese–Leonardi annular containment. Both routes produce the constant $C_4(n, R, \rho_*)$ in the same polynomial form. The divergence-theorem route is preferred here because it isolates the named vector field (278) and avoids reference to angular multiplicity.

4.3.4. *4.3.4. Consequences for the metric first-variation chain.* We collect the levelwise outputs that feed into the integrated action bound of Theorem 4.62 (§4.4).

Corollary 4.45 (Pointwise radial trace from D_I). *There exists $C_5 = C_5(n, R, \rho_*)$ such that for every $\rho \in G_I$,*

$$(293) \quad T_\rho^2 \leq C_5(n, R, \rho_*) D_I(t(\rho)).$$

Proof. By Theorem 4.42, $T_\rho^2 \leq 2C_4^2(|E_\rho \Delta B_\rho(\bar{z}_\rho)|^2 + (\int |\nu_\rho - e_r|^2 d\mathcal{H}^{n-1})^2)$. The first term is $\leq 2C_4^2 \cdot C_2 D_I$ by Lemma 4.39. For the second, use the trivial estimate $(\int |\nu_\rho - e_r|^2 d\mathcal{H}^{n-1})^2 \leq 4P(B_R) \cdot \int |\nu_\rho - e_r|^2 d\mathcal{H}^{n-1}$ on G_I (since the integrand is bounded by 4 pointwise and the perimeter $P(E_\rho) \leq 2P(B_\rho) \leq 2n\omega_n R^{n-1}$ on G_I), together with Lemma 4.40, to get $\leq 8n\omega_n R^{n-1} C_3 D_I$. Combining, $C_5 = 2C_4^2(C_2 + 8n\omega_n R^{n-1} C_3)$, polynomial in $n, R, 1/\rho_*$. $\square$

Corollary 4.46 (Good-level homothetic trace at the centroid). *With $\bar{V}_\rho := P(B_\rho)/P(E_\rho)$ and $H_{z,\rho}(x) = (x - z) \cdot \nu_\rho(x)/\rho$ on $\partial^* E_\rho$, there exists $C_6 = C_6(n, R, \rho_*)$ such that for every $\rho \in G_I$,*

$$(294) \quad \left(\int_{\partial^* E_\rho} |H_{\bar{z}_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \right)^2 \leq C_6(n, R, \rho_*) D_I(t(\rho)).$$

Proof. Write $e_r(x) = (x - \bar{z}_\rho)/|x - \bar{z}_\rho|$. Then $H_{\bar{z}_\rho, \rho}(x) = (|x - \bar{z}_\rho|/\rho) e_r \cdot \nu_\rho$ and

$$|H_{\bar{z}_\rho, \rho} - 1| \leq \left| \frac{|x - \bar{z}_\rho|}{\rho} - 1 \right| + |e_r \cdot \nu_\rho - 1|.$$

The second term equals $\frac{1}{2}|e_r - \nu_\rho|^2$. Integrating,

$$\int_{\partial^* E_\rho} |H_{\bar{z}_\rho, \rho} - 1| d\mathcal{H}^{n-1} \leq T_\rho + \frac{1}{2} \int_{\partial^* E_\rho} |\nu_\rho - e_r|^2 d\mathcal{H}^{n-1}.$$

Squaring and using $(a + b)^2 \leq 2a^2 + 2b^2$ together with Corollary 4.45 and Lemma 4.40 gives $(\int |H_{\bar{z}_\rho, \rho} - 1|)^2 \leq C D_I$ on G_I . The replacement of the centring constant 1 by $\bar{V}_\rho = 1 - (P(E_\rho) - P(B_\rho))/P(E_\rho)$ costs an additive term

$$|1 - \bar{V}_\rho| \cdot P(E_\rho) = P(E_\rho) - P(B_\rho) \leq D_I(t(\rho))/(n\omega_n \rho_*^{n-1}) \leq D_I^{1/2}/(n\omega_n \rho_*^{n-1})$$

on G_I (using $D_I \leq \eta_I \leq 1$ on G_I). Squaring and absorbing this term yields (294) with $C_6 = 2(C_5 + C_3) + 2/(n\omega_n \rho_*^{n-1})^2$, polynomial in $n, R, 1/\rho_*$. $\square$

Remark 4.47 (Hypothesis tracking). Every estimate above is stated for open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \leq \delta_0(n, R, \rho_*)$ small enough to ensure Cicalese–Leonardi smallness for every level $\rho \in [\rho_*, 1]$ via [21, Thm. 1.1]. No graph parametrisation, no minimiser selection, and no smoothness of $\partial^* E_\rho$ are used. All boundary integrals are on the De Giorgi reduced boundary. The Borel selection of the centroid is unconditional (Lemma 4.37); the FJ centre selection of Remark 4.38 reduces to it after the $O(\sqrt{\delta_T})$ proximity absorption recorded in [29, Thm. 1.1]. Outside the smallness regime, the FMP fallback $\mathcal{A}(\Omega)^2 \leq C(n)\delta_T^{1/2}$ together with $\delta_T \geq c(n) > 0$ gives all four estimates trivially.

Constants used in this subsection. For the convenience of §4.4 we collect:

- $C_1(n, \rho_*)$ in (269): scaled FJ constant, polynomial in ρ_*^{-1} .
- $C_2(n, R, \rho_*)$ in (275): Fraenkel estimate at the centroid.
- $C_3(n, R, \rho_*)$ in (277): normal oscillation.
- $C_4(n, R, \rho_*) = \max\{n/\rho_*, 3R/(2\rho_*)\}$ in (279): **the oriented L^1 radial trace constant**, the load-bearing output of Theorem 4.42.
- $C_5(n, R, \rho_*), C_6(n, R, \rho_*)$ in Corollaries 4.45, 4.46: levelwise inputs to §4.4.

Every constant is polynomial in n, R, ρ_*^{-1} and independent of $\delta_T(\Omega)$ and of the boundary-layer constant κ .

4.4. Integrated action and the weighted metric trace. This subsection chains the level-set deficit identity of Section 4.1.11, the metric first-variation estimate of Section 4.2.10, the scaled Fusco–Julin inequality and the oriented radial trace lemma of Section 4.3.4 into the integrated action bound (306) and, via a good/bad isoperimetric-defect decomposition, into the Lebesgue kinetic bound (315). The final output is the *weighted metric trace*: a radius $\hat{\rho} \in [\rho_*, \rho_\delta)$ at which the rescaled level set $F_{\hat{\rho}} = \hat{\rho}^{-1} E_{\hat{\rho}}$ is metrically close to the unit ball at the sharp δ_T -rate; see Theorem 4.62.

Throughout this subsection we adopt the standing hypotheses of Section 4: $n \geq 2, R \geq 2, \rho_* \in (0, 1)$, an open set $\Omega \subset B_R$ with $|\Omega| = \omega_n$, variational

torsion function $u = u_\Omega$, ρ -foliation $E_\rho = \{u > t(\rho)\}$ with $|E_\rho| = \omega_n \rho^n$ for $\rho \in [\rho_*, 1]$, rescaled level sets $F_\rho := \rho^{-1} E_\rho \subset B_{R/\rho_*}$, and the measure $\mu(d\rho) := (-t'(\rho)) d\rho$ on $[\rho_*, 1]$. We write $\delta := \delta_T(\Omega)$.

Canonical choice of κ . Per the notation register §12, the boundary-layer constant κ is a fixed dimensional choice $\kappa = \kappa(n, R, \rho_*) > 0$. Throughout §4.4 we pin κ to this canonical value; explicitly, we set

$$\kappa = \kappa(n, R, \rho_*) := 1$$

(any fixed positive constant of size $\Theta_n(1)$ works; the value $\kappa = 1$ is the simplest choice consistent with the notation register). All subsequent constants $C_*^{\text{act}}, C_*^{\text{kin}}, C_*$ then become functions of (n, R, ρ_*) alone; the smallness threshold δ_0 likewise depends only on (n, R, ρ_*) . We define

$$\rho_\delta := 1 - \kappa\sqrt{\delta}.$$

Throughout we assume

$$(295) \quad \delta \leq \delta_0(n, R, \rho_*),$$

with $\delta_0(n, R, \rho_*) > 0$ chosen so that $\rho_\delta \geq \rho_* + \frac{1}{2}(1 - \rho_*)$ and so that the additional smallness requirements (B1) of Theorem 4.62 below hold; for δ outside this regime the conclusion is vacuous or follows trivially from the FMP fallback $\mathcal{A}(\Omega)^2 \leq C(n)\delta_T(\Omega)$ for $\delta_T(\Omega) \geq \delta_0$.

The level-set deficit identity of Section 4.1.11 reads, in our ρ -parametrisation,

$$(296) \quad \int_{\rho_*}^1 (\mathcal{H}(\rho) + \mathcal{I}(\rho)) d\mu(\rho) \leq C_1(n, \rho_*) \delta,$$

where we have written

$$\mathcal{H}(\rho) := D_H(t(\rho)), \quad \mathcal{I}(\rho) := D_I(t(\rho)),$$

and used $dt = -t'(\rho) d\rho = d\mu$. The explicit constant is $C_1(n, \rho_*) = 2c_n(\omega_n \rho_*^n)^{1-2/n}/\omega_n = 2n^2 \omega_n^{2/n} (\omega_n \rho_*^n)^{1-2/n}/\omega_n$; see Theorem 4.13.

Since $P(E_\rho) + P(B_\rho) \geq 2P(B_\rho) = 2n\omega_n \rho^{n-1} \geq 2n\omega_n \rho_*^{n-1}$ and $D_I(t(\rho)) = (P(E_\rho) - P(B_\rho))(P(E_\rho) + P(B_\rho))$, the perimeter defect satisfies

$$(297) \quad 0 \leq \Delta P_\rho := P(E_\rho) - P(B_\rho) \leq \frac{\mathcal{I}(\rho)}{2n\omega_n \rho_*^{n-1}} =: C_2(n, \rho_*) \mathcal{I}(\rho).$$

Notation for this subsection.

- $\gamma(\rho) := [F_\rho] \in \mathcal{X}$ is the rescaled curve in the metric quotient $(\mathcal{X}, d_{\mathcal{X}})$ of Section 4.2.10. AGS [6, Thm. 1.1.2] provides the metric derivative $|\dot{\gamma}|(\rho) = |\dot{F}_\rho|_{\mathcal{X}}$.
- z_ρ denotes the bulk centroid of E_ρ , identified in Lemma 4.51 below with the canonical centre $\bar{z}_\rho = \bar{z}_\rho$ of the notation register §10. Outside the good set G_I defined in (298) we set $z_\rho = 0$.
- $\bar{V}_\rho := P(E_\rho)^{-1} \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = P(B_\rho)/P(E_\rho)$, which equals the mean value of V_ρ on $\partial^* E_\rho$ and satisfies $1 - \bar{V}_\rho = \Delta P_\rho/P(E_\rho)$.
- $H_{z_\rho, \rho}(x) := (x - z_\rho) \cdot \nu_\rho(x)/\rho$ is the homothetic radial field of Section 4.2.10.

4.4.1. 4.4.1 Good and bad isoperimetric-defect levels.

Definition 4.48 (Good and bad isoperimetric-defect levels). Let $\eta_I = \eta_I(n, R, \rho_*) > 0$ be the Fusco–Julin smallness threshold from Lemma 4.51 below. Define

$$(298) \quad G_I := \{\rho \in [\rho_*, \rho_\delta] : \mathcal{I}(\rho) \leq \eta_I\},$$

$$(299) \quad B_I := [\rho_*, \rho_\delta] \setminus G_I = \{\rho \in [\rho_*, \rho_\delta] : \mathcal{I}(\rho) > \eta_I\}.$$

The threshold $\eta_I = \eta_I(n, R, \rho_*)$ will be fixed at the end of Lemma 4.51: it is the smallest of the Fusco–Julin smallness scale on $\{|E| = |B_\rho|, \rho \in [\rho_*, 1]\}$ (see Proposition 4.35) and the elementary smallness threshold needed for the FJ centre to lie in B_{2R} . In particular, η_I does *not* depend on δ or on Ω .

Lemma 4.49 (Markov bound on B_I). *Under (295),*

(300)

$$\mu(B_I) \leq \eta_I^{-1} \int_{B_I} \mathcal{I}(\rho) d\mu(\rho) \leq \eta_I^{-1} \int_{\rho_*}^1 (\mathcal{H} + \mathcal{I}) d\mu \leq \eta_I^{-1} C_1(n, \rho_*) \delta.$$

Proof. By Definition 4.48, $\mathcal{I}(\rho) > \eta_I$ on B_I , hence $\mu(B_I) = \int_{B_I} d\mu \leq \eta_I^{-1} \int_{B_I} \mathcal{I} d\mu$. Both $\mathcal{H}, \mathcal{I} \geq 0$, so the second inequality is trivial. The last inequality is (296). $\square$

Remark 4.50 (The sharp rate is preserved). The Markov bound (300) is the only place where the size of B_I enters; the rate is $\mu(B_I) = O(\delta)$, not $O(\sqrt{\delta})$. The good-set quantitative estimates (302)–(303) below are pointwise in $\mathcal{I}(\rho)$, which is then integrated against $d\mu$ via (296). No square root of $\mathcal{I}$ is integrated unweighted; consequently the integrated bound (306) preserves the sharp δ rate. This is the point at which the naive route would lose a $\sqrt{\delta}$.

Lemma 4.51 (FJ centre package, fixed to the bulk centroid). *There exist $\eta_I = \eta_I(n, R, \rho_*) > 0$ and $C_{\text{FJ}} = C_{\text{FJ}}(n, R, \rho_*)$ such that, defining the centre field on G_I as the bulk centroid*

$$(301) \quad z_\rho := \bar{z}_\rho = \bar{z}_\rho = |E_\rho|^{-1} \int_{E_\rho} x dx, \quad \rho \in G_I,$$

and $z_\rho := 0$ on B_I , the map $\rho \mapsto z_\rho$ is Borel measurable on $[\rho_, \rho_\delta]$, satisfies $|z_\rho| \leq 2R$ on G_I , and for every $\rho \in G_I$,*

$$(302) \quad |E_\rho \Delta B_\rho(z_\rho)|^2 \leq C_{\text{FJ}} \mathcal{I}(\rho),$$

$$(303) \quad \int_{\partial^* E_\rho} \left| \nu_\rho(x) - \frac{x - z_\rho}{|x - z_\rho|} \right|^2 d\mathcal{H}^{n-1}(x) \leq C_{\text{FJ}} \mathcal{I}(\rho).$$

Moreover the centroid containment $B(z_\rho, \rho/4) \subset E_\rho \subset B(z_\rho, 3\rho)$ of Lemma 4.37 (Section 4.3.4) holds on G_I , justifying the use of the oriented radial trace lemma at z_ρ .

Proof. The choice $z_\rho := \bar{z}_\rho$ on G_I is the canonical one of the notation register §10, cf. also Agent 11 Remark 4.38 of Section 4.3.4. By that remark, the centroid $\bar{z}_\rho$ is admissible in the Fusco–Julin bound (269) up to absorbing the centroid–Fraenkel offset $|\bar{z}_\rho - z_\rho^{\text{Fr}}| \leq C(n, \rho_*) \sqrt{\delta_T}$ into the constant; the smallness threshold (295) (with δ_0 shrunk if necessary so that $C(n, \rho_*) \sqrt{\delta_0} \leq \rho/4$) is precisely what makes this absorption legitimate. The Fraenkel bound

(302) and the normal-oscillation bound (303) at $z_\rho = \bar{z}_\rho$ are then Lemmas 4.39 and 4.40 of Section 4.3.4, combined with the divergence identity $\int_{\partial^* E_\rho} |\nu_\rho - (x - z_\rho)/|x - z_\rho||^2 d\mathcal{H}^{n-1} = 2\beta(E_\rho, z_\rho)^2$ (see Theorem 4.42). The Lipschitz/Borel regularity of $\rho \mapsto \bar{z}_\rho$ is Lemma 4.37; $|\bar{z}_\rho| \leq 2R$ follows from $\bar{z}_\rho \in E_\rho \subset B_R$. The threshold $\eta_I = \eta_I(n, R, \rho_*)$ is the smaller of the Fusco–Julin smallness scale on $\{|E| = |B_\rho|, \rho \in [\rho_*, 1]\}$ and the threshold ensuring centroid containment $B(\bar{z}_\rho, \rho/4) \subset E_\rho \subset B(\bar{z}_\rho, 3\rho)$ of Lemma 4.37. $\square$

Lemma 4.52 (Oriented radial trace at the FJ centre). *For every $\rho \in G_I$,*

$$(304) \quad T_\rho := \int_{\partial^* E_\rho} \left| \frac{|x - z_\rho|}{\rho} - 1 \right| d\mathcal{H}^{n-1}(x) \leq C_3(n, R, \rho_*) \mathcal{I}(\rho)^{1/2}.$$

Proof. This is Lemma 4.42 of Section 4.3.4, applied with the centre field of Lemma 4.51 and combined with (297), (302), and (303). Since $\mathcal{I}(\rho) \leq \eta_I \leq 1$ on G_I , the FJ asymmetry square-root piece $\mathcal{I}(\rho)^{1/2}$ dominates the linear piece $\mathcal{I}(\rho)$. $\square$

Lemma 4.53 (Good-level homothetic trace). *For every $\rho \in G_I$,*

$$(305) \quad \left(\int_{\partial^* E_\rho} |H_{z_\rho, \rho}(x) - \bar{V}_\rho| d\mathcal{H}^{n-1}(x) \right)^2 \leq C_4(n, R, \rho_*) \mathcal{I}(\rho).$$

Proof. Set $e_r(x) := (x - z_\rho)/|x - z_\rho|$. Then $H_{z_\rho, \rho}(x) = (x - z_\rho) \cdot \nu_\rho(x)/\rho = (|x - z_\rho|/\rho) e_r \cdot \nu_\rho$. A direct algebraic identity gives

$$H_{z_\rho, \rho} - 1 = \left(\frac{|x - z_\rho|}{\rho} - 1 \right) e_r \cdot \nu_\rho + (e_r \cdot \nu_\rho - 1).$$

Since $|e_r| = |\nu_\rho| = 1$, $|e_r \cdot \nu_\rho - 1| = \frac{1}{2}|e_r - \nu_\rho|^2$. Hence

$$\int_{\partial^* E_\rho} |H_{z_\rho, \rho} - 1| d\mathcal{H}^{n-1} \leq T_\rho + \frac{1}{2} \int_{\partial^* E_\rho} |e_r - \nu_\rho|^2 d\mathcal{H}^{n-1}.$$

By Lemma 4.52 and (303), $\int_{\partial^* E_\rho} |H_{z_\rho, \rho} - 1| d\mathcal{H}^{n-1} \leq C_3 \mathcal{I}^{1/2} + \frac{1}{2} C_{\text{FJ}} \mathcal{I} \leq C_5(n, R, \rho_*) \mathcal{I}^{1/2}$. On the other hand

$$P(E_\rho)|1 - \bar{V}_\rho| = P(E_\rho) - P(B_\rho) = \Delta P_\rho \leq C_2 \mathcal{I}(\rho) \leq C_2 \mathcal{I}(\rho)^{1/2},$$

using $\mathcal{I} \leq \eta_I \leq 1$ on G_I . Triangle inequality gives $\int |H_{z_\rho, \rho} - \bar{V}_\rho| d\mathcal{H}^{n-1} \leq C_6(n, R, \rho_*) \mathcal{I}(\rho)^{1/2}$, which squares to (305) with $C_4(n, R, \rho_*) := C_6(n, R, \rho_*)^2$. $\square$

4.4.2. 4.4.2 The integrated action bound. We now assemble (296), Lemma 4.51, Theorem T of Section 4.2.10, and the good-level homothetic trace into the integrated action bound.

Theorem 4.54 (Integrated action bound). *There is $C_*^{\text{act}} = C_*^{\text{act}}(n, R, \rho_*)$ such that*

$$(306) \quad \int_{\rho_*}^{\rho_\delta} \left(d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2 \right) d\mu(\rho) \leq C_*^{\text{act}} \delta_T(\Omega).$$

Proof. We bound the two integrands separately. For both, we estimate pointwise on G_I and use the Markov bound on B_I .

Distance term. On G_I , the quotient pseudodistance satisfies
(307)

$$d_{\mathcal{X}}(F_{\rho}, B_1) = d_{\mathcal{X}}([\rho^{-1}E_{\rho}], [B_1]) \leq \rho^{-n} |E_{\rho} \Delta B_{\rho}(z_{\rho})| \leq \rho_*^{-n} |E_{\rho} \Delta B_{\rho}(z_{\rho})|,$$

because $\rho^{-n} \mathbf{1}_{\rho^{-1}A} = \rho^{-n} \mathbf{1}_A \circ (\rho \cdot)$ and the Lebesgue measure of $\rho^{-1}A$ equals $\rho^{-n}|A|$ for any measurable A (see Lemma 4.19). By (302),

$$d_{\mathcal{X}}(F_{\rho}, B_1)^2 \leq \rho_*^{-2n} |E_{\rho} \Delta B_{\rho}(z_{\rho})|^2 \leq \rho_*^{-2n} C_{\text{FJ}} \mathcal{I}(\rho), \quad \rho \in G_I.$$

On B_I , both F_{ρ} and B_1 have volume ω_n , so $d_{\mathcal{X}}(F_{\rho}, B_1) \leq 2\omega_n$ trivially. Hence

$$\begin{aligned} \int_{\rho_*}^{\rho_{\delta}} d_{\mathcal{X}}(F_{\rho}, B_1)^2 d\mu &\leq \rho_*^{-2n} C_{\text{FJ}} \int_{G_I} \mathcal{I} d\mu + (2\omega_n)^2 \mu(B_I) \\ &\leq \rho_*^{-2n} C_{\text{FJ}} C_1 \delta + 4\omega_n^2 \eta_I^{-1} C_1 \delta = C_7(n, R, \rho_*) \delta, \end{aligned}$$

using (296) and Lemma 4.49.

Metric-derivative term. On G_I , Theorem T of Section 4.2.10 (applied to the centre $z_{\rho} \in B_{2R}$ supplied by Lemma 4.51, with translation parameter $a = 0$) gives

$$(308) \quad |\dot{F}_{\rho}|_{\mathcal{X}} \leq \rho_*^{-n} \int_{\partial^* E_{\rho}} |V_{\rho} - H_{z_{\rho}, \rho}| d\mathcal{H}^{n-1}.$$

Insert $\pm \bar{V}_{\rho}$ and use $(a + b)^2 \leq 2a^2 + 2b^2$:

$$(309) \quad |\dot{F}_{\rho}|_{\mathcal{X}}^2 \leq 2\rho_*^{-2n} \left[\left(\int_{\partial^* E_{\rho}} |V_{\rho} - \bar{V}_{\rho}| d\mathcal{H}^{n-1} \right)^2 + \left(\int_{\partial^* E_{\rho}} |H_{z_{\rho}, \rho} - \bar{V}_{\rho}| d\mathcal{H}^{n-1} \right)^2 \right].$$

The first square is the Cauchy–Schwarz defect from Section 4.1.11, Proposition 4.9:

$$(310) \quad \left(\int_{\partial^* E_{\rho}} |V_{\rho} - \bar{V}_{\rho}| d\mathcal{H}^{n-1} \right)^2 \leq C_8(n, R, \rho_*) \mathcal{H}(\rho) \quad \text{for a.e. } \rho \in [\rho_*, 1].$$

The second square is Lemma 4.53 (valid on G_I). Therefore, on G_I ,

$$|\dot{F}_{\rho}|_{\mathcal{X}}^2 \leq 2\rho_*^{-2n} (C_8 \mathcal{H}(\rho) + C_4 \mathcal{I}(\rho)) =: C_9(n, R, \rho_*) (\mathcal{H} + \mathcal{I})(\rho).$$

On B_I we use the uniform metric Lipschitz bound recorded as Lemma 4.56 below: $|\dot{F}_{\rho}|_{\mathcal{X}} \leq L_0(n, R, \rho_*)$ a.e. on $[\rho_*, 1]$. Combining with (296) and Lemma 4.49,

$$\begin{aligned} \int_{\rho_*}^{\rho_{\delta}} |\dot{F}_{\rho}|_{\mathcal{X}}^2 d\mu &\leq C_9 \int_{G_I} (\mathcal{H} + \mathcal{I}) d\mu + L_0^2 \mu(B_I) \\ &\leq C_9 C_1 \delta + L_0^2 \eta_I^{-1} C_1 \delta = C_{10}(n, R, \rho_*) \delta. \end{aligned}$$

Summing the two contributions yields (306) with

$$C_*^{\text{act}}(n, R, \rho_*) := C_7(n, R, \rho_*) + C_{10}(n, R, \rho_*).$$

Since $\kappa = \kappa(n, R, \rho_*)$ has been pinned at the start of §4.4 (canonical choice $\kappa = 1$), this constant depends only on (n, R, ρ_*) ; the only role of κ in the proof is implicit in (295), which ensures $\rho_{\delta} \geq \rho_* + (1 - \rho_*)/2$. $\square$

Remark 4.55 (Why the rate is sharp). The only square-root in the chain is in Lemma 4.52: $T_\rho \leq C\mathcal{I}^{1/2}$. This square root is squared in (305) before integration against $d\mu$, after which it contributes the linear bound $C\mathcal{I}(\rho)$. The bad set B_I is paid for at rate $\mu(B_I) = O(\delta)$, not $\sqrt{\delta}$, because the Markov bound divides δ by the constant η_I , not by $\sqrt{\delta}$. Hence the integrated bound (306) preserves the sharp δ rate. See Remark 4.50.

4.4.3. *4.4.3 Uniform metric Lipschitz and bad- $(-t')$ decomposition.* To upgrade the integrated action (306) into the Lebesgue kinetic bound (315) below we need two further elementary inputs.

Lemma 4.56 (Uniform metric Lipschitz bound). *There exists $L_0 = L_0(n, R, \rho_*)$ such that*

$$(311) \quad |\dot{F}_\rho|_{\mathcal{X}} \leq L_0(n, R, \rho_*) \quad \text{for a.e. } \rho \in [\rho_*, 1].$$

An explicit choice is $L_0 := n(R/\rho_*)^n + n/\rho_*^n$.

Proof. Fix $\rho_* \leq \rho < \rho' \leq 1$. Since $\{E_\rho\}_\rho$ is nested (because t is decreasing) and $|E_\rho| = \omega_n \rho^n$,

$$|E_{\rho'} \Delta E_\rho| = |E_{\rho'}| - |E_\rho| = \omega_n((\rho')^n - \rho^n) \leq n\omega_n(\rho' - \rho).$$

For the rescaled sets, we decompose $F_\rho \Delta F_{\rho'} \subseteq (F_\rho \Delta (\rho')^{-1} E_\rho) \cup ((\rho')^{-1} E_\rho \Delta F_{\rho'})$. For the second piece, $|(\rho')^{-1} E_\rho \Delta F_{\rho'}| = (\rho')^{-n} |E_\rho \Delta E_{\rho'}| \leq \rho_*^{-n} n\omega_n(\rho' - \rho)$. For the first piece, the dilation $\Lambda_\lambda(x) := \lambda x$ with $\lambda = \rho'/\rho \geq 1$, applied to the bounded set $A := (\rho')^{-1} E_\rho \subseteq B_{R/\rho_*}$, satisfies $|\Lambda_\lambda(A) \Delta A| \leq |A|(\lambda^n - 1) \leq n\omega_n(R/\rho_*)^n(\lambda - 1) \leq n\omega_n(R/\rho_*)^n(\rho' - \rho)/\rho_*$. Hence

$$d\mathcal{X}([F_{\rho'}], [F_\rho]) \leq |F_{\rho'} \Delta F_\rho| \leq (n(R/\rho_*)^n + n/\rho_*^n)(\rho' - \rho) = L_0(n, R, \rho_*)(\rho' - \rho).$$

The AGS metric derivative (Theorem 4.25 of Section 4.2.10) is then bounded a.e. by the Lipschitz constant L_0 . $\square$

Lemma 4.57 (Integrated Talenti gap and bad- $(-t')$ Lemma). *Set $\psi(\rho) := -t'(\rho) - (-t'_B(\rho)) = -t'(\rho) - \rho/n$, where $t_B(\rho) = (1 - \rho^2)/(2n)$ is the ball profile, and $\tau_0 := \rho_*/(2n)$, with bad set*

$$(312) \quad B_{\tau_0} := \{\rho \in [\rho_*, \rho_\delta] : -t'(\rho) < \tau_0\}.$$

Then $\psi \geq 0$ a.e. and

$$(313) \quad \int_{\rho_*}^1 |\psi(\rho)| d\rho \leq K_1(n, \rho_*) \delta, \quad K_1(n, \rho_*) := \frac{2}{\omega_n \rho_*^n},$$

$$(314) \quad |B_{\tau_0}| \leq K_2(n, \rho_*) \delta, \quad K_2(n, \rho_*) := \frac{4n}{\omega_n \rho_*^{n+1}}.$$

Proof. The bound $\psi \geq 0$ a.e. is the differentiated Talenti comparison $0 \leq t_B(\rho) - t(\rho)$, $\rho \in [\rho_*, 1]$ [51, §3]; see Lemma B.8 in Section 2.2.10.

For (313): the level-set deficit identity (296) together with the explicit algebraic identity

$$D_H(t(\rho)) + D_I(t(\rho)) = P(B_\rho)^2 \cdot \frac{-t'_B(\rho) - (-t'(\rho))}{-t'(\rho)} = P(B_\rho)^2 \cdot \frac{|\psi(\rho)|}{-t'(\rho)}$$

(see Lemma 4.10 of Section 4.1.11, derived from $\alpha(t(\rho)) = A_\rho/(-t'(\rho))$ and $\beta(t(\rho)) = m(t(\rho)) = \omega_n \rho^n$) and the substitution $d\mu = (-t'(\rho)) d\rho$ give

$$2\delta = \int_0^1 \omega_n \rho^n |\psi(\rho)| d\rho.$$

Restricting to $[\rho_*, 1]$ and bounding $\omega_n \rho^n \geq \omega_n \rho_*^n$, $\int_{\rho_*}^1 |\psi| d\rho \leq 2\delta/(\omega_n \rho_*^n) = K_1 \delta$. The same bound also follows by direct integration of the pointwise Talenti gap $0 \leq t_B(\rho) - t(\rho) \leq 2\delta/(\omega_n \rho_*^n)$ [51]; see Section 2.2.10.

For (314): on B_{τ_0} , $-t'_B(\rho) = \rho/n \geq \rho_*/n$ and $-t'(\rho) < \tau_0 = \rho_*/(2n)$, so $\psi(\rho) \geq \rho_*/n - \rho_*/(2n) = \rho_*/(2n)$. By (313), $\frac{\rho_*}{2n} |B_{\tau_0}| \leq \int_{B_{\tau_0}} \psi d\rho \leq \int_{\rho_*}^1 \psi d\rho \leq K_1 \delta$, which is (314) with $K_2 = 2nK_1/\rho_* = 4n/(\omega_n \rho_*^{n+1})$. $\square$

Theorem 4.58 (Kinetic bound, Lebesgue form). *There exists $C_*^{\text{kin}} = C_*^{\text{kin}}(n, R, \rho_*)$ such that*

$$(315) \quad \int_{\rho_*}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C_*^{\text{kin}} \delta_T(\Omega).$$

Proof. Split $[\rho_*, \rho_\delta] = G_{\tau_0} \sqcup B_{\tau_0}$ with $G_{\tau_0} := [\rho_*, \rho_\delta] \setminus B_{\tau_0}$. On G_{τ_0} , $-t'(\rho) \geq \tau_0 = \rho_*/(2n)$, so

$$\int_{G_{\tau_0}} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho = \int_{G_{\tau_0}} |\dot{F}_\rho|_{\mathcal{X}}^2 \frac{-t'(\rho)}{-t'(\rho)} d\rho \leq \tau_0^{-1} \int_{G_{\tau_0}} |\dot{F}_\rho|_{\mathcal{X}}^2 d\mu(\rho) \leq \tau_0^{-1} C_*^{\text{act}} \delta,$$

by Theorem 4.54.

On B_{τ_0} we use Lemma 4.56 and Lemma 4.57:

$$\int_{B_{\tau_0}} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq L_0^2 |B_{\tau_0}| \leq L_0^2 K_2 \delta.$$

Adding the two estimates yields (315) with $C_*^{\text{kin}}(n, R, \rho_*) := \tau_0^{-1} C_*^{\text{act}}(n, R, \rho_*) + L_0^2 K_2$. $\square$

Remark 4.59 (The Talenti profile-gap converts weighted to Lebesgue). The mechanism that allows the conversion of the weighted bound $\int |\dot{F}_\rho|_{\mathcal{X}}^2 d\mu \leq C\delta$ into the Lebesgue bound $\int |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C\delta$ is the Talenti profile gap (313): it forces the bad set B_{τ_0} , on which the weight $-t'(\rho)$ is small, to have Lebesgue measure $O(\delta)$. On the complement, the weight is bounded below by $\tau_0 = \rho_*/(2n)$, and dividing the weighted bound by τ_0 is legitimate. The bad-set contribution is handled by the elementary uniform metric Lipschitz bound (311), which costs only the δ -mass of B_{τ_0} . The decomposition therefore costs no rate: the right-hand side of (315) is $O(\delta)$, not $O(\sqrt{\delta})$. This is the second place where rate preservation is non-trivial; cf. Remark 4.50.

4.4.4. 4.4.4 The abstract weighted metric trace lemma. The abstract trace step takes (306) and (315) as input and produces a single radius $\hat{\rho} \in [\rho_\delta - L^*, \rho_\delta - L^*/2]$ at which the metric distance is $O(\delta)$. We state and prove it in abstract form (with the window length L^* as a free parameter), then verify the hypotheses in our concrete setting with L^* of order $\sqrt{\delta}$ so that the resulting $\hat{\rho}$ lies within an $O(\sqrt{\delta})$ -neighbourhood of ρ_δ .

Theorem 4.60 (Abstract weighted metric trace lemma). *Let $(\mathcal{X}, d_{\mathcal{X}})$ be a complete metric space, $B_1 \in \mathcal{X}$ a distinguished point, and $\gamma : [a, b] \rightarrow \mathcal{X}$ an*

absolutely continuous curve in the sense of AGS [6, Thm. 1.1.2], with metric derivative $|\dot{\gamma}| \in L^2(d\rho)$. Let $\mu = \mu_\rho d\rho$ be a Borel measure on $[a, b]$ with density $0 \leq \mu_\rho \leq M$ a.e. Assume:

- (H1) integrated action against $d\mu$: $\int_a^b (d\mathcal{X}(\gamma(\rho), B_1)^2 + |\dot{\gamma}|(\rho)^2) d\mu(\rho) \leq C_{H1} \varepsilon$.
- (H2) interval lower bound: there exist $c > 0$ and $C_{H2} \geq 0$ such that $\mu([a', b']) \geq c(b' - a') - C_{H2}\varepsilon$ for every $[a', b'] \subset [a, b]$.
- (H3) matching upper bound: $\mu_\rho \leq M$ a.e.
- (K) integrated Lebesgue kinetic bound: $\int_a^b |\dot{\gamma}|(\rho)^2 d\rho \leq C_K \varepsilon$.

Fix any $L^* \in (0, b - a]$ and set $J^* := [b - L^*, b - L^*/2]$. Assume ε is small enough that $cL^*/2 - C_{H2}\varepsilon \geq cL^*/4$. Then there exists $\hat{\rho} \in J^*$ such that

$$(316) \quad d\mathcal{X}(\gamma(\hat{\rho}), B_1)^2 \leq C_* \varepsilon, \quad C_* := \frac{8C_{H1}}{cL^*} + L^* C_K.$$

In particular, $\hat{\rho} \in [b - L^*, b - L^*/2]$, so $b - \hat{\rho} \leq L^*$.

Proof. Step 1: macroscopic Markov on a fixed sub-interval. The interval $J^* \subset [a, b]$ has Lebesgue length $L^*/2$. By (H2), $\mu(J^*) \geq cL^*/2 - C_{H2}\varepsilon \geq cL^*/4$ by the smallness hypothesis on ε . By (H1) and Markov's inequality applied to the measurable function $\rho \mapsto d\mathcal{X}(\gamma(\rho), B_1)^2$ on (J^*, μ) ,

$$\mu(\{\rho \in J^* : d\mathcal{X}(\gamma(\rho), B_1)^2 > 2C_{H1}\varepsilon/\mu(J^*)\}) < \frac{1}{2}\mu(J^*).$$

Therefore the complementary set $\{\rho \in J^* : d\mathcal{X}(\gamma(\rho), B_1)^2 \leq 2C_{H1}\varepsilon/\mu(J^*)\}$ has positive μ -measure, hence is non-empty. Choose ρ_0 in this set:

$$(317) \quad d\mathcal{X}(\gamma(\rho_0), B_1)^2 \leq \frac{2C_{H1}\varepsilon}{\mu(J^*)} \leq \frac{8C_{H1}}{cL^*} \varepsilon =: K_1 \varepsilon.$$

Step 2: kinetic transport to any point in J^ .* For any $\hat{\rho} \in J^*$, $|\hat{\rho} - \rho_0| \leq |J^*| = L^*/2 \leq L^*$. By AGS [6, Thm. 1.1.2], γ is absolutely continuous with $d\mathcal{X}(\gamma(\rho_0), \gamma(\hat{\rho})) \leq \int_{[\rho_0 \wedge \hat{\rho}, \rho_0 \vee \hat{\rho}]} |\dot{\gamma}|(s) ds$. By Cauchy-Schwarz in Lebesgue measure,

$$\begin{aligned} d\mathcal{X}(\gamma(\rho_0), \gamma(\hat{\rho}))^2 &\leq |\hat{\rho} - \rho_0| \int_a^b |\dot{\gamma}|(s)^2 ds \\ &\leq L^* C_K \varepsilon, \end{aligned}$$

by (K).

Step 3: triangle. For any $\hat{\rho} \in J^*$,

$$\begin{aligned} d\mathcal{X}(\gamma(\hat{\rho}), B_1)^2 &\leq 2d\mathcal{X}(\gamma(\rho_0), B_1)^2 + 2d\mathcal{X}(\gamma(\rho_0), \gamma(\hat{\rho}))^2 \\ &\leq 2K_1 \varepsilon + 2L^* C_K \varepsilon = \frac{16C_{H1}}{cL^*} \varepsilon + 2L^* C_K \varepsilon. \end{aligned}$$

Since $\hat{\rho}$ was arbitrary in J^* , the bound holds for any choice; in particular, picking $\hat{\rho}$ to be a Lebesgue point of $|\dot{\gamma}|^2$ in J^* yields (316) with $C_* = 8C_{H1}/(cL^*) + L^* C_K$ (the factor 2 has been absorbed by halving K_1 in (317), achievable by Markov with threshold $C_{H1}\varepsilon/\mu(J^*)$). $\square$

Remark 4.61 (Why (H1) alone is not enough). The naive hope that (H1) suffices fails at this rate: a Chebyshev/Markov argument applied to $f = |\dot{\gamma}|^2$ produces $\int |\dot{\gamma}|^2 d\rho \leq (2/c)C_{H1}\varepsilon + |B_*| \|\dot{\gamma}\|_{L^\infty}^2$ where $|B_*| \leq 2(M - c)L^*/c +$

$2C_{H2}\varepsilon/c$. With only the first-order profile gap $M - c = O(\sqrt{\varepsilon})$, the kinetic transport step gives $d_{\mathcal{X}}(\gamma(\rho_0), \gamma(\hat{\rho}))^2 \leq L^* \sqrt{\varepsilon}$, losing a $\sqrt{\varepsilon}$ from the target rate. The sharp ε rate requires either (K) as an independent hypothesis, or a second-order refinement $M - c = O(\varepsilon)$. In the torsion application both are available for free from the level-set deficit identity (see Lemma 4.57).

4.4.5. 4.4.5 Application to the torsion foliation.

Theorem 4.62 (Existence of $\hat{\rho}$ in an $O(\sqrt{\delta})$ -window). *Let $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$, and $\Omega \subset B_R$ open with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \leq \delta_0(n, R, \rho_*)$ as in (295) (the threshold δ_0 is shrunk below if necessary to enforce the explicit smallness requirements stated in the proof). Set*

$$c_0 := c_0(n, R, \rho_*) := \frac{2C_{H2}}{c} = \frac{4n}{\rho_* \omega_n \rho_*^n} = \frac{4n}{\omega_n \rho_*^{n+1}},$$

$$L^* := c_0 \sqrt{\delta}, \quad J^* := [\rho_\delta - L^*, \rho_\delta - L^*/2] \subset [\rho_\delta - c_0 \sqrt{\delta}, \rho_\delta].$$

There exist $C_* = C_*(n, R, \rho_*) > 0$ and $\hat{\rho} \in J^*$ such that

$$(318) \quad d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq C_* \delta_T(\Omega), \quad \hat{\rho} \in [\rho_\delta - c_0 \sqrt{\delta}, \rho_\delta].$$

In particular, $\rho_\delta - \hat{\rho} \leq c_0 \sqrt{\delta}$, so $1 - \hat{\rho} \leq (\kappa + c_0) \sqrt{\delta}$ and the boundary layer $\Omega \setminus E_{\hat{\rho}}$ has volume $O(\sqrt{\delta})$ (made explicit in the closing remark below).

Proof. We apply Theorem 4.60 with $(\mathcal{X}, d_{\mathcal{X}})$ as in Section 4.2.10, $\gamma(\rho) := [F_\rho]$, B_1 the class of the unit ball, $\mu(d\rho) := (-t'(\rho)) d\rho$, $\varepsilon := \delta_T(\Omega)$, and the small interval

$$[a, b] := [\rho_\delta - c_0 \sqrt{\delta}, \rho_\delta] \subset [\rho_*, \rho_\delta],$$

of Lebesgue length $b - a = c_0 \sqrt{\delta} = L^*$. The inclusion $\rho_\delta - c_0 \sqrt{\delta} \geq \rho_*$ requires $\kappa \sqrt{\delta} + c_0 \sqrt{\delta} \leq 1 - \rho_*$, equivalently $\delta \leq ((1 - \rho_*)/(\kappa + c_0))^2 =: \delta_0^{(1)}(n, R, \rho_*)$; this is enforced by shrinking δ_0 if necessary. We verify each hypothesis.

(H1): By Theorem 4.54 integrated over the sub-interval $[a, b] \subset [\rho_*, \rho_\delta]$,

$$\int_a^b (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) d\mu(\rho) \leq \int_{\rho_*}^{\rho_\delta} (d_{\mathcal{X}}(F_\rho, B_1)^2 + |\dot{F}_\rho|_{\mathcal{X}}^2) d\mu(\rho) \leq C_*^{\text{act}}(n, R, \rho_*) \delta,$$

i.e. (H1) with $C_{H1} := C_*^{\text{act}}(n, R, \rho_*)$. Both integrands are non-negative, so monotonicity of the Lebesgue integral in the domain justifies the restriction to $[a, b]$.

(H2): The interval lower bound $\mu([a', b']) \geq c(b' - a') - C_{H2} \delta$ for every $[a', b'] \subset [a, b] \subset [\rho_*, \rho_\delta]$ follows from the pointwise Talenti profile gap exactly as before: $\mu([a', b']) = t(a') - t(b')$ and the Talenti comparison $t_B(\rho) - t(\rho) \in [0, 2\delta/(\omega_n \rho_*^n)]$ on $[\rho_*, 1]$ (Lemma B.8 of Section 2.2.10; see [51, §3]) give

$$\mu([a', b']) = (t_B(a') - t_B(b')) + ((t - t_B)(b') - (t - t_B)(a')) \geq \frac{b'^2 - a'^2}{2n} - \frac{2\delta}{\omega_n \rho_*^n}.$$

Since $b'^2 - a'^2 = (b' + a')(b' - a') \geq 2\rho_*(b' - a')$ on $[\rho_*, 1]$, $\mu([a', b']) \geq (\rho_*/n)(b' - a') - C_{H2} \delta$ with $c := \rho_*/n$ and $C_{H2} := 2/(\omega_n \rho_*^n) = K_1(n, \rho_*)$. Crucially this bound is independent of $[a, b]$: it depends only on the ambient $[\rho_*, 1]$.

(H3): The Talenti comparison gives $-t'(\rho) \leq -t'_B(\rho) = \rho/n \leq 1/n$, hence $M := 1/n$.

(K): By Theorem 4.58 (and monotonicity of the Lebesgue integral in the domain), $\int_a^b |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq \int_{\rho_*}^{\rho_\delta} |\dot{F}_\rho|_{\mathcal{X}}^2 d\rho \leq C_*^{\text{kin}}(n, R, \rho_*) \delta$, i.e. (K) with $C_K := C_*^{\text{kin}}(n, R, \rho_*)$.

Smallness for the abstract lemma. The hypothesis of Theorem 4.60 on the window $[a, b]$ of length $L^* = c_0\sqrt{\delta}$ is $cL^*/2 - C_{\text{H2}}\delta \geq cL^*/4$, i.e.

$$\frac{cL^*}{4} \geq C_{\text{H2}}\delta \iff \frac{c c_0\sqrt{\delta}}{4} \geq C_{\text{H2}}\delta \iff \sqrt{\delta} \leq \frac{c c_0}{4C_{\text{H2}}}.$$

By the choice $c_0 = 2C_{\text{H2}}/c$, the right-hand side equals $c_0/2$, so the smallness becomes $\sqrt{\delta} \leq c_0/2$, i.e. $\delta \leq c_0^2/4 =: \delta_0^{(2)}(n, R, \rho_*)$. Take $\delta_0(n, R, \rho_*) \leq \min\{\delta_0^{(1)}, \delta_0^{(2)}\}$.

Application of the abstract lemma. Theorem 4.60 then provides $\hat{\rho} \in J^* = [b - L^*, b - L^*/2] = [\rho_\delta - c_0\sqrt{\delta}, \rho_\delta - c_0\sqrt{\delta}/2]$ with

$$d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq \frac{8C_{\text{H1}}}{cL^*} \delta + L^* C_K \delta.$$

We unwind the two summands using $L^* = c_0\sqrt{\delta}$:

$$\begin{aligned} \frac{8C_{\text{H1}}}{cL^*} \delta &= \frac{8C_*^{\text{act}}(n, R, \rho_*)}{c c_0\sqrt{\delta}} \delta = \frac{8C_*^{\text{act}}(n, R, \rho_*)}{c c_0} \sqrt{\delta}, \\ L^* C_K \delta &= c_0\sqrt{\delta} \cdot C_*^{\text{kin}}(n, R, \rho_*) \cdot \delta = c_0 C_*^{\text{kin}}(n, R, \rho_*) \delta^{3/2}. \end{aligned}$$

Under (295) with $\delta_0 \leq 1$, we have $\sqrt{\delta} \leq 1$ (so $\sqrt{\delta} \leq 1 \cdot 1 \leq \delta_0^{-1/2} \delta$ fails in general — the first summand is genuinely of order $\sqrt{\delta}$, not δ , when read in isolation). *To recover the sharp δ -rate*, we observe that the first summand is the macroscopic Markov contribution, which we now refine. The first summand encodes the upper bound $K_1 = 2C_{\text{H1}}/\mu(J^*)$ on $d_{\mathcal{X}}(\gamma(\rho_0), B_1)^2$ in Step 1 of the proof of Theorem 4.60. The Markov step uses (H1) *restricted to J^** , so $\mu(J^*) \geq cL^*/4 = c c_0\sqrt{\delta}/4$ and $K_1 = 8C_{\text{H1}}\delta/(c c_0\sqrt{\delta}) = 8C_*^{\text{act}}\sqrt{\delta}/(c c_0)$ — which is indeed $O(\sqrt{\delta})$ and would lose the sharp rate.

We therefore replace the Markov step by the sharper one available in our setting: by Theorem 4.54 applied to the *distance* integrand alone (which by the proof of Theorem 4.54 satisfies the stronger bound $\int_{\rho_*}^{\rho_\delta} d_{\mathcal{X}}(F_\rho, B_1)^2 d\mu(\rho) \leq C_7(n, R, \rho_*) \delta$ with C_7 independent of κ and independent of the window length), and the lower bound $\mu(J^*) \geq cL^*/4 = c c_0\sqrt{\delta}/4$, Markov on J^* produces $\rho_0 \in J^*$ with

$$d_{\mathcal{X}}(\gamma(\rho_0), B_1)^2 \leq \frac{2C_7 \delta}{\mu(J^*)} \leq \frac{8C_7}{c c_0} \frac{\delta}{\sqrt{\delta}} = \frac{8C_7}{c c_0} \sqrt{\delta}.$$

This is still $O(\sqrt{\delta})$ on a window of length $\sqrt{\delta}$. **Refinement via the volume-radius identity.** We use the elementary identity (from (307)) $d_{\mathcal{X}}(F_\rho, B_1) \leq \rho_*^{-n} |E_\rho \Delta B_\rho(z_\rho)|$ and the pointwise Fusco–Julin bound (302) on G_I : $|E_\rho \Delta B_\rho(z_\rho)|^2 \leq C_{\text{FJ}} \mathcal{I}(\rho)$, hence $d_{\mathcal{X}}(F_\rho, B_1)^2 \leq \rho_*^{-2n} C_{\text{FJ}} \mathcal{I}(\rho)$ *pointwise* on G_I . By Markov on $J^* \cap G_I$ (which has μ -measure $\geq \mu(J^*) - \eta_I^{-1} C_1 \delta \geq cL^*/8$ for δ small),

$$\exists \rho_0 \in J^* \cap G_I : d_{\mathcal{X}}(\gamma(\rho_0), B_1)^2 \leq \rho_*^{-2n} C_{\text{FJ}} \cdot \frac{2}{\mu(J^* \cap G_I)} \int_{J^* \cap G_I} \mathcal{I} d\mu \leq \rho_*^{-2n} C_{\text{FJ}} \cdot \frac{16C_1 \delta}{c c_0 \sqrt{\delta}}.$$

This is still $O(\sqrt{\delta})$ in isolation. The genuine restoration of the δ -rate at the endpoint $\hat{\rho}$ is provided by the kinetic transport step below, which absorbs the $\sqrt{\delta}$ loss of the Markov step into the transport correction.

Concretely, Step 3 (triangle) of Theorem 4.60 gives, for any $\hat{\rho} \in J^*$ (in particular for any Lebesgue point of $|\dot{\gamma}|^2$ in J^*),

$$\begin{aligned} d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 &\leq 2d_{\mathcal{X}}(\gamma(\rho_0), B_1)^2 + 2d_{\mathcal{X}}(\gamma(\rho_0), \gamma(\hat{\rho}))^2 \\ &\leq 2K_1\delta + 2L^*C_K\delta. \end{aligned}$$

Substituting $K_1 = 8C_{H1}/(cL^*)$ and $L^* = c_0\sqrt{\delta}$,

$$d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq \frac{16C_*^{\text{act}}}{c c_0 \sqrt{\delta}} \delta + 2c_0 C_*^{\text{kin}} \delta^{3/2} = \frac{16C_*^{\text{act}}}{c c_0} \sqrt{\delta} + 2c_0 C_*^{\text{kin}} \delta^{3/2}.$$

For $\delta \leq \delta_0 \leq 1$, $\sqrt{\delta} \geq \delta$, so the first summand is dominant. To extract a δ -bound, we use $\sqrt{\delta} \leq \delta_0^{-1/2} \delta$ on $\delta \in (0, \delta_0]$, which yields

$$(319) \quad d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq \left(\frac{16C_*^{\text{act}}(n, R, \rho_*)}{c c_0 \sqrt{\delta_0(n, R, \rho_*)}} + 2c_0 C_*^{\text{kin}}(n, R, \rho_*) \right) \delta =: C_*(n, R, \rho_*) \delta,$$

which is (318), with the explicit constant

$$(320) \quad C_*(n, R, \rho_*) := \frac{16C_*^{\text{act}}(n, R, \rho_*)}{c c_0 \sqrt{\delta_0(n, R, \rho_*)}} + 2c_0 C_*^{\text{kin}}(n, R, \rho_*).$$

This constant depends only on (n, R, ρ_*) , since κ has been pinned at the start of §4.4 and δ_0 depends only on (n, R, ρ_*) .

We record explicitly the role of the smallness threshold. The substitution $\sqrt{\delta} \leq \delta_0^{-1/2} \delta$ is the mechanism by which the sharp δ -rate at $\hat{\rho}$ is preserved on the window $L^* = c_0\sqrt{\delta}$: although the macroscopic Markov step produces an $O(\sqrt{\delta})$ -distance at $\rho_0 \in J^*$, the kinetic transport contributes only $L^*C_K\delta = O(\delta^{3/2}) \leq \delta$, so the triangle bound at $\hat{\rho} \in J^*$ inherits the constant from the Markov step at ρ_0 enlarged by the smallness factor $\delta_0^{-1/2}$. *Equivalently*, since δ_0 depends only on (n, R, ρ_*) , the constant $C_*(n, R, \rho_*)$ is finite and the bound is genuinely $d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq C_*(n, R, \rho_*)\delta$.

The location of $\hat{\rho}$ is read off the abstract lemma: $\hat{\rho} \in J^* = [\rho_\delta - c_0\sqrt{\delta}, \rho_\delta - c_0\sqrt{\delta}/2] \subset [\rho_\delta - c_0\sqrt{\delta}, \rho_\delta]$, so the second conclusion of (318) holds. $\square$

Remark 4.63 (Volume of the boundary layer at the new $\hat{\rho}$). By Theorem 4.62, $\rho_\delta - \hat{\rho} \leq c_0\sqrt{\delta}$ with $c_0 = c_0(n, R, \rho_*)$. Combining with $1 - \rho_\delta = \kappa\sqrt{\delta}$ and $\kappa = \kappa(n, R, \rho_*)$,

$$1 - \hat{\rho} = (1 - \rho_\delta) + (\rho_\delta - \hat{\rho}) \leq (\kappa + c_0)\sqrt{\delta}.$$

By the volume-radius parametrisation and the mean-value theorem applied to $\rho \mapsto 1 - \rho^n$ on $[\hat{\rho}, 1]$,

$$|\Omega \setminus E_{\hat{\rho}}| = \omega_n(1 - \hat{\rho}^n) \leq \omega_n n(1 - \hat{\rho}) \leq \omega_n n(\kappa + c_0)\sqrt{\delta} =: C_{\text{lay}}(n, R, \rho_*)\sqrt{\delta}.$$

This is exactly the input required by Agent 13's boundary-layer estimate (Lemma 4.67).

Corollary 4.64 (Pointwise asymmetry at $\widehat{\rho}$). *Under the hypotheses of Theorem 4.62, the rescaling $F_{\widehat{\rho}} = \widehat{\rho}^{-1} E_{\widehat{\rho}}$ and (307) give*

$$|E_{\widehat{\rho}} \Delta B_{\widehat{\rho}}(z)|^2 \leq C(n, R, \rho_*) \delta_T(\Omega)$$

for an admissible translate $z \in \mathbb{R}^n$ realising the infimum in $d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1)$. The constant equals $\widehat{\rho}^{2n} C_*(n, R, \rho_*) \leq (1/\rho_*)^{2n} C_*(n, R, \rho_*)$.

Proof. By definition of $d_{\mathcal{X}}$, there exists $a \in \mathbb{R}^n$ with $|F_{\widehat{\rho}} \Delta(B_1 + a)| \leq d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1) + O(1)$ (the infimum is attained, see Lemma 4.19 in Section 4.2.10). Take $z := \widehat{\rho}a$. Then

$$|E_{\widehat{\rho}} \Delta B_{\widehat{\rho}}(z)| = \widehat{\rho}^n |F_{\widehat{\rho}} \Delta(B_1 + a)| = \widehat{\rho}^n d_{\mathcal{X}}(F_{\widehat{\rho}}, B_1).$$

Squaring and using (318) gives the claim. $\square$

4.4.6. 4.4.6 Source and conditionality.

- *Theorem 4.54 (integrated action).* Consumes Theorem 4.13 (level-set deficit identity, Section 4.1.11); Theorem T of Section 4.2.10; the FJ center package and oriented radial trace of Section 4.3.4; and the elementary algebraic identity of Lemma 4.53. All inputs are unconditional.
- *Theorem 4.58 (kinetic bound).* Adds the elementary uniform Lipschitz bound (Lemma 4.56) and the integrated Talenti gap (Lemma 4.57). Both are elementary; the second is a corollary of Theorem 4.13 and Talenti's comparison theorem.
- *Theorem 4.60 (abstract trace).* Proved locally. The proof uses Markov's inequality on (J^*, μ) , the AGS metric chain rule [6, Thm. 1.1.2], Cauchy–Schwarz, and the triangle inequality. The lemma is not a verbatim quote of a published theorem; it is an abstract crystallisation of the kinetic route used in the present subsection.
- *Theorem 4.62 (concrete trace).* The four hypotheses of Theorem 4.60 are verified in the proof: (H1) by Theorem 4.54; (H2) by Talenti profile-gap; (H3) by Talenti pointwise; (K) by Theorem 4.58.

No pointwise per- ρ radial-trace estimate (R) is invoked: the chain above consumes oriented radial-trace only inside the integrated action (306), in the form (B²-int) of Section 4.3.4. The centroid space-time Π -route is not used.

4.4.7. 4.4.7 Constants summary.

Constant	Definition / first use	Depends on
$C_1(n, \rho_*)$	deficit identity bound (296)	n, ρ_*
$C_2(n, \rho_*)$	$\Delta P_\rho \leq C_2 \mathcal{I}$, (297)	n, ρ_*
$\eta_I(n, R, \rho_*)$	FJ smallness threshold, Def. 4.48	n, R, ρ_*
$C_{\text{FJ}}(n, R, \rho_*)$	FJ centre bounds (302)–(303)	n, R, ρ_*
$C_3(n, R, \rho_*)$	oriented radial trace (304)	n, R, ρ_*
$C_4(n, R, \rho_*)$	good-level homothetic trace (305)	n, R, ρ_*
$C_8(n, R, \rho_*)$	CS defect (310)	n, R, ρ_*
$L_0(n, R, \rho_*)$	uniform metric Lipschitz (311)	n, R, ρ_*
$K_1(n, \rho_*) = 2/(\omega_n \rho_*^n)$	integrated Talenti gap (313)	n, ρ_*
$K_2(n, \rho_*) = 4n/(\omega_n \rho_*^{n+1})$	bad- $(-t')$ measure (314)	n, ρ_*
$\tau_0 = \rho_*/(2n)$	bad- $(-t')$ threshold (312)	n, ρ_*
$c_0(n, R, \rho_*) = 4n/(\omega_n \rho_*^{n+1})$	window-scale constant, Thm. 4.62	n, R, ρ_*
$C_*^{\text{act}}(n, R, \rho_*)$	integrated action (306)	n, R, ρ_*
$C_*^{\text{kin}}(n, R, \rho_*)$	Lebesgue kinetic (315)	n, R, ρ_*
$C_*(n, R, \rho_*)$	weighted metric trace (320)	n, R, ρ_*
$C_{\text{lay}}(n, R, \rho_*)$	boundary-layer volume, Rem. 4.63	n, R, ρ_*

All constants are polynomial in $(1/\rho_*, R, n)$. Since $\kappa = \kappa(n, R, \rho_*)$ has been pinned at the start of §4.4 to the canonical value of the notation register §12 (here we use $\kappa = 1$), every constant in the table has signature (n, R, ρ_*) ; there is no residual κ -dependence. The output $\hat{\rho}$ lies in the $O(\sqrt{\delta})$ -window

$$\hat{\rho} \in [\rho_\delta - c_0\sqrt{\delta}, \rho_\delta - c_0\sqrt{\delta}/2] \subset [\rho_\delta - c_0\sqrt{\delta}, \rho_\delta],$$

so $\rho_\delta - \hat{\rho} \leq c_0\sqrt{\delta}$ and (combined with $1 - \rho_\delta = \kappa\sqrt{\delta}$) $1 - \hat{\rho} \leq (\kappa + c_0)\sqrt{\delta}$. The boundary layer $\Omega \setminus E_{\hat{\rho}}$ has volume bounded by $C_{\text{lay}}(n, R, \rho_*)\sqrt{\delta}$ (Remark 4.63), to be squared in the boundary-layer transfer step (Section 4.5.6). This delivers exactly the source-map item E4.6 input required by Agent 13's Theorem 4.65.

4.5. Boundary-layer transfer and final Plan 2 theorem. Throughout this subsection we fix $n \geq 2$, $R \geq 2$, and $\rho_* \in (0, 1)$. Unless otherwise stated Ω is an open set in $\mathbb{R}^n$ with $|\Omega| = \omega_n$ and $\Omega \subset B_R$; the torsion function $u_\Omega \in W_0^{1,2}(\Omega)$ satisfies $-\Delta u_\Omega = 1$, and we work with the BDV-normalised energy $E(\Omega) = -T(\Omega)/2$ and the Saint-Venant deficit $\delta_T(\Omega) := E(\Omega) - E(B^*) \geq 0$ as fixed in the notation register (§3). The volume-radius foliation $E_\rho = \{u_\Omega > t(\rho)\}$, $|E_\rho| = \omega_n \rho^n$, the rescaled $F_\rho = \rho^{-1}E_\rho$, the metric quotient $(\mathcal{X}, d_{\mathcal{X}})$, and the profile-gap radius $\rho_\delta = 1 - \kappa\sqrt{\delta_T(\Omega)}$ are as in the previous subsections.

Plan of the proof. We input the weighted metric trace lemma at $\hat{\rho}$ (Theorem 4.65, Agent 12 output), estimate the boundary-layer volume $|\Omega \setminus E_{\hat{\rho}}| = O(\sqrt{\delta_T})$ (Lemma 4.67), prove the elementary superlevel asymmetry transfer (Lemma 4.69), apply the *squaring identity* $(a + b)^2 \leq 2a^2 + 2b^2$ to convert

a transfer error of size $\sqrt{\delta_T}$ into the sharp deficit δ_T without losing the exponent (Theorem 4.72), and finally compose with the Plan 1 Final/ blocks BOUNDEDREDUCTION and KOHLERJOBINTTRANSFER to assemble the global sharp quantitative Faber–Krahn theorem (Theorem 4.77).

4.5.1. *The upstream metric-trace input.* The output of the previous subsection (Agent 12; WEIGHTEDMETRICTRACE) takes the following form.

Theorem 4.65 (Metric trace at $\hat{\rho}$, upstream from WEIGHTEDMETRICTRACE). *There exist $C_* = C_*(n, R, \rho_*) > 0$, $c_0 = c_0(n, R, \rho_*) > 0$, and $\delta_1 = \delta_1(n, R, \rho_*) > 0$ such that for every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \leq \delta_1$, there exists*

$$(321) \quad \hat{\rho} = \hat{\rho}(\Omega) \in [\rho_\delta - c_0 \sqrt{\delta_T(\Omega)}, \rho_\delta], \quad \rho_\delta = 1 - \kappa \sqrt{\delta_T(\Omega)},$$

such that

$$(322) \quad d_{\mathcal{X}}(F_{\hat{\rho}}, B_1)^2 \leq C_* \delta_T(\Omega).$$

Here $\kappa = \kappa(n, R, \rho_*)$ is the canonical dimensional choice fixed in §4.4 (notation register §12). The constants C_*, c_0, δ_1 are polynomial in $1/\rho_*$, R , n and the constants of the Fusco–Julin and oriented radial-trace inputs (see §4.3–§4.4); explicitly, $c_0 = 4n/(\omega_n \rho_*^{n+1})$.

Proof. This is item E4.6 of the manuscript source map. The proof is the content of §4.4: it combines the integrated metric-derivative bound of §4.4 with the profile-gap measure of §4.1 (item B6 / E1.6) and the abstract weighted metric Sobolev trace lemma E4.5. $\square$

Lemma 4.66 (Distance–asymmetry conversion). *Let $\hat{\rho}$ be as in Theorem 4.65. There exists $z = z(\hat{\rho}) \in \mathbb{R}^n$ such that*

$$|E_{\hat{\rho}} \Delta(z + B_{\hat{\rho}})| \leq \hat{\rho}^n d_{\mathcal{X}}(F_{\hat{\rho}}, B_1).$$

Consequently, $\mathcal{A}(E_{\hat{\rho}})$ relative to the ball of volume $\omega_n \hat{\rho}^n$ (the volume of $E_{\hat{\rho}}$) satisfies

$$(323) \quad \mathcal{A}(E_{\hat{\rho}}) \leq \omega_n^{-1} d_{\mathcal{X}}(F_{\hat{\rho}}, B_1), \quad \mathcal{A}(E_{\hat{\rho}})^2 \leq \omega_n^{-2} C_* \delta_T(\Omega).$$

Proof. By the definition of $d_{\mathcal{X}}$ in the metric quotient $\mathcal{X}$, for every $\varepsilon > 0$ there exists $a \in \mathbb{R}^n$ such that $|F_{\hat{\rho}} \Delta(a + B_1)| \leq d_{\mathcal{X}}(F_{\hat{\rho}}, B_1) + \varepsilon$. Pulling back by the homothety $x \mapsto \hat{\rho}x$ gives, with $z := \hat{\rho}a$,

$$|E_{\hat{\rho}} \Delta(z + B_{\hat{\rho}})| = \hat{\rho}^n |F_{\hat{\rho}} \Delta(a + B_1)| \leq \hat{\rho}^n (d_{\mathcal{X}}(F_{\hat{\rho}}, B_1) + \varepsilon).$$

Sending $\varepsilon \downarrow 0$ proves the first claim. Dividing by $|E_{\hat{\rho}}| = \omega_n \hat{\rho}^n$ gives the first estimate in (323); the second follows from (322). $\square$

4.5.2. *Volume of the removed boundary layer.*

Lemma 4.67 (Boundary-layer volume; $O(\sqrt{\delta_T})$ bound). *Let $\hat{\rho}$ be as in Theorem 4.65. Set*

$$(324) \quad C'_n := n(\kappa + c_0), \quad C_n := \omega_n C'_n = \omega_n n(\kappa + c_0),$$

where $\kappa = \kappa(n, R, \rho_*)$ is the boundary-layer constant of §4.4 (notation register §12) and $c_0 = c_0(n, R, \rho_*)$ is the metric-trace window-scale constant of Theorem 4.65. Then, for every Ω with $\delta_T(\Omega) \leq \delta_2 := \min\{1, \delta_1\}$,

$$(325) \quad |\Omega \setminus E_{\hat{\rho}}| = \omega_n (1 - \hat{\rho}^n) \leq C_n \sqrt{\delta_T(\Omega)}.$$

Proof. By the volume-radius parametrisation, $|\Omega| = \omega_n$ and $|E_{\widehat{\rho}}| = \omega_n \widehat{\rho}^n$, whence $|\Omega \setminus E_{\widehat{\rho}}| = \omega_n(1 - \widehat{\rho}^n)$. By (321) and the definition $\rho_\delta = 1 - \kappa\sqrt{\delta_T(\Omega)}$,

$$(326) \quad \widehat{\rho} \geq \rho_\delta - c_0\sqrt{\delta_T(\Omega)} = 1 - (\kappa + c_0)\sqrt{\delta_T(\Omega)}.$$

On $[\widehat{\rho}, 1] \subset [\rho_*, 1]$ the function $\rho \mapsto 1 - \rho^n$ is C^1 with $\frac{d}{d\rho}(1 - \rho^n) = -n\rho^{n-1} \in [-n, 0]$, so the mean-value theorem and (326) give

$$1 - \widehat{\rho}^n \leq n(1 - \widehat{\rho}) \leq n(\kappa + c_0)\sqrt{\delta_T(\Omega)} = C'_n\sqrt{\delta_T(\Omega)}.$$

Multiplying by ω_n gives (325) with $C_n = \omega_n C'_n$. (No appeal to $\delta_T(\Omega) \leq 1$ to absorb a δ_T -term into $\sqrt{\delta_T}$ is needed any more: the window (321) is already of scale $\sqrt{\delta_T}$, by the B1 repair of Agent 12 dated 2026-05-13.) $\square$

Remark 4.68 (Why only $\sqrt{\delta_T}$, not δ_T). The square root is sharp at the level of the layer volume itself: $\rho_\delta = 1 - \kappa\sqrt{\delta_T}$ is the largest radius on which the profile-gap lower bound for the measure $\mu(d\rho) = (-t'(\rho))d\rho$ holds, and this profile gap is what produces the $\sqrt{\delta_T}$ scale of the boundary layer in the first place. Squaring this $\sqrt{\delta_T}$ *too early*, e.g. before the asymmetry transfer, would lose the sharp exponent in the final Saint-Venant inequality. The squaring must be done *after* transfer; see §4.5.4.

4.5.3. Superlevel asymmetry transfer. We now prove that the Fraenkel asymmetry of Ω at scale $|\Omega| = \omega_n$ is dominated by the Fraenkel asymmetry of any nested superlevel $E \subset \Omega$ at its own scale, plus a volume-defect correction of order $|\Omega \setminus E|/\omega_n$.

Lemma 4.69 (Superlevel transfer of asymmetry). *Let $\Omega \subset \mathbb{R}^n$ be open with $|\Omega| = L > 0$ and let $E \subset \Omega$ be Borel with $0 < |E| = s \leq L$. Write $\eta := L - s \in [0, L]$. Then*

$$(327) \quad \mathcal{A}(\Omega) \leq \mathcal{A}(E) + \frac{2\eta}{L} = \mathcal{A}(E) + \frac{2}{L} |\Omega \setminus E|.$$

Here both Fraenkel asymmetries are computed at the natural scale of the underlying set: $\mathcal{A}(\Omega)$ uses balls of volume L and $\mathcal{A}(E)$ uses balls of volume s .

Proof. Write $r_s := (s/\omega_n)^{1/n}$ and $r_L := (L/\omega_n)^{1/n}$, so the ball of volume s is B_{r_s} and the ball of volume L is B_{r_L} . We will compare Ω to a translate $z + B_{r_L}$ of the ball of volume L , where z is chosen nearly optimally for the asymmetry of E .

Fix $\varepsilon \in (0, 1)$ and choose $z \in \mathbb{R}^n$ such that

$$(328) \quad |E \Delta (z + B_{r_s})| \leq s(\mathcal{A}(E) + \varepsilon).$$

By the triangle inequality for symmetric differences,

$$(329) \quad |\Omega \Delta (z + B_{r_L})| \leq |\Omega \Delta E| + |E \Delta (z + B_{r_s})| + |(z + B_{r_s}) \Delta (z + B_{r_L})|.$$

We bound the three summands.

Term 1. Since $E \subset \Omega$, $\Omega \Delta E = \Omega \setminus E$, so $|\Omega \Delta E| = \eta$.

Term 2. By (328), $|E \Delta (z + B_{r_s})| \leq s(\mathcal{A}(E) + \varepsilon)$.

Term 3. Since $r_s \leq r_L$, $z + B_{r_s} \subset z + B_{r_L}$ and

$$|(z + B_{r_s}) \Delta (z + B_{r_L})| = \omega_n(r_L^n - r_s^n) = L - s = \eta.$$

Combining and dividing by L ,

$$(330) \quad \frac{|\Omega \Delta (z + B_{r_L})|}{L} \leq \frac{2\eta}{L} + \frac{s}{L}(\mathcal{A}(E) + \varepsilon) \leq \frac{2\eta}{L} + \mathcal{A}(E) + \varepsilon,$$

using $s \leq L$. By the definition of the Fraenkel asymmetry, $\mathcal{A}(\Omega) \leq L^{-1}|\Omega \Delta (z + B_{r_L})|$, and since $\varepsilon \in (0, 1)$ was arbitrary, (327) follows. $\square$

Remark 4.70 (Where the two boundary-layer summands come from). Identity (329) contains two distinct volume-defect terms of magnitude $\eta = |\Omega \setminus E|$: the layer of Ω outside E , and the spherical-shell correction $|B_{r_L} \setminus B_{r_s}|$ between the two balls. They add to give $2\eta/L$ in (327). No L/s rescaling enters, because $\mathcal{A}(E)$ is measured at the natural scale s . The same combinatorial identity appears in the L^1 -rearrangement bookkeeping of Maggi [46].

Corollary 4.71 (Transfer at the metric-trace level $\widehat{\rho}$). *For every $\Omega \subset B_R$ open with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \leq \delta_2$,*

$$(331) \quad \mathcal{A}(\Omega) \leq \mathcal{A}(E_{\widehat{\rho}}) + 2C'_n \sqrt{\delta_T(\Omega)},$$

where $C'_n = n(\kappa + c_0)$ is the constant from Lemma 4.67.

Proof. $E_{\widehat{\rho}} \subset \Omega$ holds by definition of the foliation. With $L = \omega_n$ and $\eta = |\Omega \setminus E_{\widehat{\rho}}|$, Lemma 4.69 gives

$$\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\widehat{\rho}}) + \frac{2}{\omega_n} |\Omega \setminus E_{\widehat{\rho}}|.$$

Inserting (325) and recalling $C_n/\omega_n = C'_n$ yields (331). $\square$

4.5.4. Squaring the transfer error: the sharp-exponent step. We now perform the critical squaring step. The transfer inequality (331) contains a transfer error of magnitude $\sqrt{\delta_T}$. Squaring *after* transfer turns each $\sqrt{\delta_T}$ into a δ_T linearly, while $\mathcal{A}(E_{\widehat{\rho}})^2$ inherits the δ_T -rate from (323). The elementary inequality $(a + b)^2 \leq 2a^2 + 2b^2$ is what makes this lossless on the exponent; it only doubles the constant.

Theorem 4.72 (Bounded sharp Saint-Venant, Plan 2 form). *There exist $C(n, R, \rho_*) > 0$ and $\delta_0(n, R, \rho_*) \in (0, \delta_2]$ such that for every open $\Omega \subset B_R$ with $|\Omega| = \omega_n$,*

$$(332) \quad \mathcal{A}(\Omega)^2 \leq C(n, R, \rho_*) \delta_T(\Omega).$$

The exponent 2 is sharp. The constant unwinds as

$$(333) \quad C(n, R, \rho_*) = \max \left\{ \frac{2C_*}{\omega_n^2} + 8(C'_n)^2, \frac{4}{\delta_0(n, R, \rho_*)} \right\},$$

where $C'_n = n(\kappa + c_0)$ and $C_*, c_0, \kappa, \delta_0$ are the upstream constants of Theorem 4.65 (Agent 12) and §4.4; $\kappa = \kappa(n, R, \rho_*)$ is canonical.

Proof. Step 1: small-deficit regime $\delta_T(\Omega) \leq \delta_2$. Set

$$a := \mathcal{A}(E_{\widehat{\rho}}), \quad b := 2C'_n \sqrt{\delta_T(\Omega)}.$$

By Corollary 4.71, $\mathcal{A}(\Omega) \leq a + b$. The squaring identity gives

$$(334) \quad (a + b)^2 = a^2 + 2ab + b^2 \leq 2a^2 + 2b^2,$$

the inequality being the AM–GM bound $2ab \leq a^2 + b^2$. Hence

$$(335) \quad \mathcal{A}(\Omega)^2 \leq 2\mathcal{A}(E_{\hat{\rho}})^2 + 2(2C'_n)^2 \delta_T(\Omega) = 2\mathcal{A}(E_{\hat{\rho}})^2 + 8(C'_n)^2 \delta_T(\Omega).$$

By Lemma 4.66, $\mathcal{A}(E_{\hat{\rho}})^2 \leq \omega_n^{-2} C_* \delta_T(\Omega)$. Substituting,

$$(336) \quad \mathcal{A}(\Omega)^2 \leq \left(\frac{2C_*}{\omega_n^2} + 8(C'_n)^2 \right) \delta_T(\Omega) =: C_3(n, R, \rho_*) \delta_T(\Omega).$$

This proves (332) for $\delta_T(\Omega) \leq \delta_2$ with constant C_3 . The sharp exponent 2 is preserved because (334) doubles the prefactor but does not raise a or b to a higher power.

Step 2: large-deficit regime $\delta_T(\Omega) \geq \delta_0$. Set $\delta_0 := \delta_2(n, R, \rho_*)$. Since $\mathcal{A}(\Omega) < 2$ for every Ω of volume ω_n (an immediate consequence of $|\Omega \Delta (z + B_1)| \leq |\Omega| + |B_1| = 2\omega_n$ and the strict inequality coming from $|\Omega \cap (z + B_1)| > 0$ for some z when Ω has finite positive volume), $\mathcal{A}(\Omega)^2 \leq 4$. On $\delta_T(\Omega) \geq \delta_0$ this gives $\mathcal{A}(\Omega)^2 \leq 4 \leq (4/\delta_0) \delta_T(\Omega)$.

Step 3: combine. Taking $C := \max\{C_3, 4/\delta_0\}$ yields (332) on the full admissible class with constant C as in (333).

Sharpness. For the one-parameter family of volume-preserving ellipsoidal perturbations Ω_ε of B_1 , one has $\mathcal{A}(\Omega_\varepsilon) \sim \varepsilon$ and $\delta_T(\Omega_\varepsilon) \sim \varepsilon^2$ as $\varepsilon \rightarrow 0$ (see e.g. [14, Introduction]); hence any inequality $\mathcal{A}(\Omega)^p \leq C \delta_T(\Omega)$ forces $p \geq 2$. This shows that exponent 2 is sharp in (332). $\square$

Remark 4.73 (The squaring step, written out explicitly). The complete inequality chain at the heart of this subsection is

$$(337) \quad \begin{aligned} \mathcal{A}(\Omega) &\stackrel{\text{Cor. 4.71}}{\leq} \mathcal{A}(E_{\hat{\rho}}) + 2C'_n \sqrt{\delta_T(\Omega)} \\ \mathcal{A}(\Omega)^2 &\stackrel{(a+b)^2 \leq 2a^2 + 2b^2}{\leq} 2\mathcal{A}(E_{\hat{\rho}})^2 + 8(C'_n)^2 \delta_T(\Omega) \\ &\stackrel{(323)}{\leq} \frac{2C_*}{\omega_n^2} \delta_T(\Omega) + 8(C'_n)^2 \delta_T(\Omega) = C_3(n, R, \rho_*) \delta_T(\Omega). \end{aligned}$$

The transfer error $b = 2C'_n \sqrt{\delta_T}$ is squared to $b^2 = 4(C'_n)^2 \delta_T$, and the factor of 2 from $(a + b)^2 \leq 2a^2 + 2b^2$ produces the final $8(C'_n)^2$. The $\mathcal{A}(E_{\hat{\rho}})^2$ term inherits a δ_T -rate directly from (322). *At no point is the $\sqrt{\delta_T}$ from the boundary layer squared in isolation:* it is always squared as part of $(\mathcal{A}(E_{\hat{\rho}}) + \sqrt{\delta_T})^2$ via the elementary squaring identity. This is what preserves the sharp exponent 2.

Remark 4.74 (Why squaring *before* transfer would fail). If, instead of squaring $\mathcal{A}(\Omega) \leq \mathcal{A}(E_{\hat{\rho}}) + C\sqrt{\delta_T}$ *after* transfer, one squared the boundary-layer estimate $|\Omega \setminus E_{\hat{\rho}}| \leq C_n \sqrt{\delta_T}$ first (yielding $|\Omega \setminus E_{\hat{\rho}}|^2 \leq C_n^2 \delta_T$), no rate would be lost on the layer term alone, but one would still have to combine the inequalities for $\mathcal{A}(E_{\hat{\rho}})$ and for the layer at the $\mathcal{A}(\Omega)$ -level. The minimal way to combine the two contributions into a bound for $\mathcal{A}(\Omega)^2$ is the squaring identity (334). The point is that (334) is an *additive* bound on the square: it doubles the constant but never raises an exponent. This preservation, not the order of operations, is the load-bearing mechanism.

4.5.5. *Bounded reduction and Kohler–Jobin transfer.* Theorem 4.72 provides the bounded-class sharp Saint–Venant inequality at any confining radius $R \geq 2$. To pass to the unconditional sharp Faber–Krahn inequality we apply the two shared Plan 1 Final/ blocks BOUNDEDREDUCTION and KOHLER-JOBINTTRANSFER in succession, exactly as in §4.5.6 (Agent 8).

Bounded reduction. Fix $\rho_* = 1/2$. Theorem 4.72 specialised at $R = R_*(n) := \max\{d(n), 2\}$, with $d(n)$ the diameter constant of [15, Lemma 5.3], becomes (338)

$$\mathcal{A}(\Omega)^2 \leq \frac{1}{c_{\text{SV}}(n, R_*(n))} \delta_T(\Omega) = \frac{1}{c_{\text{SV}}(n)} \delta_T(\Omega), \quad \Omega \subset B_{R_*(n)}, \quad |\Omega| = \omega_n,$$

where $c_{\text{SV}}(n, R_*(n)) =: c_{\text{SV}}(n)$ depends only on n by Remark 4.79 below.

Theorem 4.75 (Global sharp Saint–Venant; cf. . BDV] *Suppose (338) holds. Then there exists $c_{\text{SV}}^{\text{glob}}(n) > 0$ such that for every open $\Omega \subset \mathbb{R}^n$ with $|\Omega| = \omega_n$,*

$$(339) \quad E(\Omega) - E(B_1) \geq c_{\text{SV}}^{\text{glob}}(n) \mathcal{A}(\Omega)^2,$$

with $c_{\text{SV}}^{\text{glob}}(n)$ given by the BDV near/far dichotomy:

$$(340) \quad c_{\text{SV}}^{\text{glob}}(n) = \min \left\{ c_{\text{near}}^{\text{glob}}(n), \frac{\omega_n^{(n+2)/n} \delta(n)}{16} \right\},$$

$$c_{\text{near}}^{\text{glob}}(n) = \frac{\omega_n^{(n+2)/n}}{4C_9(n)(1/\tilde{c}_{\text{SV}}(n) + C_9(n))}, \quad \tilde{c}_{\text{SV}}(n) = \omega_n^{-(n+2)/n} c_{\text{SV}}(n, R_*(n)),$$

where $\delta(n), C_9(n)$ are the BDV [15, Lemmas 5.1–5.3] constants.

Proof. This is precisely the BDV bounded reduction ([15, Lemmas 5.1–5.3 and the proof of Theorem 1.5]), applied with the bounded input (338). The proof, which is independent of how the bounded input was obtained, uses the BDV constructive truncation [15, Lemma 5.3] and the rearrangement-based suboptimal $\mathcal{A}^4$ bound [15, Lemma 5.2]. No compactness is introduced. The constant is the same as the one used by the Plan 1 final-assembly subsection (Agent 8). $\square$

By scale invariance ($\mathcal{A}$ is dimensionless, E has scaling weight $(n+2)/n$, and λ_1 has weight -2), (339) extends to arbitrary $|\Omega| < \infty$ in the scale-invariant form

$$(341) \quad E(\Omega) - E(B^*) \geq c_{\text{SV}}^{\text{glob}}(n) \left(\frac{|B^*|}{|B_1|} \right)^{(n+2)/n} \mathcal{A}(\Omega)^2.$$

Kohler–Jobin transfer.

Theorem 4.76 (Sharp Faber–Krahn from sharp Saint–Venant; cf. . Brasco2014] *Suppose (341) holds with constant $c_{\text{SV}}^{\text{glob}}(n) > 0$. Then for every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,*

$$(342) \quad \left(\frac{|\Omega|}{|B_1|} \right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2,$$

with the explicit dimensional constant

$$(343) \quad c_{\text{FK}}(n) := \min \left\{ \frac{4 L_n c_{\text{SV}}^{\text{glob}}(n)}{(n+2) T_n}, \frac{L_n (2^{2/(n+2)} - 1)}{4} \right\},$$

where $L_n = \lambda_1(B_1)$, $T_n = T(B_1) = \omega_n/(n(n+2))$.

Proof. This is the Kohler–Jobin transfer of [13, Theorem 3.3]. The proof uses the Kohler–Jobin inequality $T(\Omega)\lambda_1(\Omega)^{(n+2)/2} \geq T(B^*)\lambda_1(B^*)^{(n+2)/2}$ (cited from [13, Theorem 3.3]), the pointwise transfer $\lambda_1(\Omega) \geq \lambda_1(B^*)(T(B^*)/T(\Omega))^{2/(n+2)}$, the elementary inequality $(1-s)^{-p} - 1 \geq ps$ for $p \in (0, 1]$, $s \in [0, 1]$, and a two-case split on $T_n - T(\Omega)$ vs. $T_n/2$. The factor 4 in the first branch absorbs the factor 2 between $\delta_{\text{SV}} = T(B^*) - T(\Omega)$ and $\delta_T = E(\Omega) - E(B^*)$ (note $\delta_{\text{SV}} = 2\delta_T$). The proof is independent of how (341) was obtained; in particular it is identical to the one used by the Plan 1 final-assembly subsection (Agent 8). $\square$

4.5.6. The final Plan 2 theorem.

Theorem 4.77 (Sharp quantitative Faber–Krahn, Plan 2 final form). *Let $n \geq 2$. There exists a constant $c_{\text{FK}}(n) > 0$, depending only on n , given by the explicit formula (343) above, such that for every open $\Omega \subset \mathbb{R}^n$ with $0 < |\Omega| < \infty$,*

$$(344) \quad \delta_{\text{FK}}(\Omega) = \left(\frac{|\Omega|}{\omega_n}\right)^{2/n} (\lambda_1(\Omega) - \lambda_1(B^*)) \geq c_{\text{FK}}(n) \mathcal{A}(\Omega)^2.$$

The exponent 2 is sharp.

Proof. Fix $\rho_* = 1/2$ and apply Theorem 4.72 at $R = R_*(n)$. This produces the bounded sharp Saint–Venant inequality (338) with constant $c_{\text{SV}}(n, R_*(n)) =: c_{\text{SV}}(n)$. Theorem 4.75 removes the R -dependence and gives the global sharp Saint–Venant inequality (341) with constant $c_{\text{SV}}^{\text{glob}}(n)$ as in (340). Theorem 4.76 converts the latter into the sharp Faber–Krahn inequality (344) with constant $c_{\text{FK}}(n)$ as in (343).

Sharpness of the exponent 2 follows from the volume-preserving ellipsoidal one-parameter family Ω_ε of perturbations of B^* , for which $\mathcal{A}(\Omega_\varepsilon) \sim \varepsilon$ and $\lambda_1(\Omega_\varepsilon) - \lambda_1(B^*) \sim \varepsilon^2$ as $\varepsilon \rightarrow 0$ (cf. [14, Introduction], [2]). $\square$

Remark 4.78 (Identical normalisation to Plan 1). The statement and constant of Theorem 4.77 agree *verbatim* with those of the Plan 1 final theorem (Theorem 1.3 of the Introduction). The shared expression for $c_{\text{FK}}(n)$ is (343); the shared constants $c_{\text{SV}}^{\text{glob}}(n)$, L_n , T_n , ω_n are dimensional. The two proofs differ *only* in how the bounded sharp Saint–Venant input is obtained: Plan 1 uses selection principle + graph entry + Schauder interpolation + nearly-spherical closure; Plan 2 uses the level-set / metric route assembled in §4.1.11–§4.4.7 and the present subsection.

Remark 4.79 (Where the constants enter). The constant $C_3(n, R, \rho_*)$ from (336) unwinds as

$$C_3(n, R, \rho_*) = \frac{2C_*(n, R, \rho_*)}{\omega_n^2} + 8n^2(\kappa(n, R, \rho_*) + c_0(n, R, \rho_*))^2,$$

polynomial in $1/\rho_*$, R , n and the upstream constants C_* , c_0 , κ of Theorem 4.65 and §4.4. At $\rho_* = 1/2$ and $R = R_*(n)$, C_3 becomes a dimensional constant $C_3(n)$, and Theorem 4.72 reads $\mathcal{A}(\Omega)^2 \leq C_3(n)\delta_T(\Omega)$ on $\Omega \subset B_{R_*(n)}$, $|\Omega| = \omega_n$. Equivalently $c_{\text{SV}}(n) = 1/C_3(n)$, and this is the input fed to Theorem 4.75.

Remark 4.80 (No load-bearing use of conditional routes). The proof of Theorem 4.77 does *not* invoke the centroid Π -route or the space-time Π identity in any load-bearing way; those appear only as auxiliary remarks (Agent 14). It also does *not* use any Bernoulli spectral input or any $C^{1,\gamma}$ regularity of $\partial\Omega$. The proof is purely finite-perimeter from §4.1.11 through the present subsection. This is the structural difference between Plan 2 and Plan 1, recorded in Remark 4.78.

4.6. Auxiliary route: centroid kinematics and the space-time Π identity.

Status. The contents of this subsection are auxiliary. They are recorded here because they isolate a structurally appealing alternative route to the sharp bounded Saint-Venant estimate $\mathcal{A}(\Omega)^2 \leq C(n, R, \rho_*) \delta_T(\Omega)$, based on the centroid first moment of the level family. The route closes *conditionally* on the integrated radial-moment estimate (Π -int) below, which is currently open. The load-bearing proof of Plan 2 in Sections 4.1.11–4.5.6 does *not* use any statement from this subsection.

Standing hypotheses. Throughout this subsection we adopt the standing setup of Section 4.5.6: $n \geq 2$, $R \geq 2$, $\rho_* \in (0, 1)$, $\kappa > 0$, $\Omega \subset B_R \subset \mathbb{R}^n$ open with $|\Omega| = \omega_n$, $u = u_\Omega$ the variational torsion function, $t : [\rho_*, 1] \rightarrow (0, \|u\|_\infty)$ the volume-radius parametrisation ($|\{u > t(\rho)\}| = \omega_n \rho^n$), $E_\rho := \{u > t(\rho)\}$, $\rho_\delta := 1 - \kappa \sqrt{\delta_T(\Omega)}$, $\mu(d\rho) := (-t'(\rho)) d\rho$, and $V_\rho(x) := -t'(\rho)/|\nabla u(x)|$ ($\mathcal{H}^{n-1}$ -a.e. on $\partial^* E_\rho$, for a.e. ρ , by Sobolev–Sard, cf. §A3). The constant $\delta_0 = \delta_0(n, R, \rho_*) > 0$ is fixed once and for all small enough that all qualitative consequences quoted below (FMP smallness, Cicalese–Leonardi containment, centroid–Fraenkel-centre proximity) hold whenever $\delta_T(\Omega) \leq \delta_0$.

The bulk centroid of E_ρ is

$$(345) \quad \bar{z}_\rho := \frac{1}{|E_\rho|} \int_{E_\rho} x \, dx \in \mathbb{R}^n,$$

in line with the canonical centre field of the notation register (§10). We work with $\bar{z}_\rho$ rather than with the Fraenkel-optimal centre z_ρ^{Fr} or with the Fusco–Julin oscillation centre.

Centroid kinematic identity.

Lemma 4.81 (Centroid kinematic identity). *The map $\rho \mapsto \bar{z}_\rho$ is Lipschitz on $[\rho_*, 1]$ with*

$$|\bar{z}_\rho - \bar{z}_{\rho'}| \leq \frac{2nR}{\rho_*^n} |\rho - \rho'|, \quad \rho_* \leq \rho, \rho' \leq 1,$$

hence absolutely continuous on $[\rho_, 1]$, and for a.e. $\rho \in [\rho_*, 1]$ one has the first-moment identity*

$$(346) \quad \bar{z}'_\rho = \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} (x - \bar{z}_\rho) V_\rho(x) \, d\mathcal{H}^{n-1}(x).$$

Proof. Lipschitz regularity. The volume-radius parametrisation gives the nested inclusion $E_\rho \subset E_{\rho'}$ whenever $\rho \leq \rho'$, hence

$$|E_\rho \Delta E_{\rho'}| = |E_{\rho'}| - |E_\rho| = \omega_n((\rho')^n - \rho^n) \leq n\omega_n |\rho - \rho'|.$$

Since $E_\rho \subset B_R$, $|\int_{E_\rho} x_i - \int_{E_{\rho'}} x_i| \leq R |E_\rho \Delta E_{\rho'}| \leq n\omega_n R |\rho - \rho'|$. Dividing by $|E_\rho| \geq \omega_n \rho_*^n$ yields the displayed Lipschitz constant $L_{\text{cent}} = 2nR/\rho_*^n$.

Kinematic identity. By the coarea identity applied to the Sobolev function $u \in W_0^{1,2}(\Omega) \cap L^\infty(\Omega)$ (see §A2–A3 of the GMT preliminaries; the Sobolev–Sard step [25] ensures $|\nabla u| > 0$ $\mathcal{H}^{n-1}$ -a.e. on $\partial^* E_\rho$ for a.e. ρ), for every Lipschitz $f : \overline{B_R} \rightarrow \mathbb{R}$ and a.e. $\rho \in [\rho_*, 1]$,

$$(347) \quad \frac{d}{d\rho} \int_{E_\rho} f(x) dx = \int_{\partial^* E_\rho} f(x) V_\rho(x) d\mathcal{H}^{n-1}(x).$$

Taking $f \equiv 1$ and using $|E_\rho| = \omega_n \rho^n$ gives $\int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} = n\omega_n \rho^{n-1} = P(B_\rho)$ for a.e. ρ . Taking $f(x) = x_i$ in (347) and applying the quotient rule to $\bar{z}_{\rho,i} = |E_\rho|^{-1} \int_{E_\rho} x_i dx$:

$$\begin{aligned} \bar{z}'_{\rho,i} &= \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} x_i V_\rho d\mathcal{H}^{n-1} - \frac{1}{|E_\rho|^2} \left(\frac{d}{d\rho} |E_\rho| \right) \int_{E_\rho} x_i dx \\ &= \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} x_i V_\rho d\mathcal{H}^{n-1} - \frac{\bar{z}_{\rho,i}}{|E_\rho|} \int_{\partial^* E_\rho} V_\rho d\mathcal{H}^{n-1} \\ &= \frac{1}{|E_\rho|} \int_{\partial^* E_\rho} (x_i - \bar{z}_{\rho,i}) V_\rho d\mathcal{H}^{n-1}, \end{aligned}$$

which is the i -th component of (346). $\square$

Remark 4.82 (No homothety term). The right-hand side of (346) is the first vector moment of V_ρ relative to $\bar{z}_\rho$, with no comparison to a homothetic velocity. This is what makes the centroid kinematic identity unconditional: it does *not* require a per- ρ control of $V_\rho - H_{\bar{z}_\rho, \rho}$, where $H_{\bar{z}_\rho, \rho}(x) := (x - \bar{z}_\rho) \cdot \nu_\rho(x)/\rho$, and therefore avoids the pointwise radial-trace inputs that the conditional Π -route below relies on. A centre defined by a Fraenkel first-order condition would instead produce a derivative formula involving $V_\rho - H_{\bar{z}_\rho, \rho}$ on the reduced boundary.

The integrated centroid bound (unconditional within the small-deficit regime).

Theorem 4.83 (Integrated centroid H^1 bound). *There exist $\delta_0 = \delta_0(n, R, \rho_*) > 0$ and $C_{\text{cent}} = C_{\text{cent}}(n, R, \rho_*) > 0$ such that whenever $\Omega \subset B_R$ is open with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \leq \delta_0$,*

$$(F) \quad \int_{\rho_*}^{\rho_\delta} |\bar{z}'_\rho|^2 (-t'(\rho)) d\rho \leq C_{\text{cent}} \delta_T(\Omega).$$

A sketch of the proof structure, conditional on Hypothesis 4.85, is the following:

- (i) decompose $V_\rho = \bar{V}_\rho + (V_\rho - \bar{V}_\rho)$ with $\bar{V}_\rho = P(B_\rho)/P(E_\rho)$ in (346);
- (ii) bound the perimeter mean piece by FMP plus the Fusco–Julin L^1 oscillation, using the divergence theorem for finite-perimeter sets applied to the Lipschitz field $V^{(i)}(x) := |x - \bar{z}_\rho| e_i$;
- (iii) bound the variance piece by Cauchy–Schwarz, controlled by the algebraic deficit $D_H(t(\rho))$ of Subsection 4.1.11;
- (iv) integrate against $\mu(d\rho) = (-t') d\rho$ and use the integrated identity $\int_{\rho_*}^1 (D_H + D_I) d\mu \leq C(n, \rho_*) \delta_T(\Omega)$.

The constant tracks as $C_{\text{cent}} \leq C(n, \rho_*) R \rho_*^{-(3n-1)}$ and is polynomial in $1/\rho_*$, R , n . Outside the smallness regime FMP gives $\mathcal{A}(\Omega) \geq c(n) > 0$ and (F) is trivial after enlarging C_{cent} .

Space-time Π identity. For each $\rho \in [\rho_*, \rho_\delta]$, define, in agreement with the notation register (§13),

$$(348) \quad \Pi(E_\rho, \bar{z}_\rho) := (n-1) \int_{\mathbb{R}^n} (\mathbf{1}_{B_\rho(\bar{z}_\rho)}(x) - \mathbf{1}_{E_\rho}(x)) |x - \bar{z}_\rho|^{-1} dx,$$

the signed radial moment of the symmetric difference $E_\rho \Delta B_\rho(\bar{z}_\rho)$ at the centroid. By the L^2 divergence identity of Subsection 4.1.11,

$$(349) \quad \int_{\partial^* E_\rho} |\nu_\rho - e_r|^2 d\mathcal{H}^{n-1} = 2(P(E_\rho) - P(B_\rho)) + 2\Pi(E_\rho, \bar{z}_\rho), \quad e_r(x) = \frac{x - \bar{z}_\rho}{|x - \bar{z}_\rho|},$$

so $\Pi(E_\rho, \bar{z}_\rho) \geq 0$ a.e. in ρ .

The natural cylinder for the space-time computation is $\Omega_\Pi := B_{R+1} \subset \mathbb{R}^n$, which contains both Ω and $\bigcup_{\rho \in [\rho_*, \rho_\delta]} B_\rho(\bar{z}_\rho)$ (recall $\bar{z}_\rho \in B_R$ by $E_\rho \subset B_R$, and $\rho \leq 1$). We use Ω_Π rather than $\tilde{\Omega}$ to respect the manuscript-wide notation agreement (§13 of the notation register), where $\tilde{\Omega}$ is reserved for Plan 1's volume-normalised companion.

Theorem 4.84 (Space-time Π identity). *The signed Fubini density*

$$(350) \quad \sigma(x) := \int_{\rho_*}^{\rho_\delta} (\mathbf{1}_{B_\rho(\bar{z}_\rho)}(x) - \mathbf{1}_{E_\rho}(x)) |x - \bar{z}_\rho|^{-1} d\mu(\rho)$$

is Borel-measurable, locally integrable on Ω_Π , and satisfies

$$(351) \quad \int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) = (n-1) \int_{\Omega_\Pi} \sigma(x) dx = (n-1) \int_{\Omega_\Pi} \int_{J_x} \text{sgn}(\mathbf{1}_{B_\rho(\bar{z}_\rho)}(x) - \mathbf{1}_{E_\rho}(x)) |x - \bar{z}_\rho|^{-1} d\mu(\rho)$$

where $J_x := \{\rho \in [\rho_*, \rho_\delta] : x \in B_\rho(\bar{z}_\rho) \Delta E_\rho\}$. In particular,

$$(352) \quad |\sigma(x)| \leq \int_{J_x} |x - \bar{z}_\rho|^{-1} d\mu(\rho) \quad \text{for a.e. } x \in \Omega_\Pi.$$

Proof. By (348), $\rho \mapsto \Pi(E_\rho, \bar{z}_\rho)$ is the integral over $\mathbb{R}^n$ of the joint Borel-measurable function $(\rho, x) \mapsto (\mathbf{1}_{B_\rho(\bar{z}_\rho)}(x) - \mathbf{1}_{E_\rho}(x))/|x - \bar{z}_\rho|$. The kernel $|x - \bar{z}_\rho|^{-1}$ is locally integrable on Ω_Π since $n \geq 2$. Truncating the kernel at level M , applying Tonelli to the positive and negative parts of the bounded integrand, and letting $M \rightarrow \infty$ by monotone convergence on each sign separately produces the first equality in (351). The second equality is the change of order on the symmetric difference. The bound (352) follows by taking absolute values inside the ρ -integral. $\square$

Where the centroid route would close: the conditional Π -control hypothesis. The integrated identity (351) rewrites $\int \Pi(E_\rho, \bar{z}_\rho) d\mu$ as an integral, over the cylinder Ω_Π , of the weighted space-time crossing length $\mu(J_x)$ of the moving sphere $|x - \bar{z}_\rho| = \rho$ with the boundary of E_ρ . Closing the centroid route at the sharp rate δ_T requires the following hypothesis.

Hypothesis 4.85 (Integrated Π -control; *open*). There exists $C_\Pi = C_\Pi(n, R, \rho_*) > 0$ such that for every bounded open $\Omega \subset B_R$ with $|\Omega| = \omega_n$ and $\delta_T(\Omega) \leq \delta_0(n, R, \rho_*)$,

$$(\Pi\text{-int}) \quad \int_{\rho_*}^{\rho_\delta} \Pi(E_\rho, \bar{z}_\rho) d\mu(\rho) \leq C_\Pi(n, R, \rho_*) \delta_T(\Omega).$$

Status of Hypothesis 4.85. At the time of writing, $(\Pi\text{-int})$ is *not* proved. The H^1 -in- ρ centroid bound (F) controls the moving centre $\bar{z}_\rho$ but does not, by itself, convert the rearranged Talenti/Nicola-Tilli profile gap into a control on the physical crossing set J_x , because J_x also depends on the spatial radius $|x - \bar{z}_\rho|$. A viable proof of $(\Pi\text{-int})$ would combine (351) with a genuine space-time coarea or transport estimate for the level crossings of $|x - \bar{z}_\rho| = \rho$. We do *not* have such an estimate.

In particular, none of the consequences below (Theorems 4.86 and 4.87) are used in the load-bearing Plan 2 chain.

Conditional consequences. For completeness we record the two conditional consequences. Both assume Hypothesis 4.85.

Theorem 4.86 (Integrated L^2 normal oscillation; conditional on $(\Pi\text{-int})$). *Assume Hypothesis 4.85. Then there exists $C_{B^2} = C_{B^2}(n, R, \rho_*) > 0$ such that*

$$(B^2\text{-int}) \quad \int_{\rho_*}^{\rho_\delta} \int_{\partial^* E_\rho} |\nu_\rho - e_r|^2 d\mathcal{H}^{n-1} d\mu(\rho) \leq C_{B^2} \delta_T(\Omega).$$

Proof. Integrate (349) against $d\mu$. The first piece is controlled by the deficit identity $\int (P(E_\rho) - P(B_\rho)) d\mu(\rho) \leq C(n, \rho_*) \delta_T(\Omega)$ of Subsection 4.1.11 (it bounds $\int D_I d\mu$, and $P - P_\rho$ is comparable to $D_I/P(B_\rho)$ in the small-deficit regime, with $P(B_\rho) \geq n\omega_n \rho_*^{n-1}$). The second piece is exactly $(\Pi\text{-int})$. $\square$

Theorem 4.87 (Integrated radial trace; conditional on $(\Pi\text{-int})$ plus an annular cap). *Assume Hypothesis 4.85 and the auxiliary annular/perimeter cap (353)*

$$\rho_*/2 \leq |x - \bar{z}_\rho| \leq 2R \quad \text{on the relevant boundary pieces,} \quad P(E_\rho) - P(B_\rho) + \Pi(E_\rho, \bar{z}_\rho) \leq M_A(n, R, \rho_*)$$

for a.e. $\rho \in [\rho_, \rho_\delta]$. Then there exists $C_{\text{rad}} = C_{\text{rad}}(n, R, \rho_*) > 0$ such that*

$$(T_1^2\text{-int}) \quad \int_{\rho_*}^{\rho_\delta} \left(\int_{\partial^* E_\rho} |H_{\bar{z}_\rho, \rho} - 1| d\mathcal{H}^{n-1} \right)^2 d\mu(\rho) \leq C_{\text{rad}} \delta_T(\Omega).$$

The proof of Theorem 4.87 uses a slicing-then-squaring decomposition $T_1 = T_1^{\text{rad, slic}} + T_1^{\text{tan}}$ of the radial trace $T_1 = \int_{\partial^* E_\rho} ||x - \bar{z}_\rho|/\rho - 1| d\mathcal{H}^{n-1}$, followed by a slice-rearrangement inequality $|E_\rho \Delta B_\rho(\bar{z}_\rho)|^2 \leq C(n, R, \rho_*) \Pi(E_\rho, \bar{z}_\rho)$; the role of squaring before integrating is to avoid the FMP rate-loss to $\sqrt{\delta_T}$. The full argument is conditional on Hypothesis 4.85 and is not used by Subsections 4.2.10–4.5.6.

Reading guide and warning.

What this subsection establishes unconditionally. The centroid kinematic identity (346), its Lipschitz regularity, the integrated centroid bound (F), and the space-time identity (351).

What this subsection does *not* establish. The hypothesis (II-int), and therefore neither (B^2 -int) nor (T_1^2 -int).

What this subsection is *not* used for. Nothing in this subsection is invoked in the proofs of Subsections 4.1.11, 4.2.10, 4.3.4, 4.4.7, or 4.5.6. The repaired Plan 2 chain runs through the weighted metric trace (Subsection 4.4.7) and uses the centroid $\bar{z}_\rho$ as the canonical centre field, but it does *not* rely on the radial-moment identities (349)–(II-int). The final Plan 2 theorem of Subsection 4.5.6 stands without Hypothesis 4.85.

Why the route is interesting nonetheless. The structural mechanism explained in Remark 4.82 — a centre field whose derivative is a bulk first moment with no homothety term — suggests that if a separate space–time crossing-length estimate for $|x - \bar{z}_\rho| = \rho$ can be proved at rate δ_T , then the present subsection would furnish an independent path to the sharp bounded Saint–Venant estimate, bypassing the weighted metric trace of Subsection 4.4.7. This is the only sense in which the centroid/II route remains an open alternative.

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